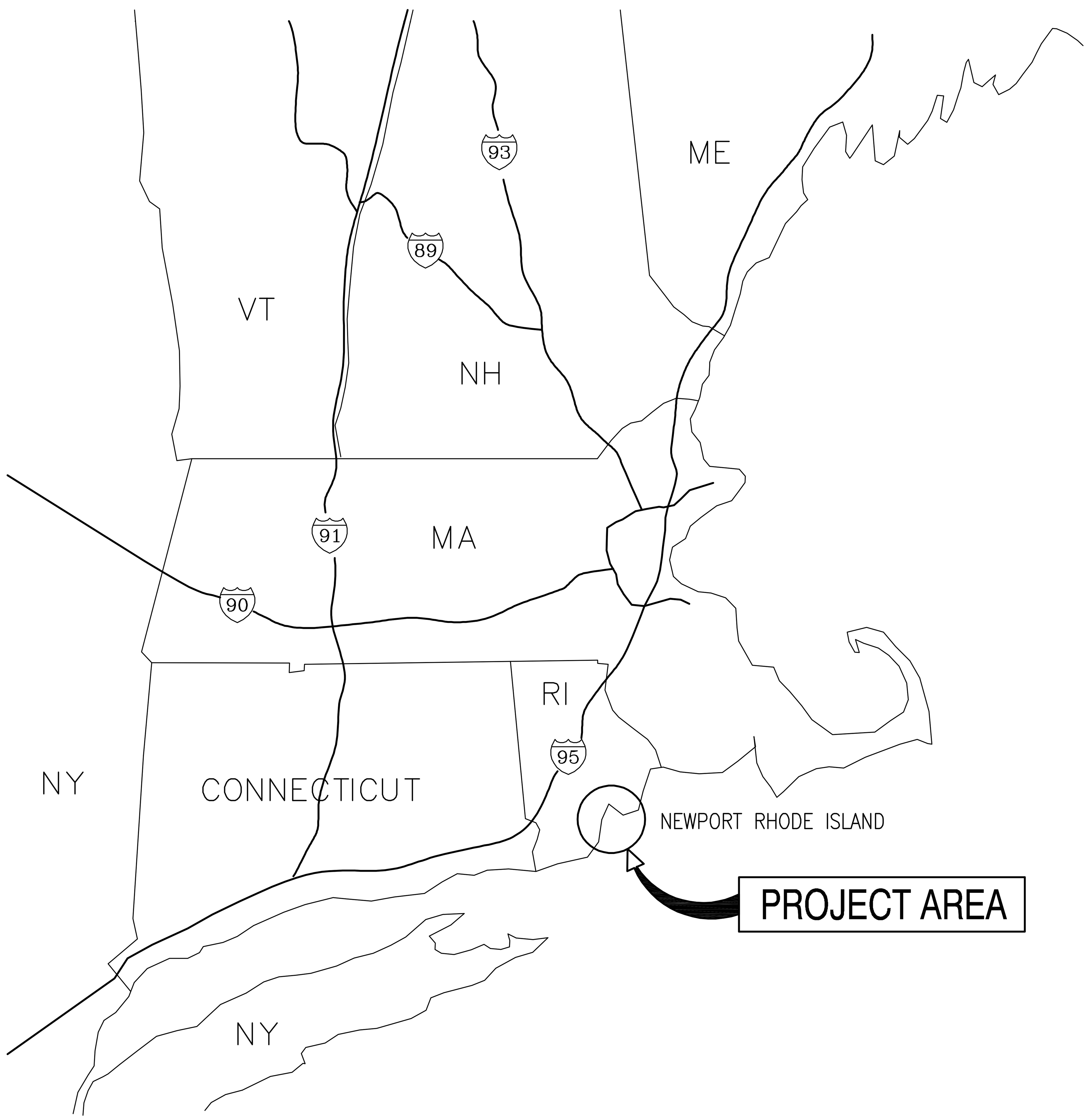
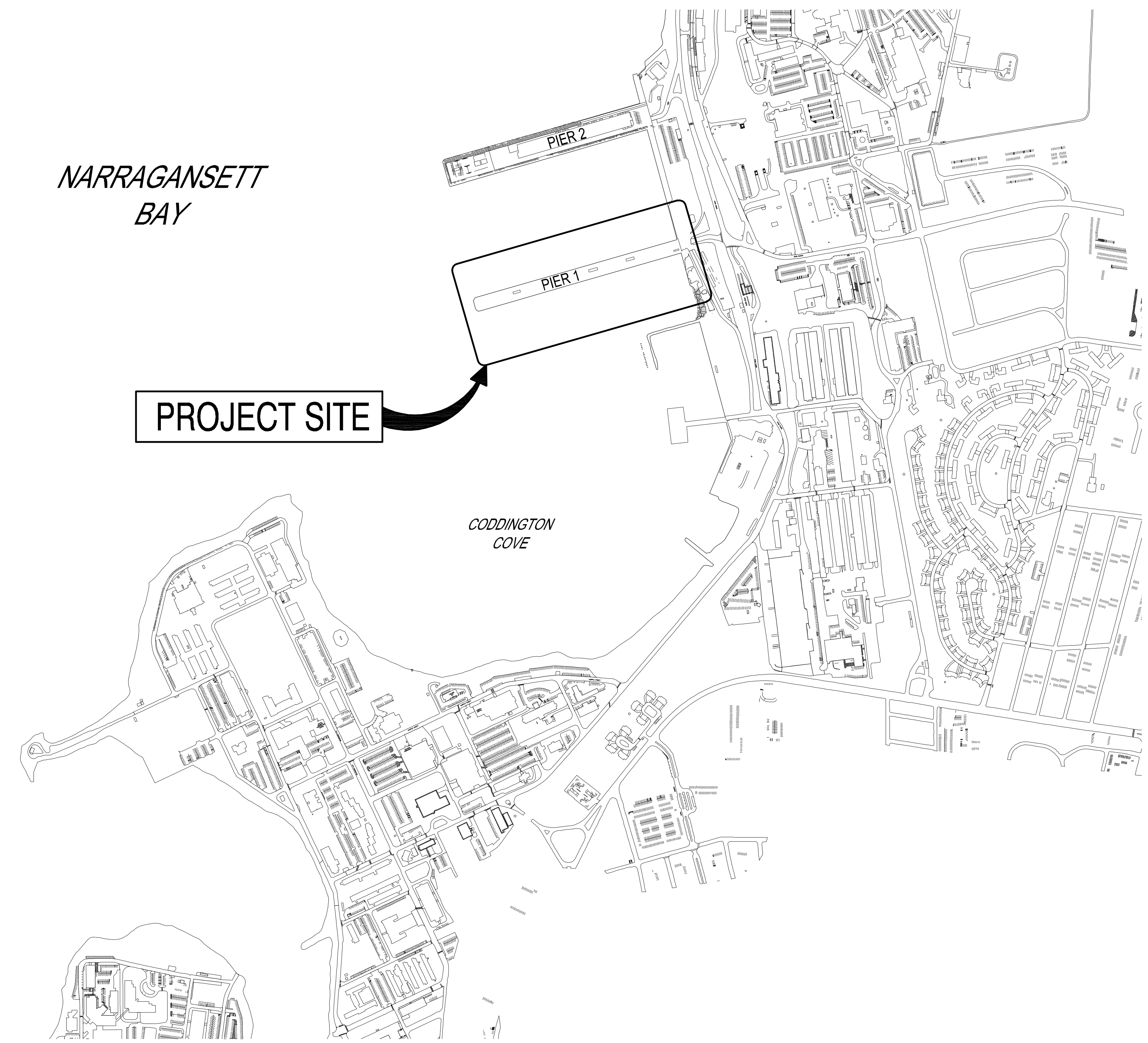
USCG OCR HOMEPORT INITIATIVE
OPC PIER
NAVAL STATION NEWPORT
NEWPORT, RHODE ISLAND



VICINITY MAP
SCALE: NTS

PLAN NORTH



LOCATION MAP
SCALE: NTS

PLAN NORTH

SYM	DESCRIPTION
DATE	APPR

NAVFAC

VICTOR M. GOVONI, CIVIL, No. 41734, REGISTERED PROFESSIONAL ENGINEER, COMMONWEALTH OF MASSACHUSETTS

SEAL

Stantec

A/E INFO

APPROVED James E. Donohue, R.A.
FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES CHG	DRW HBF	CHK MN
PM/DM		MRI/CWJ
BRANCH MANAGER		NEK
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
TITLE SHEET

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936570
SHEET 1 OF 247

G-001

DRAWFORM REVISION: 25 AUGUST 2020

LIST OF SHEETS

GENERAL SHEET INDEX

SHEET NO.		NAVFAC DWG. NO.	SHEET TITLE
NO.	SHEET		
1	G-001	12936570	TITLE SHEET
2	G-002	12936571	INDEX OF DRAWINGS - SHEET 1 OF 2
3	G-003	12936572	INDEX OF DRAWINGS - SHEET 2 OF 2
4	G-004	12936573	GENERAL ABBREVIATIONS
5	G-101	12936574	OVERVIEW EXISTING SITE CONTROLS
6	G-102	12936575	GENERAL SITE PLAN
7	G-103	12936576	GENERAL ARRANGEMENT PLAN
8	G-104	12936577	GENERAL ARRANGEMENT SECTION
9	G-105	12936578	CONTRACTOR ACCESS ROUTE PLAN
10	G-106	12936579	MOORING ARRANGEMENT PLAN
11	G-107	12936580	BATHYMETRIC SURVEY PLAN

HAZARDOUS MATERIALS SHEET INDEX

SHEET NO.		NAVFAC DWG. NO.	SHEET TITLE
NO.	SHEET		
12	H-101	12936581	HAZARDOUS MATERIALS NOTES & PLAN

SURVEY SHEET INDEX

SHEET NO.		NAVFAC DWG. NO.	SHEET TITLE
NO.	SHEET		
13	V-001	12936582	CODDINGTON COVE EXISTING CONDITIONS OVERVIEW
14	V-101	12936583	EXISTING CONDITIONS - PIER 2 LANDSIDE
15	V-102	12936584	EXISTING CONDITIONS - PIER 1 LANDSIDE
16	V-103	12936585	EXISTING CONDITIONS - RETENTION POND
17	V-104	12936586	EXISTING CONDITIONS - NOAA BUILDING
18	V-105	12936587	EXISTING CONDITIONS - PIER 1 SHEET 1 OF 5
19	V-106	12936588	EXISTING CONDITIONS - PIER 1 SHEET 2 OF 5
20	V-107	12936589	EXISTING CONDITIONS - PIER 1 SHEET 3 OF 5
21	V-108	12936590	EXISTING CONDITIONS - PIER 1 SHEET 4 OF 5
22	V-109	12936591	EXISTING CONDITIONS - PIER 1 SHEET 5 OF 5

GEOTECHNICAL SHEET INDEX

SHEET NO.		NAVFAC DWG. NO.	SHEET TITLE
NO.	SHEET		
23	B-101	12936592	ESTIMATED TOP OF WEATHERED ROCK CONTOUR PLAN
24	B-601	12936593	SOIL BORING LOGS - SHEET 1 OF 6
25	B-602	12936594	SOIL BORING LOGS - SHEET 2 OF 6
26	B-603	12936595	SOIL BORING LOGS - SHEET 3 OF 6
27	B-604	12936596	SOIL BORING LOGS - SHEET 4 OF 6
28	B-605	12936597	SOIL BORING LOGS - SHEET 5 OF 6
29	B-606	12936598	SOIL BORING LOGS - SHEET 6 OF 6

DREDGE SHEET INDEX

SHEET NO.		NAVFAC DWG. NO.	SHEET TITLE
NO.	SHEET		
30	CN101	12936599	OVERALL DREDGING PLAN (BASE BID)
31	CN301	12936600	DREDGING SECTIONS - SHEET 1 OF 3 (BASE BID)
32	CN302	12936601	DREDGING SECTIONS - SHEET 2 OF 3 (BASE BID)
33	CN303	12936602	DREDGING SECTIONS - SHEET 3 OF 3 (BASE BID)

CIVIL SHEET INDEX

SHEET NO.		NAVFAC DWG. NO.	SHEET TITLE
NO.	SHEET		
34	C-001	12936603	CIVIL LEGEND, ABBREVIATIONS & GENERAL NOTES (BASE BID)
35	C-002	12936604	GENERAL NOTES (BASE BID)
36	CD101	12936605	SITE DEMOLITION PLAN (BASE BID)
37	CD102	12936606	UTILITY DEMOLITION PLAN - PIER 1 LAND SIDE (BASE BID)
38	C-101	12936607	SITE PLAN (BASE BID)
39	C-102	12936608	UTILITY PLAN - PIER 1 LAND SIDE (BASE BID)
40	C-103	12936613	COMPOSITE UTILITY LAYOUT PLAN - PIER 1 LANDSIDE (BASE BID)
41	C-201	12936610	UTILITY PROFILES - SHEET 1 OF 2
42	C-202	12936611	UTILITY PROFILES - SHEET 2 OF 2 (BASE BID)
43	C-501	12936612	EROSION CONTROL NOTES & DETAILS (BASE BID)
44	C-502	12936613	SITE DETAILS - SHEET 1 OF 6 (BASE BID)
45	C-503	12936614	SITE DETAILS - SHEET 2 OF 6
46	C-504	12936615	SITE DETAILS - SHEET 3 OF 6
47	C-505	12936616	SITE DETAILS - SHEET 4 OF 6
48	C-506	12936617	SITE DETAILS - SHEET 5 OF 6
49	C-507	12936618	SITE DETAILS - SHEET 6 OF 6

STRUCTURAL SHEET INDEX

SHEET NO.		NAVFAC DWG. NO.	SHEET TITLE
NO.	SHEET		
50	S-001	12936619	STRUCTURAL ABBREVIATION (BASE BID)
51	S-002	12936620	STRUCTURAL GENERAL NOTES - SHEET 1 OF 2 (BASE BID)
52	S-003	12936621	STRUCTURAL GENERAL NOTES - SHEET 2 OF 2 (BASE BID)
53	S-004	12936622	STRUCTURAL DESIGN LOADS AND STANDARD DETAILS - SHEET 1 OF 2
54	S-005	12936623	STRUCTURAL DESIGN LOADS AND STANDARD DETAILS - SHEET 2 OF 2
55	SD001	12936624	STRUCTURAL DEMOLITION NOTES (BASE BID)
56	SD101	12936625	OVERALL STRUCTURAL DEMOLITION PLAN (BASE BID)
57	SD102	12936626	PIER 1 DEMOLITION PLAN
58	SD103	12936627	S45N BULKHEAD DEMOLITION PLAN
59	SD104	12936628	S45S BULKHEAD DEMOLITION PLAN (BASE BID)
60	SD201	12936629	S45N BULKHEAD DEMOLITION ELEVATION
61	SD202	12936630	TEMPORARY SHEETING BULKHEAD ELEVATION
62	SD301	12936631	TYPICAL SECTION FOR PIER 1 DEMOLITION (BASE BID)
63	SD302	12936632	TYPICAL SECTION FOR BULKHEAD DEMOLITION
64	SD303	12936633	SECTION OF DEMOLITION AT S45N BULKHEAD (BASE BID)
65	SD501	12936634	MISCELLANEOUS DEMOLITION DETAILS
66	SD601	12936635	PIER 1 AND S45N BULKHEAD DEMOLITION PHOTOGRAPHS (BASE BID)
67	SD602	12936636	S45N BULKHEAD DEMOLITION PHOTOGRAPHS (BASE BID)
68	S-101	12936637	OVERALL BULKHEAD PLAN (BASE BID)
69	S-102	12936638	GENERAL DECK PLAN
70	S-103	12936639	PILE PLAN - SHEET 1 OF 7
71	S-104	12936640	PILE PLAN - SHEET 2 OF 7
72	S-105	12936641	PILE PLAN - SHEET 3 OF 7
73	S-106	12936642	PILE PLAN - SHEET 4 OF 7
74	S-107	12936643	PILE PLAN - SHEET 5 OF 7
75	S-108	12936644	PILE PLAN - SHEET 6 OF 7
76	S-109	12936645	PILE PLAN - SHEET 7 OF 7
77	S-110	12936646	FRAMING PLAN - SHEET 1 OF 7
78	S-111	12936647	FRAMING PLAN - SHEET 2 OF 7
79	S-112	12936648	FRAMING PLAN - SHEET 3 OF 7
80	S-113	12936649	FRAMING PLAN - SHEET 4 OF 7
81	S-114	12936650	FRAMING PLAN - SHEET 5 OF 7
82	S-115	12936651	FRAMING PLAN - SHEET 6 OF 7
83	S-116	12936652	FRAMING PLAN - SHEET 7 OF 7
84	S-117	12936653	CONCRETE PLANKS PLAN - SHEET 1 OF 7
85	S-118	12936654	CONCRETE PLANKS PLAN - SHEET 2 OF 7
86	S-119	12936655	CONCRETE PLANKS PLAN - SHEET 3 OF 7
87	S-120	12936656	CONCRETE PLANKS PLAN - SHEET 4 OF 7
88	S-121	12936657	CONCRETE PLANKS PLAN - SHEET 5 OF 7
89	S-122	12936658	CONCRETE PLANKS PLAN - SHEET 6 OF 7
90	S-123	12936659	CONCRETE PLANKS PLAN - SHEET 7 OF 7
91	S-124	12936660	DECK PLAN - SHEET 1 OF 7
92	S-125	12936661	DECK PLAN - SHEET 2 OF 7
93	S-126	12936662	DECK PLAN - SHEET 3 OF 7
94	S-127	12936663	DECK PLAN - SHEET 4 OF 7
95	S-128	12936664	DECK PLAN - SHEET 5 OF 7
96	S-129	12936665	DECK PLAN - SHEET 6 OF 7
97	S-130	12936666	DECK PLAN - SHEET 7 OF 7
98	S-131	12936667	FENDER PILE PLAN - SHEET 1 OF 6
99	S-132	12936668	FENDER PILE PLAN - SHEET 2 OF 6

STRUCTURAL SHEET INDEX

SHEET NO.		NAVFAC DWG. NO.	SHEET TITLE
NO.	SHEET		
100	S-133	12936669	FENDER PILE PLAN - SHEET 3 OF 6
101	S-134	12936670	FENDER PILE PLAN - SHEET 4 OF 6
102	S-135	12936671	FENDER PILE PLAN - SHEET 5 OF 6
103	S-136	12936672	FENDER PILE PLAN - SHEET 6 OF 6
104	S-137	12936673	DECK DRAINAGE PLAN - SHEET 1 OF 2
105	S-138	12936674	DECK DRAINAGE PLAN - SHEET 2 OF 2
106	S-201	12936675	PIER SOUTHERN ELEVATION - SHEET 1 OF 3
107	S-202	12936676	PIER SOUTHERN ELEVATION - SHEET 2 OF 3
108	S-203	12936677	PIER SOUTHERN ELEVATION - SHEET 3 OF 3
109	S-204	12936678	PIER NORTHERN ELEVATION - SHEET 1 OF 3
110	S-205	12936679	PIER NORTHERN ELEVATION - SHEET 2 OF 3
111	S-206	12936680	PIER NORTHERN ELEVATION - SHEET 3 OF 3
112	S-207	12936681	REPAIRED S45N BULKHEAD ELEVATION & PIER 1 SECTION
113	S-208	12936682	S45S BULKHEAD ELEVATION (BASE BID)
114	S-301	12936683	PIER LONGITUDINAL SECTIONS - SHEET 1 OF 2
115	S-302	12936684	PIER LONGITUDINAL SECTIONS - SHEET 2 OF 2
116	S-303	12936685	PIER ABUTMENT & APPROACH SLAB LONGITUDINAL SECTION
117	S-304	12936686	LONGITUDINAL SECTIONS AT UTILITY TRENCH
118	S-305	12936687	LONGITUDINAL SECTION CROSS UTILITY TRENCH AT ABUTMENT
119	S-306	12936688	PIER ABUTMENT & UTILITY TRENCH TRANSVERSE SECTION
120	S-307	12936689	PIER TRANSVERSE SECTIONS - SHEET 1 OF 5
121	S-308	12936690	PIER TRANSVERSE SECTIONS - SHEET 2 OF 5
122	S-309	12936691	PIER TRANSVERSE SECTIONS - SHEET 3 OF 5
123	S-310	12936692	PIER TRANSVERSE SECTIONS - SHEET 4 OF 5
124	S-311	12936693	PIER TRANSVERSE SECTIONS - SHEET 5 OF 5
125	S-401	12936694	ABUTMENT & LANDSIDE FRAMING PART PLAN
126	S-402	12936699	OUTBOARD SUBSTATION PART PLAN
127	S-403	12936700	INBOARD SUBSTATION PART PLAN
128	S-404	12936697	S45S BULKHEAD PART PLAN & SECTION (BASE BID)
129	S-501	12936698	TURBIDITY CURTAIN NOTES & DETAILS (BASE BID)
130	S-502	12936699	PILE DETAILS - CONCRETE-FILLED STEEL PIPE PILES
131	S-503	12936704	PILE DETAILS - STEEL PILES
132	S-504	12936701	CONCRETE PILE CAP DETAILS - SHEET 1 OF 3
133	S-505	12936706	CONCRETE PILE CAP DETAILS - SHEET 2 OF 3
134	S-506	12936703	CONCRETE PILE CAP DETAILS - SHEET 3 OF 3
135	S-507	12936708	REINFORCED BOLLARD FOUNDATION DETAILS
136	S-508	12936705	REINFORCEMENT DETAILS FOR CONCRETE BLOCKOUT AT BENT 43 - SHEET 1 OF 2
137	S-509	12936706	REINFORCEMENT DETAILS FOR CONCRETE BLOCKOUT AT BENT 43 - SHEET 2 OF 2
138	S-510	12936707	CONCRETE DUCTBANK REINFORCEMENT DETAILS
139	S-511	12936708	REINFORCEMENT DETAILS
140	S-512	12936709	CONCRETE BEAM DETAILS
141	S-513	12936710	TYPICAL PIER DUCTBANK DETAILS
142	S-514	12936711	LANDSIDE PIPE PILES CONNECTION DETAILS
143	S-515	12936712	PRECAST CONCRETE PLANK DETAILS
144	S-516	12936713	CONDUITS AND FITTINGS AT EXPANSION JOINT
145	S-517	12936714	PRECAST MANHOLE DETAILS
146	S-518	12936715	UTILITY TRENCH COVER DETAILS
147	S-519	12936716	FENDER SYSTEM DETAILS - SHEET 1 OF 2
148	S-520	12936717	FENDER SYSTEM DETAILS - SHEET 2 OF 2
149	S-521	12936718	PIPE SUPPORT DETAILS
150	S-522	12936719	MISCELLANEOUS SECTIONS & DETAILS - SHEET 1 OF 2
151	S-523	12936720	MISCELLANEOUS SECTIONS & DETAILS - SHEET 2 OF 2
152	S-524	12936721	S45N BULKHEAD MODIFICATION DETAILS - SHEET 1 OF 2
153	S-525	12936722	S45N BULKHEAD MODIFICATION DETAILS - SHEET 2 OF 2
154	S-526	12936723	S45S BULKHEAD DETAILS - SHEET 1 OF 3 (BASE BID)
155	S-527	12936724	S45S BULKHEAD DETAILS - SHEET 2 OF 3 (BASE BID)
156	S-528	12936725	S45S BULKHEAD DETAILS - SHEET 3 OF 3 (BASE BID)
157	S-529	12936730	AIR COMPRESSOR FOUNDATION & STEEL FRAME PART PLAN
158	S-530	12936727	AIR COMPRESSOR DETAILS
159	S-531	12936728	AIR COMPRESSOR FOUNDATION DETAILS
160	S-532	12936729	AIR COMPRESSOR STAIRS ACCESS DETAILS - SHEET 1 OF 2
161	S-533	12936730	AIR COMPRESSOR STAIRS ACCESS DETAILS - SHEET 2 OF 2
162	S-534	12936731	15KV SWITCHGEAR STRUCTURE DETAILS
163	S-601	12936732	STEEL PILE SCHEDULE
164	S-602	12936733	FENDER PILE SCHEDULE - SHEET 1 OF 2
165	S-603	12936734	FENDER PILE SCHEDULE - SHEET 2 OF 2

SYM	DESCRIPTION	DATE	APPR

NAVFAC

VICTOR M. GOVONI CIVIL No. 41734 REGISTERED PROFESSIONAL ENGINEER COMMONWEALTH OF MASSACHUSETTS

SEAL

Stantec

A/E INFO

APPROVED

James E. Donahue, R.A.

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

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INDEX OF DRAWINGS - SHEET 1 OF 2

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936571

SHEET 2 OF 247

G-002

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MECHANICAL SHEET INDEX

SHEET NO. NO.	SHEET	NAVFAC DWG. NO.	SHEET TITLE
166	M-001	12936735	MECHANICAL LEGEND, ABBREVIATIONS & GENERAL NOTES
167	M-101	12936736	MECHANICAL PLAN - SHEET 1 OF 7
168	M-102	12936737	MECHANICAL PLAN - SHEET 2 OF 7
169	M-103	12936738	MECHANICAL PLAN - SHEET 3 OF 7
170	M-104	12936739	MECHANICAL PLAN - SHEET 4 OF 7
171	M-105	12936740	MECHANICAL PLAN - SHEET 5 OF 7
172	M-106	12936741	MECHANICAL PLAN - SHEET 6 OF 7
173	M-107	12936742	MECHANICAL PLAN - SHEET 7 OF 7
174	M-301	12936743	MECHANICAL LONGITUDINAL SECTIONS - SHEET 1 OF 2
175	M-302	12936744	MECHANICAL LONGITUDINAL SECTIONS - SHEET 2 OF 2
176	M-303	12936745	MECHANICAL TRANSVERSE SECTIONS
177	M-401	12936746	MECHANICAL COMPRESSED AIR ENCLOSURE - PART PLAN
178	M-402	12936747	REDUCED PRESSURE BACKFLOW PREVENTER DETAILS
179	M-501	12936748	NORTHERN MECHANICAL ENCLOSURE DETAILS
180	M-502	12936749	SOUTHERN MECHANICAL ENCLOSURE DETAILS
181	M-503	12936750	PIER HEATED ENCLOSURE - SANITARY AND FRESH WATER
182	M-504	12936751	PIPE SUPPORT DETAILS - SHEET 1 OF 2
183	M-505	12936752	PIPE SUPPORT DETAILS - SHEET 2 OF 2
184	M-506	12936753	MECHANICAL ENCLOSURE VFD PUMP CONTROLLER DETAILS
185	M-601	12936754	MECHANICAL PIPE SCHEDULES
186	M-701	12936755	COMPRESSED AIR SCHEMATIC (BID OPTION)
187	M-702	12936756	COMPRESSED AIR HEAT RECOVERY SCHEMATIC
188	M-703	12936757	POTABLE FRESH WATER (FW) SCHEMATIC
189	M-901	12936758	MECHANICAL UTILITIES ISOMETRIC VIEWS

ELECTRICAL SHEET INDEX

SHEET NO. NO.	SHEET	NAVFAC DWG. NO.	SHEET TITLE
190	E-001	12936759	ELECTRICAL LEGEND, ABBREVIATIONS & GENERAL NOTES
191	E-101	12936760	ELECTRICAL DECK PLAN - SHEET 1 OF 7
192	E-102	12936761	ELECTRICAL DECK PLAN - SHEET 2 OF 7
193	E-103	12936762	ELECTRICAL DECK PLAN - SHEET 3 OF 7
194	E-104	12936763	ELECTRICAL DECK PLAN - SHEET 4 OF 7
195	E-105	12936764	ELECTRICAL DECK PLAN - SHEET 5 OF 7
196	E-106	12936765	ELECTRICAL DECK PLAN - SHEET 6 OF 7
197	E-107	12936766	ELECTRICAL DECK PLAN - SHEET 7 OF 7
198	E-108	12936767	ELECTRICAL SITE PLAN - PIER 2
199	E-109	12936768	ELECTRICAL SITE PLAN - PIER 1
200	E-110	12936769	ELECTRICAL SITE PLAN - NOAA BUILDING
201	E-111	12936770	ELECTRICAL SITE PLAN - BUILDING 76
202	E-112	12936771	CATHODIC PROTECTION PILE PLAN
203	E-113	12936772	GROUNDING PLAN - PIER WEST
204	E-114	12936773	GROUNDING PLAN - PIER EAST
205	E-115	12936774	LIGHTNING PROTECTION PLAN - PIER WEST
206	E-116	12936775	LIGHTNING PROTECTION PLAN - PIER EAST
207	E-401	12936776	SUBSTATION P1W PART PLAN & ELEVATION
208	E-402	12936777	SUBSTATION P1E PART PLAN & ELEVATION
209	E-403	12936778	SUBSTATION GROUNDING PART PLANS
210	E-404	12936779	SUBSTATION P1W VAULT PART PLAN & ELEVATION
211	E-405	12936780	SUBSTATION P1E VAULT PART PLAN & ELEVATION
212	E-501	12936781	MANHOLE & DUCTBANK DETAILS
213	E-502	12936782	15KV SWITCHGEAR STRUCTURE DETAILS
214	E-503	12936783	POWER MOUND DETAILS - SHEET 1 OF 2
215	E-504	12936784	POWER MOUND DETAILS - SHEET 2 OF 2
216	E-505	12936785	METERING SYSTEM DETAILS - SHEET 1 OF 3
217	E-506	12936786	METERING SYSTEM DETAILS - SHEET 2 OF 3
218	E-507	12936787	METERING SYSTEM DETAILS - SHEET 3 OF 3
219	E-508	12936788	LUMINAIRE SCHEDULE & DETAILS
220	E-509	12936789	HIGH MAST LIGHTING DETAILS
221	E-510	12936790	SUBSTATION P1E SWITCHGEAR ELEVATIONS
222	E-511	12936791	SUBSTATION P1W SWITCHGEAR ELEVATIONS
223	E-512	12936792	MISCELLANEOUS DETAILS - SHEET 1 OF 2
224	E-513	12936793	MISCELLANEOUS DETAILS - SHEET 2 OF 2
225	E-514	12936794	HEAT TRACE CABINET & DETAILS
226	E-601	12936795	PRIMARY & DEMOLITION ONE LINE DIAGRAM
227	E-602	12936796	PIER ONE LINE DIAGRAM - EAST
228	E-603	12936797	PIER ONE LINE DIAGRAM - WEST
229	E-604	12936798	COMMUNICATIONS ONE LINE DIAGRAM
230	E-605	12936799	MANHOLE SCHEDULE & FOLDOUT DIAGRAMS
231	E-606	12936800	MINI POWER CENTER PANEL SCHEDULES
232	E-607	12936801	SWITCHGEAR BREAKER CONTROL DIAGRAM
233	E-608	12936802	HEAT TRACE ONE LINE DIAGRAM & SCHEDULES
234	E-609	12936803	LIGHTING CONTROL DIAGRAMS
235	E-610	12936804	SUBSTATION P1E SWITCHGEAR SCHEDULE
236	E-611	12936805	SUBSTATION P1W SWITCHGEAR SCHEDULE
237	E-612	12936806	SUBSTATION P1E PANEL SCHEDULES
238	E-613	12936807	SUBSTATION P1W PANEL SCHEDULES
239	E-614	12936808	FEEDER SCHEDULE - SHEET 1 OF 2
240	E-615	12936809	FEEDER SCHEDULE - SHEET 2 OF 2

ELECTRICAL SUBSTATION 1 SHEET INDEX

SHEET NO. NO.	SHEET	NAVFAC DWG. NO.	SHEET TITLE
241	ES101	12936810	ELECTRICAL SITE PLAN - SUBSTATION 1
242	ES102	12936811	SUBSTATION 1 PLAN VIEW
243	ES501	12936812	BREAKER 152-11 ELEVATION VIEW
244	ES601	12936813	SUBSTATION 1 SINGLE LINE DIAGRAM
245	ES602	12936814	BREAKER 152-11 SCHEMATIC DIAGRAM
246	ES603	12936815	BREAKER 152-11 DIGITAL CONTROL DIAGRAM
247	ES604	12936816	BREAKER 152-11 COMMUNICATIONS DIAGRAM

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COMMONWEALTH OF MASSACHUSETTS
VICTOR M. GOVONI
CIVIL
No. 41734
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INDEX OF DRAWINGS - SHEET 2 OF 2

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SHEET 3 OF 247

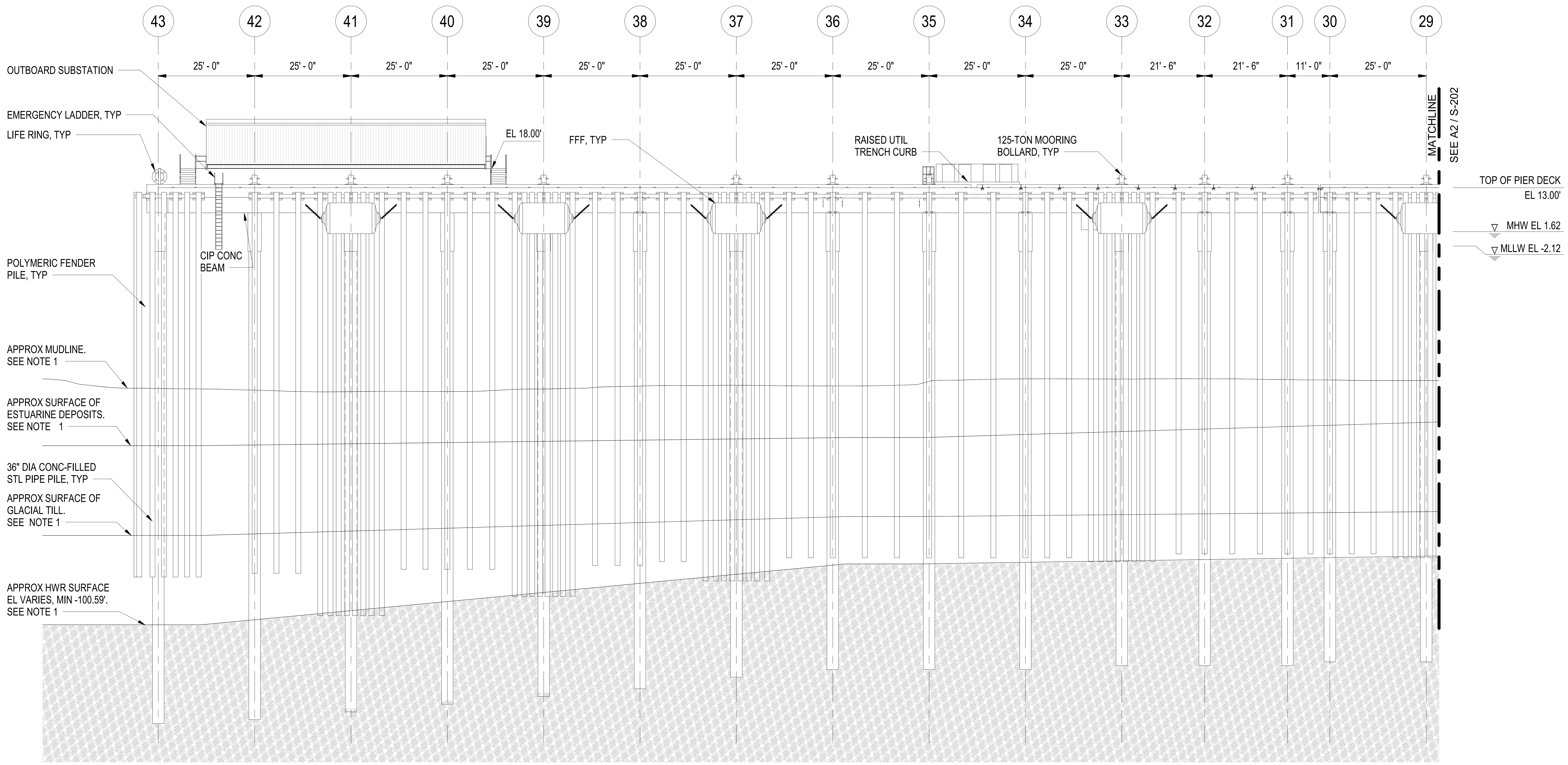
G-003

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 5:33:19 PM G-003
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

GENERAL SHEET NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA.
2. SEE ATTACHMENT A IN THE SPECIFICATIONS FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.



A2 **PIER ELEVATION BENT 43 TO BENT 29 - SOUTH**

SCALE: 1/16" = 1'-0"

1/16" = 1' - 0" 0 5' 10' 20' 30' 60'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW SA CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER SOUTHERN ELEVATION - SHEET 1 OF 3

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936675

SHEET 106 OF 247

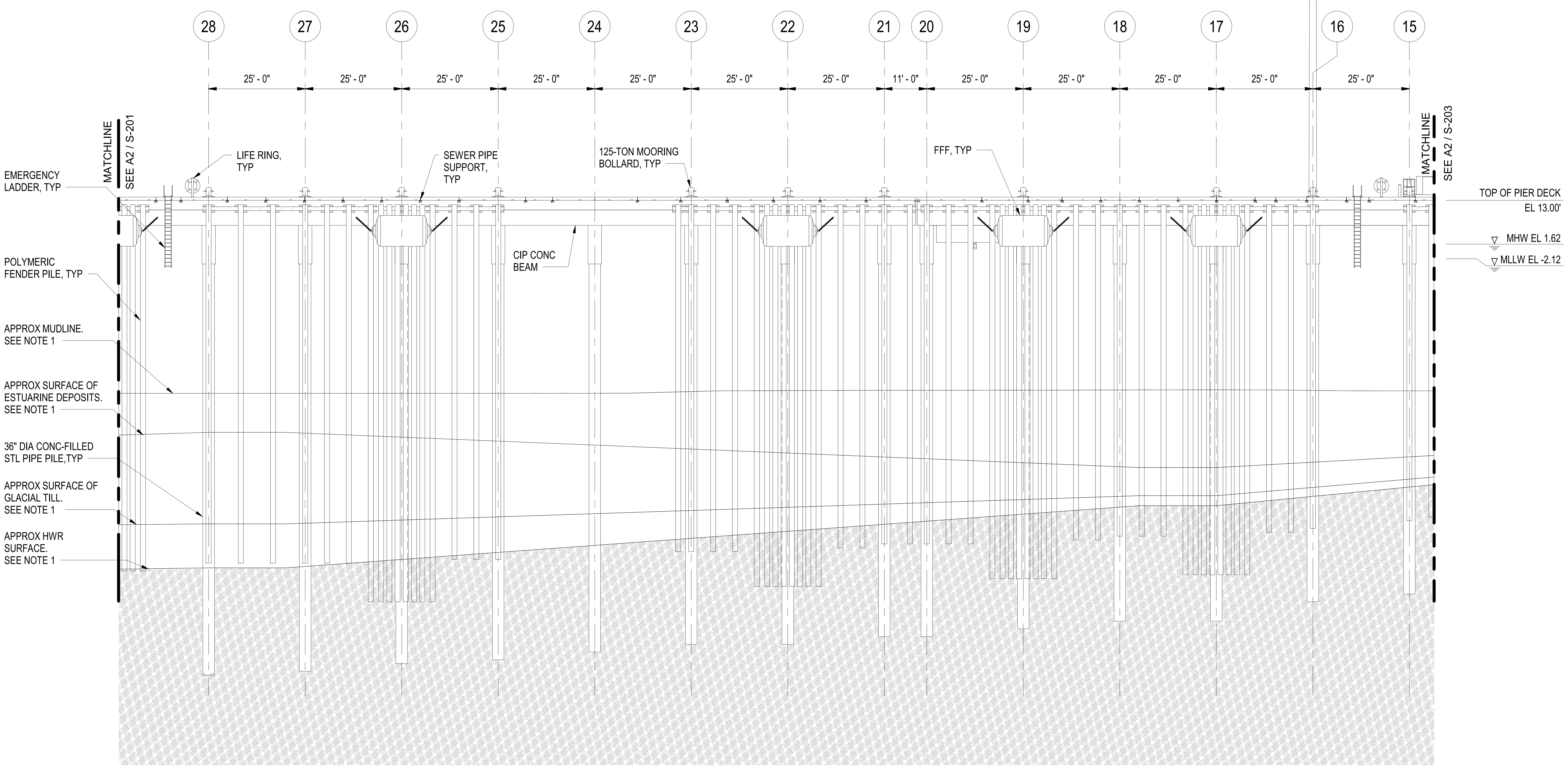
S-201

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:35:43 PM S-201
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

GENERAL SHEET NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA.
2. SEE ATTACHMENT A IN THE SPECIFICATIONS FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.



A2 PIER ELEVATION BENT 29 TO BENT 15 - SOUTH

SCALE: 1/16" = 1'-0"

1/16" = 1' - 0"   0 5' 10' 20' 30' 60'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

APPROVED
James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES JZ	DRW SA	CHK MJF
PM/DM		MRI/CWJ
BRANCH MANAGER		NEK
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER SOUTHERN ELEVATION - SHEET 2 OF 3

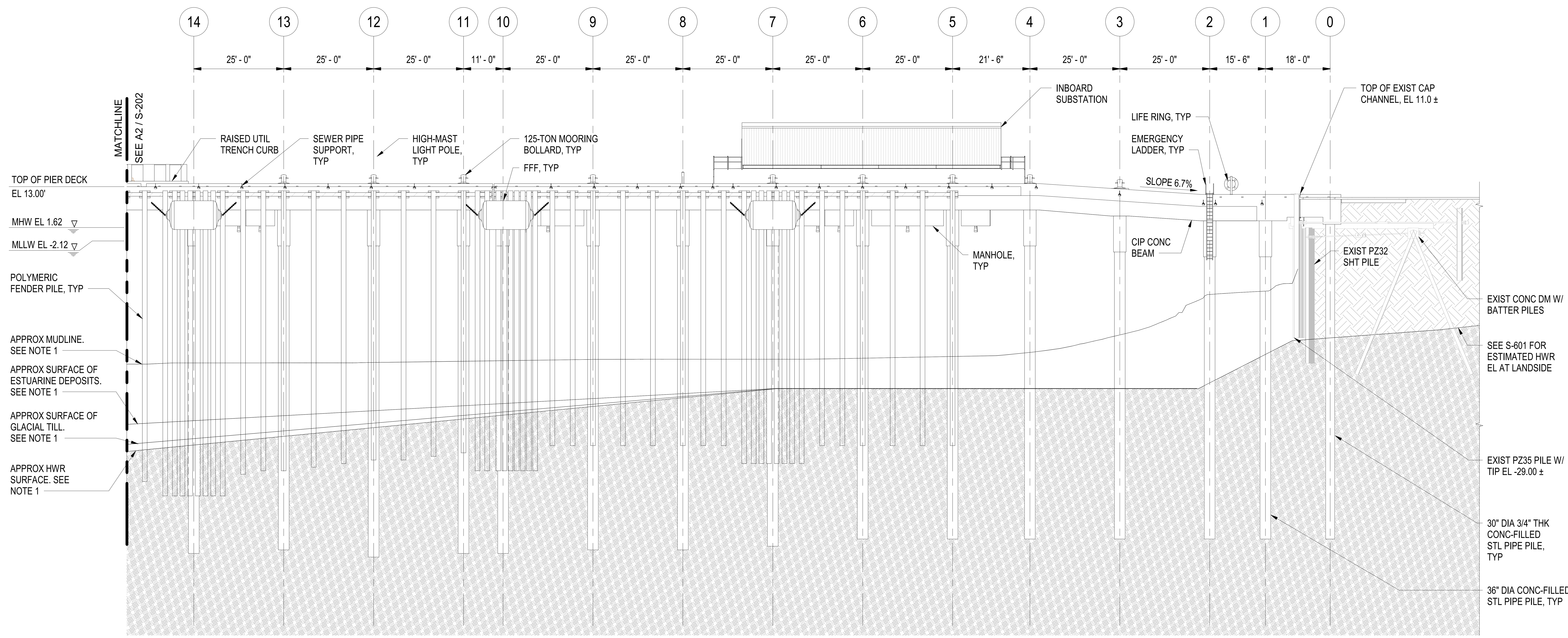
SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936676
SHEET 107 OF 247

S-202

DRAWFORM REVISION: 25 AUGUST 2020

GENERAL SHEET NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA.
2. SEE ATTACHMENT A IN THE SPECIFICATIONS FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.
3. BASE STEEL FRAMING FOR AC ENCLOSURE NOT SHOWN FOR CLARITY.
4. SEE NOTE 5 ON S-109 FOR PILE CONSTRUCTION WITHIN EXISTING RIPRAP.
5. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR LONGITUDINAL SLOPE, TYPICAL.



A2 PIER ELEVATION BENT 14 TO BENT 0 - SOUTH

SCALE: 1/16" = 1'-0"

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA
SEAL

Stantec

A/E INFO

APPROVED
James E. Donahue, R.A.
FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES JZ	DRW SA	CHK MJF
PM/DM		MRI/CWJ
BRANCH MANAGER		NEK
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER SOUTHERN ELEVATION - SHEET 3 OF 3

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936677
SHEET 108 OF 247

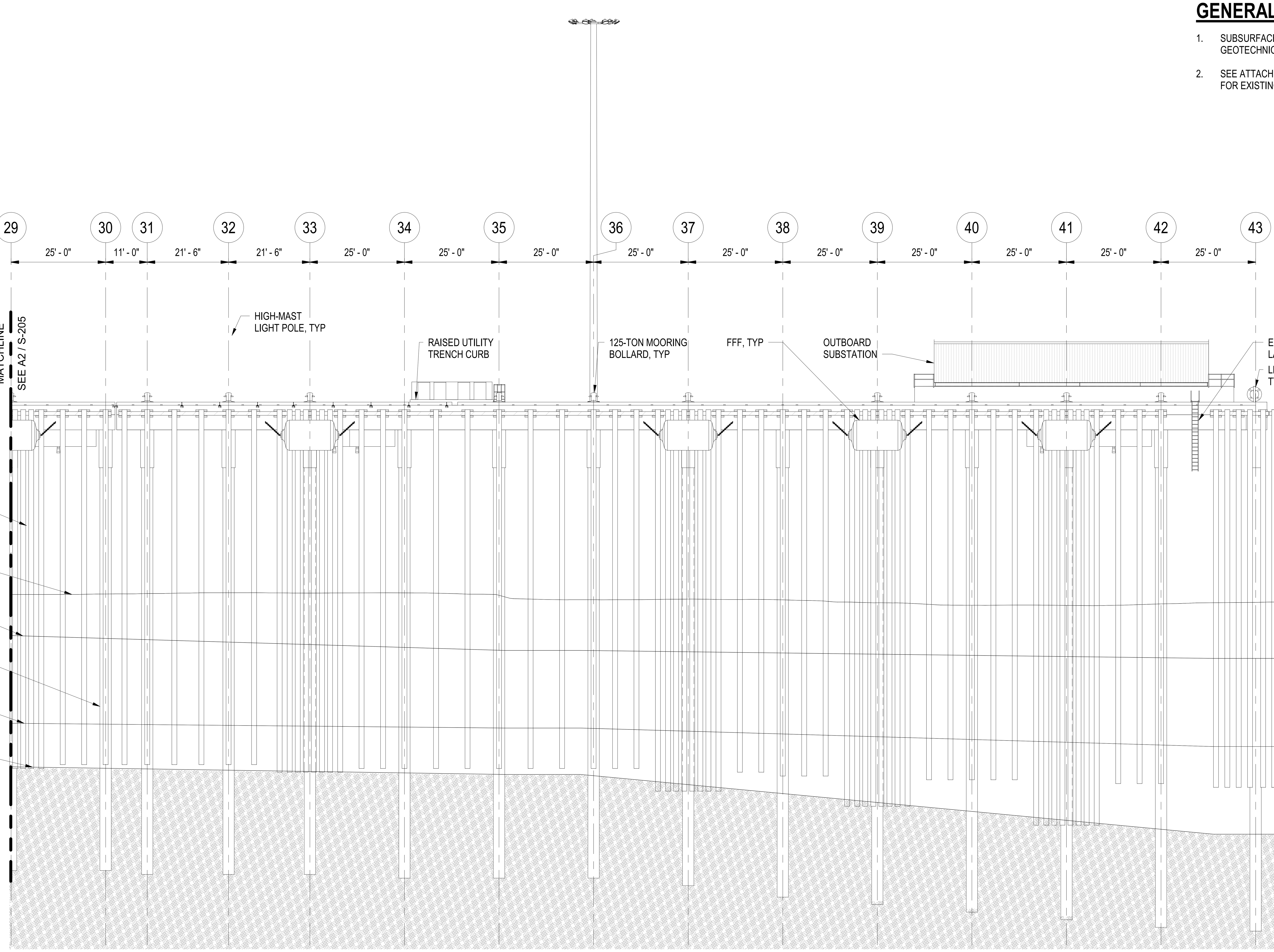
S-203

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:35:49 PM S-203
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

GENERAL SHEET NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA.
2. SEE ATTACHMENT A IN THE SPECIFICATIONS FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.



A2 PIER ELEVATION BENT 29 TO BENT 43 - NORTH

SCALE: 1/16" = 1'-0"

1/16" = 1' - 0" 0 5' 10' 20' 30' 60'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW SA CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER NORTHERN ELEVATION - SHEET 1 OF 3

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936678

SHEET 109 OF 247

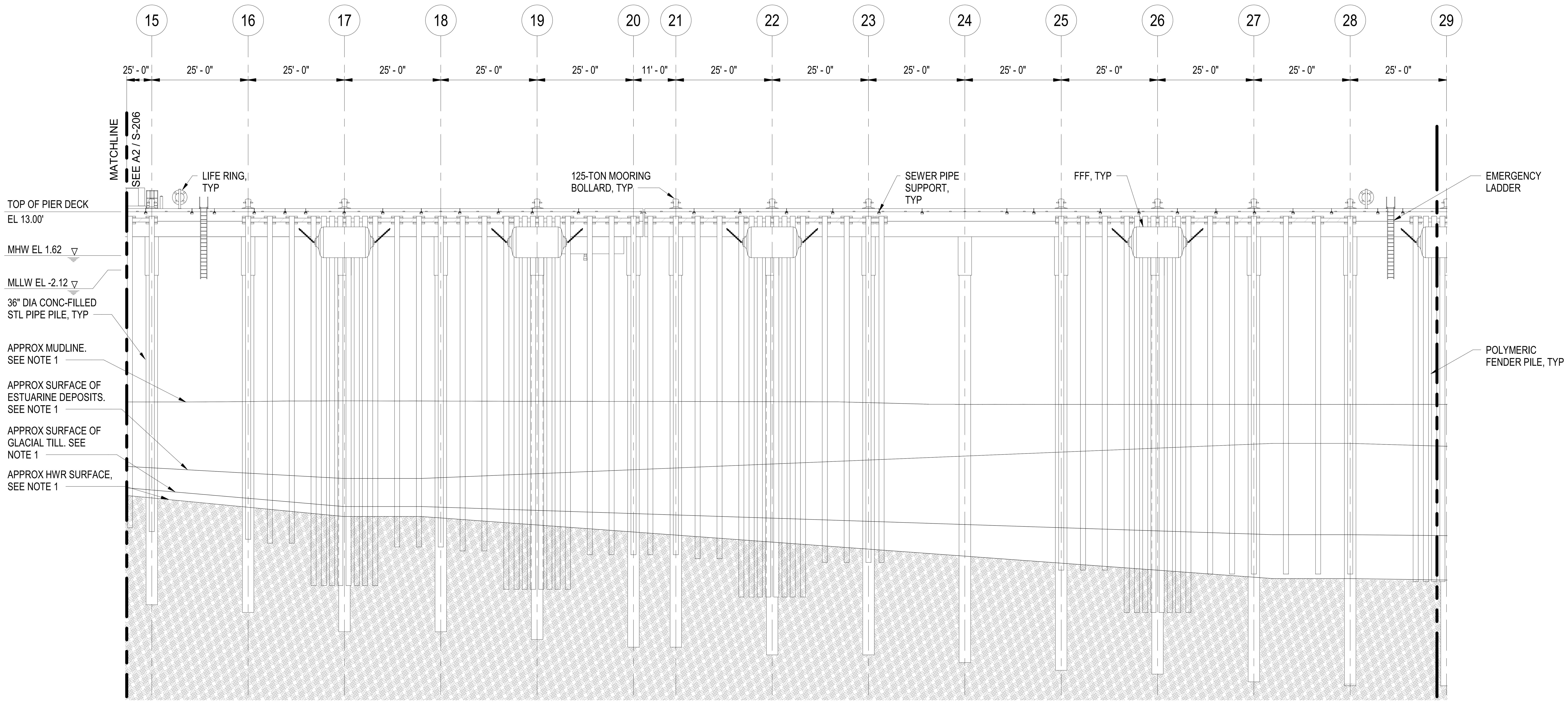
S-204

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:35:51 PM S-204
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

GENERAL SHEET NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA.
2. SEE ATTACHMENT A IN THE SPECIFICATIONS FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.



A2 PIER ELEVATION BENT 15 TO BENT 29 - NORTH

SCALE: 1/16" = 1'-0"

1/16" = 1' - 0" 0 5' 10' 20' 30' 60'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW SA CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

PIER NORTHERN ELEVATION - SHEET 2 OF 3

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936679

SHEET 110 OF 247

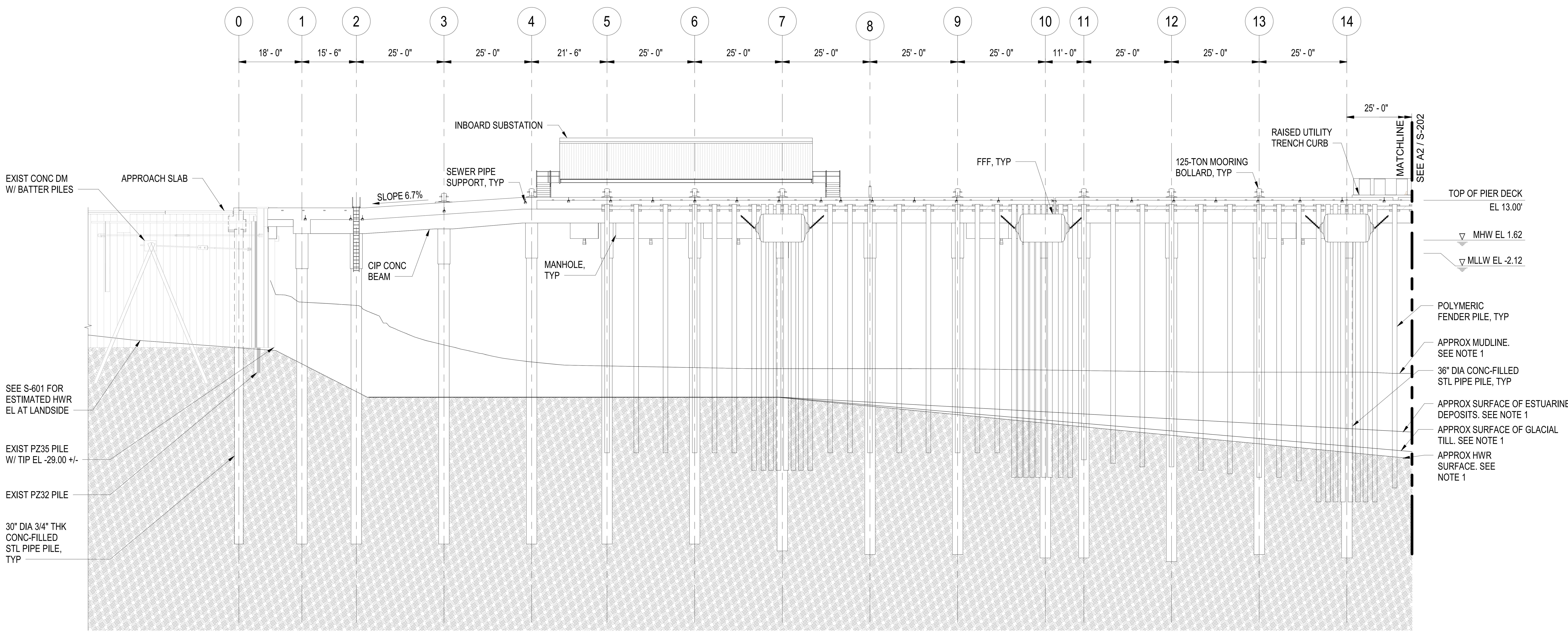
S-205

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:35:53 PM S-205
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

GENERAL SHEET NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA.
2. SEE ATTACHMENT A IN THE SPECIFICATIONS FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.
3. BASE STEEL FRAMING FOR AC ENCLOSURE NOT SHOWN FOR CLARITY.
4. SEE NOTE 5 ON S-109 FOR PILE CONSTRUCTION WITHIN EXISTING RIPRAP.
5. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR LONGITUDINAL SLOPE, TYPICAL.



A2 **PIER ELEVATION BENT 0 TO BENT 14 - NORTH**

SCALE: 1/16" = 1'-0"

1/16" = 1' - 0" 0 5' 10' 20' 30' 60'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW SA CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT
NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

PIER NORTHERN ELEVATION - SHEET 3 OF 3

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936680

SHEET 111 OF 247

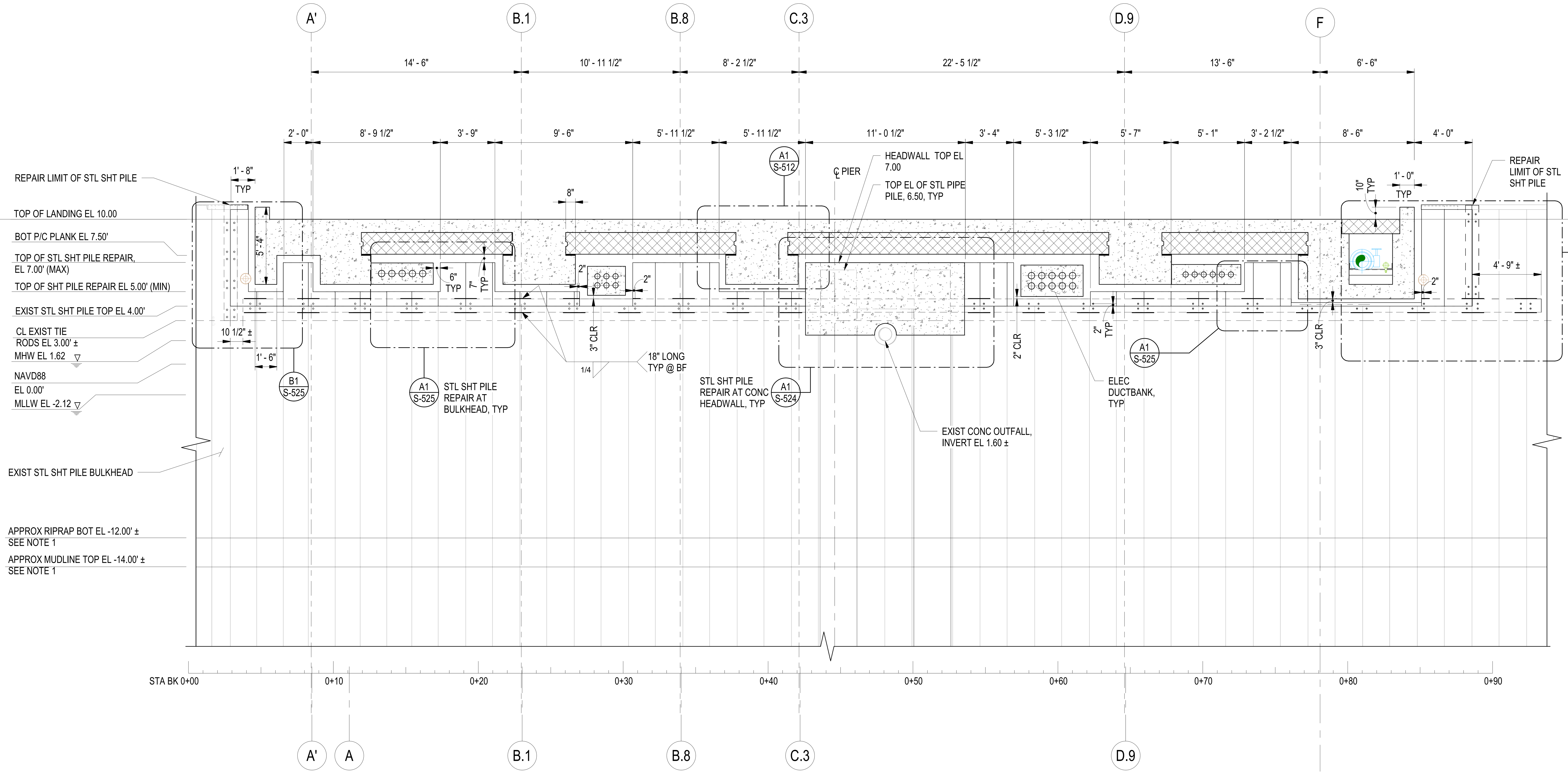
S-206

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:35:56 PM S-206
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA.



A2 S45N REPAIRED BULKHEAD ELEVATION & PIER 1 SECTION

SCALE: 1/4" = 1'-0" (S-130)

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

Stantec

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW KCJ CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

REPAIRED S45N BULKHEAD ELEVATION & PIER 1 SECTION

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

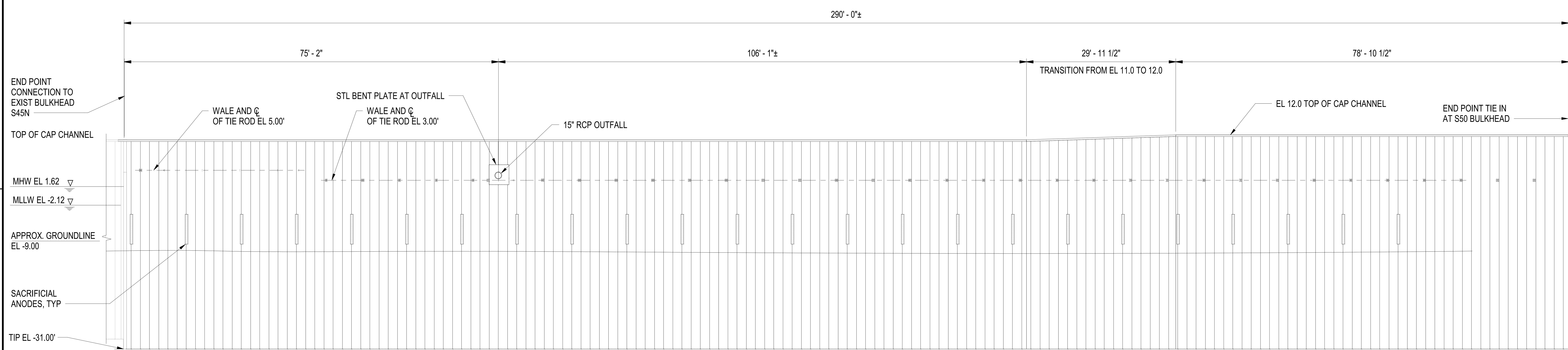
NAVFAC DRAWING NO. 12936681

SHEET 112 OF 247

S-207

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:35:57 PM S-207
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt



B2 ELEVATION BULKHEAD S45S

SCALE: 3/32" = 1'-0" (S-404)

NOTES:

1. SACRIFICIAL ANODES TO BE INSTALLED ON BOTH SIDES OF STEEL SHEET PILE BULKHEAD. ANODES TO BE INSTALLED FOR EVERY THREE PAIRS OF STEEL SHEETING.
2. PROVIDE MINIMUM 3" CONTINUITY WELD AT THE TOP OF EACH SHEET PILE KNUCKLE.

1" = 10' 0 5' 10' 20' 40'

SYM | DESCRIPTION | DATE | APPR

NAVFAC

VICTOR M. GOYONI CIVIL No. 41734 COMMONWEALTH OF MASSACHUSETTS REGISTERED PROFESSIONAL ENGINEER

Stantec

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES MJF DRW KCJ CHK JZ
PM/DM MRI/CWJ
BRANCH MANAGER NEK
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
S45S BULKHEAD ELEVATION (BASE BID)

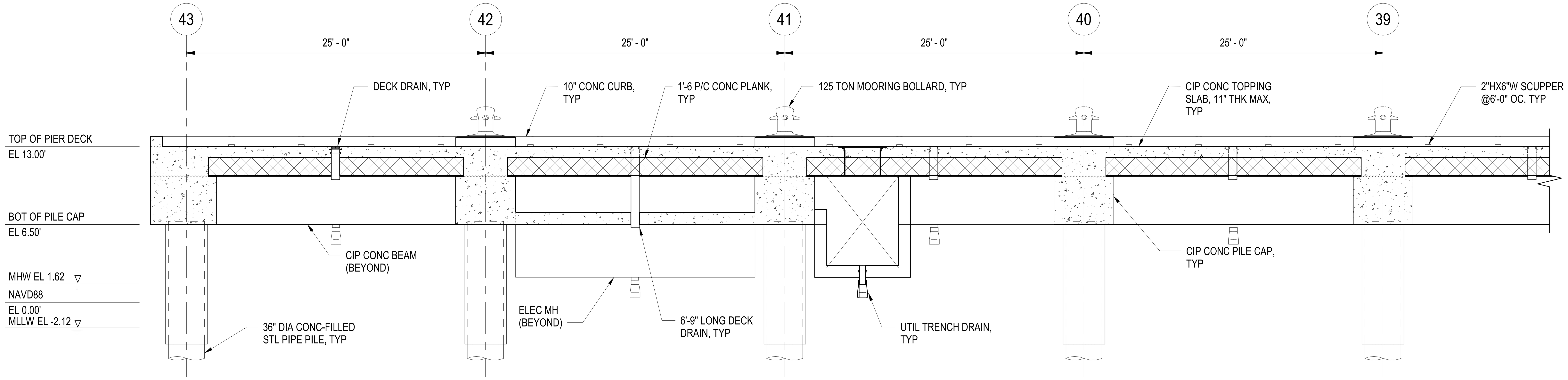
SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936682
SHEET 113 OF 247
S-208

DRAWFORM REVISION: 25 AUGUST 2020

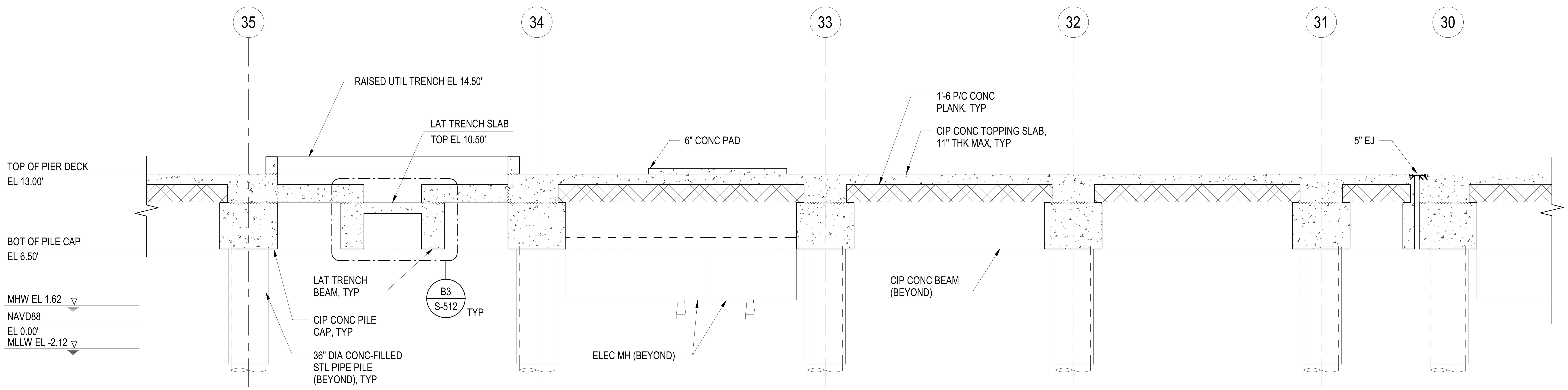
3/2/2026 7:35:58 PM S-208
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

NOTES:

1. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR LONGITUDINAL SLOPE, TYPICAL.



C2 **PIER SECTION BETWEEN BENTS 43 & 39**
SCALE: 3/16" = 1'-0" (S-103, S-110, S-117, S-124)



A2 **PIER SECTION AT RAISED UTILITY TRENCH & EXPANSION JOINT**
SCALE: 3/16" = 1'-0" (S-104, S-111, S-118, S-125)

3/16" = 1' - 0" 0 5' 10' 20'

SYM	DESCRIPTION	DATE	APPR

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

Stantec

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES JZ DRW KM CHK MJF
PM/DM MRI/CWJ
BRANCH MANAGER NEK
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER LONGITUDINAL SECTIONS - SHEET 1 OF 2

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936683
SHEET 114 OF 247
S-301

DRAWFORM REVISION: 25 AUGUST 2020

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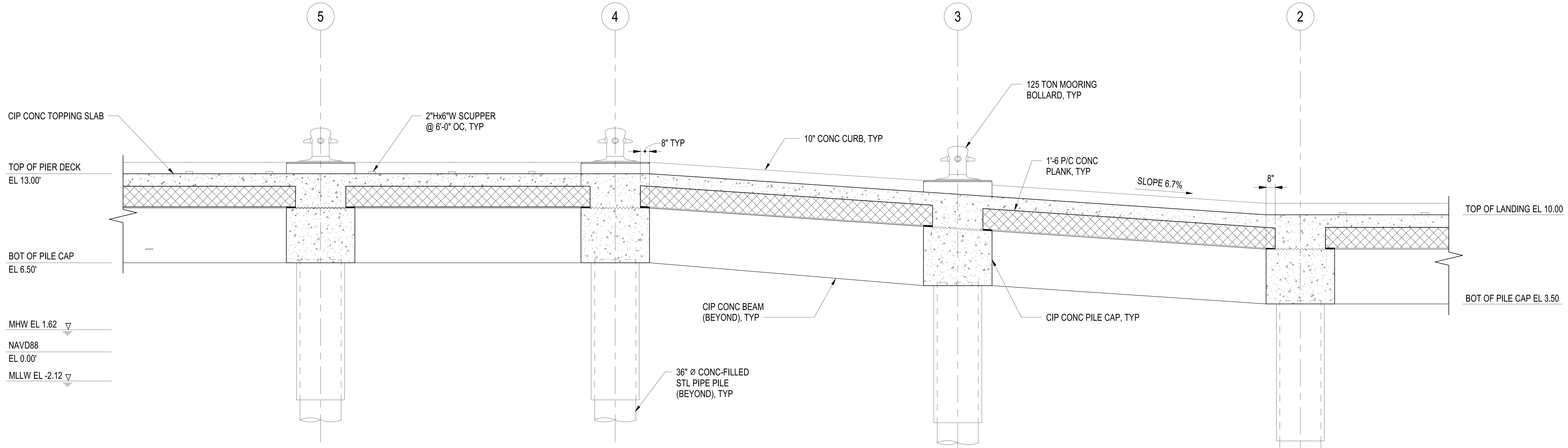
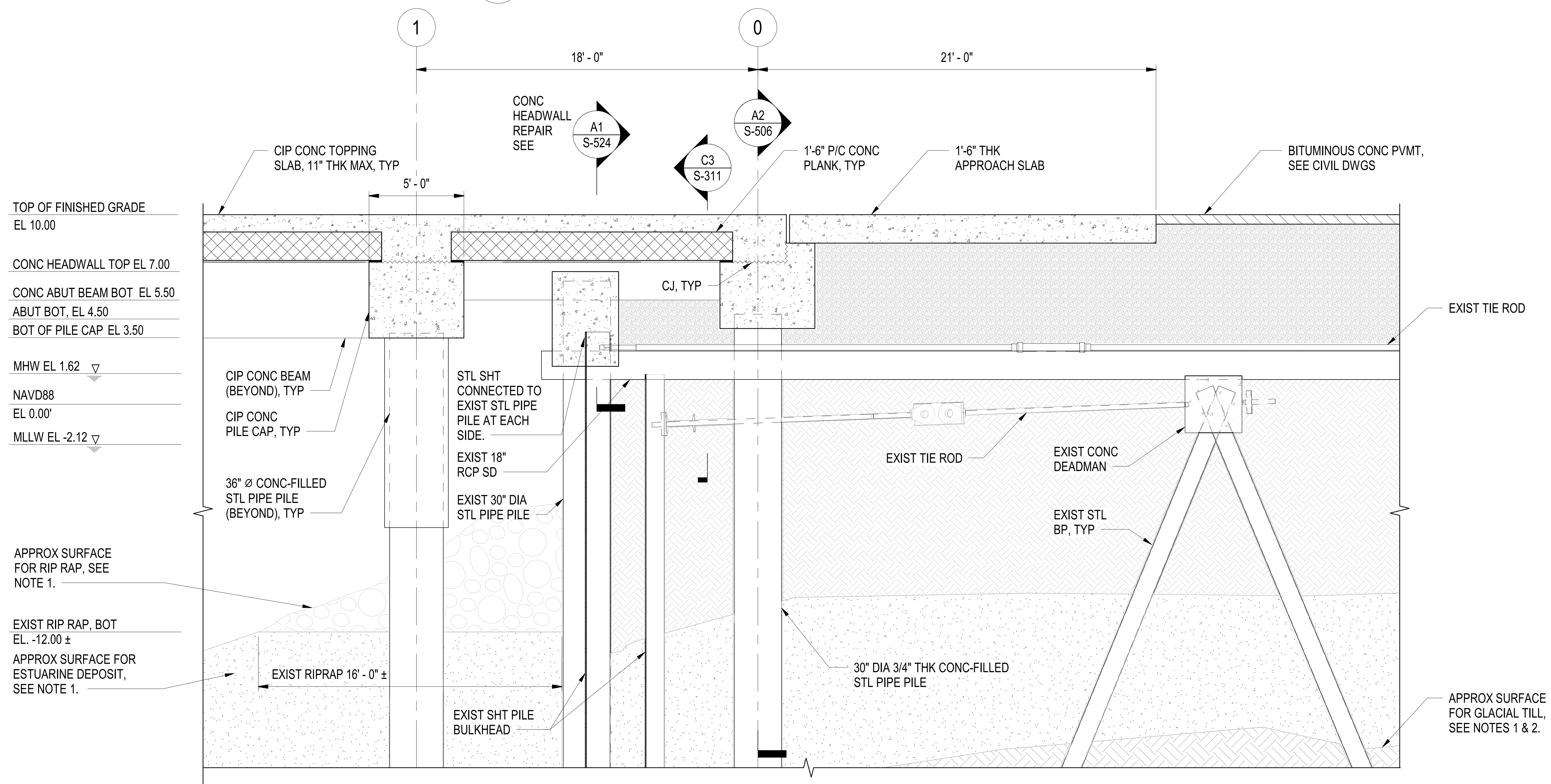
S-301



C2 PIER SECTION AT RAMP

SCALE: 1/4" = 1'-0" (S-108, S-115, S-122, S-129)



A2 PIER SECTION AT BULKHEAD & CONCRETE HEADWALL

SCALE: 1/4" = 1'-0" (S-109, S-116, S-123, S-130)

NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA. SEE ATTACHMENT A FOR ADDITIONAL INFORMATION ON AS-BUILT CONDITIONS.
2. SEE NOTE 5 ON S-109 FOR PILE CONSTRUCTION WITHIN EXISTING RIPRAP.
3. SEE NOTE 6 ON DRAWING S-109.
4. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR LONGITUDINAL SLOPE, TYPICAL.

LEGEND:

- EXISTING FILL
- EXISTING GLACIAL TILL
- EXISTING ESTUARINE DEPOSIT
- EXISTING AASHTO #57 CONSOLIDATED CRUSHED STONE

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

APPROVED James E. Donohue, R.A.

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ | DRW KM | CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER LONGITUDINAL SECTIONS - SHEET 2 OF 2

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936684

SHEET 115 OF 247

S-302

DRAWFORM REVISION: 25 AUGUST 2020

S-302
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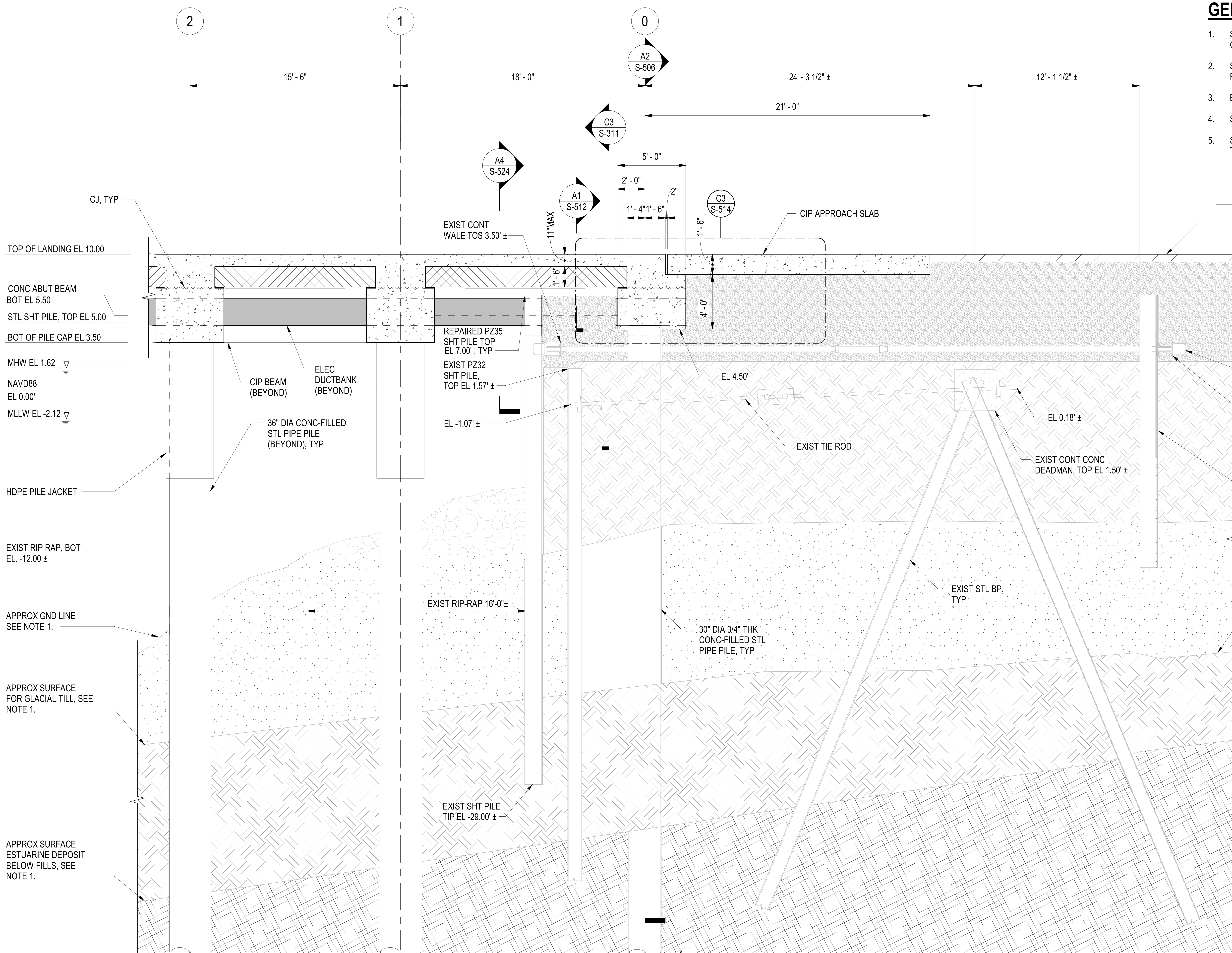
GENERAL SHEET NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA.
2. SEE ATTACHMENT A IN THE SPECIFICATIONS FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.
3. BASE STEEL FRAMING FOR AC ENCLOSURE NOT SHOWN FOR CLARITY.
4. SEE NOTE 5 ON S-109 FOR PILE CONSTRUCTION WITHIN EXISTING RIPRAP.
5. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR LONGITUDINAL SLOPE, TYPICAL.



A3 **TYPICAL SECTION AT BULKHEAD**

SCALE: 1/4" = 1'-0" (S-109) (S-116 ,S-123, S-130)

LEGEND:

- EXISTING AASHTO #57 CONSOLIDATED CRUSHED STONE
- EXISTING FILL
- EXISTING ESTUARINE DEPOSIT
- EXISTING GLACIAL TILL
- EXISTING HIGHLY WEATHERED ROCK

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

Stantec

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES	JZ	DRW	KM	CHK	MJF
PM/DM					MRI/CWJ
BRANCH MANAGER					NEK
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER ABUTMENT & APPROACH SLAB LONGITUDINAL SECTION

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936685
SHEET 116 OF 247

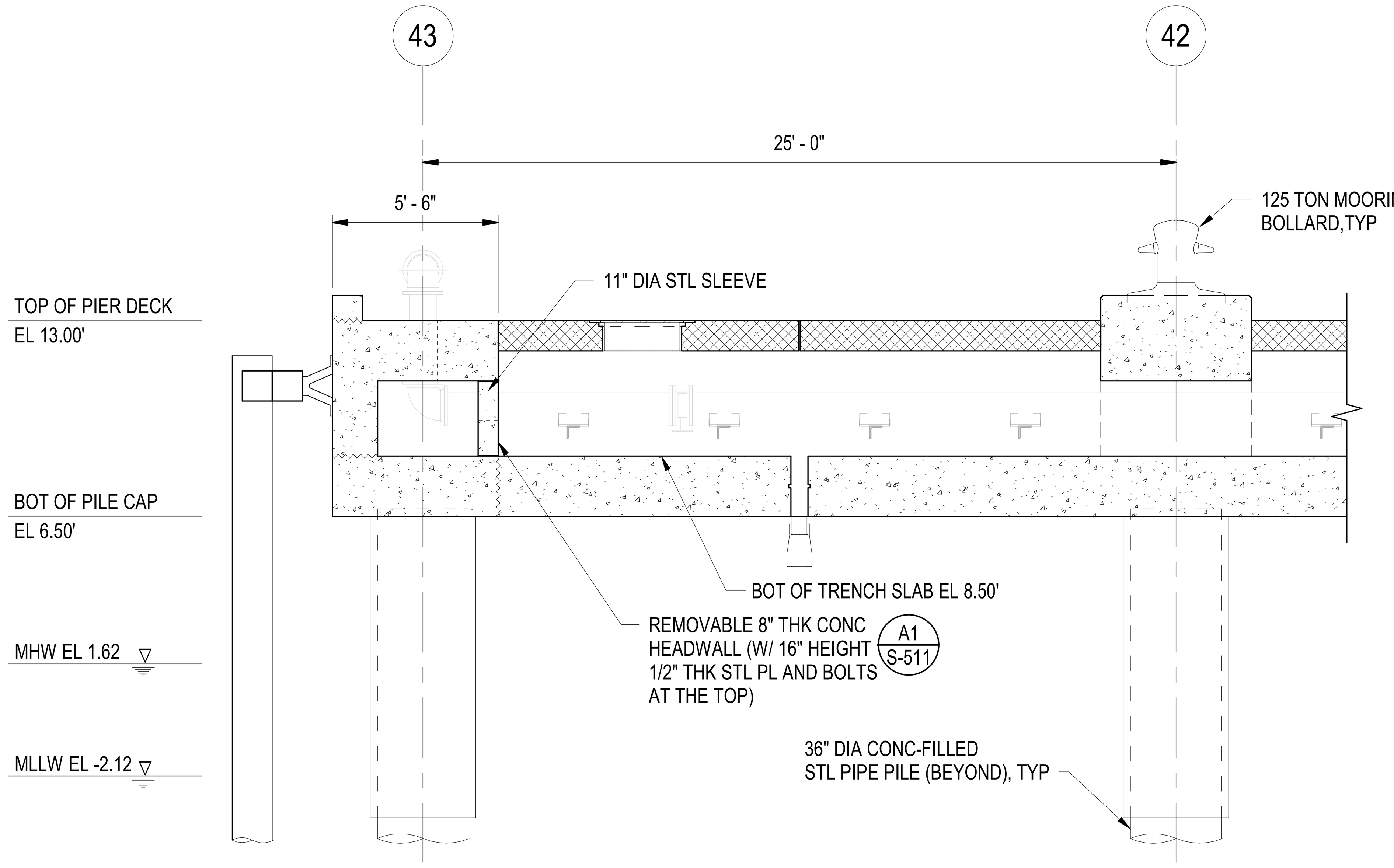
S-303

DRAWFORM REVISION: 20 AUGUST 2020

3/2/2026 7:36:07 PM S-303
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

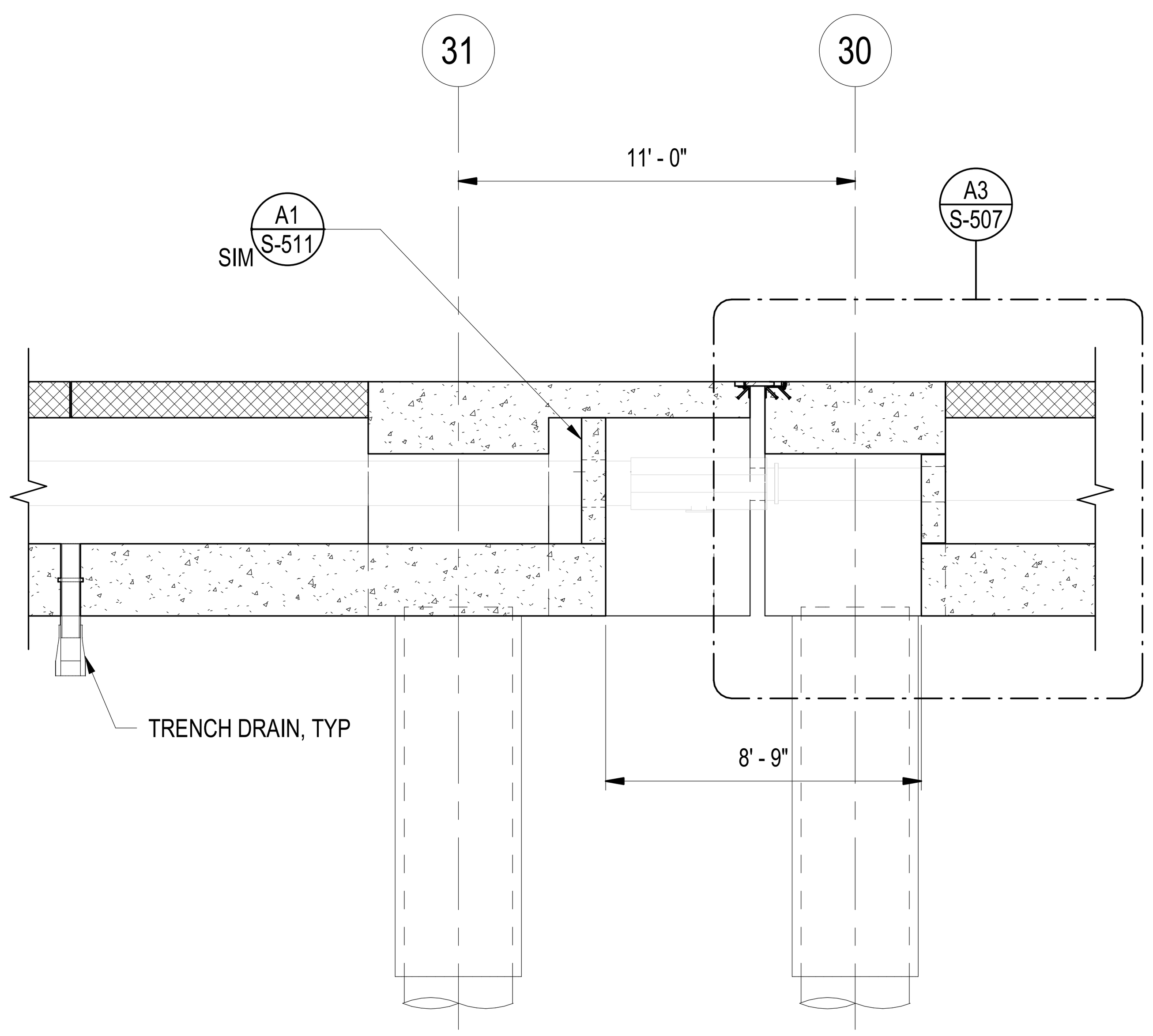
1 2 3 4 5

D



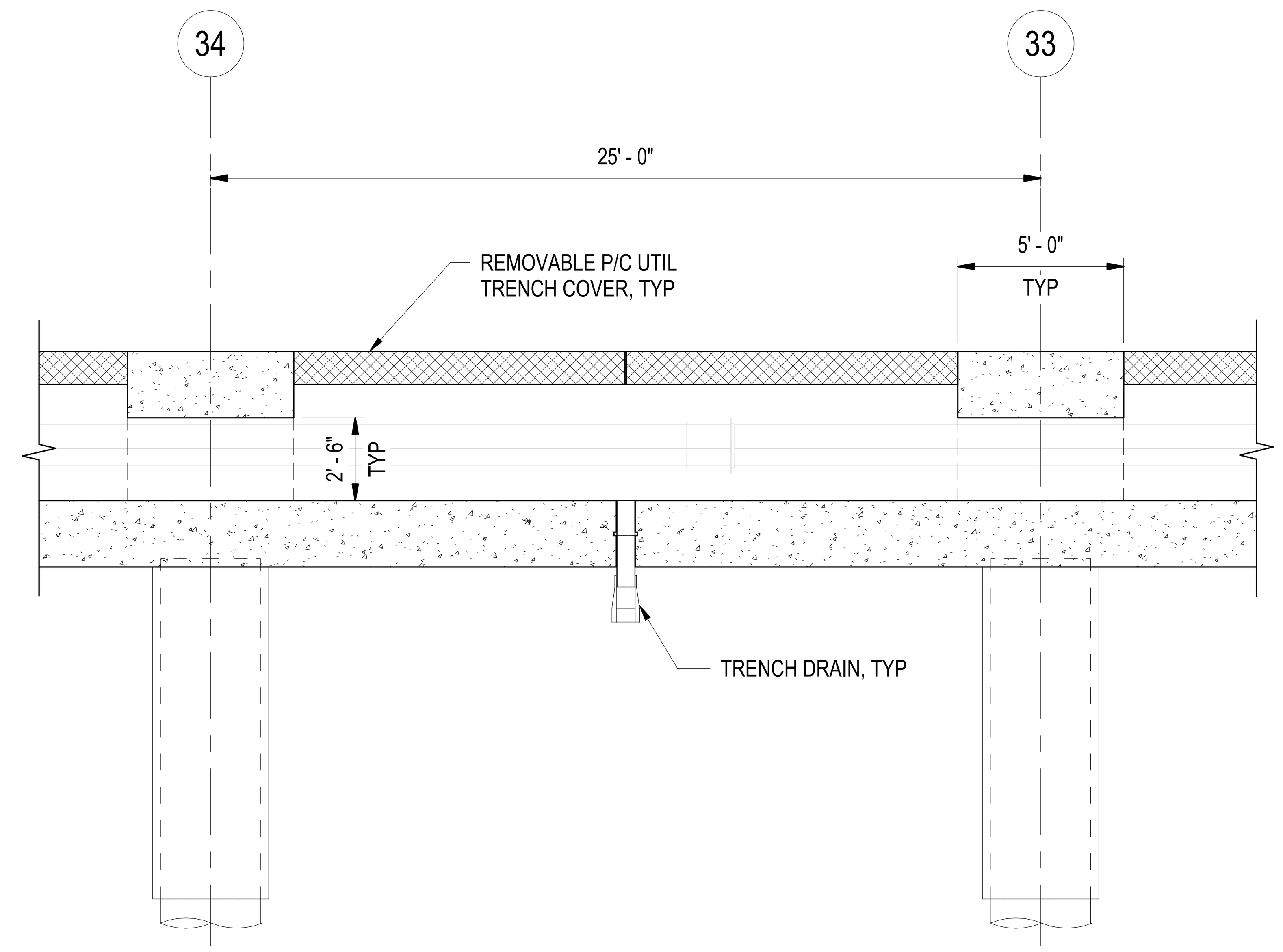
C1 END OF TRENCH SECTION

SCALE: 1/4" = 1'-0" (S-103)



C3 TYPICAL TRENCH SECTION AT EXPANSION JOINT

SCALE: 1/4" = 1'-0" (S-104, S-304)

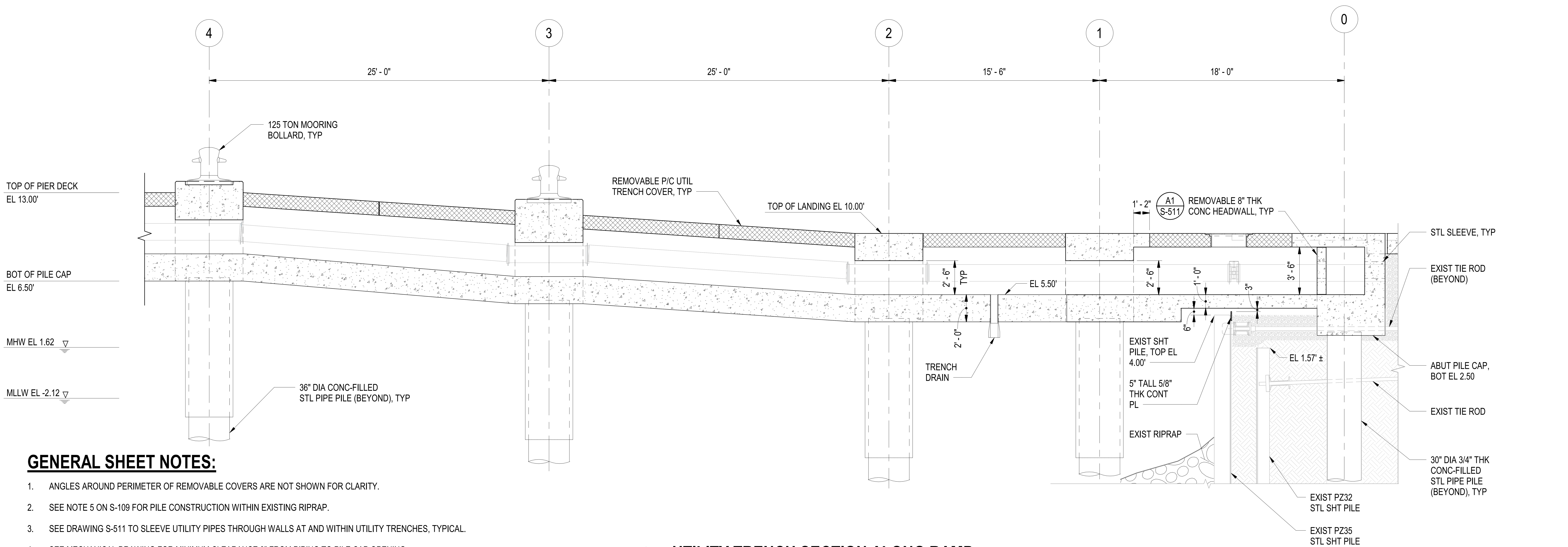


C4 TYPICAL TRENCH SECTION

SCALE: 1/4" = 1'-0" (S-104)

C

B



A

GENERAL SHEET NOTES:

1. ANGLES AROUND PERIMETER OF REMOVABLE COVERS ARE NOT SHOWN FOR CLARITY.
2. SEE NOTE 5 ON S-109 FOR PILE CONSTRUCTION WITHIN EXISTING RIPRAP.
3. SEE DRAWING S-511 TO SLEEVE UTILITY PIPES THROUGH WALLS AT AND WITHIN UTILITY TRENCHES, TYPICAL.
4. SEE MECHANICAL DRAWING FOR MINIMUM CLEARANCE 3" FROM PIPING TO PILE CAP OPENING.
5. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR LONGITUDINAL SLOPE, TYPICAL.

A3 UTILITY TRENCH SECTION ALONG RAMP

SCALE: 1/4" = 1'-0" (S-108, S-115, S-122, S-129)

1/4" = 1' - 0" 0 1' 5' 10' 15'

1 2 3 4 5

SYM | DESCRIPTION | DATE | APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW KM CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
LONGITUDINAL SECTIONS AT UTILITY TRENCH

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936686

SHEET 117 OF 247

S-304

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3/2/2026 7:36:09 PM S-304
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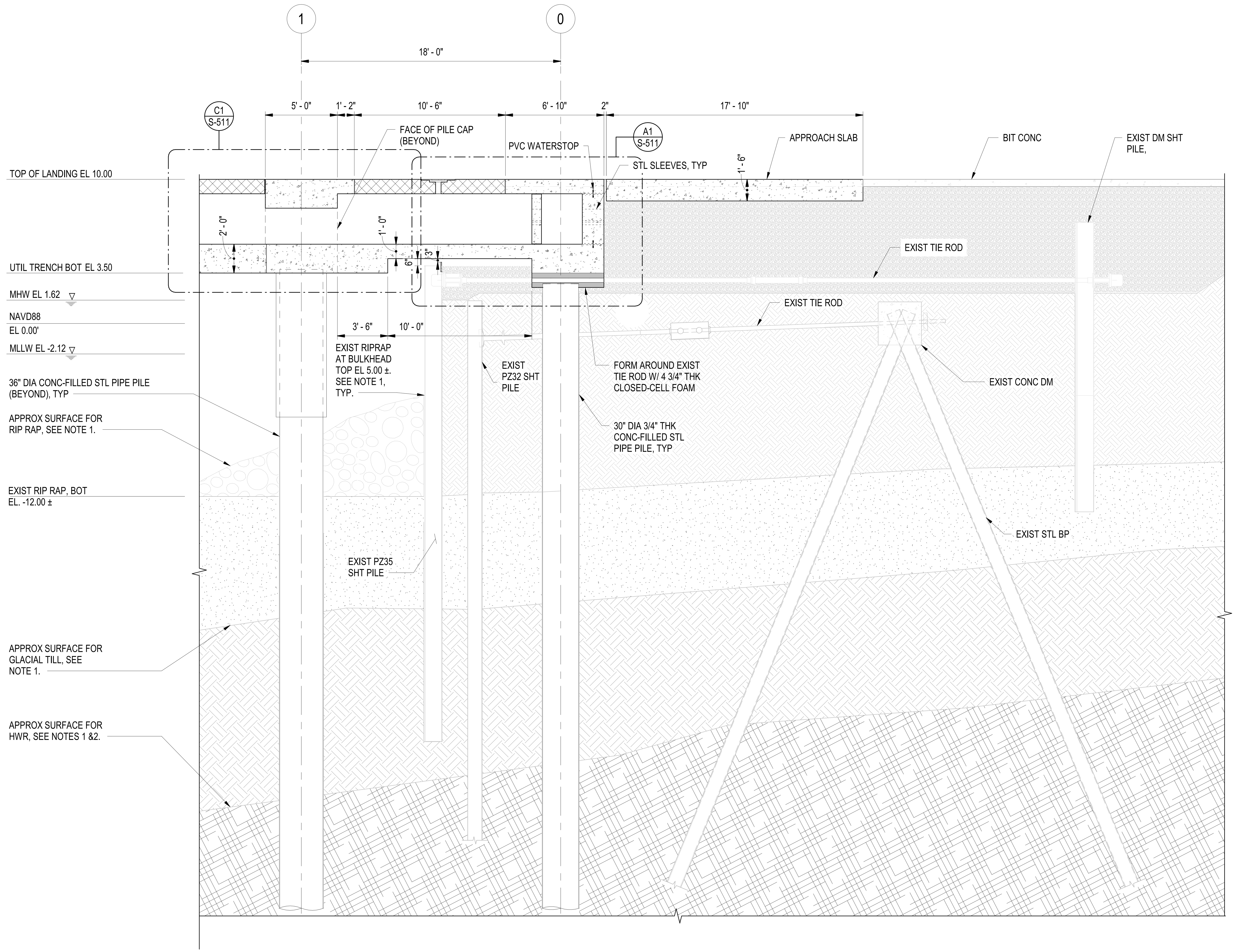

A3 **SECTION THROUGH UTILITY TRENCH AT ABUTMENT**

SCALE: 1/4" = 1'-0" (S-109, S-116, S-123, S-130)

GENERAL SHEET NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA. SEE ATTACHMENT A FOR ADDITIONAL INFORMATION ON AS-BUILT CONDITIONS.
2. L ANGLES AROUND PERIMETER OF REMOVABLE COVERS NOT SHOWN FOR CLARITY.
3. SEE NOTE 5 ON S-109 FOR PILE CONSTRUCTION WITHIN EXISTING RIPRAP.
4. SEE DRAWING S-511 TO SLEEVE UTILITY PIPES THROUGH WALLS AT AND WITHIN UTILITY TRENCHES, TYPICAL.
5. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR LONGITUDINAL SLOPE, TYPICAL.

LEGEND:

- EXISTING AASHTO #57 CONSOLIDATED CRUSHED STONE
- EXISTING FILL
- EXISTING ESTUARINE DEPOSIT
- EXISTING GLACIAL TILL
- EXISTING HIGHLY WEATHERED ROCK

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

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A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ	DRW KM	CHK MJF

PM/DM	MRI/CWJ
BRANCH MANAGER	NEK
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
LONGITUDINAL SECTION CROSS UTILITY TRENCH AT ABUTMENT

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936687

SHEET 118 OF 247

S-305

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:36:10 PM S-305
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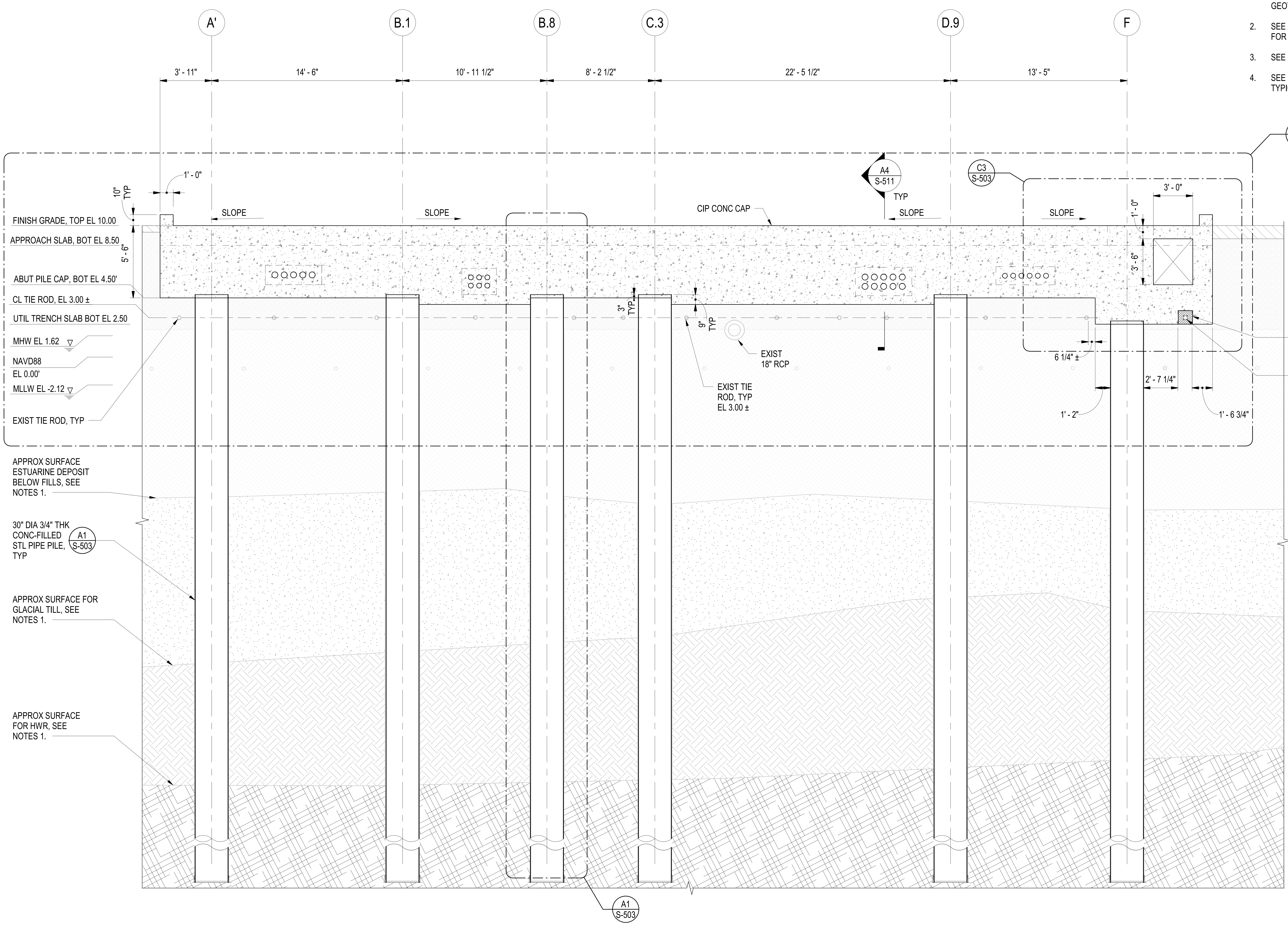
GENERAL SHEET NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA.
2. SEE ATTACHMENT A IN THE SPECIFICATIONS FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.
3. SEE NOTE 5 ON S-109 FOR PILE CONSTRUCTION WITHIN EXISTING RIPRAP.
4. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR TRANSVERSE SLOPE, TYPICAL.



A3 **PIER ABUTMENT SECTION**
SCALE: 1/4" = 1'-0" (S-109) (S-116, S-123, S-130)

LEGEND:

- AASHTO #57 CONSOLIDATED CRUSHED STONE
- EXISTING FILL
- EXISTING ESTUARINE DEPOSIT
- EXISTING GLACIAL TILL
- EXISTING HIGHLY WEATHERED ROCK

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES JZ / DRW KM / CHK MJF	
PM/DM	MRI/CWJ
BRANCH MANAGER	NEK
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

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NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
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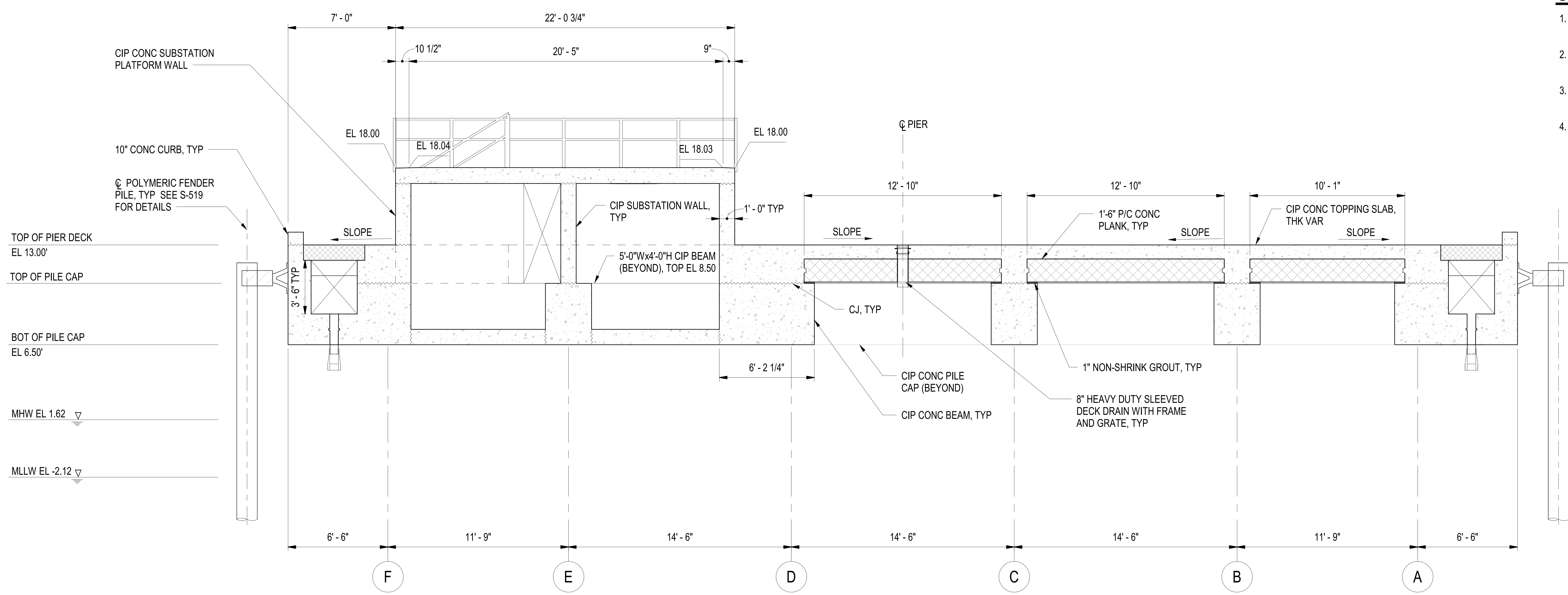
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER ABUTMENT & UTILITY TRENCH TRANSVERSE SECTION

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936688
SHEET 119 OF 247
S-306

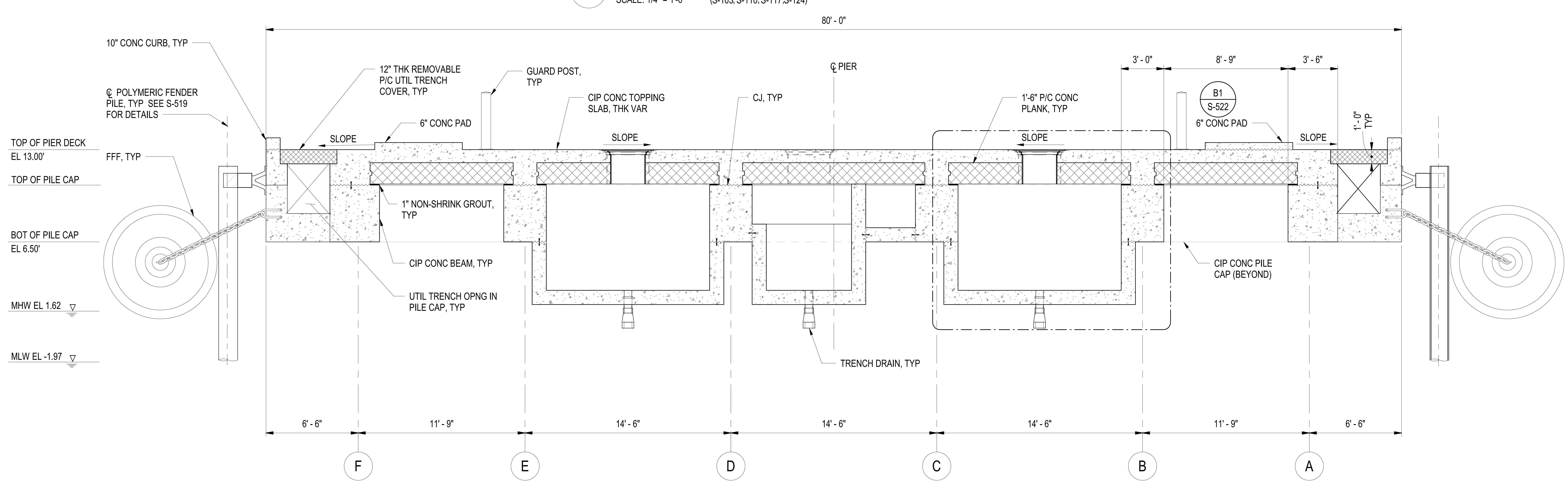
GENERAL SHEET NOTES:

1. PREFABRICATED SWITCH GEAR HOUSE ENCLOSURE NOT SHOWN FOR CLARITY.
2. SEE DRAWING S-402 FOR ADDITIONAL DETAILS FOR SUBSTATION.
3. STEEL ANGLES AROUND PERIMETER OF REMOVABLE COVERS NOT SHOWN FOR CLARITY.
4. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR TRANSVERSE SLOPE, TYPICAL.



C3 **SECTION AT PILE CAP BETWEEN BENT LINES 43 & 42**

SCALE: 1/4" = 1'-0" (S-103, S-110, S-117,S-124)



A3 **SECTION BETWEEN BENT LINES 34 & 33**

SCALE: 1/4" = 1'-0" (S-104, S-111, S-118,S-125)

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

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A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW KM CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER TRANSVERSE SECTIONS - SHEET 1 OF 5

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936689

SHEET 120 OF 247

S-307

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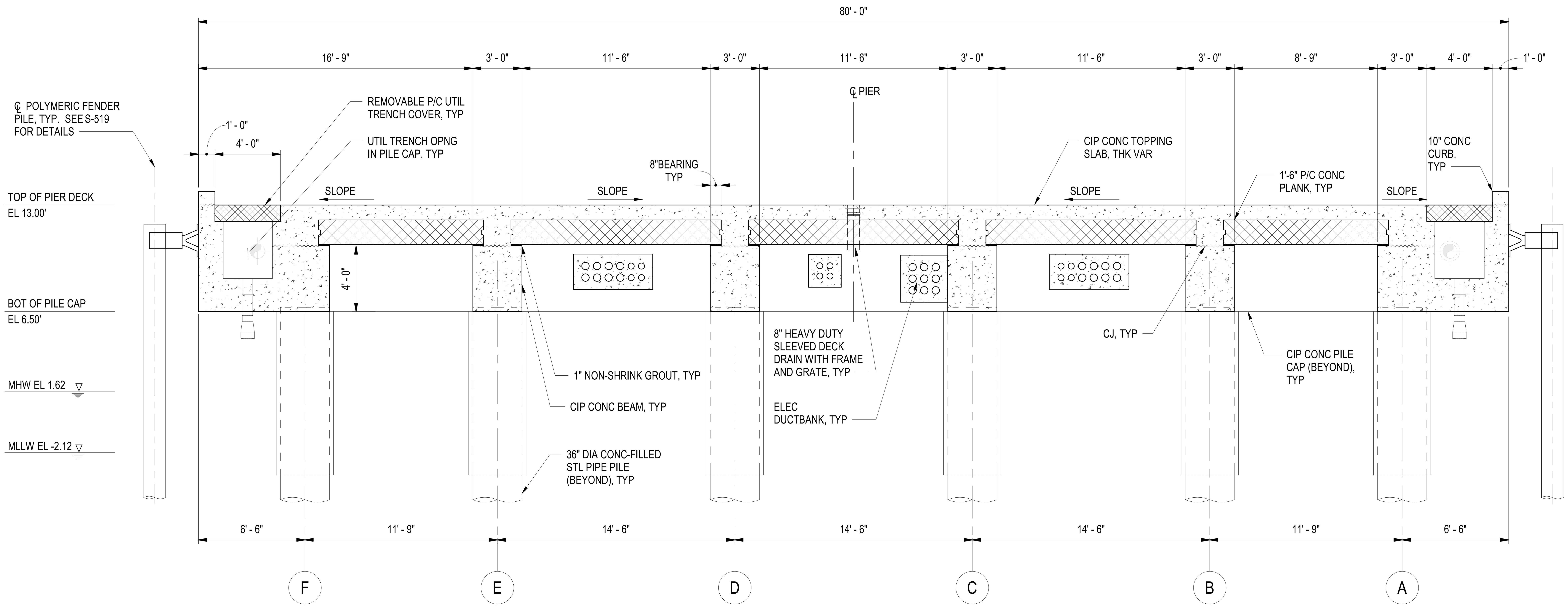
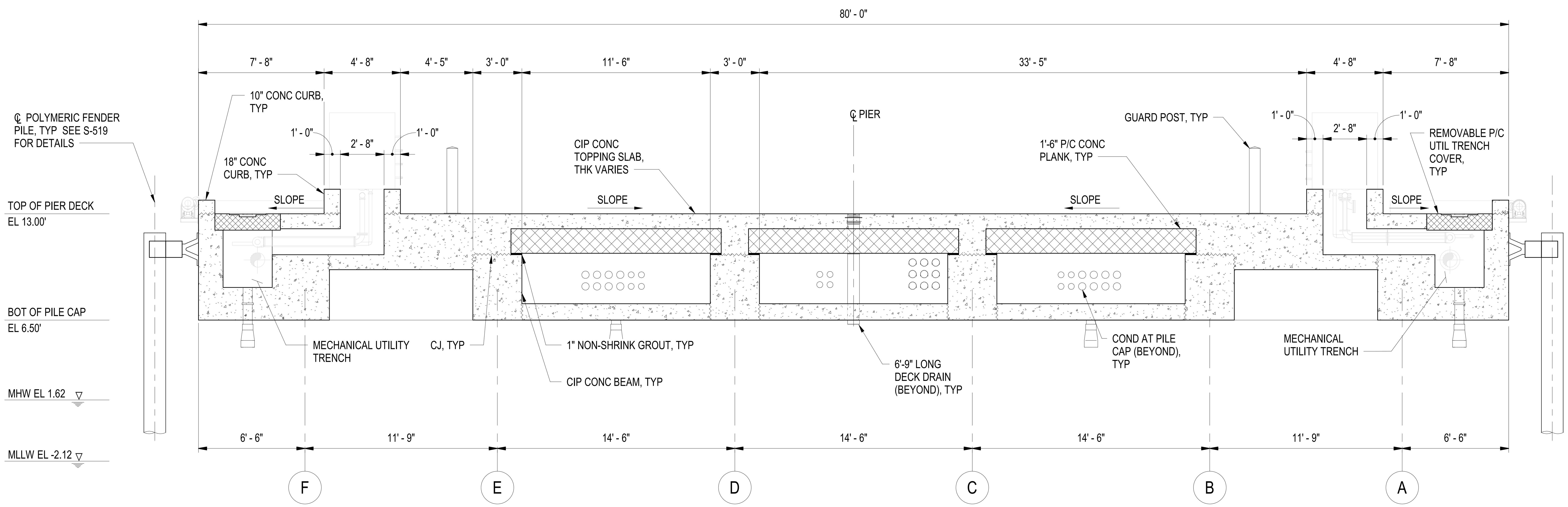
GENERAL SHEET NOTES:

1. L ANGLES AROUND PERIMETER OF REMOVABLE COVERS NOT SHOWN FOR CLARITY.
2. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR TRANSVERSE SLOPE, TYPICAL.



C3 **TYPICAL PIER SECTION**

SCALE: 1/4" = 1'-0" (S-104) (S-111, S-118, S-125)



A3 **TYPICAL SECTION THROUGH RAISED UTILITY TRENCH**

SCALE: 1/4" = 1'-0" (S-104) (S-111, S-118, S-125)

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

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ACTIVITY

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SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW KM CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER TRANSVERSE SECTIONS - SHEET 2 OF 5

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936690

SHEET 121 OF 247

S-308

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:36:14 PM S-308
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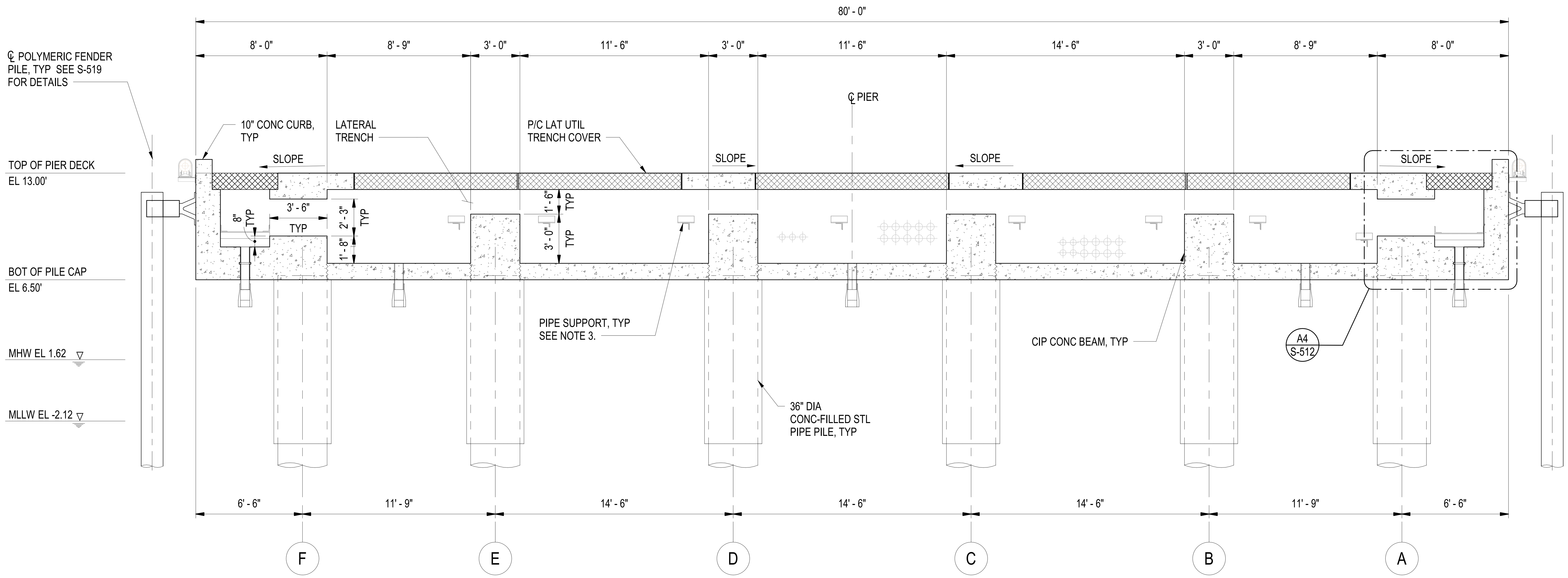
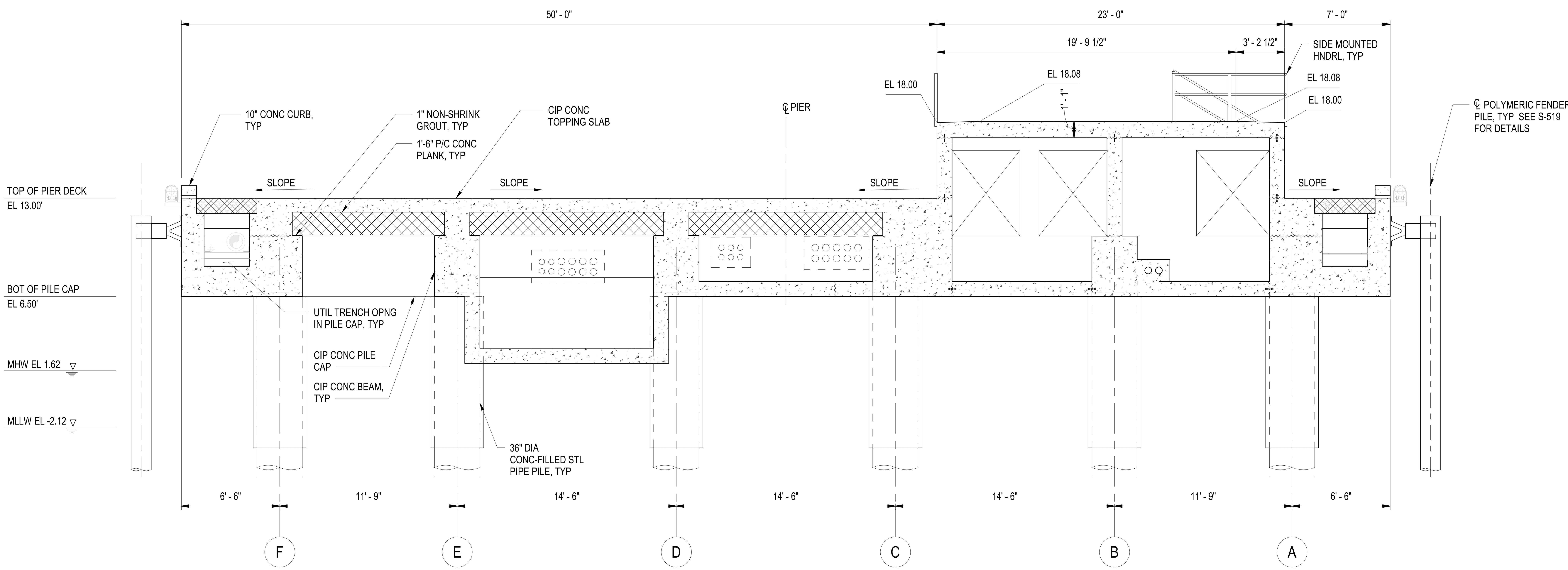
GENERAL SHEET NOTES:

1. L ANGLES AROUND PERIMETER OF REMOVABLE COVERS NOT SHOWN FOR CLARITY.
2. PREFAB ENCLOSURE NOT SHOWN FOR CLARITY.
3. SEE MECHANICAL DRAWINGS FOR DETAILS ON PIPE SUPPORTS, TYPICAL.
4. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR TRANSVERSE SLOPE, TYPICAL.



C3 SECTION AT LATERAL TRENCH
SCALE: 1/4" = 1'-0" (S-108) (S-115, S-122, S-129)



A3 PIER SECTION AT INBOARD SUBSTATION
SCALE: 1/4" = 1'-0" (S-108, S-115, S-122, S-129)

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM | DESCRIPTION | DATE | APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
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SATISFACTORY TO DATE 06 FEB 2026
DES JZ DRW KM CHK MJF
PM/DM MRI/CWJ
BRANCH MANAGER NEK
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER TRANSVERSE SECTIONS - SHEET 3 OF 5

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936691
SHEET 122 OF 247
S-309
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3/2/2026 7:36:16 PM S-309
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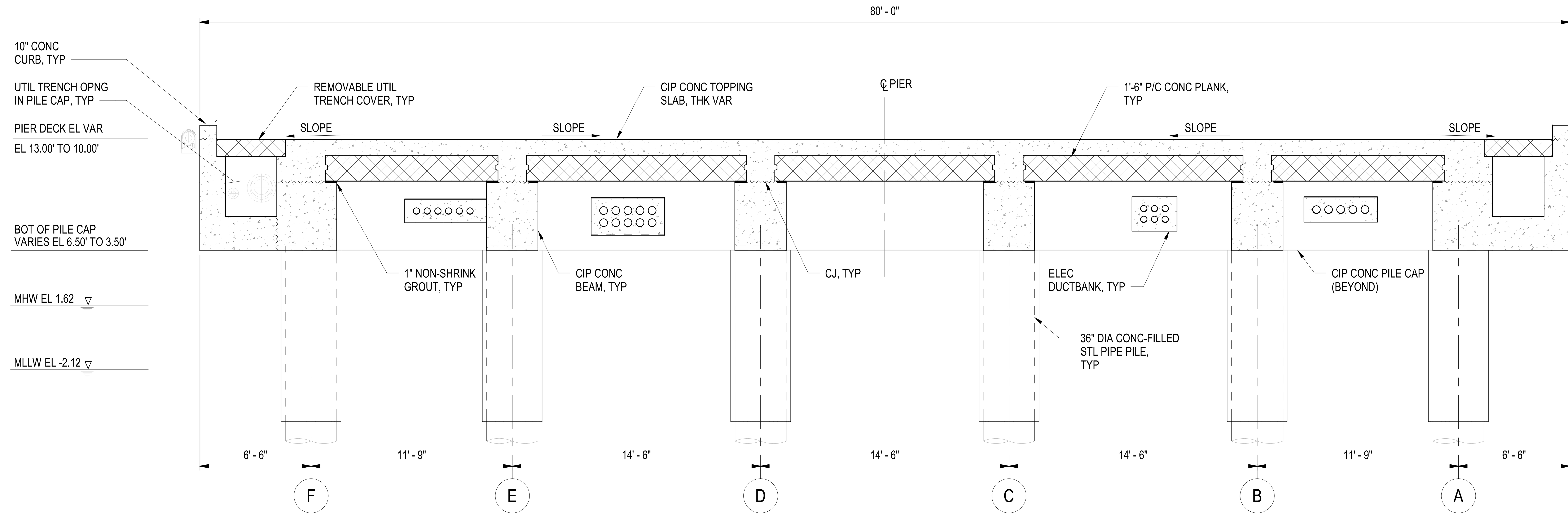
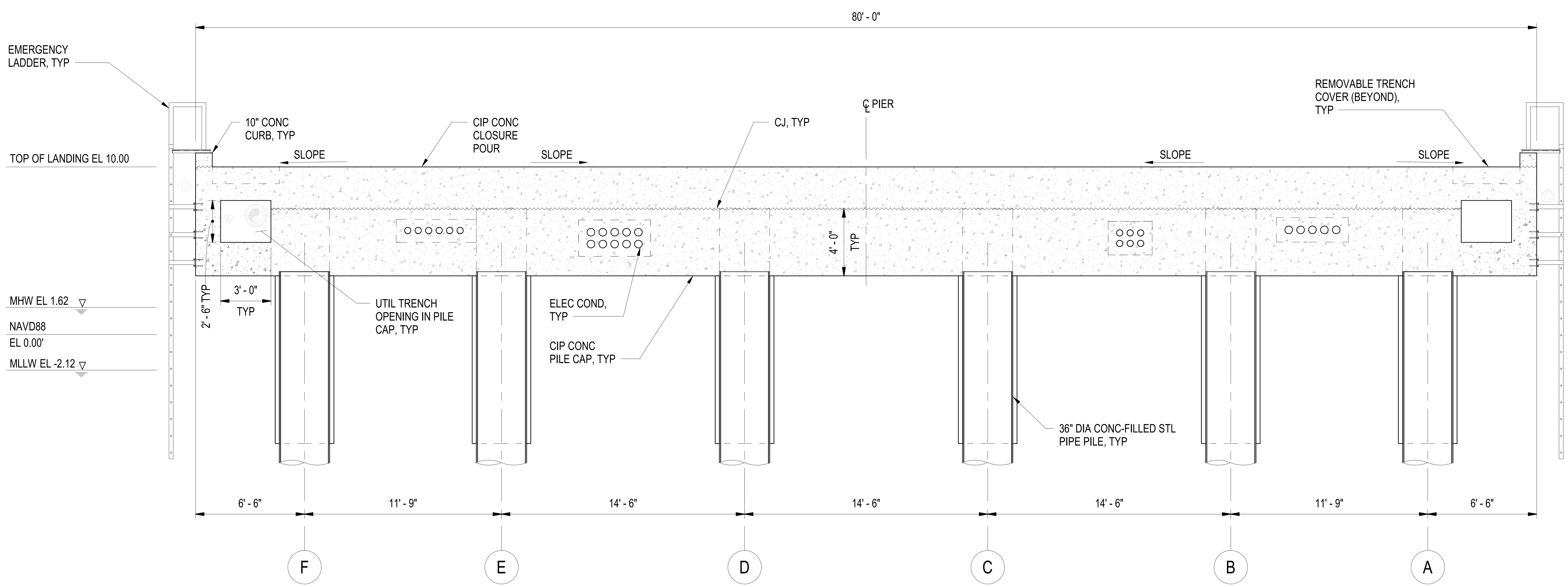
GENERAL SHEET NOTES:

1. L ANGLES AROUND PERIMETER OF REMOVABLE COVERS NOT SHOWN FOR CLARITY.
2. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR TRANSVERSE SLOPE, TYPICAL.



C3 **SECTION AT RAMP BETWEEN BENT LINES 3 & 2**

SCALE: 1/4" = 1'-0" (S-109) (S-116, S-123, S-130)



A3 **SECTION AT BOTTOM OF RAMP & BENT LINE 2**

SCALE: 1/4" = 1'-0" (S-109) (S-116, S-123, S-130)

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

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ACTIVITY

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SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW KM CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIER TRANSVERSE SECTIONS - SHEET 4 OF 5

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936692

SHEET 123 OF 247

S-310

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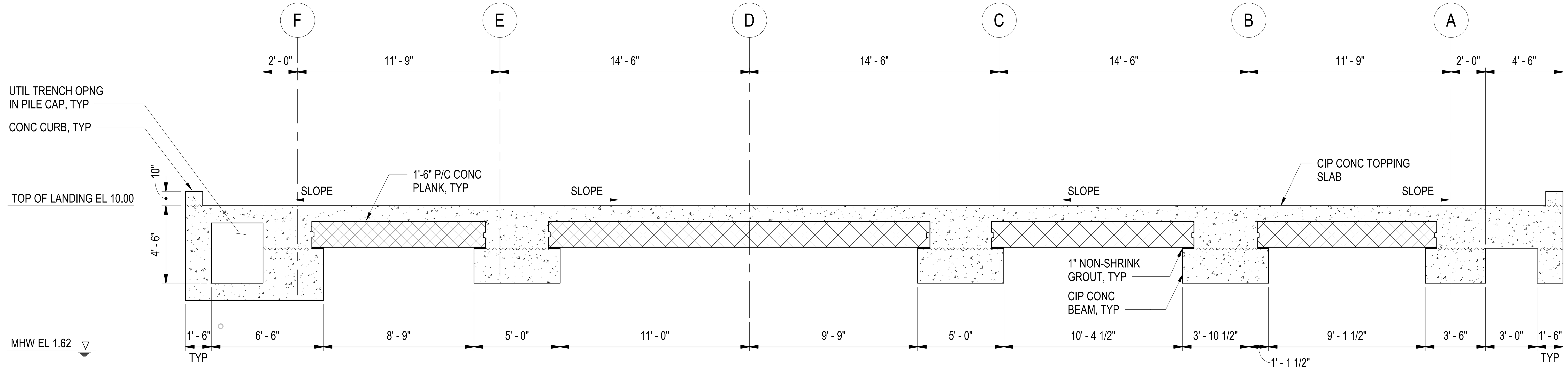
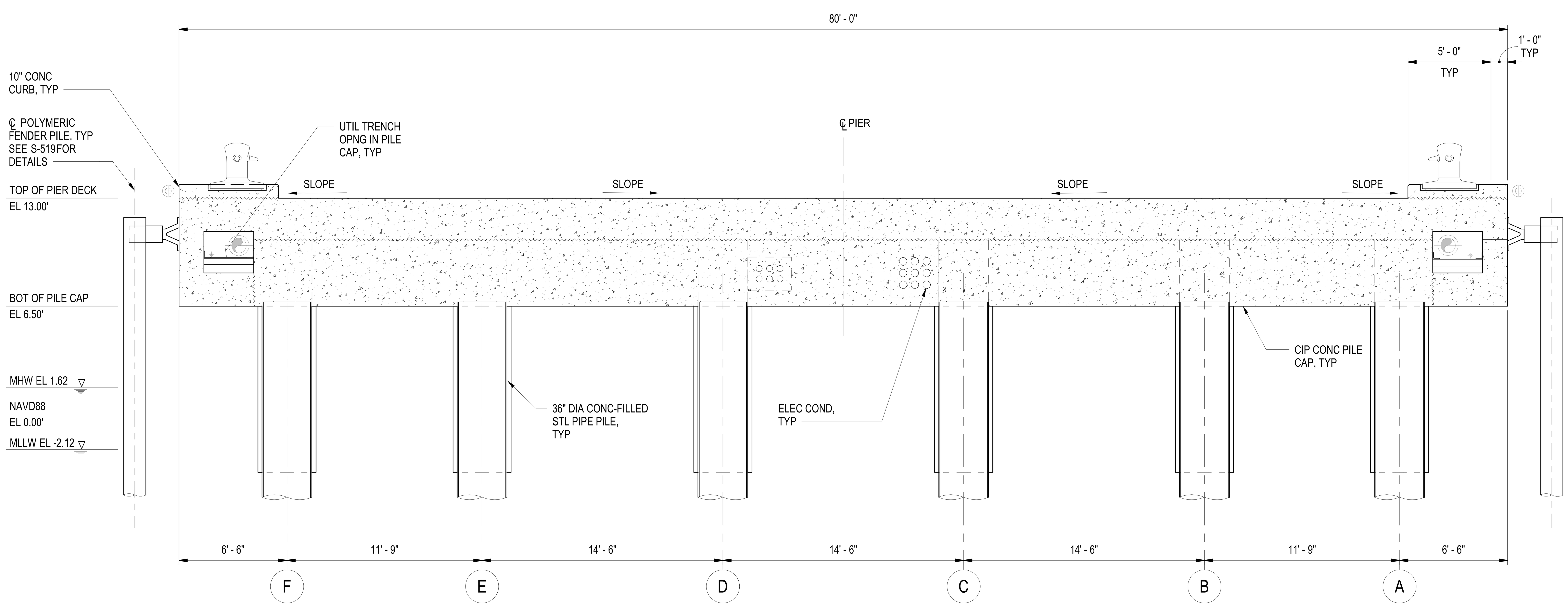


C3 SECTION AT PILE CAP BETWEEN BULKHEAD & BENT LINE 0

SCALE: 1/4" = 1'-0" (S-109, S-116, S-123, S-130, S-302, S-303)



A3 PIER SECTION AT MOORING BOLLARD ABOVE UTILITY TRENCH

SCALE: 1/4" = 1'-0" (S-105, S-112, S-119, S-126)

1/4" = 1' - 0" 0 1' 5' 10' 15'

GENERAL SHEET NOTES:

1. ANGLES AROUND PERIMETER OF REMOVABLE COVERS NOT SHOWN FOR CLARITY.
2. SEE MECHANICAL DRAWINGS FOR DETAILS ON SANITARY SEWER LINE.
3. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR TRANSVERSE SLOPE, TYPICAL.

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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DES JZ DRW KM CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

PIER TRANSVERSE SECTIONS - SHEET 5 OF 5

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936693

SHEET 124 OF 247

S-311

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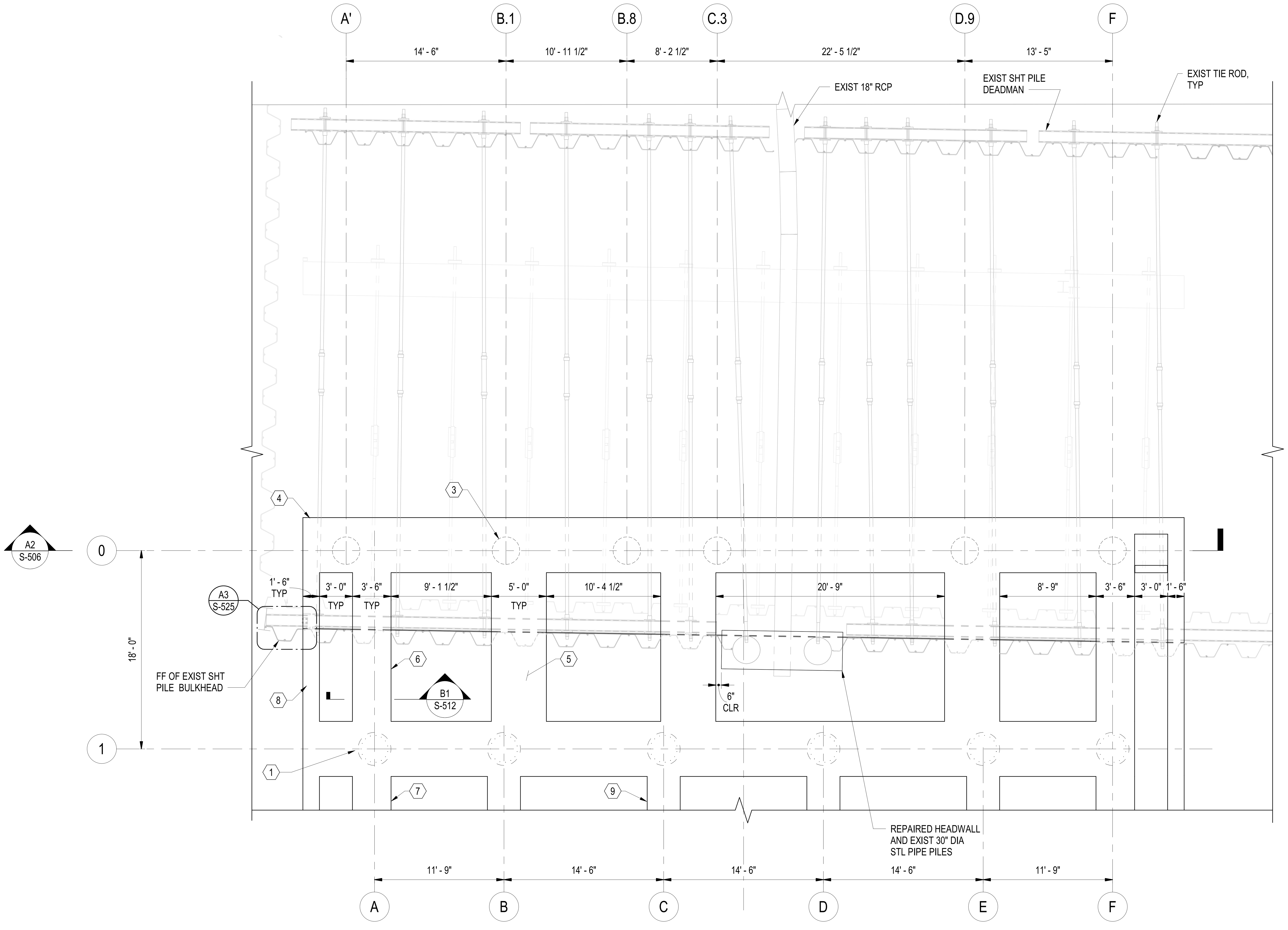
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

GENERAL SHEET NOTES:

1. CONCRETE PIER DECK NOT SHOWN FOR CLARITY.
2. PROVIDE SLEEVE WITH PROPER DIAMETER FOR UTILITY PIPES AT AND WITHIN UTILITY TRENCHES. SEE DRAWING S-511 FOR DETAILS.

SHEET KEYNOTES:

1. 36" DIAMETER CONCRETE-FILLED STEEL PIPE PILE, SEE DRAWING S-502.
3. 30" DIAMETER x 3/4" THICK STEEL PIPE PILE (6 TOTAL), SEE DRAWING S-503 TYPICAL.
4. 5'-0"x5'-6" CONCRETE PILE CAP, SEE SHEET S-311.
5. 5'-0" WIDE BY 2'-0" DEEP CAST-IN-PLACE CONCRETE BEAM (TOTAL 2), TYPICAL.
6. 3'-6" WIDE BY 2'-0" DEEP CAST-IN-PLACE CONCRETE BEAM (TOTAL 2), TYPICAL.
7. 3'-6" WIDE BY 4'-0" DEEP CAST-IN-PLACE CONCRETE BEAM, TYPICAL.
8. 1'-6" WIDE BY 2'-0" DEEP CAST-IN-PLACE CONCRETE BEAM LOCATED BETWEEN BENTS 0 & 1 ONLY.
9. 3'-0" WIDE BY 4'-0" DEEP CAST-IN-PLACE CONCRETE BEAM, TYPICAL.



A3 ABUTMENT FRAMING PLAN

SCALE: 3/16" = 1'-0"

PLAN NORTH

LEGEND:

#	BENT DESIGNATION
A	PILE ROW DESIGNATION
	36" Ø PILE BELOW
	30" Ø PILE BELOW

3/16" = 1' - 0" 0 5' 10' 20'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

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DES JZ DRW KCJ CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT
NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

ABUTMENT & LANDSIDE FRAMING PART PLAN

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936694

SHEET 125 OF 247

S-401

DRAWFORM REVISION: 25 AUGUST 2020

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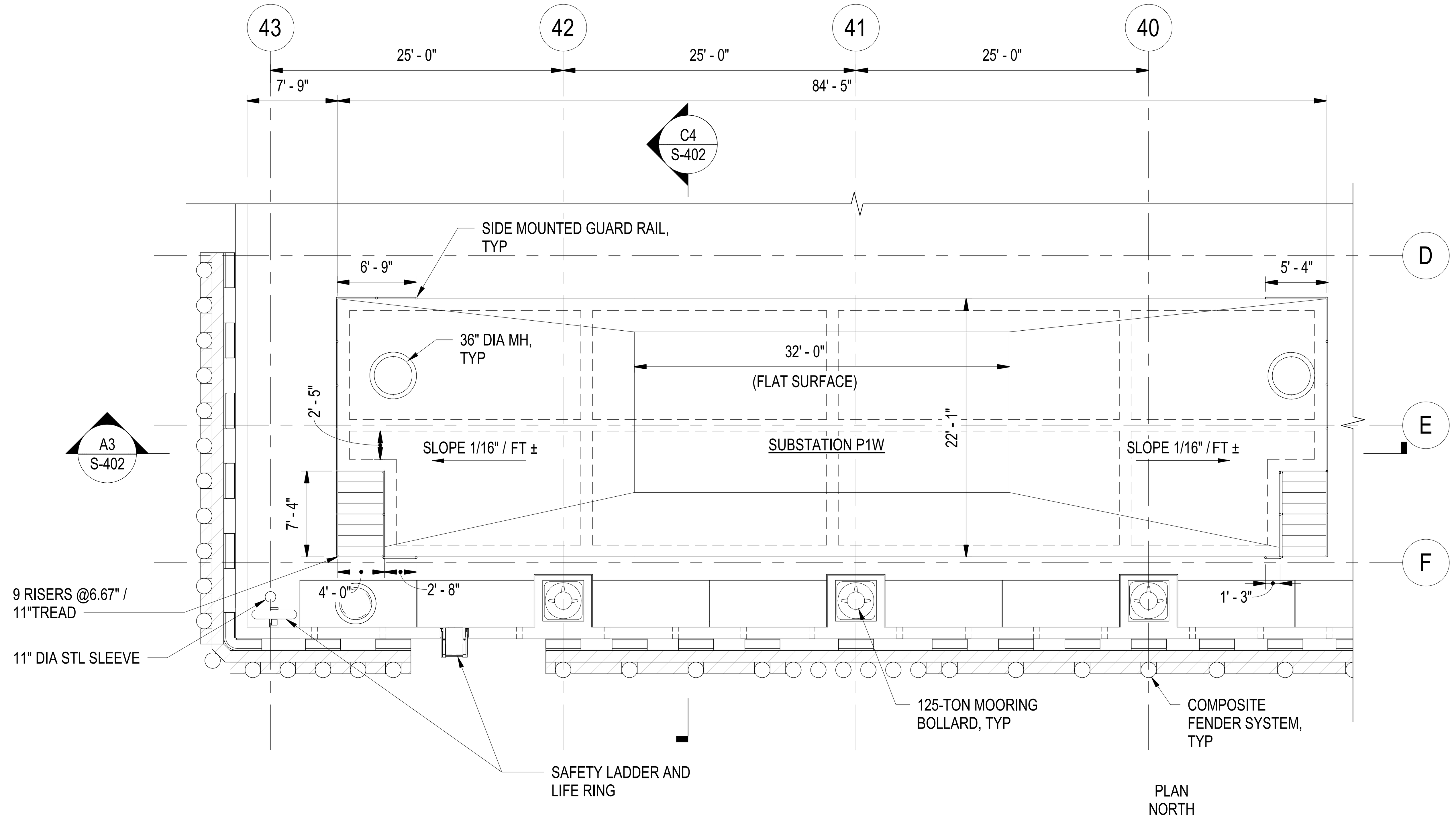
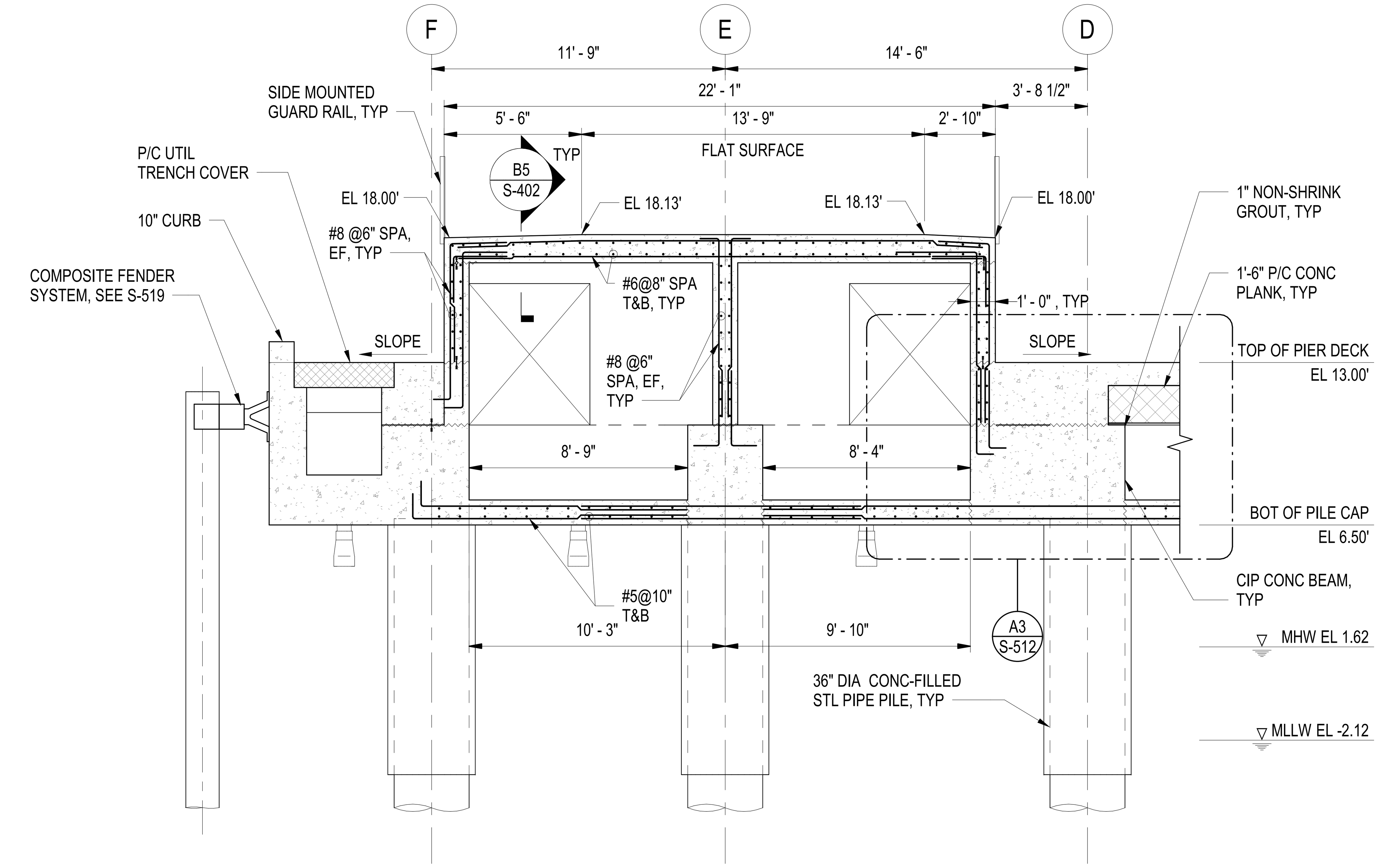
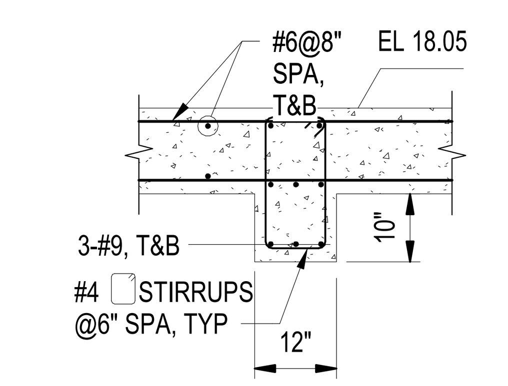


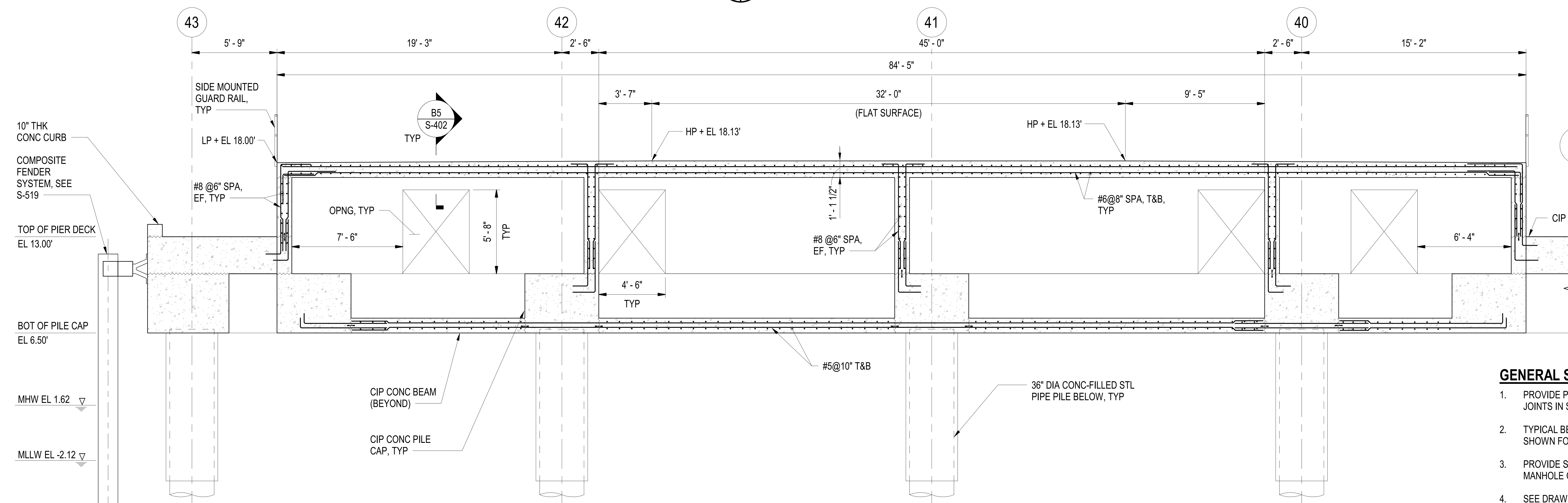
C2 OUTBOARD SUBSTATION PLATFORM PLAN
SCALE: 1/8" = 1'-0"



C4 OUTBOARD SUBSTATION PLATFORM SECTION
SCALE: 1/4" = 1'-0"



B5 DETAIL
SCALE: 1/2" = 1'-0" (S-402, S-403)



A3 OUTBOARD SUBSTATION LONGITUDINAL SECTION
SCALE: 1/4" = 1'-0"

GENERAL SHEET NOTES:

1. PROVIDE PVC WATERSTOPS AT ALL CONSTRUCTION JOINTS IN SUBSTATION PLATFORM WALLS AND SLABS.
2. TYPICAL BEAM AND PILE CAP REINFORCEMENT NOT SHOWN FOR CLARITY.
3. PROVIDE SUPPLEMENTAL REINFORCEMENT AT MANHOLE OPENINGS AS SHOWN ON S-517.
4. SEE DRAWING S-522 FOR ADDITIONAL REINFORCEMENT AT STAIRS AND HAND RAIL DETAILS.
5. PREFABRICATED SUBSTATION ENCLOSURE AND SWITCH GEAR HOUSE ARE NOT SHOWN FOR CLARITY.

1/2" = 1'-0" 0 6" 1' 2' 3' 4' 5'
1/8" = 1'-0" 0 5' 10' 15' 30'
1/4" = 1'-0" 0 1' 5' 10' 15'

SYM | DESCRIPTION | DATE | APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES JZ	DRW KCJ	CHK MJF	
PM/DM			MRI/CWJ
BRANCH MANAGER			NEK
CHIEF ENG/ARCH			JWJ
FIRE PROTECTION			HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
CI CORE - HAMPTON ROADS
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
OUTBOARD SUBSTATION PART PLAN

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936699
SHEET 126 OF 247
S-402

DRAWFORM REVISION: 25 AUGUST 2020

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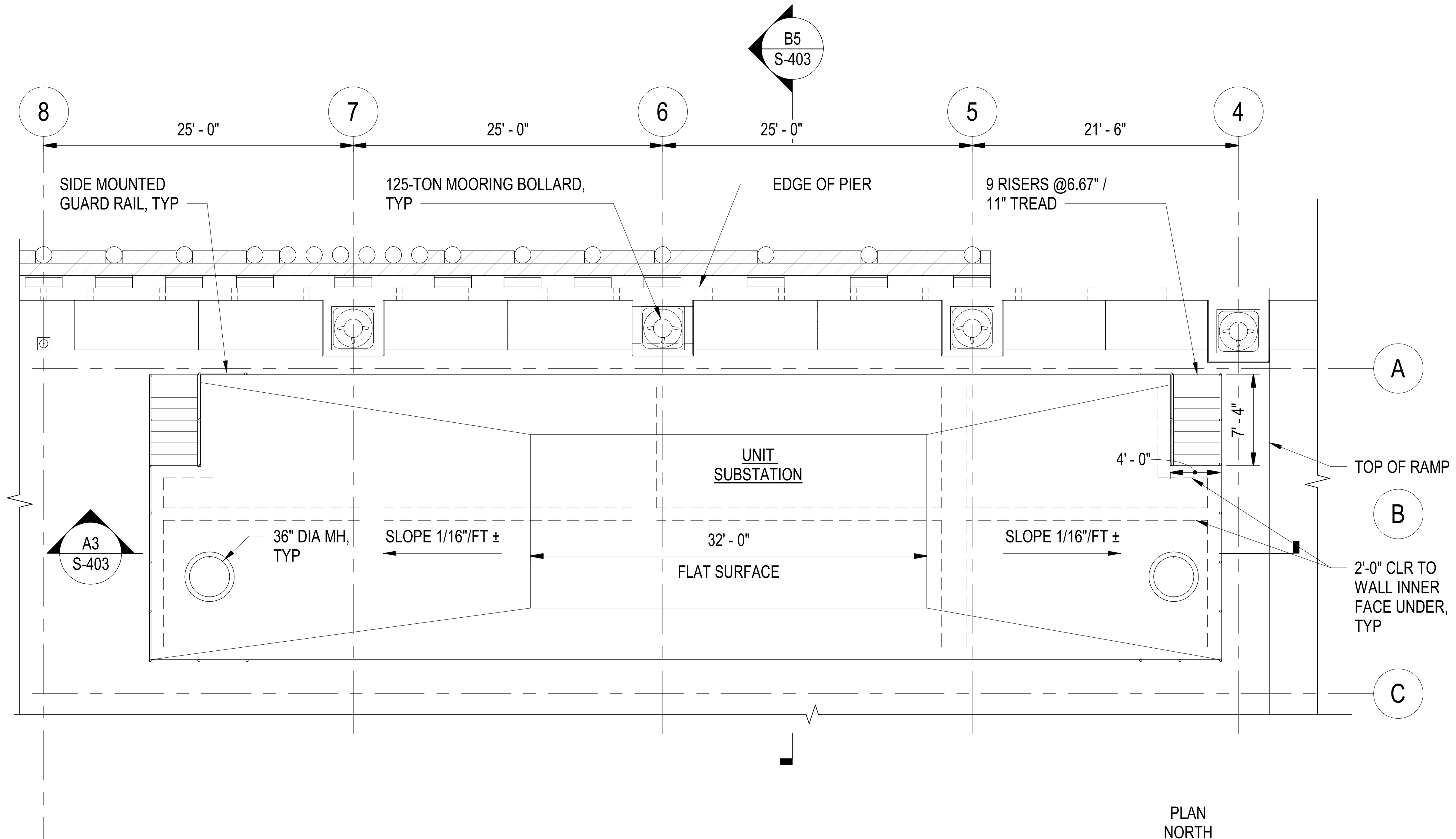
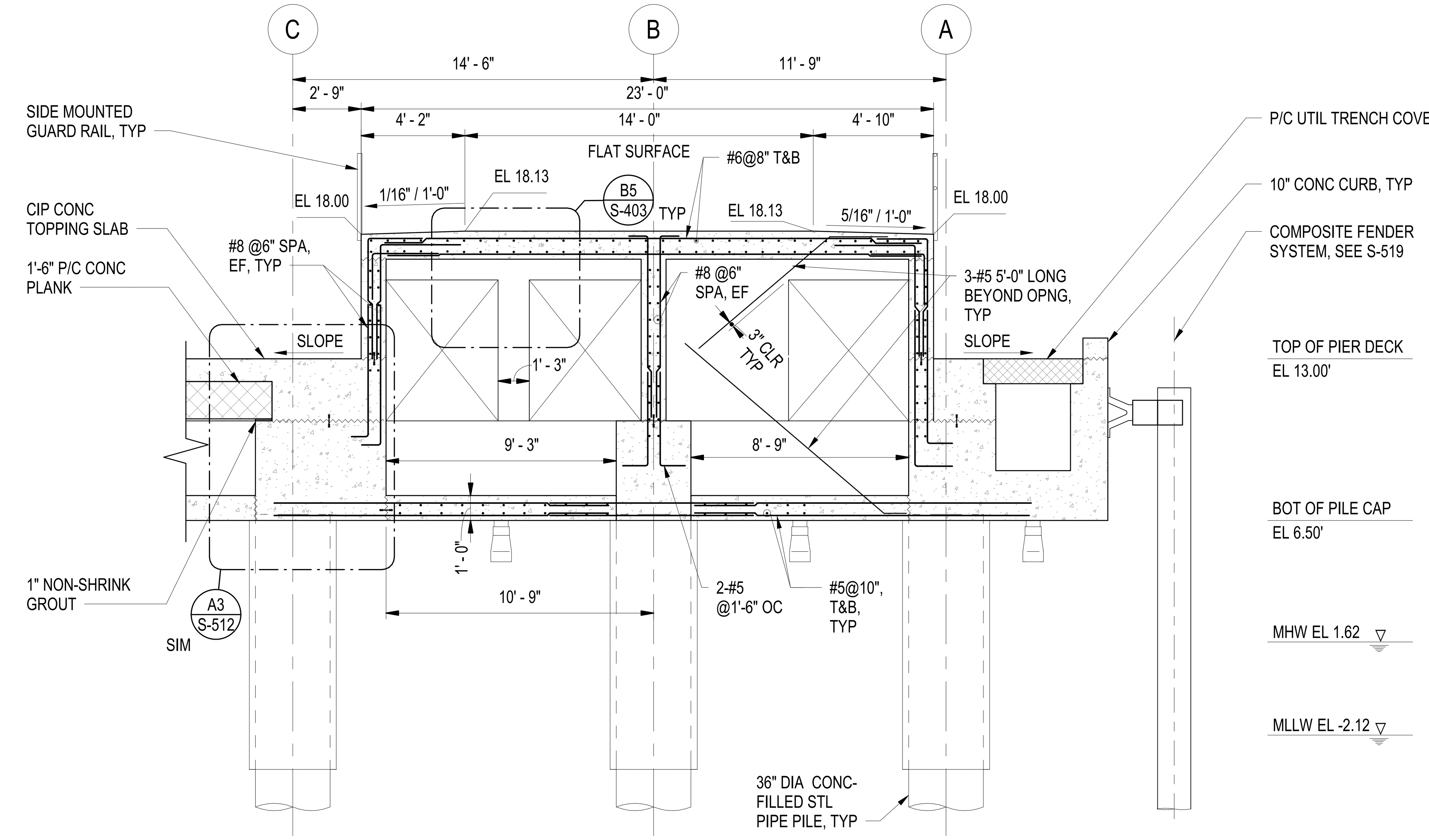
S-402



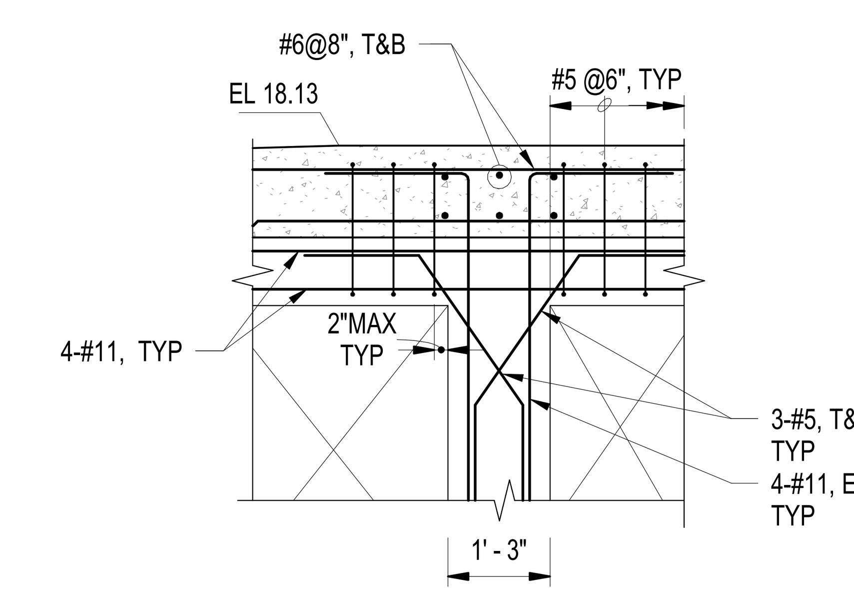
C2 INBOARD SUBSTATION PLATFORM PLAN

SCALE: 1/8" = 1'-0"



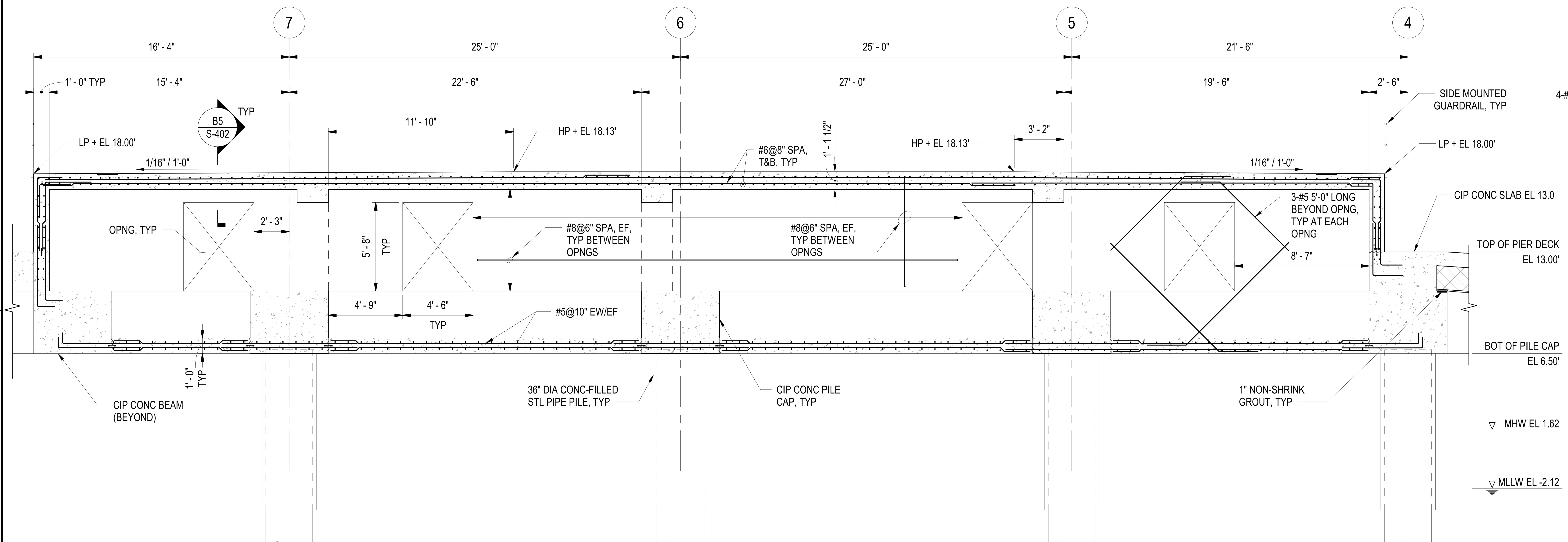
C4 INBOARD SUBSTATION PLATFORM SECTION

SCALE: 1/4" = 1'-0"



B5 REBAR DETAIL

SCALE: 1/2" = 1'-0" (S-403)



A3 INBOARD SUBSTATION PLATFORM LONGITUDINAL SECTION

SCALE: 1/4" = 1'-0"

GENERAL SHEET NOTES:

1. PROVIDE PVC WATERSTOPS AT ALL CONSTRUCTION JOINTS IN SUBSTATION PLATFORM WALLS AND SLABS.
2. TYPICAL BEAM AND PILE CAP REINFORCEMENT NOT SHOWN FOR CLARITY.
3. PROVIDE SUPPLEMENTAL REINFORCEMENT AT MANHOLE OPENINGS AS SHOWN ON S-517.
4. SEE DRAWING S-522 FOR ADDITIONAL REINFORCEMENT AT STAIRS AND HAND RAIL DETAILS.
5. PREFABRICATED SUBSTATION ENCLOSURE AND SWITCH GEAR HOUSE ARE NOT SHOWN FOR CLARITY.

1/2" = 1'-0" 0 6" 1' 2' 3' 4' 5'

1/4" = 1'-0" 0 1' 5' 10' 15'

1/8" = 1'-0" 0 5' 10' 15' 30'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW KCJ CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
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NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
INBOARD SUBSTATION PART PLAN

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936700

SHEET 127 OF 247

S-403

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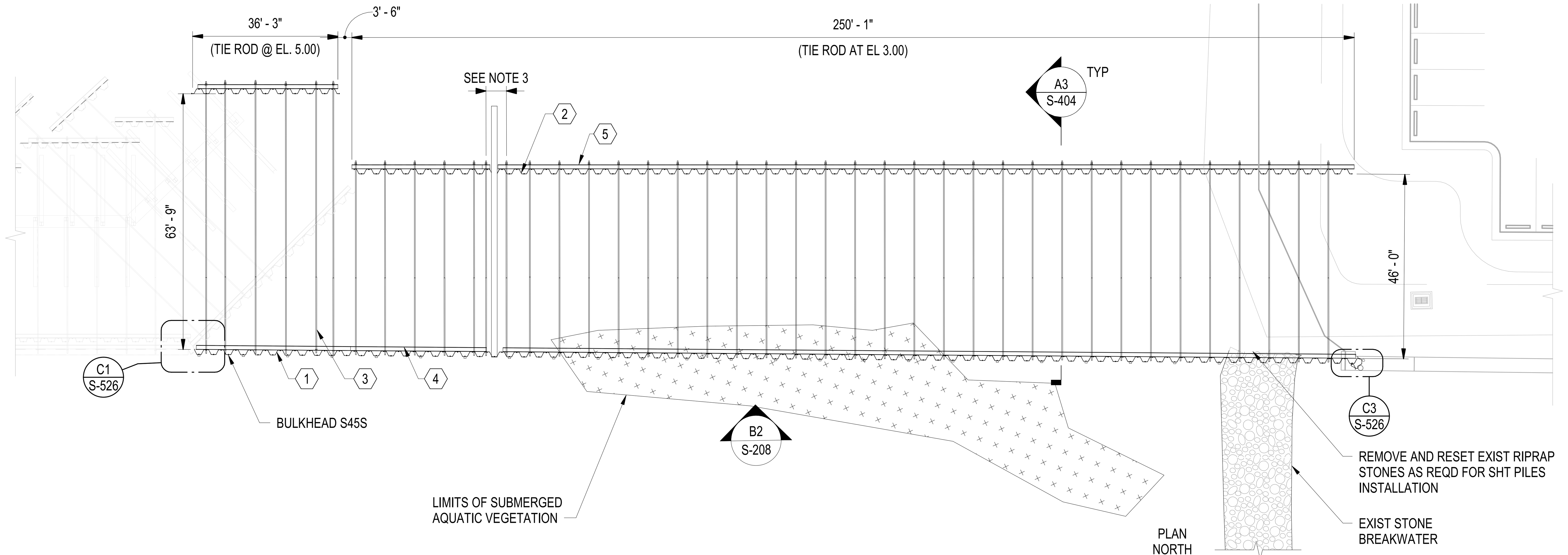
S-403

GENERAL SHEET NOTES:

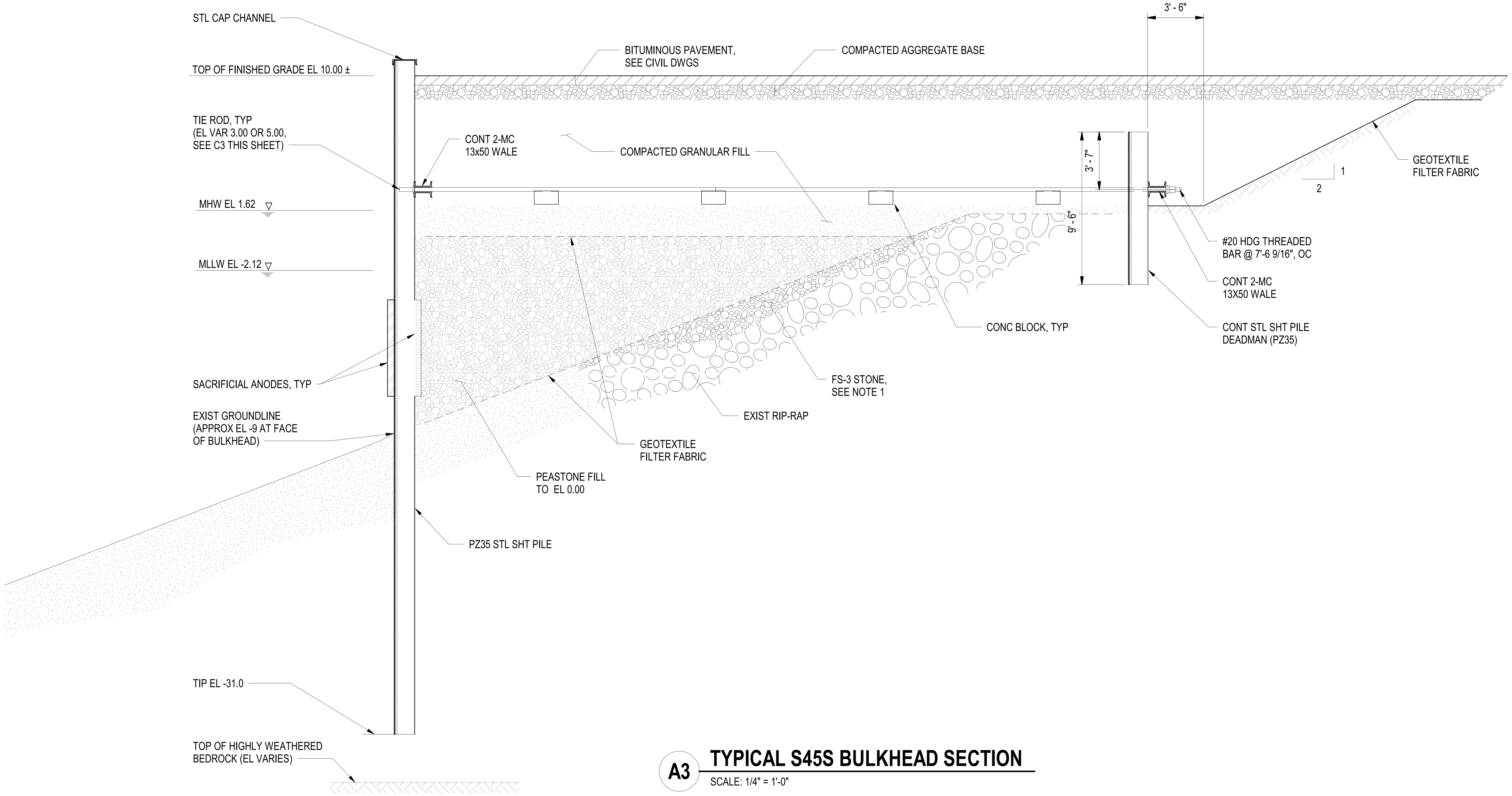
1. EXISTING RIP-RAP STONE VOIDS TO BE COVERED WITH FS-3 TYPE STONE TO CREATE A LEVEL SURFACE FOR GEOTEXTILE FABRIC.
2. ROCK WITHIN THE LIMITS OF THE STEEL SHEET PILE DRIVING LINE MUST BE TEMPORARILY REMOVED AND PUT BACK IN PLACE AFTER BULKHEAD INSTALLATION.
3. PROVIDE A SUPPLEMENTAL TIE ROD ON EACH SIDE OF THE OUTFALL.

SHEET KEYNOTES:

1. PZ35 STEEL SHEET PILE.
2. PZ35 STEEL SHEET PILE DEADMAN.
3. TIE ROD @ 7' - 6 9/16", TYP
4. CONTINUOUS BULKHEAD WALE.
5. CONTINUOUS DEADMAN WALE.



C3 **S45S BULKHEAD FRAMING PART PLAN**
SCALE: 1" = 20'-0"



A3 **TYPICAL S45S BULKHEAD SECTION**
SCALE: 1/4" = 1'-0"

1" = 20' — 0, 10', 20', 40', 80'

1/4" = 1' - 0" — 0, 1', 5', 10', 15'

SYM	DESCRIPTION	DATE	APPR

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COMMONWEALTH OF MASSACHUSETTS — VICTOR M. GOYONI — CIVIL — No. 41734 — REGISTERED PROFESSIONAL ENGINEER

SEAL

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

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DES MJF DRW KCJ CHK JZ

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
S45S BULKHEAD PART PLAN & SECTION (BASE BID)

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936697

SHEET 128 OF 247

S-404

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MATERIALS:

1. TURBIDITY CURTAIN BARRIERS MUST BE ORANGE IN COLOR IN ORDER TO ATTRACT THE ATTENTION OF NEARBY BOATERS.
2. THE TURBIDITY CURTAIN FABRIC MUST MEET THE MINIMUM REQUIREMENTS NOTED IN TABLE 1 ON THIS DRAWING.
3. SEAMS IN THE TURBIDITY CURTAIN FABRIC MUST BE EITHER VULCANIZED WELDED OR SEWN, AND MUST DEVELOP THE FULL STRENGTH OF THE FABRIC.
4. FLOTATION DEVICES MUST BE FLEXIBLE, BUOYANT UNITS CONTAINED IN AN INDIVIDUAL FLOTATION SLEEVE OR COLLAR ATTACHED TO THE CURTAIN. BUOYANCY PROVIDED BY THE FLOTATION UNITS MUST BE SUFFICIENT TO SUPPORT THE WEIGHT OF THE CURRENT AND MAINTAIN A FREEBOARD OF AT LEAST 3 INCHES ABOVE THE WATER SURFACE LEVEL AS INDICATED IN THE TURBIDITY CURTAIN DETAIL ON THIS DRAWING.
5. LOAD LINES MUST BE FABRICATED INTO THE TOP AND BOTTOM OF ALL FLOATING TURBIDITY CURTAINS. THE TOP LOAD LINE MUST CONSIST OF WOVEN WEBBING OR VINYL-SHEATHED STEEL CABLE AND MUST HAVE A BREAK STRENGTH IN EXCESS OF 10,000 POUNDS. THE SUPPLEMENTAL (BOTTOM) LOAD LINE MUST CONSIST OF A CHAIN INCORPORATED INTO THE BOTTOM HEM OF THE CURTAIN OF SUFFICIENT WEIGHT TO SERVE AS BALLAST TO HOLD THE CURTAIN IN A VERTICAL POSITION. ADDITIONAL ANCHORAGE MUST BE PROVIDED AS NECESSARY. THE LOAD LINES MUST HAVE SUITABLE CONNECTING DEVICES WHICH DEVELOP THE FULL BREAKING STRENGTH FOR CONNECTING TO LOAD LINES IN ADJACENT SECTIONS (SEE TURBIDITY CURTAIN DETAIL ON THIS DRAWING).
6. BOTTOM ANCHORS ARE REQUIRED. BOTTOM ANCHORS MUST BE SUFFICIENT TO HOLD THE CURTAIN IN THE SAME POSITION RELATIVE TO THE BOTTOM OF THE WATERCOURSE WITHOUT INTERFERING WITH THE ACTION OF THE CURTAIN. THE ANCHOR MAY DIG INTO THE BOTTOM (GRAPPLING HOOK, PLOW, OR FLUKE TYPE) OR MAY BE WEIGHTED (MUSHROOM TYPE), AND SHOULD BE ATTACHED TO A FLOATING ANCHOR BUOY VIA AN ANCHOR LINE. THE ANCHOR LINE WOULD THEN RUN FROM THE BUOY TO THE TOP LOAD LINE OF THE CURTAIN. THESE LINES MUST CONTAIN ENOUGH SLACK TO ALLOW THE BUOY AND CURTAIN TO FLOAT FREELY WITH A WATER SURFACE ELEVATION INCREASE FROM THE MEAN LOWER LOW WATER (MLLW) ELEVATION TO THE MEAN HIGHER HIGH WATER (MHHW) ELEVATION WITHOUT PULLING THE BUOY OR CURTAIN DOWN. THESE LINES MUST BE CHECKED REGULARLY TO MAKE SURE THEY DO NOT BECOME ENTANGLED WITH DEBRIS. ANCHOR SPACING WILL VARY WITH CURRENT VELOCITY AND POTENTIAL WIND AND WAVE ACTION, THEREFORE THE MANUFACTURER'S RECOMMENDATIONS SHOULD BE FOLLOWED. SEE ORIENTATION OF EXTERNAL ANCHORS AND ANCHOR BUOYS AS SHOWN IN FIGURE 1 ON THIS DRAWING FOR INSTALLATION.

INSTALLATION:

1. THE CURTAIN SHOULD NEVER TOUCH THE BOTTOM. A MINIMUM 1 FOOT "GAP" SHOULD EXIST BETWEEN THE WEIGHTED LOWER END OF THE SKIRT AND THE BOTTOM AT MLLW. MOVEMENT OF THE LOWER SKIRT OVER THE BOTTOM DUE TO CURRENT OR ELEVATION FLUCTUATION ON THE FLOTATION SYSTEM MAY FAN AND STIR SEDIMENTS ALREADY SETTLED OUT.
2. WHERE PRACTICABLE TURBIDITY CURTAINS SHOULD BE LOCATED PARALLEL TO THE DIRECTION OF FLOW OF A MOVING BODY OF WATER.
3. WHEN SIZING THE LENGTH OF A FLOATING CURTAIN, ALLOW AN ADDITIONAL 10 TO 20 PERCENT VARIANCE TO STRAIGHT LINE MEASUREMENTS. THIS WILL ALLOW FOR MEASURING ERRORS, MAKE INSTALLING EASIER AND REDUCE STRESS FROM POTENTIAL WAVE ACTION DURING HIGH WINDS.
4. AN ATTEMPT SHOULD BE MADE TO AVOID AN EXCESSIVE AMOUNT OF JOINTS IN THE CURTAIN. A MINIMUM CONTINUOUS SPAN OF 50 FEET BETWEEN JOINTS IS REQUIRED.
5. FOR STABILITY REASONS, A MAXIMUM SPAN OF 100 FEET BETWEEN JOINTS (ANCHOR OR STAKE LOCATIONS) IS REQUIRED. IF SPACINGS EXCEEDING THIS ARE ALLOWED BY THE MANUFACTURER, DATA MUST BE SUBMITTED FOR REVIEW.
6. THE ENDS OF THE CURTAIN (BOTH FLOATING UPPER AND WEIGHTED LOWER) SHOULD EXTEND WELL UNDER THE EXISTING STRUCTURE TO BE REMOVED. THE ENDS SHOULD BE SECURED FIRMLY TO FULLY ENCLOSE THE AREA WHERE SEDIMENT MAY ENTER THE WATER.
7. TYPICAL ALIGNMENTS OF TURBIDITY CURTAINS CAN BE SEEN IN FIGURE 2 ON THIS DRAWING. THE NUMBER AND SPACING OF EXTERNAL ANCHORS MAY VARY DEPENDING ON CURRENT VELOCITIES AND POTENTIAL WIND AND WAKE ACTION. THE MANUFACTURER'S RECOMMENDATIONS SHOULD BE FOLLOWED.
8. IN RIVERS OR IN OTHER MOVING WATER, IT IS IMPORTANT TO SET ALL THE CURTAIN ANCHOR POINTS. CARE MUST BE TAKEN TO ENSURE THAT ANCHOR POINTS ARE OF SUFFICIENT HOLDING POWER TO RETAIN THE CURTAIN UNDER THE EXISTING CURRENT CONDITIONS, PRIOR TO PUTTING THE FURLED CURTAIN INTO THE WATER. AGAIN, ANCHOR BUOYS SHOULD BE EMPLOYED ON ALL ANCHORS TO PREVENT THE CURRENT FROM SUBMERGING THE FLOTATION AT THE ANCHOR POINTS.
9. WHEN THE ANCHORS ARE SECURE, THE FURLED CURTAIN SHOULD BE SECURED TO THE UPSTREAM ANCHOR POINT AND THEN SEQUENTIALLY ATTACHED TO EACH NEXT DOWNSTREAM ANCHOR POINT UNTIL THE ENTIRE CURTAIN IS IN POSITION. AT THIS POINT, AND BEFORE UNFURLING, THE "LAY" OF THE CURTAIN SHOULD BE ASSESSED AND ANY NECESSARY ADJUSTMENTS MADE TO THE ANCHORS. FINALLY, WHEN THE LOCATION IS ASCERTAINED TO BE AS DESIRED, THE FURLING LINES SHOULD BE CUT TO ALLOW THE SKIRT TO DROP.
10. ALWAYS ATTACH ANCHOR LINES TO THE FLOTATION DEVICE, NOT TO THE BOTTOM OF THE CURTAIN. THE ANCHORING LINE ATTACHED TO THE FLOTATION DEVICE ON THE DOWNSTREAM SIDE WILL PROVIDE SUPPORT FOR THE CURTAIN. ATTACHING THE ANCHORS TO THE BOTTOM OF THE CURTAIN COULD CAUSE PREMATURE FAILURE OF THE CURTAIN DUE TO THE STRESSES IMPARTED ON THE MIDDLE SECTION OF THE CURTAIN.

REMOVAL:

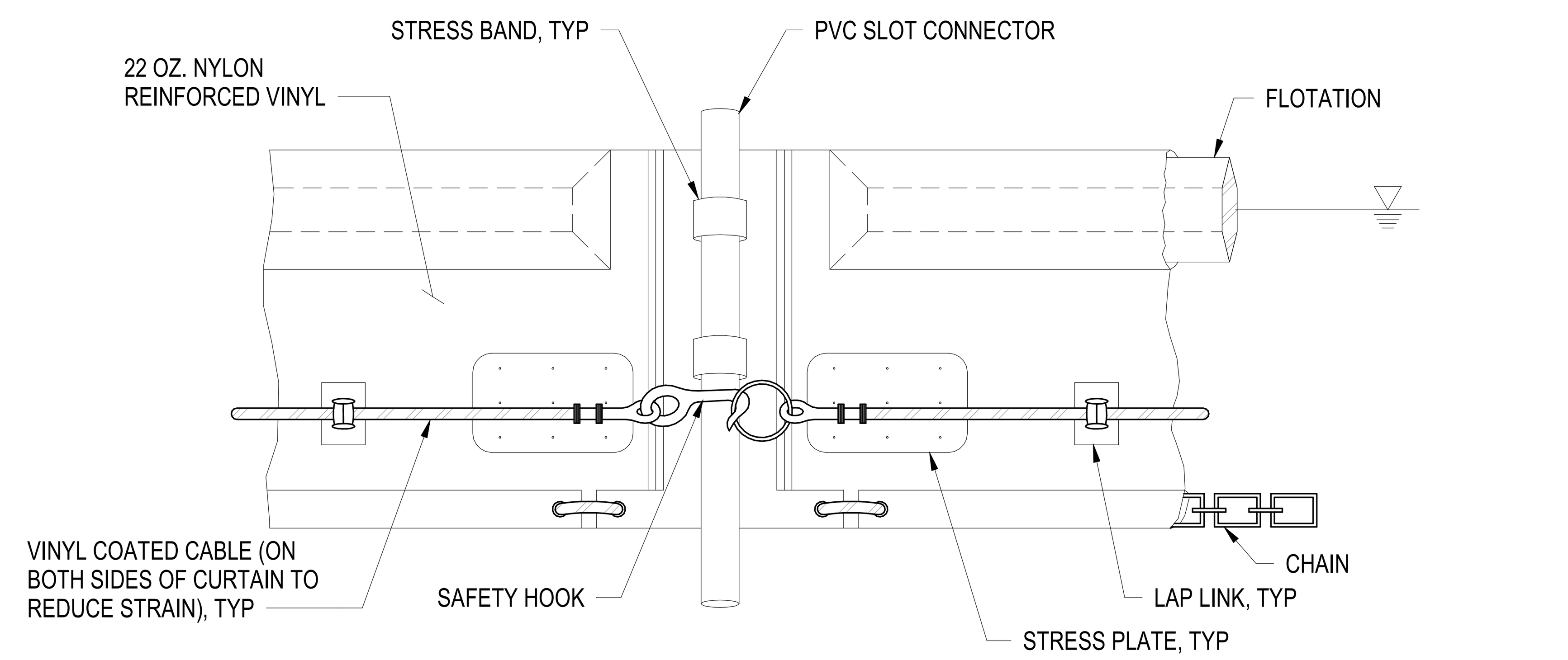
1. CARE SHOULD BE TAKEN TO PROTECT THE SKIRT FROM DAMAGE AS THE TURBIDITY CURTAIN IS DRAGGED FROM THE WATER.
2. IF THE CURTAIN IS TO BE REUSED AT THE SITE, THE AREA SELECTED TO BRING THE CURTAIN ASHORE SHOULD BE FREE OF SHARP ROCKS, BROKEN CEMENT, DEBRIS, ETC. SO AS TO MINIMIZE DAMAGE WHEN HAULING THE CURTAIN. ANY DAMAGE TO THE CURTAIN MUST BE REPAIRED AS SPECIFIED.
3. IF THE CURTAIN HAS A DEEP SKIRT, IT CAN BE FURTHER PROTECTED BY RUNNING A SMALL BOAT ALONG ITS LENGTH WITH A CREW INSTALLING FURLING LINES BEFORE ATTEMPTING TO REMOVE THE CURTAIN FROM THE WATER.

MAINTENANCE:

1. THE CONTRACTOR MUST BE RESPONSIBLE FOR MAINTENANCE OF THE TURBIDITY CURTAIN FOR THE DURATION OF THE PROJECT IN ORDER TO ENSURE THE CONTINUOUS PROTECTION OF THE WATERWAY.
2. SHOULD REPAIRS TO THE GEOTEXTILE FABRIC BECOME NECESSARY, REPAIR KITS AVAILABLE FROM THE ORIGINAL MANUFACTURER MUST BE USED. MANUFACTURER'S INSTRUCTIONS MUST BE FOLLOWED TO ENSURE THE ADEQUACY OF THE REPAIR.
3. WHEN THE CURTAIN IS NO LONGER REQUIRED, THE CURTAIN AND RELATED COMPONENTS MUST BE REMOVED IN SUCH A MANNER AS TO MINIMIZE TURBIDITY. REMAINING SEDIMENT MUST BE SUFFICIENTLY SETTLED BEFORE REMOVING THE CURTAIN.

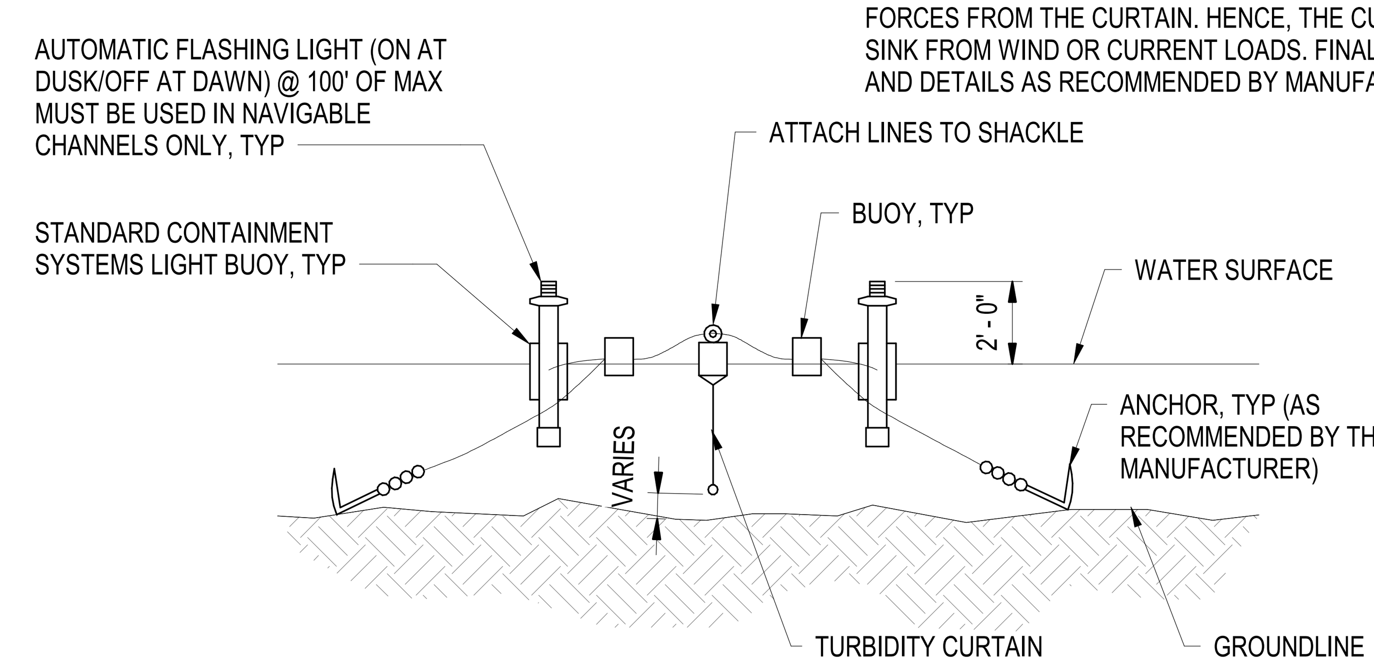
TABLE 1 - FABRIC PROPERTIES

PHYSICAL PROPERTY	REQUIREMENT
THICKNESS (MILS)	45
WEIGHT (OZ./SY)	22
TENSILE STRENGTH (LB)	300
UV INHIBITOR	REQUIRED

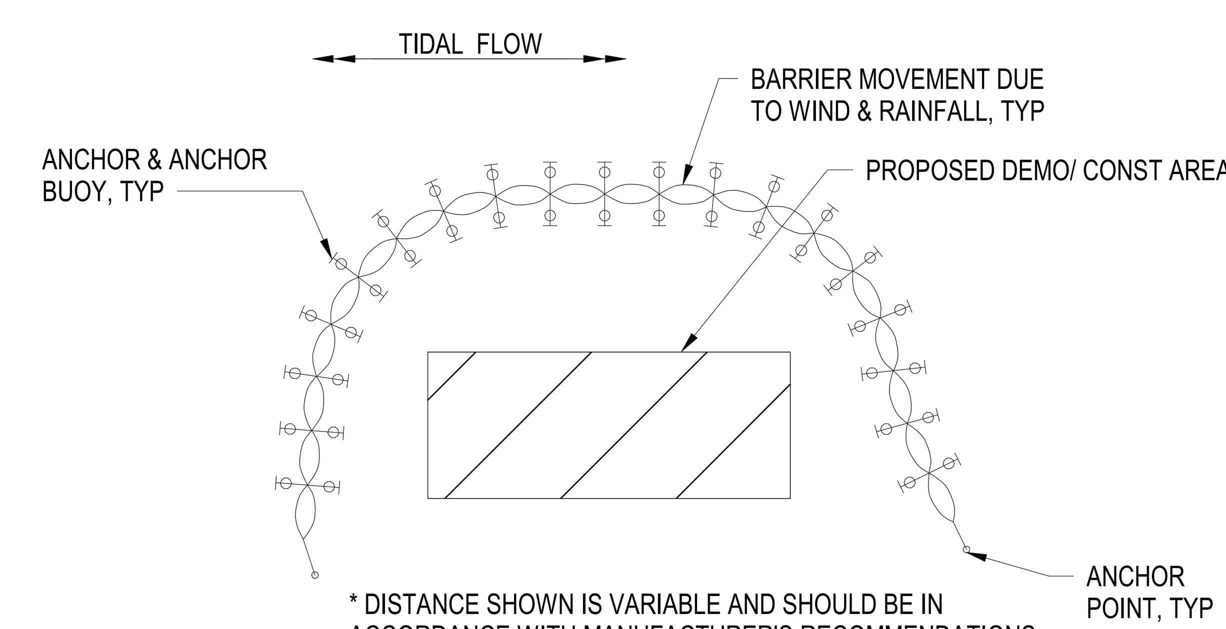


C3 TURBIDITY CURTAIN DETAIL
NOT TO SCALE

NOTE:
ANCHORING WITH BUOYS, AS SHOWN, REMOVES ALL VERTICAL FORCES FROM THE CURTAIN. HENCE, THE CURTAIN WILL NOT SINK FROM WIND OR CURRENT LOADS. FINAL CONFIGURATION AND DETAILS AS RECOMMENDED BY MANUFACTURER.



B3 FIGURE 1 - SECTION
SCALE: 1/4" = 1'-0"



* DISTANCE SHOWN IS VARIABLE AND SHOULD BE IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.

A3 FIGURE 2 - PLAN
NOT TO SCALE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA
SEAL

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A/E INFO

APPROVED
James E. Donohue, R.A.
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES JZ	DRW SA	CHK MJF
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BRANCH MANAGER		NEK
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
TURBIDITY CURTAIN NOTES & DETAILS (BASE BID)

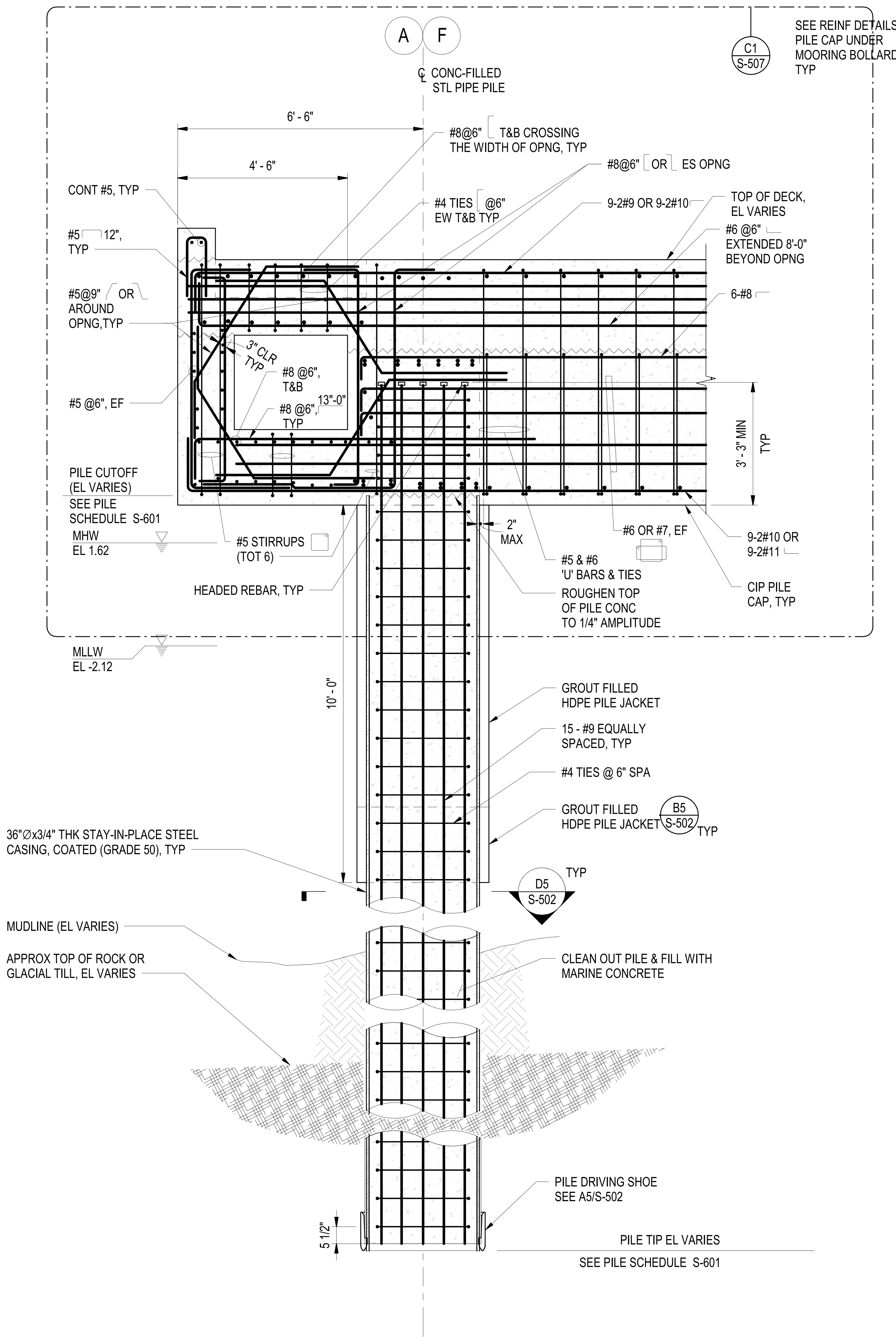
SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936698
SHEET 129 OF 247

S-501

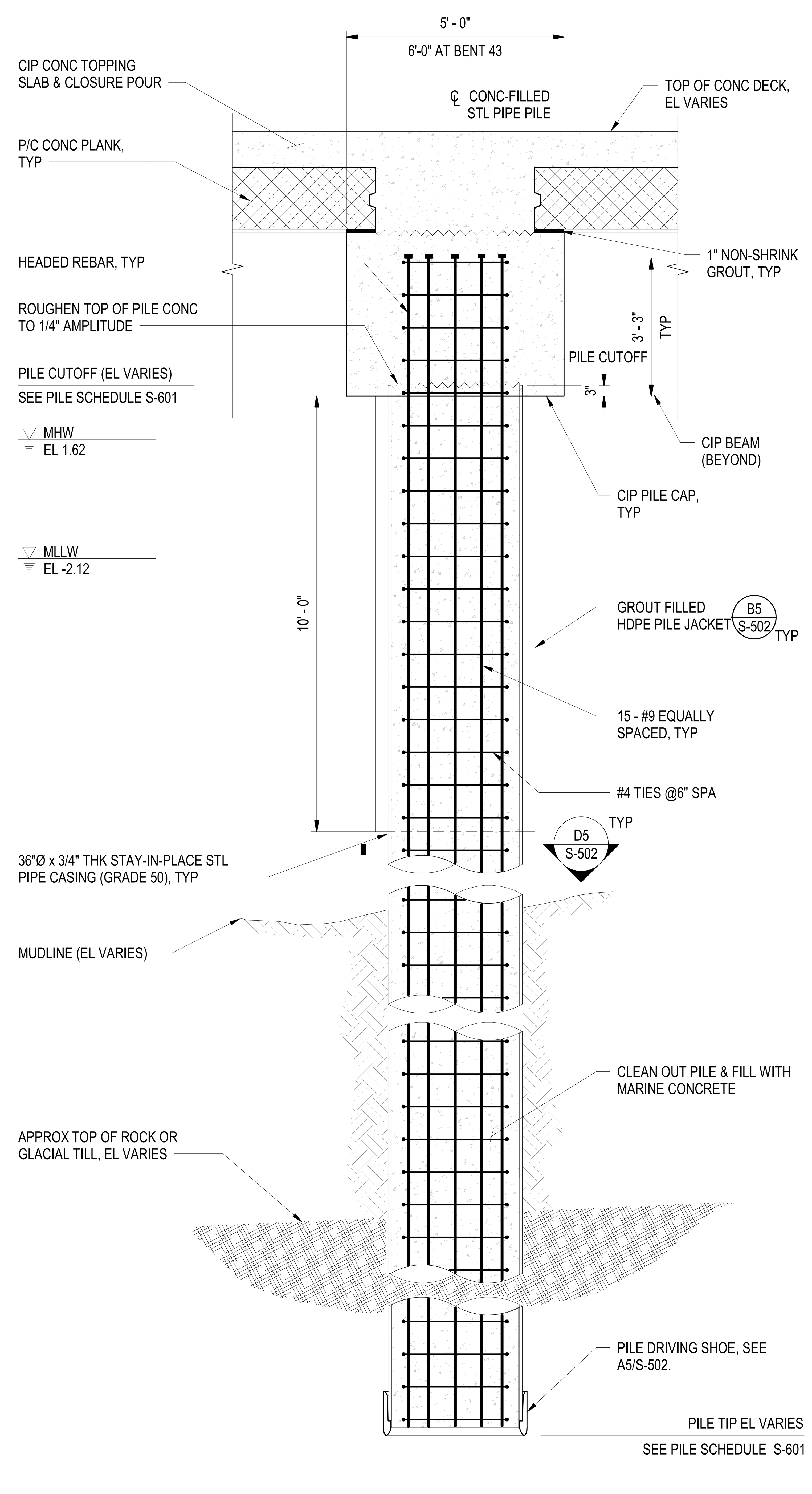
DRAWFORM REVISION: 25 AUGUST 2020

GENERAL SHEET NOTES:

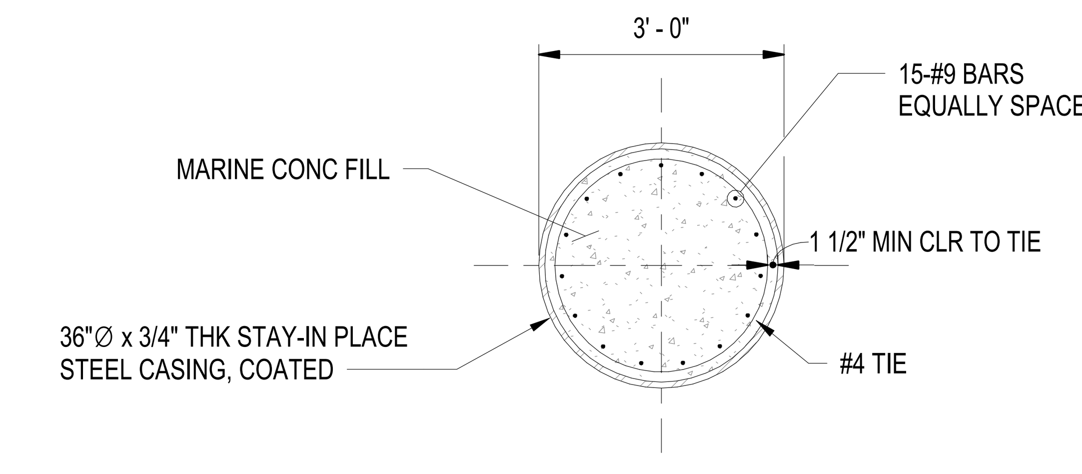
1. ATTACH DRIVING SHOE TO PIPE PILE PER MANUFACTURER RECOMMENDATION.
2. SEE DWG E-112 FOR ANODE LAYOUT.
3. NOT ALL REINFORCEMENT ARE SHOWN IN PILE CAP FOR CLARITY. SEE DRAWING S-5104 FOR ADDITIONAL INFORMATION ON REINFORCEMENT IN PILE CAP.



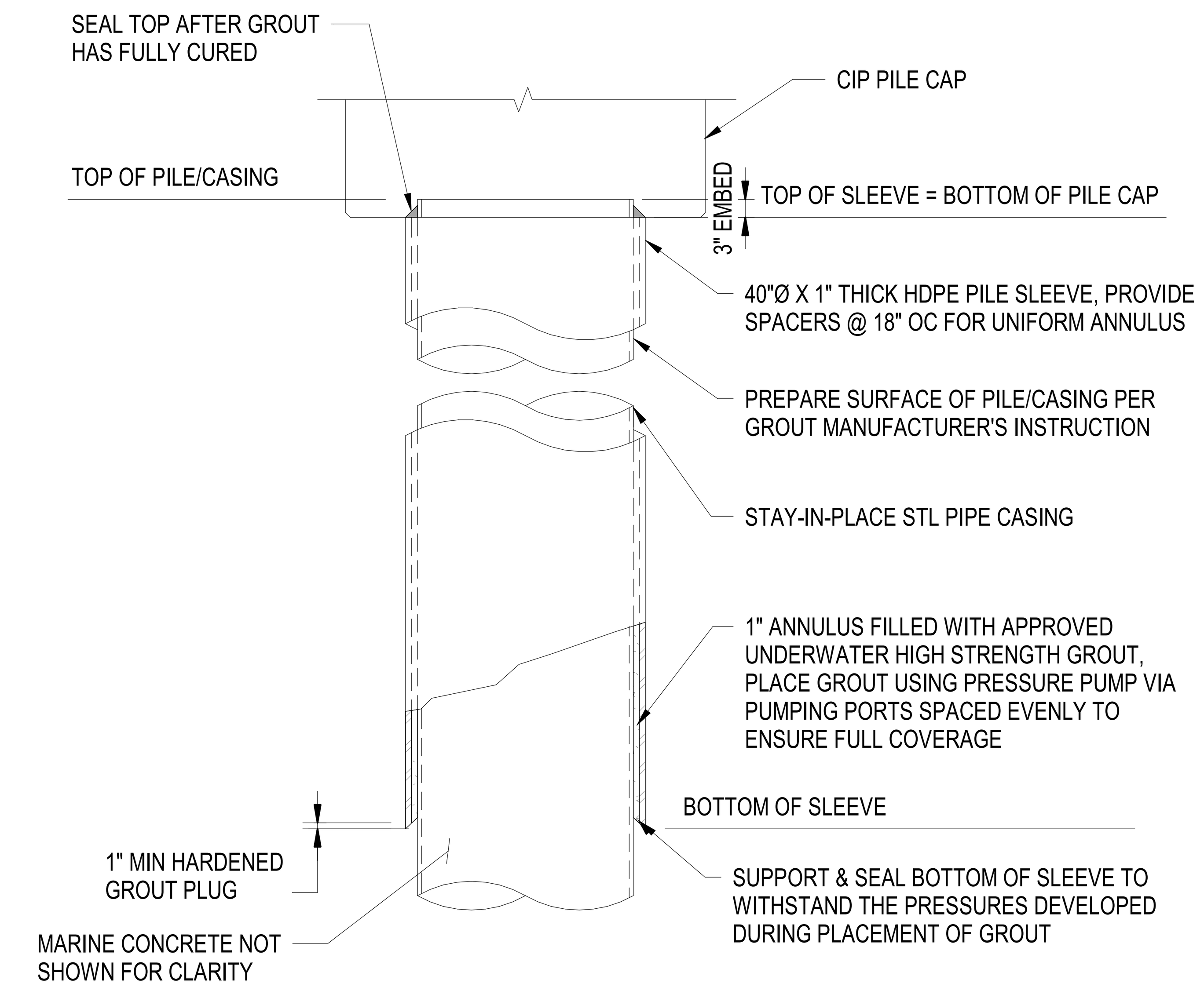
A1 36" DIAMETER STEELPIPE PILES AT UTILITY TRENCH
SCALE: 1/2" = 1'-0" (S-508)



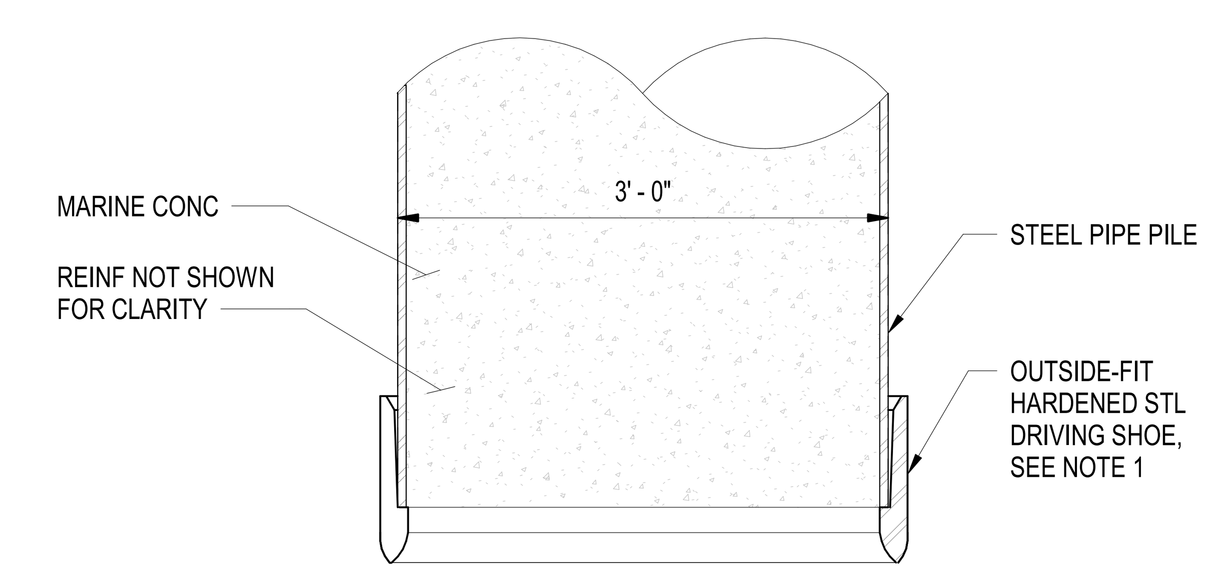
A3 36" DIAMETER STEELPIPE PILES BELOW PILE CAP
SCALE: 1/2" = 1'-0"



D5 STEEL PIPE PILE
SCALE: 1/2" = 1'-0"



B5 HDPE PIPE JACKET
SCALE: 1/2" = 1'-0" S-503



A5 PILE DRIVING SHOE
SCALE: 1" = 1'-0" S-503

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

1" = 1' - 0" 0 6" 1' 2' 3'

SYM	DESCRIPTION	DATE	APPR

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BRANCH MANAGER		NEK
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

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NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PILE DETAILS - CONCRETE-FILLED STEEL PIPE PILES

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936699
SHEET 130 OF 247
S-502

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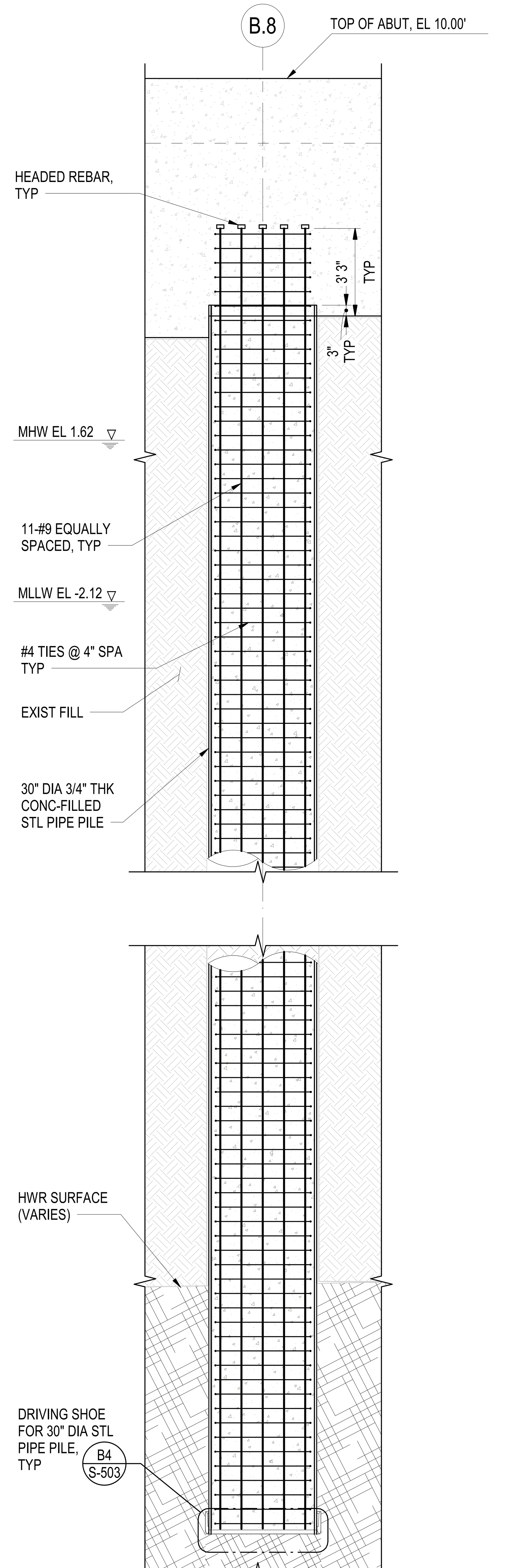
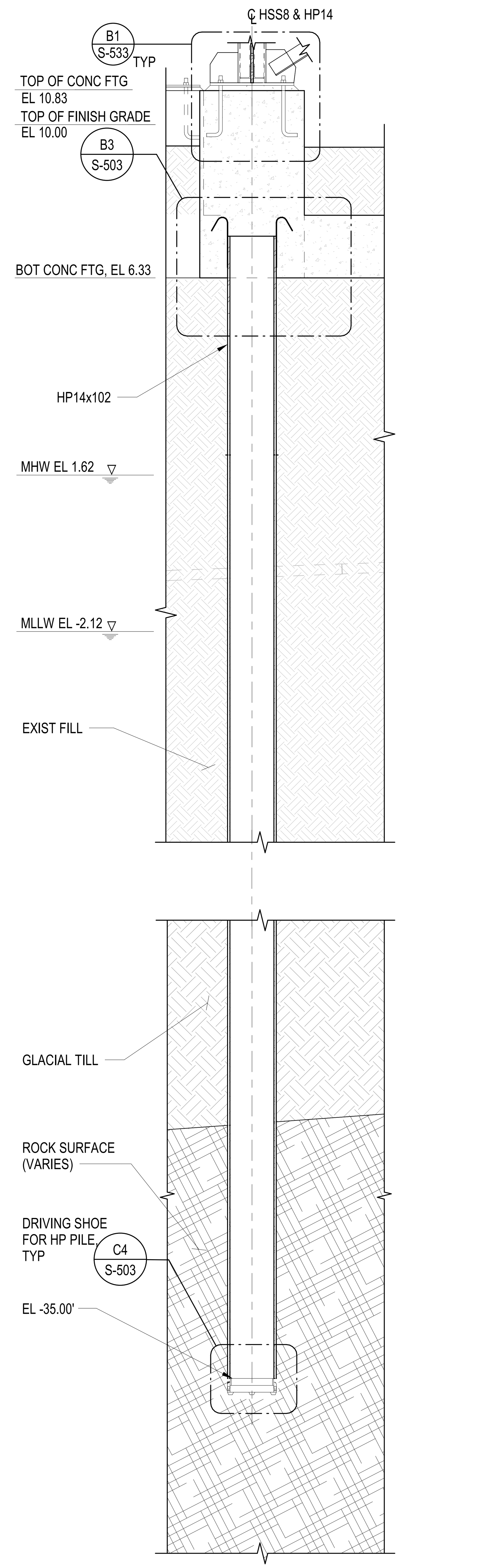
GENERAL SHEET NOTES:

1. DRILL HOLE WITH PROPER DIAMETERS (1/8" LARGER THAN REBAR DIAMETER AS TYPICAL) TO PULL THROUGH REINFORCEMENT AS TYPICAL.
2. DRILL STEEL PILE TO PULL REINFORCEMENT THROUGH. DO NOT USE TORCH ON ANY STEEL PILE.
3. ATTACH DRIVING SHOE TO HP PILE PER MANUFACTURER RECOMMENDATION.



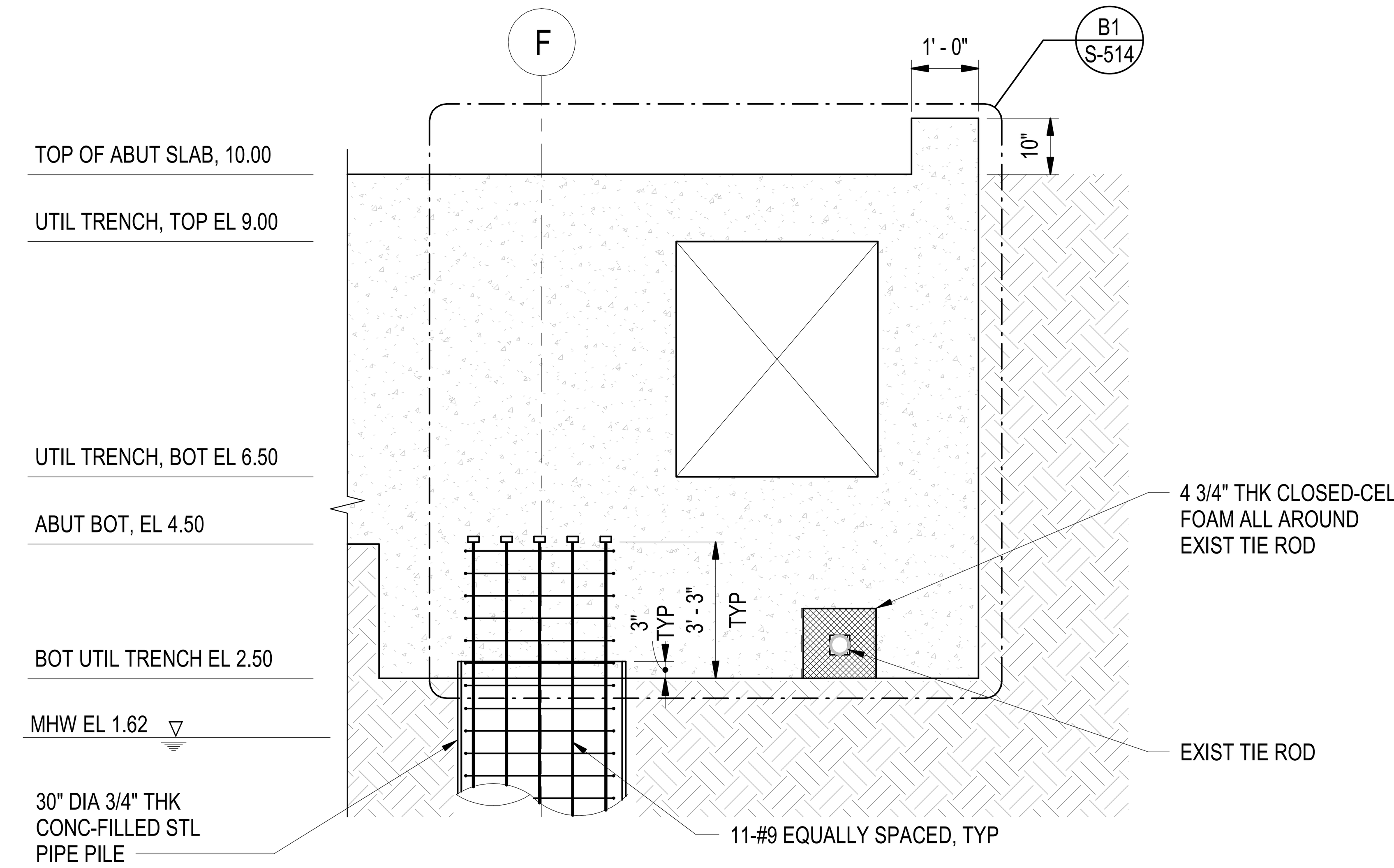
A1 TYPICAL DETAIL FOR 30" DIAMETER STEEL PIPE PILES

SCALE: 1/2" = 1'-0" (S-306)



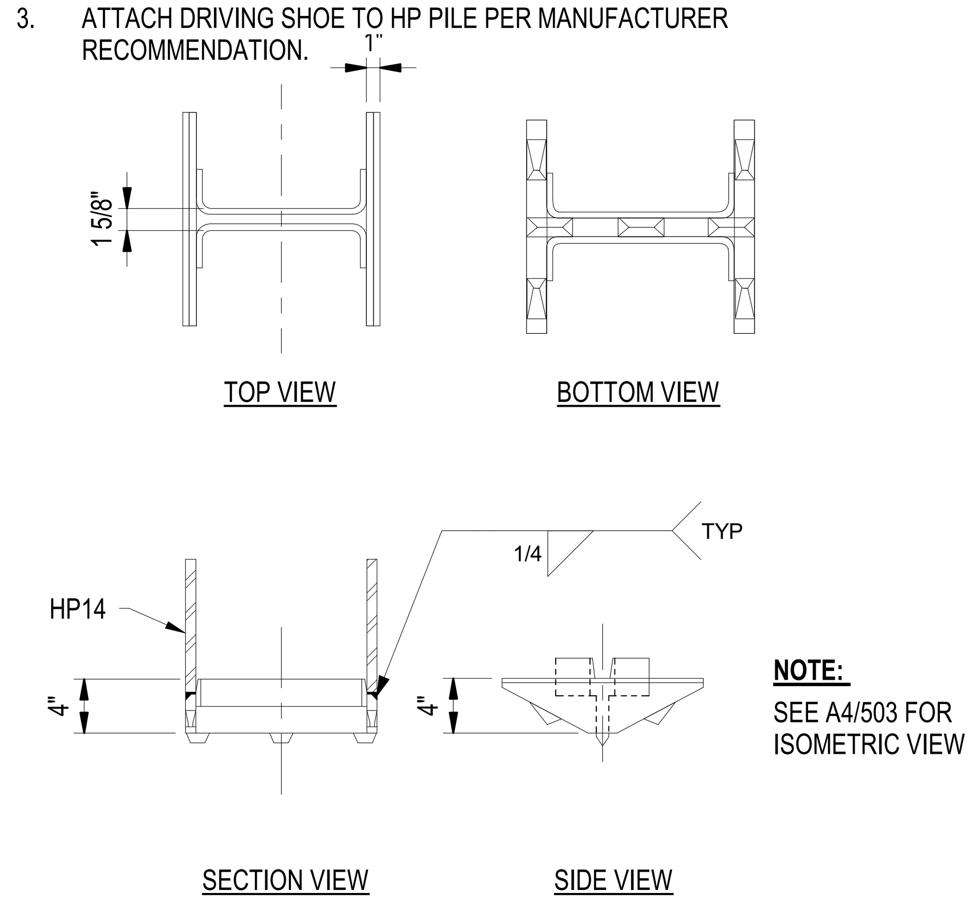
A2 HP PILES AT AIR COMPRESSOR ENCLOSURE

SCALE: 1/2" = 1'-0" (S-530)



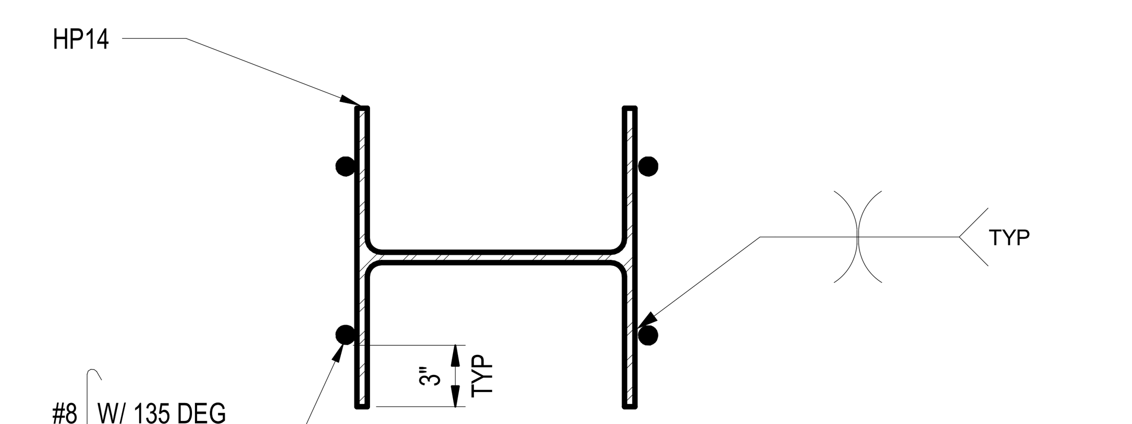
C3 BOTTOM OF UTILITY TRENCH AT LANDSIDE

SCALE: 1/2" = 1'-0" (S-306)



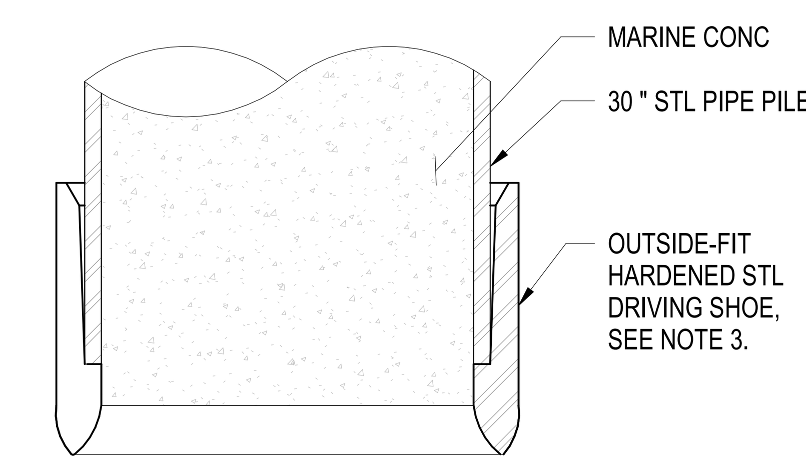
C4 TYPICAL PILE SHOE FOR HP PILES

SCALE: 1" = 1'-0"



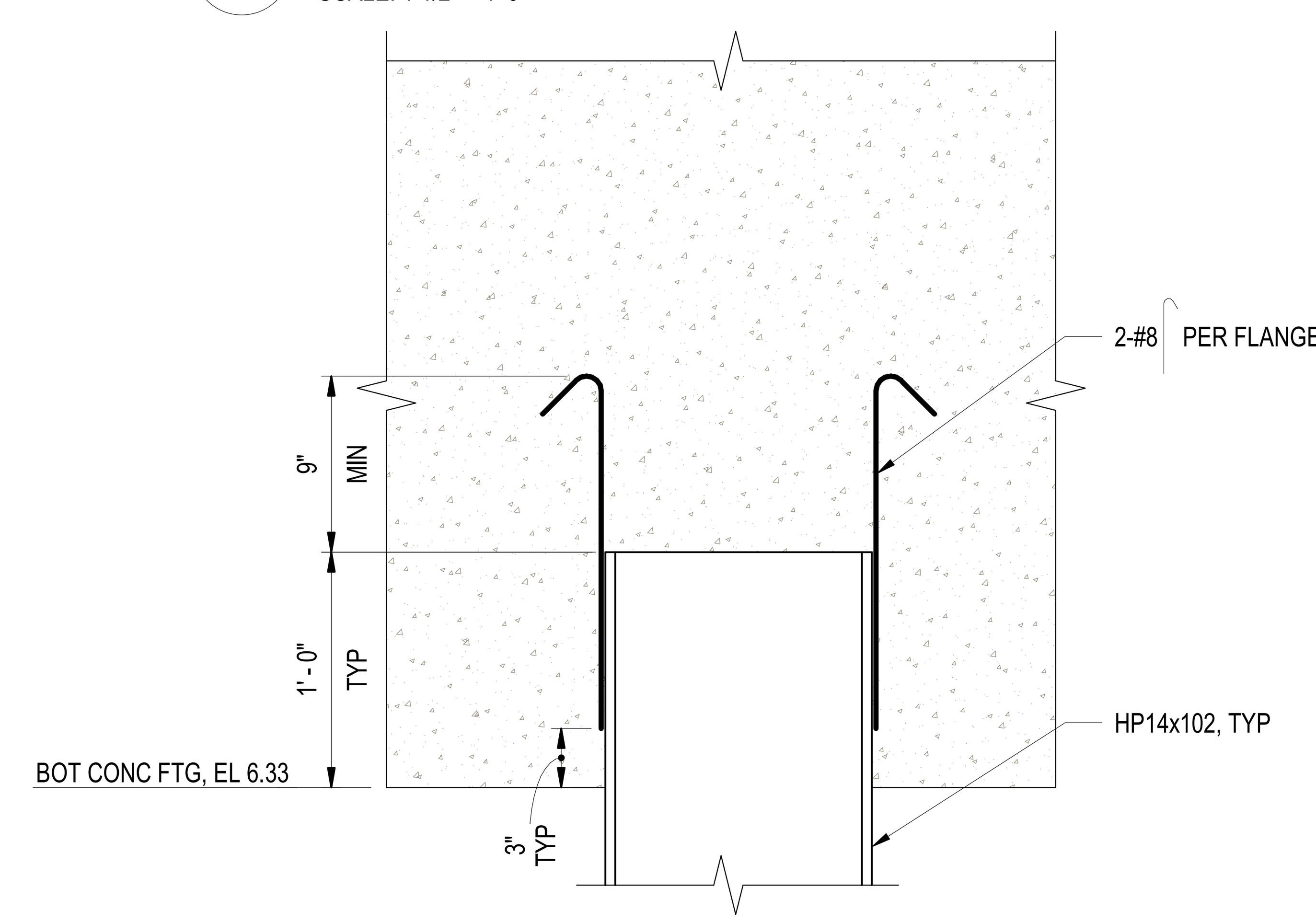
B3 TYPICAL PILE HEAD CONNECTION TO CONCRETE

SCALE: 1 1/2" = 1'-0"



B4 DRIVING SHOE AT 30" DIAMETER STEELPIPE PILES

SCALE: 1" = 1'-0" (S-503)



A3 TYPICAL PILE HEAD CONNECTION TO CONCRETE - SECTION

SCALE: 1 1/2" = 1'-0" (S-534)

A4 ISOMETRIC VIEW OF HP DRIVING SHOE

SCALE: 1" = 1'-0"

1/2" = 1'-0" 0 6" 1' 2' 3' 4' 5'

1" = 1'-0" 0 6" 1' 2' 3'

1 1/2" = 1'-0" 0 3" 6" 9" 1' 2'

SYM	DESCRIPTION	DATE	APPR	

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

APPROVED James E. Donahue, R.A.

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES	JZ	DRW	AFC	CHK	MJF
PM/DM					MRI/CWJ
BRANCH MANAGER					NEK
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PILE DETAILS - STEEL PILES

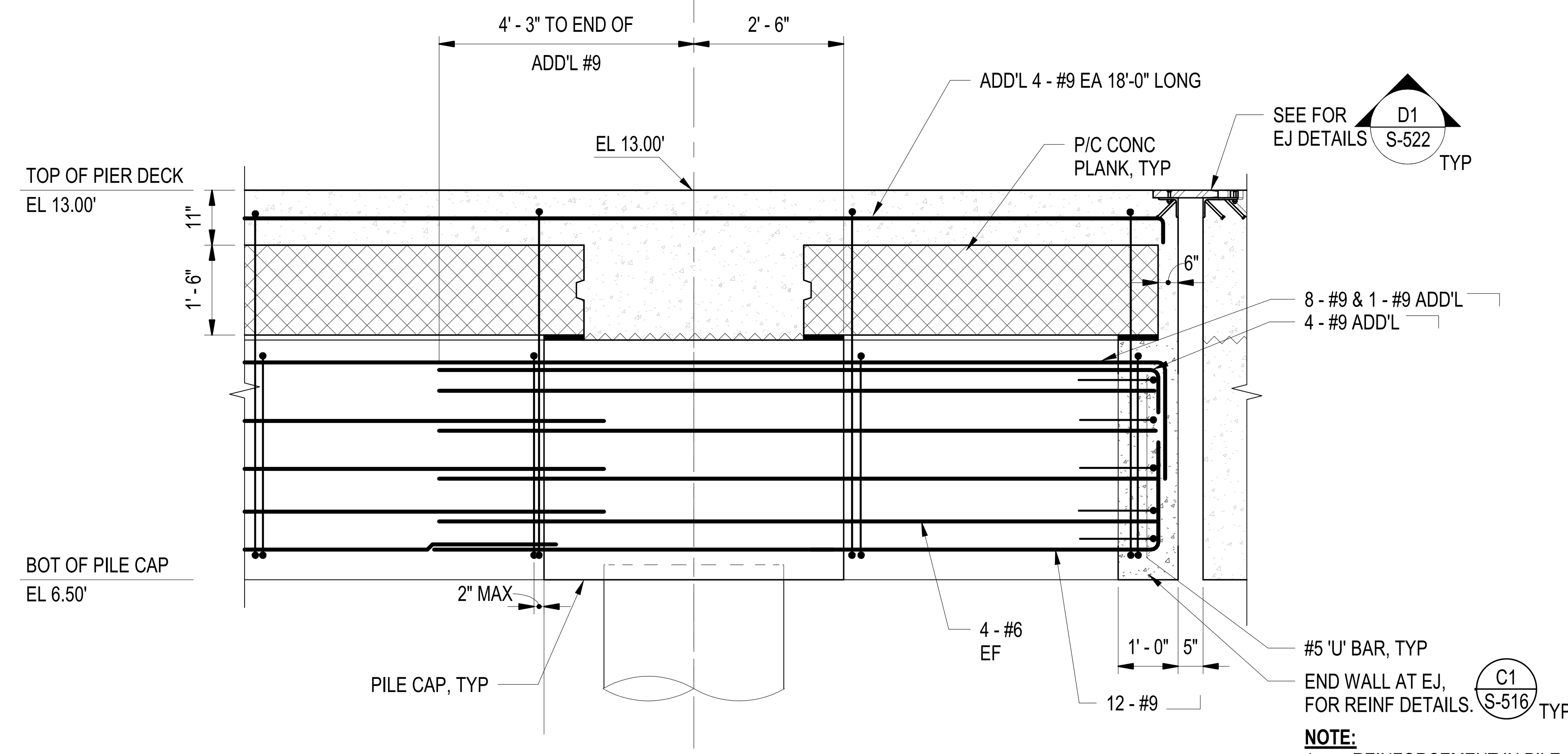
SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936704
SHEET 131 OF 247

S-503

DRAWFORM REVISION: 25 AUGUST 2020

1 2 3 4 5

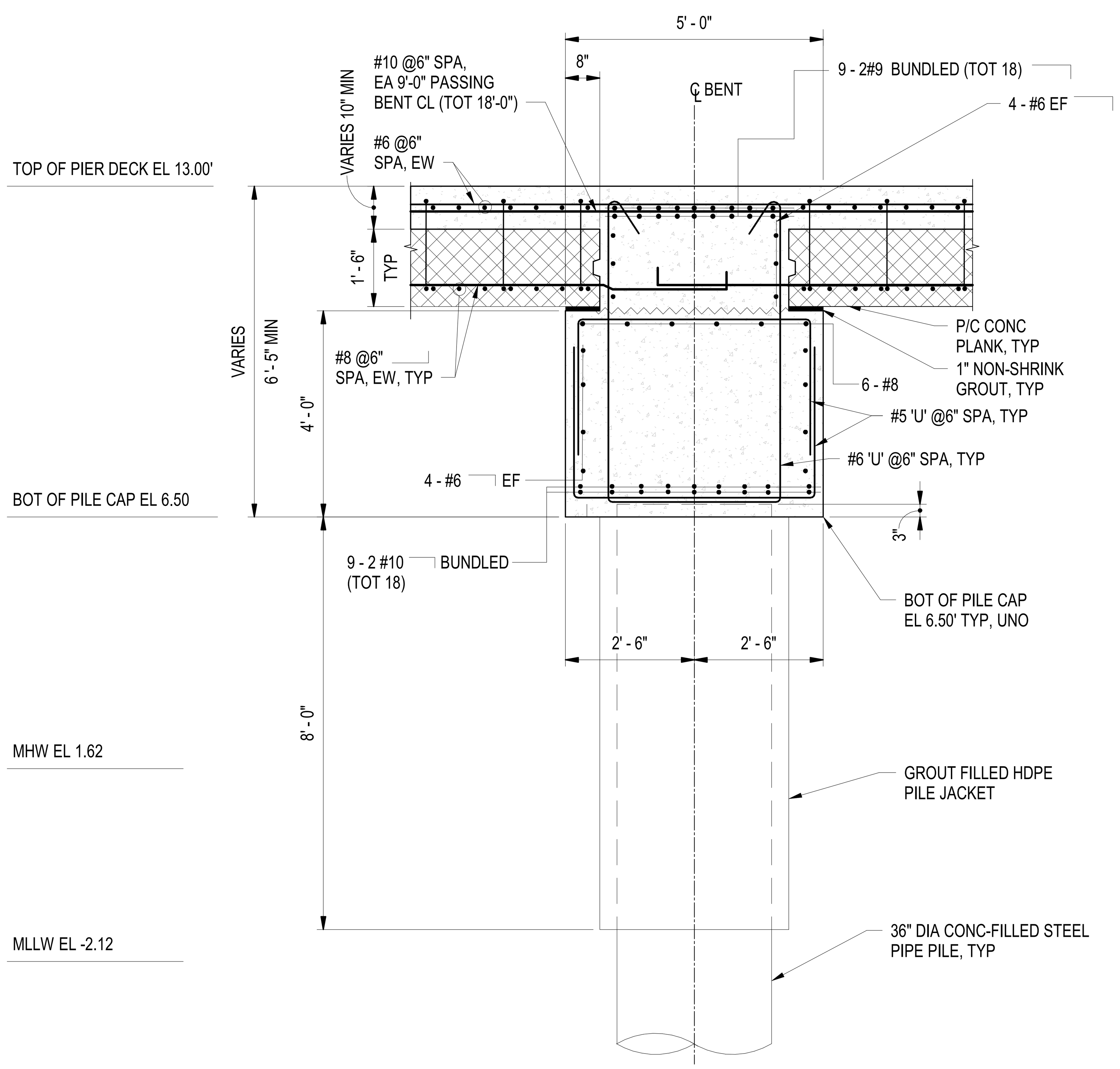
D C B A



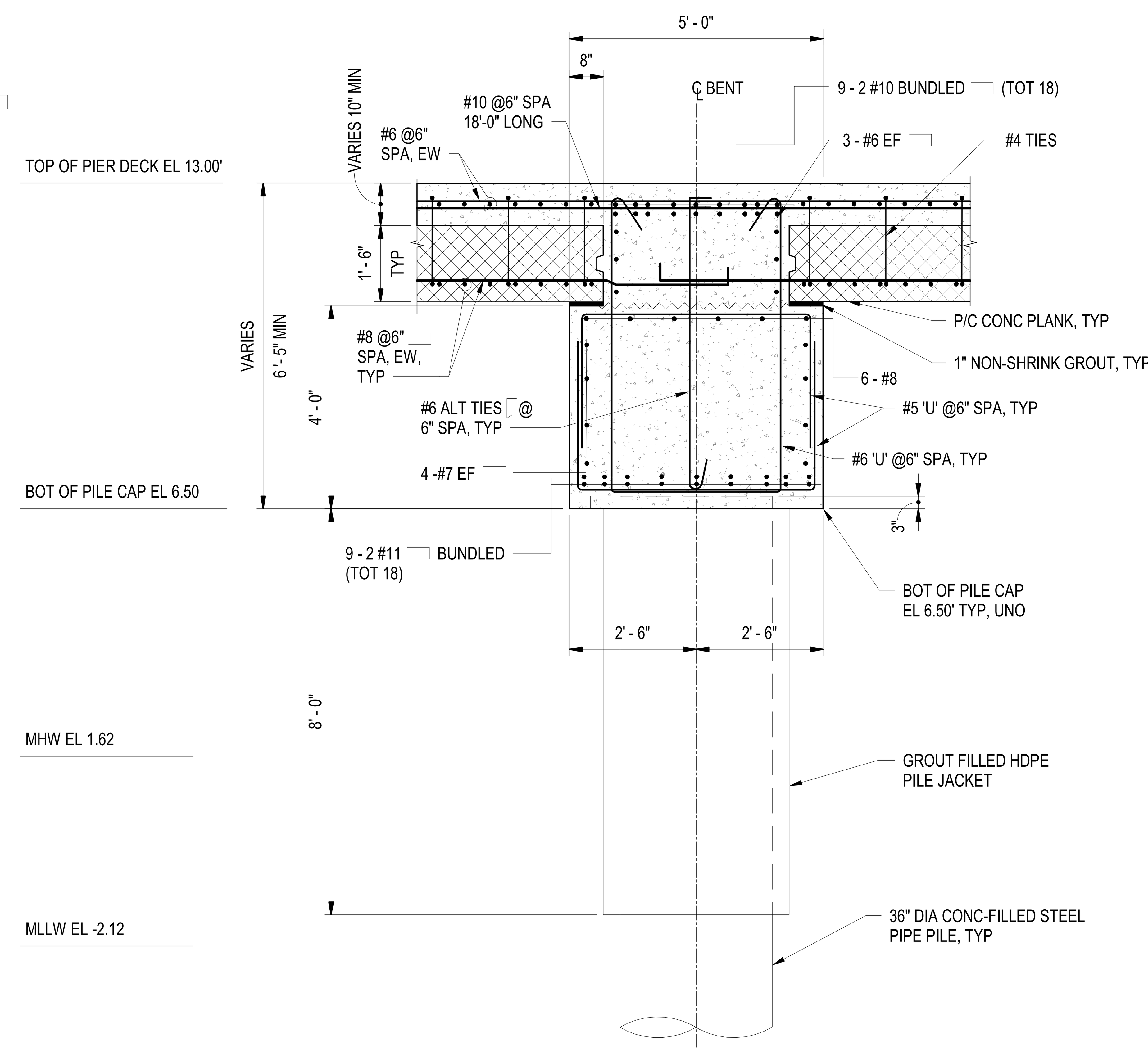
NOTE:
1. REINFORCEMENT IN PILE CAP, BEAM, AND TOPPING SLAB ARE NOT SHOWN FOR CLARITY.
2. SEE DRAWING S-516 FOR ADDITIONAL REINFORCEMENT.

C3 CANTILEVERED BEAM SECTION AT EXPANSION JOINT
SCALE: 1/2" = 1'-0" (S-120)

NOTES:
1. SEE DRAWING S-515 FOR ADDITIONAL INFORMATION ON REINFORCEMENT IN PRECAST PLANKS.



A1 TYPICAL PILE CAP SECTION AT BENTS 1-6, 12-16, & 22-43
SCALE: 1/2" = 1'-0" (S-508)



A3 TYPICAL PILE CAP SECTION AT BENTS 7-11 & 17-21
SCALE = 1/2" = 1'-0" (S-504)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES	JZ	DRW	AFC	CHK	MJF
PM/DM					MRI/CWJ
BRANCH MANAGER					NEK
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
CONCRETE PILE CAP DETAILS - SHEET 1 OF 3

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936701
SHEET 132 OF 247

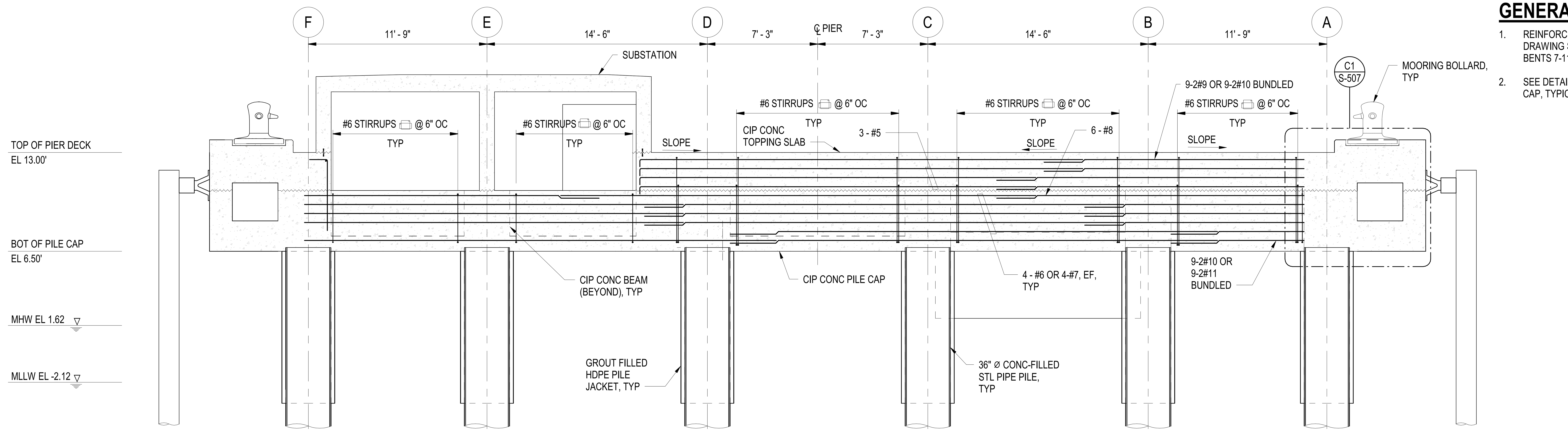
S-504

DRAWFORM REVISION: 26 AUGUST 2020

3/2/2026 7:36:32 PM S-504
Autodesk Docs://2241015255_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

GENERAL SHEET NOTES:

1. REINFORCEMENT SHOWN IS FOR TYPICAL PILE CAPS, SEE DRAWING S-504 FOR REINFORCEMENT IN CAPS BETWEEN BENTS 7-11 AND 17-21.
2. SEE DETAILS FOR REINFORCEMENT NOT SHOWN IN PILE CAP, TYPICAL.

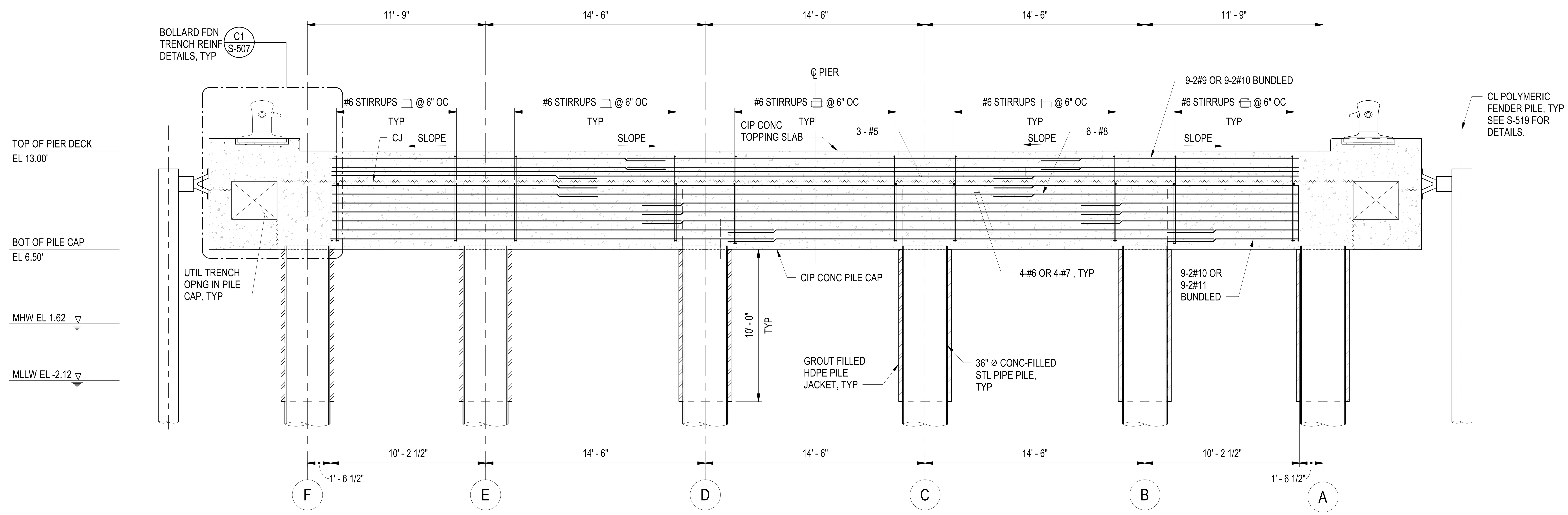


NOTE:

PREFAB SUBSTATION ENCLOSURE NOT SHOWN FOR CLARITY

C2 TYPICAL PILE CAP REINFORCEMENT AT BENTS 42-39 (BENTS 9-4 SIMILAR)

SCALE: 1/4" = 1'-0" (S-517)



A2 TYPICAL PILE CAP REINFORCEMENT AT BENTS 39-10 (BENTS 3-2 SIMILAR)

SCALE: 1/4" = 1'-0"

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

Stantec

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW AFC CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT
NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

CONCRETE PILE CAP DETAILS - SHEET 2 OF 3

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

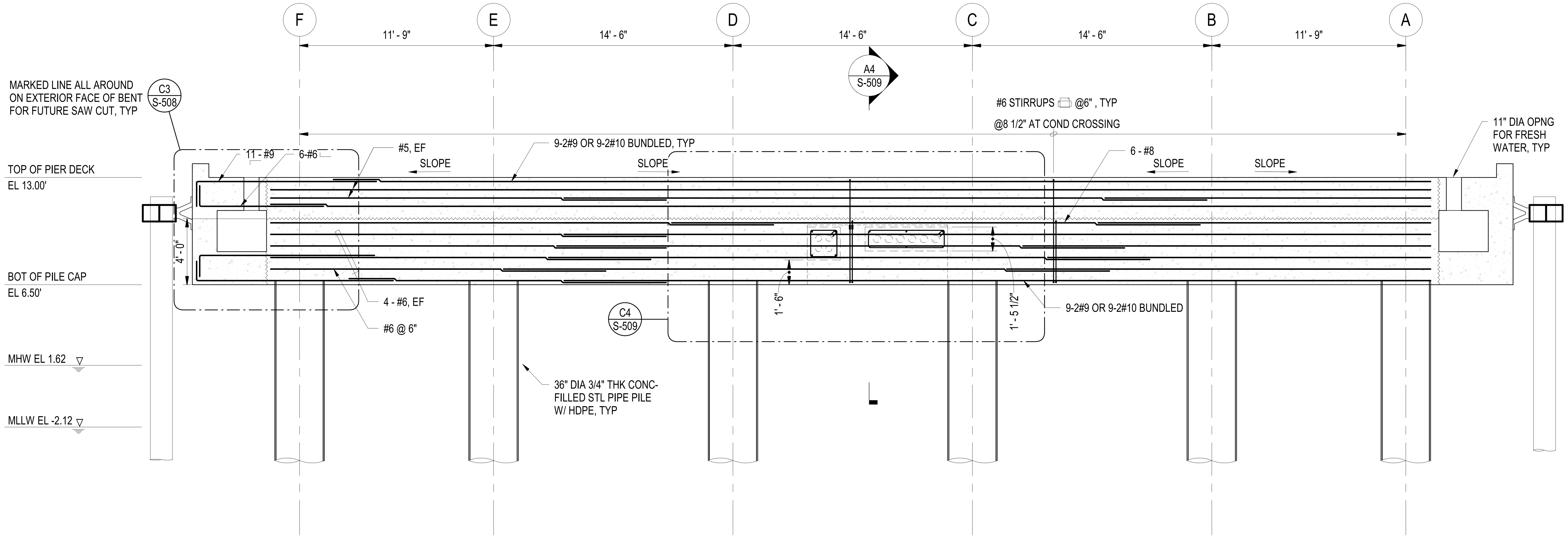
NAVFAC DRAWING NO. 12936706

SHEET 133 OF 247

S-505

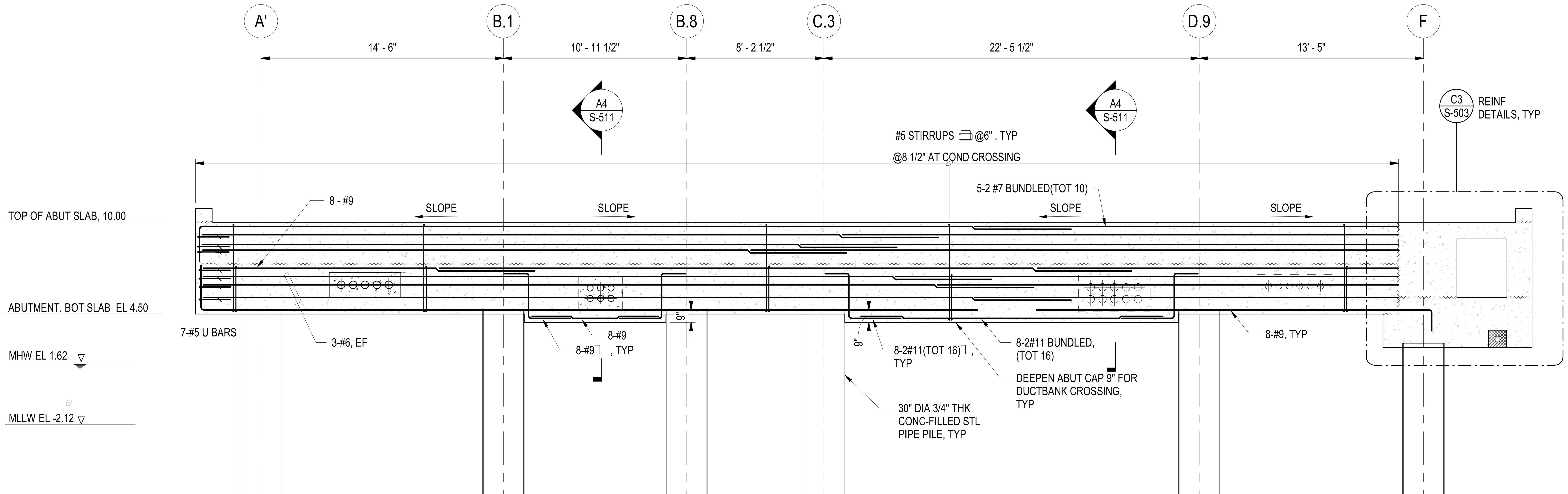
DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:36:34 PM S-505 Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt



C2 PILE CAP REINFORCEMENT AT BENT 43

SCALE: 1/4" = 1'-0"



A2 TRANSVERSE SECTION AT BENT LINE 0

SCALE: 1/4" = 1'-0" (S-302, S-306)

NOTES:

1. SEE DETAILS FOR REINFORCEMENT NOT SHOWN IN PILE CAP, TYPICAL.

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA
SEAL

Stantec

A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES	JZ	DRW	AFC	CHK	MJF
PM/DM					MRI/CWJ
BRANCH MANAGER					NEK
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT
NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

CONCRETE PILE CAP DETAILS - SHEET 3 OF 3

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936703
SHEET 134 OF 247

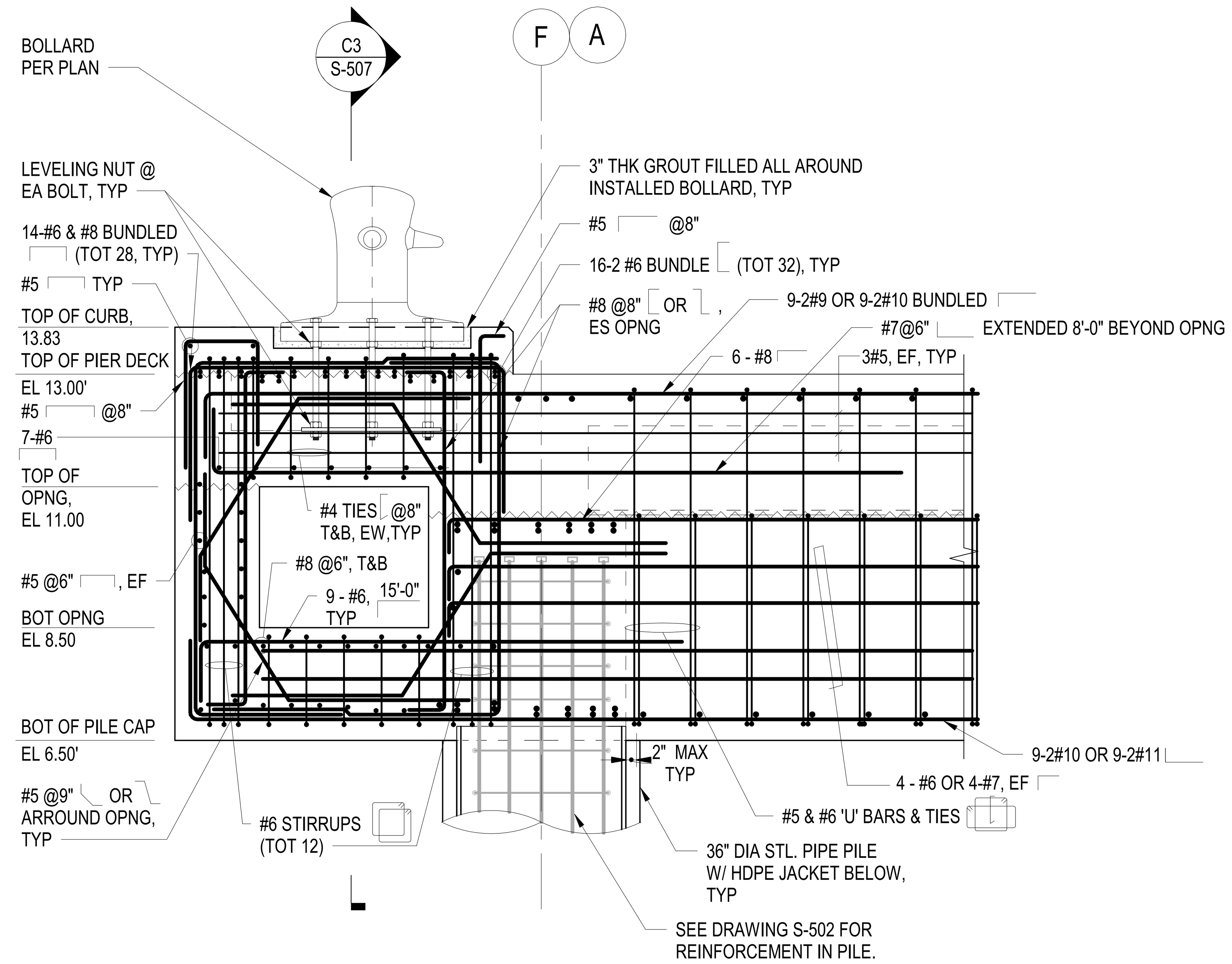
S-506

DRAWFORM REVISION: 25 AUGUST 2020

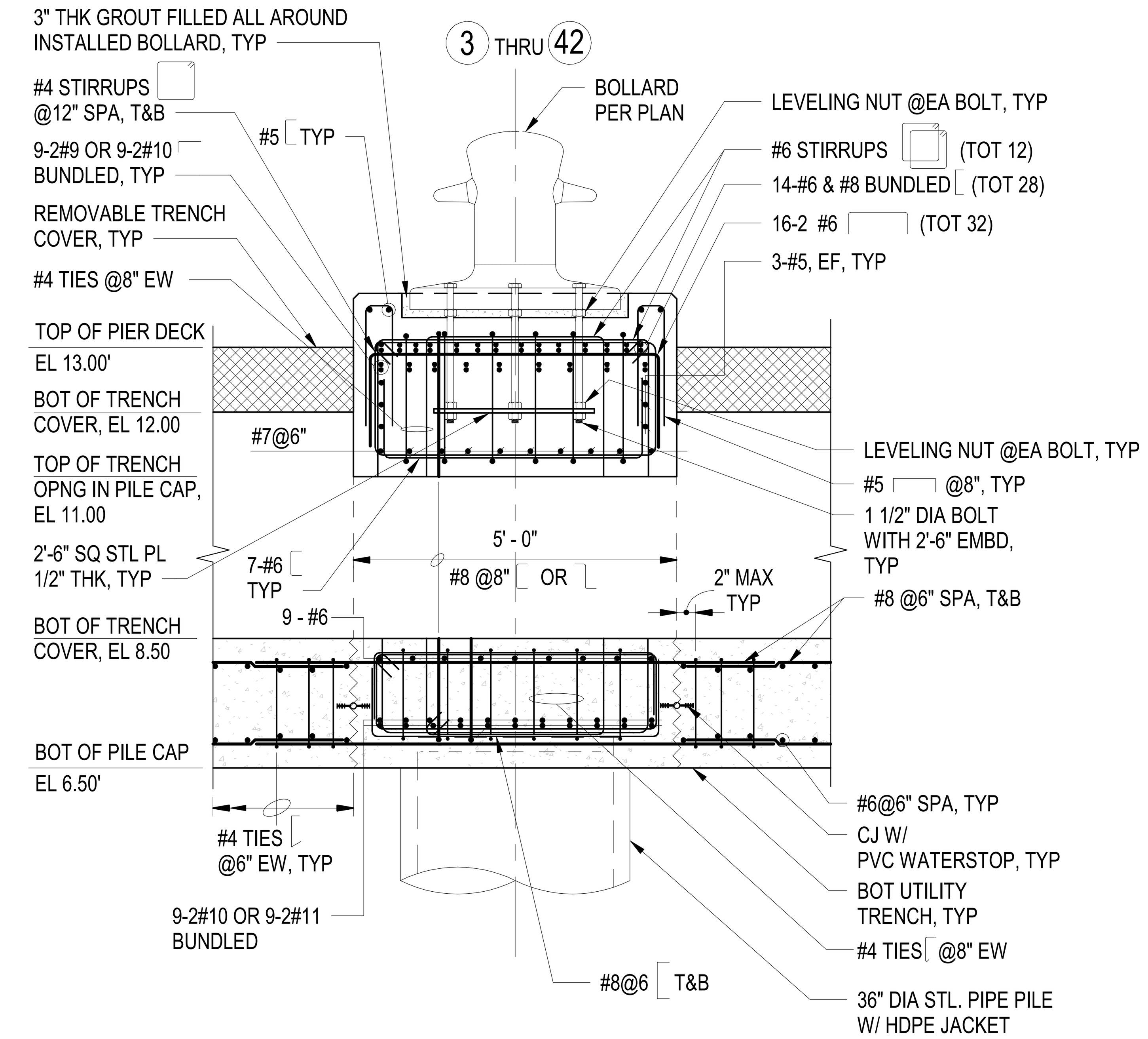
3/2/2026 7:36:35 PM
S-506
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

GENERAL SHEET NOTES:

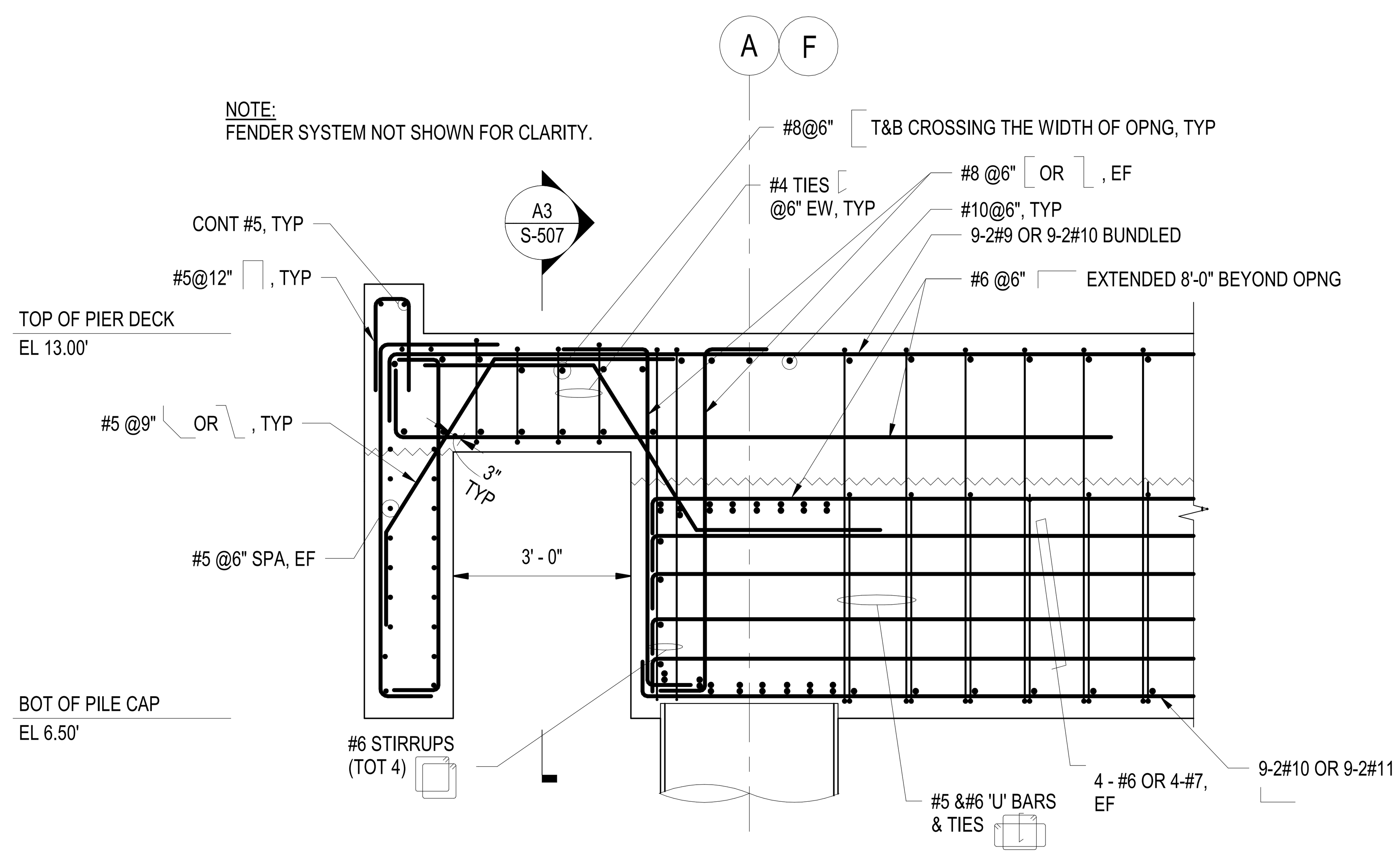
1. FOR ADDITIONAL REINFORCEMENT DETAILS IN PILE CAP SEE S-504 & S-506.
2. FOR REINFORCEMENT DETAILS ON PILE CAP WITHOUT BOLLARD, SEE DRAWING S-502.
3. SEE DRAWING S-511 TO SLEEVE UTILITY PIPES THROUGH HEAD WALLS WITHIN UTILITY TRENCHES, TYPICAL.
4. SEE DRAWING S-512 FOR ADDITIONAL INFORMATION ON REINFORCEMENT.



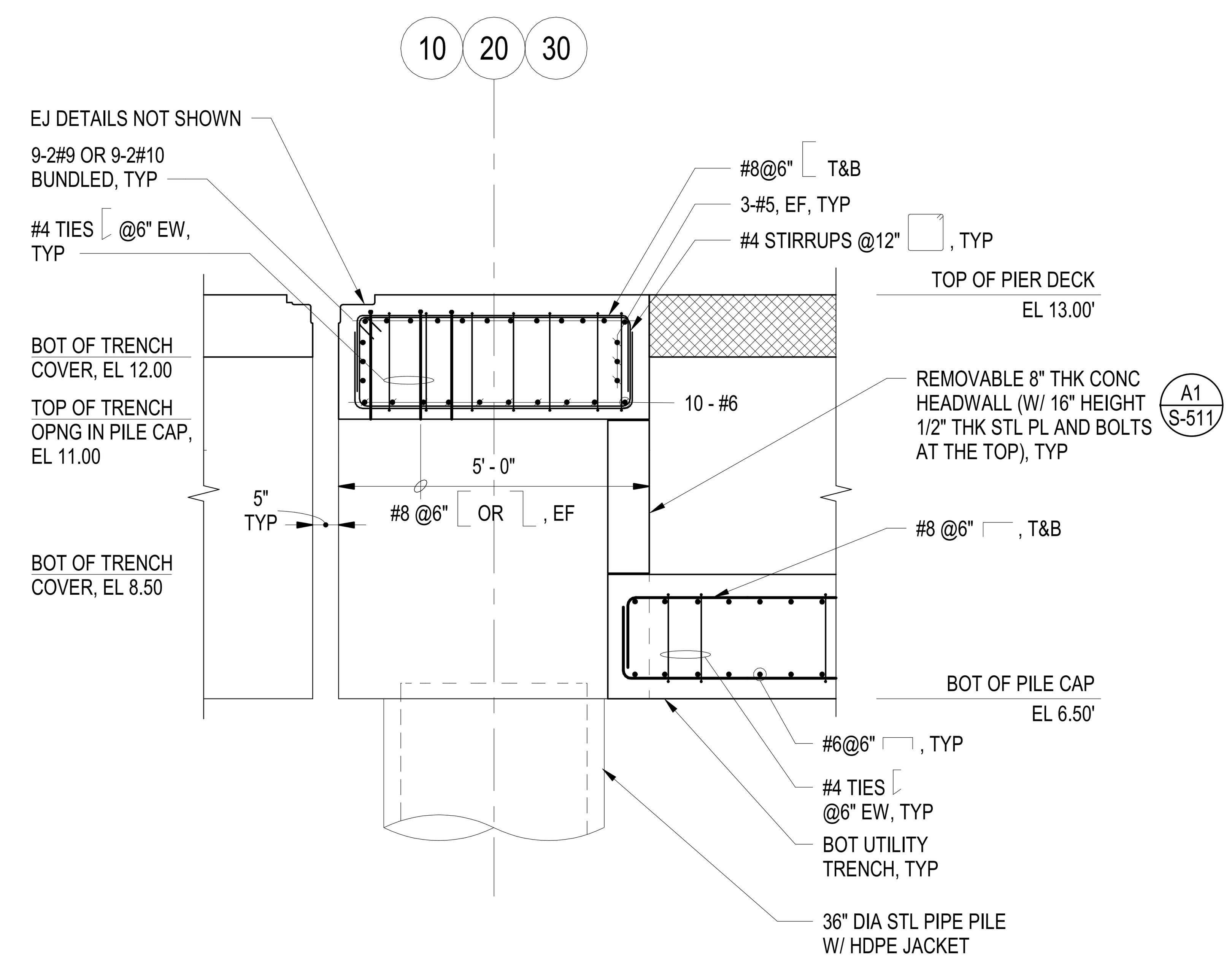
C1 **TYPICAL DETAILS FOR BOLLARD FOUNDATION AT BENTS 3 THROUGH 42**
SCALE: 1/2" = 1'-0" (S-502, S-505)



C3 **TYPICAL SECTION FOR BOLLARD FOUNDATION**
SCALE: 1/2" = 1'-0"



A1 **TYPICAL DETAILS FOR BENTS AT EXPANSION JOINT**
SCALE: 1/2" = 1'-0" (S-111)



A3 **TYPICAL SECTION AT EXPANSION JOINT**
SCALE: 1/2" = 1'-0" (S-111, S-304)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

SYM	DESCRIPTION
DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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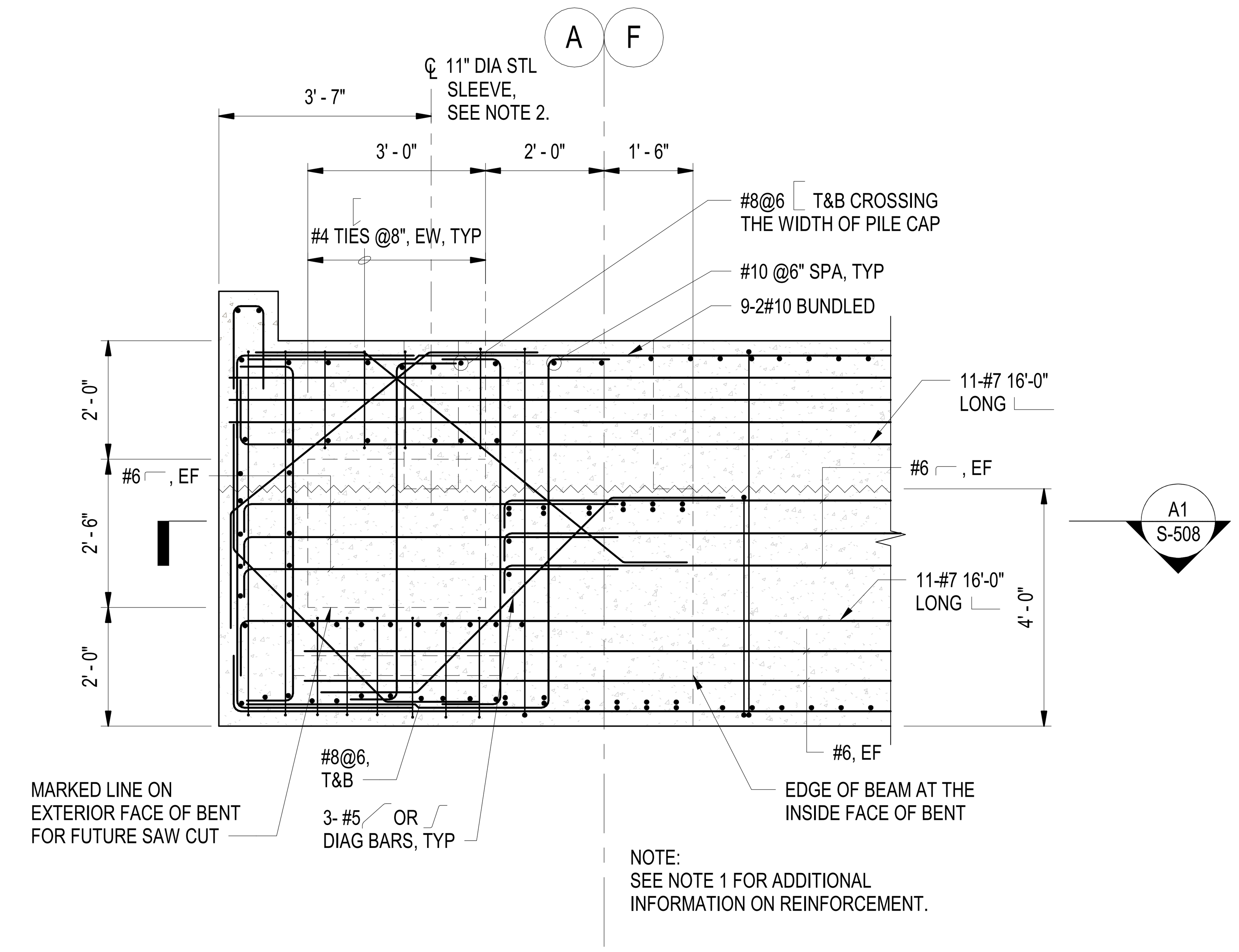
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES JZ DRW AFC CHK MJF
PM/DM MRI/CWJ
BRANCH MANAGER NEK
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
REINFORCED BOLLARD FOUNDATION DETAILS

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936708
SHEET 135 OF 247
S-507

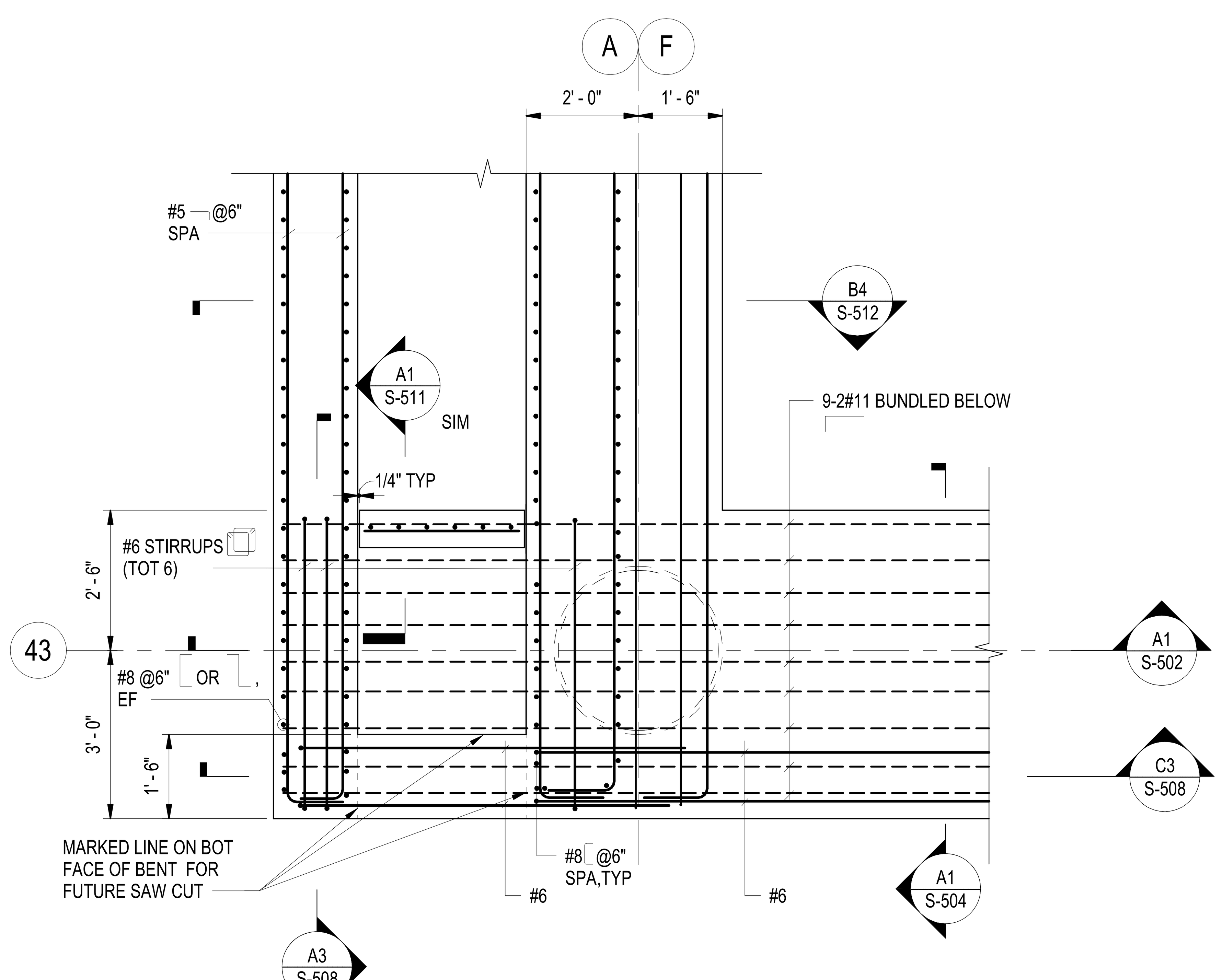
GENERAL SHEET NOTES:

1. SEE DRAWINGS S-502 & S-504 FOR ADDITIONAL REINFORCEMENT INFORMATION IN PILE CAP.
2. SEE DRAWING S-517 FOR ADDITIONAL REINFORCEMENT AROUND 11" DIAMETER STEEL SLEEVE, TYPICAL. BEND REINFORCING STEEL WHEN REQUIRED.
3. SEE DRAWING S S-502 AND S-512 FOR ADDITIONAL ENFORCEMENT ON REINFORCEMENT.



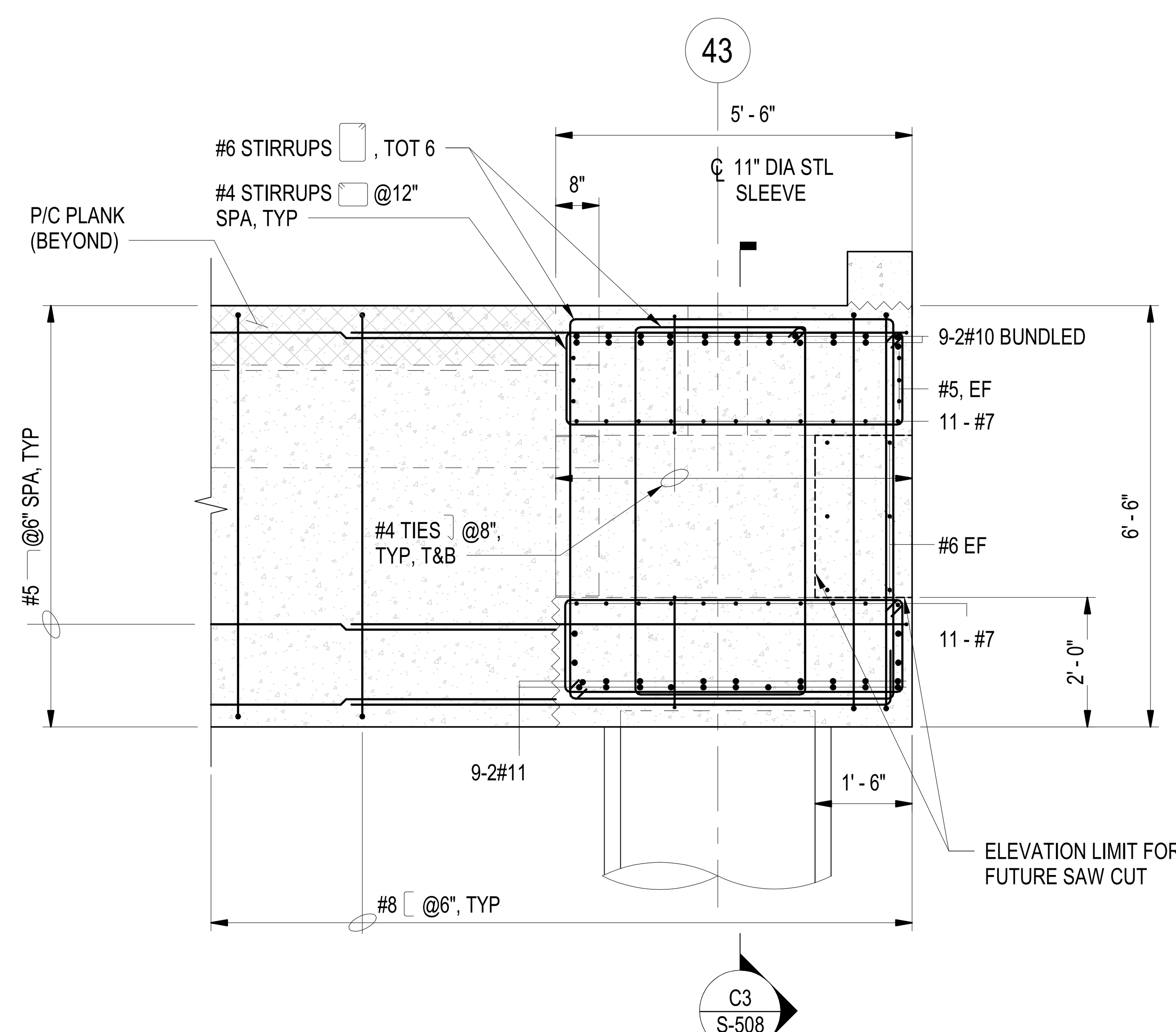
C3 **REINFORCEMENT DETAIL AT BLOCK OUT**

SCALE: 1/2" = 1'-0" (S-506)



A1 **REINFORCEMENT DETAILS AT BENT 43**

SCALE: 1/2" = 1'-0" (S-110)



A3 **REINFORCEMENT DETAILS AT ELECTRICAL BLOCK OUT**

SCALE: 1/2" = 1'-0" (S-117)

1/2" = 1'-0" 0 6" 1' 2' 3' 4' 5'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

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A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES	JZ	DRW	AFC	CHK	MJF

PM/DM	MRI/CWJ
BRANCH MANAGER	NEK
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

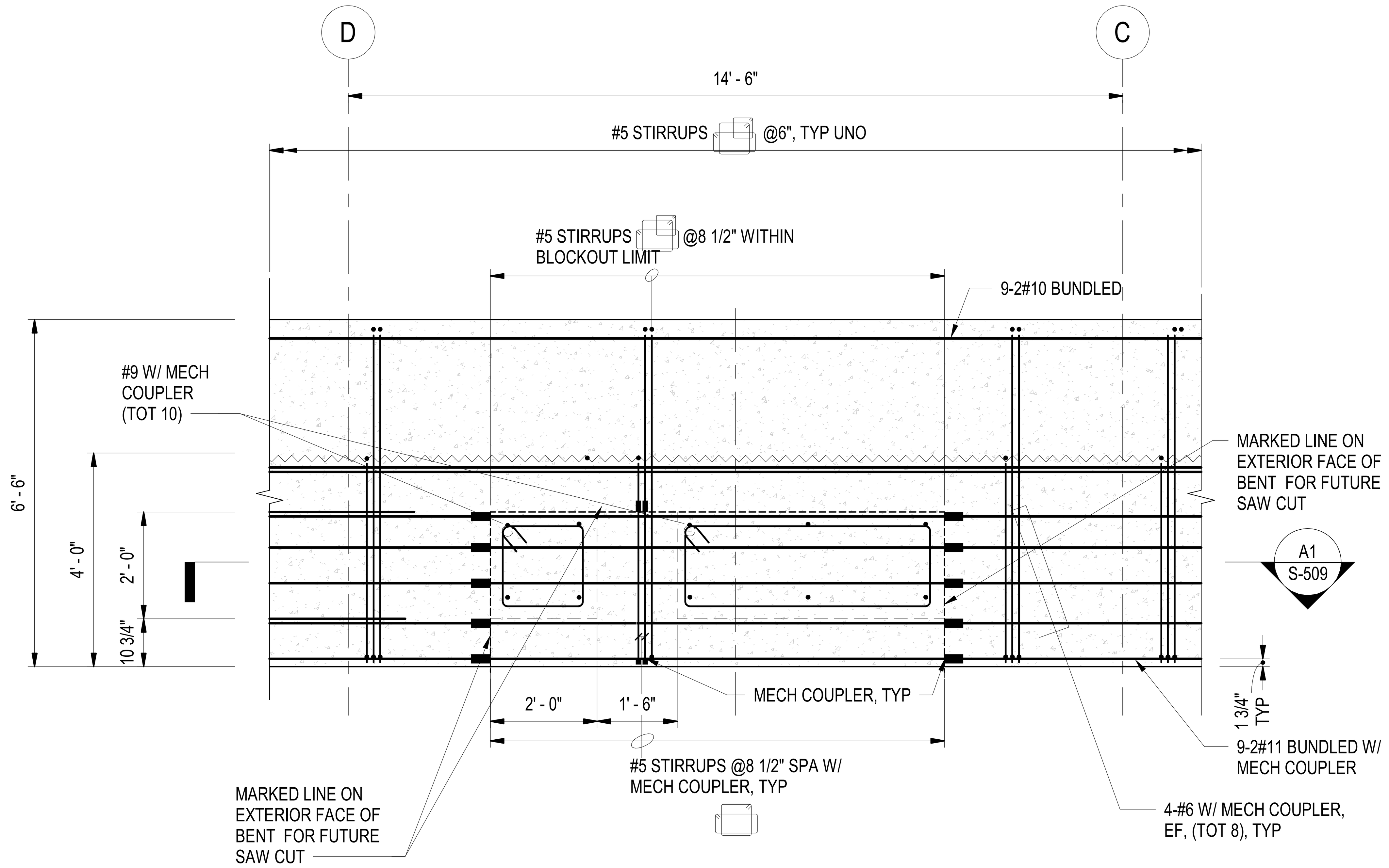
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
CI CORE - HAMPTON ROADS
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
REINFORCEMENT DETAILS FOR CONCRETE BLOCKOUT AT BENT 43 - SHEET 1 OF 2

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936705
SHEET 136 OF 247

S-508

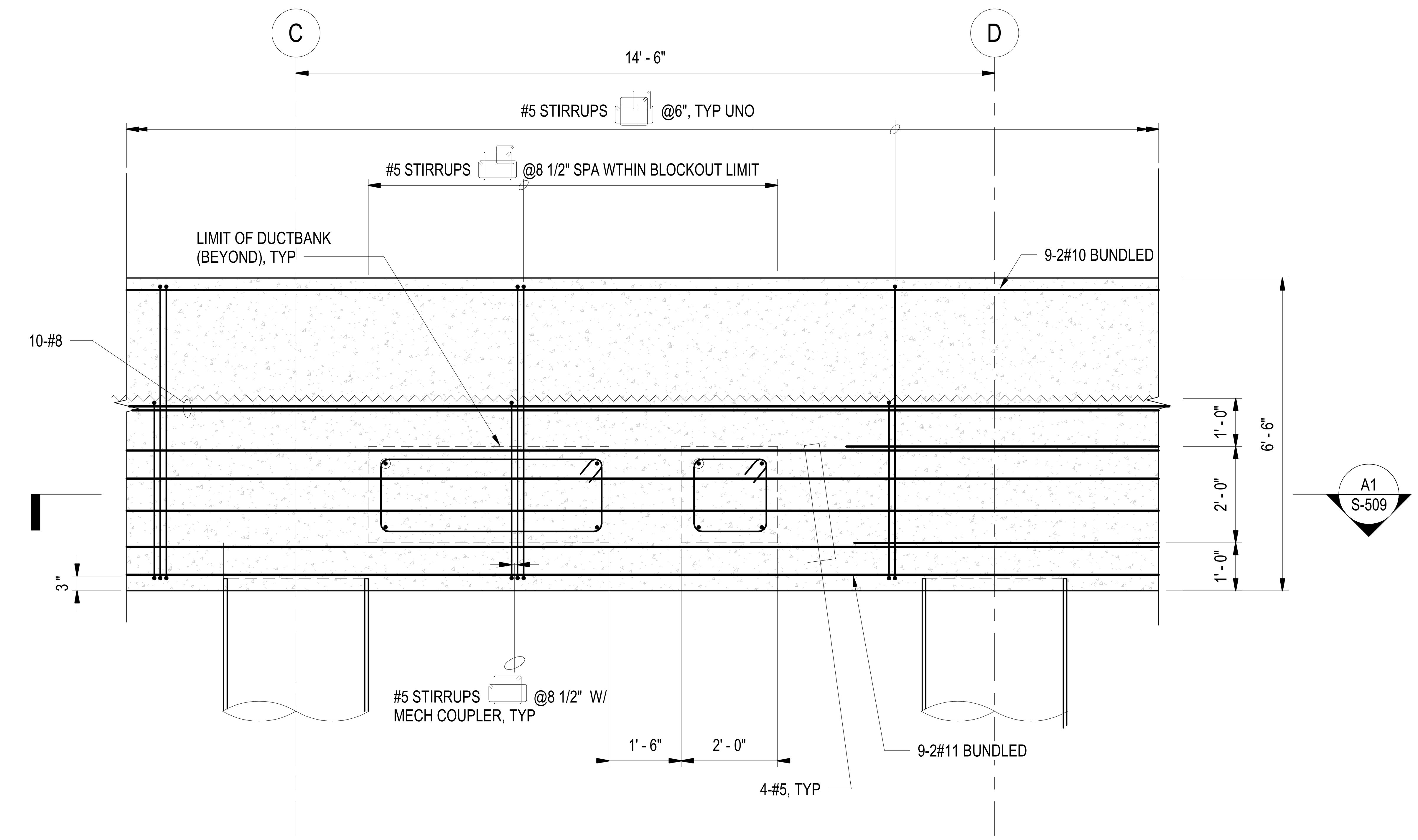
DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:36:39 PM S-508
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt



C1 REINFORCEMENT DETAILS AT BLOCK OUT

SCALE: 1/2" = 1'-0" (S-506)

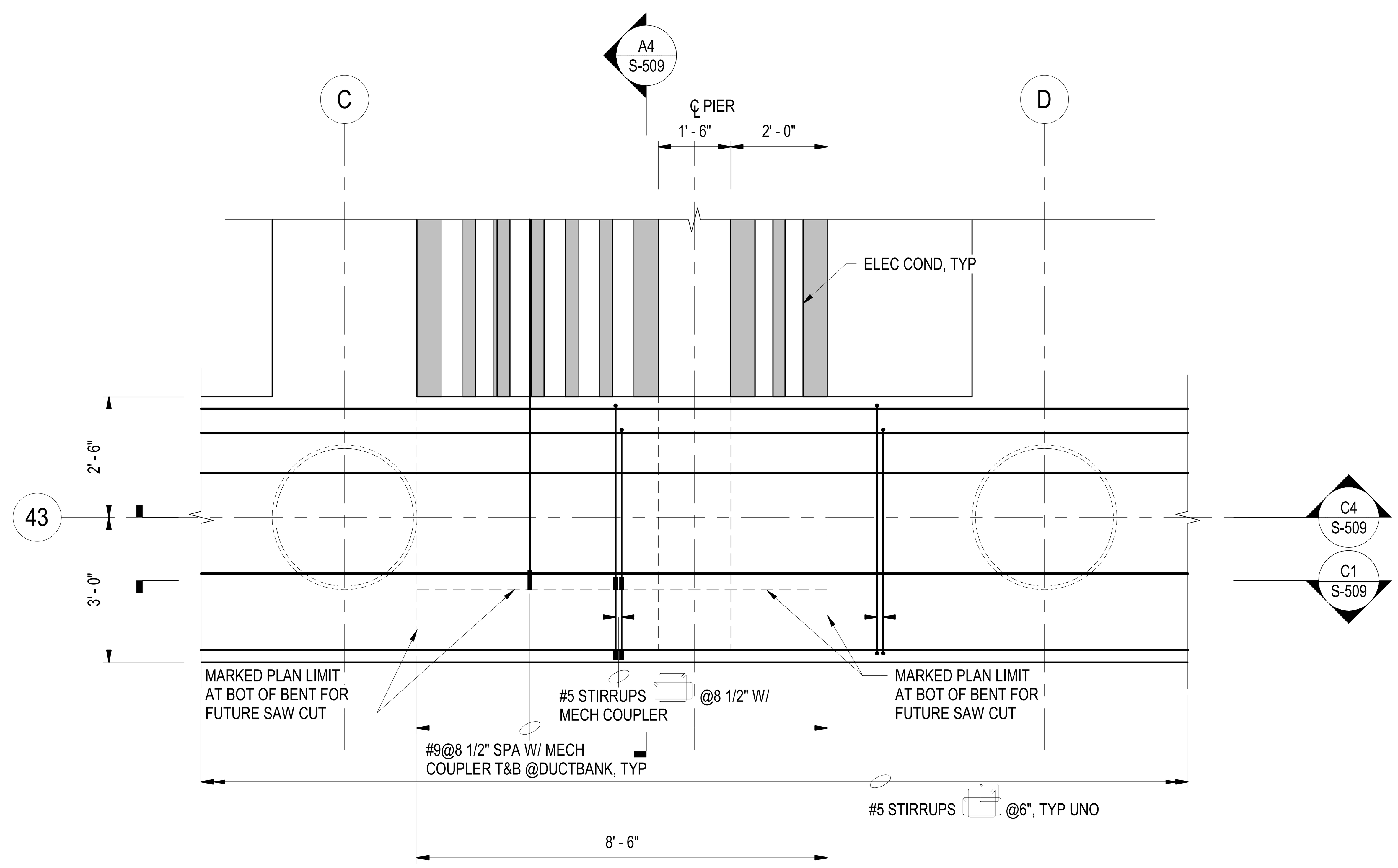


C4 REINFORCEMENT DETAILS BEYOND BLOCK OUT

SCALE: 1/2" = 1'-0" (S-506)

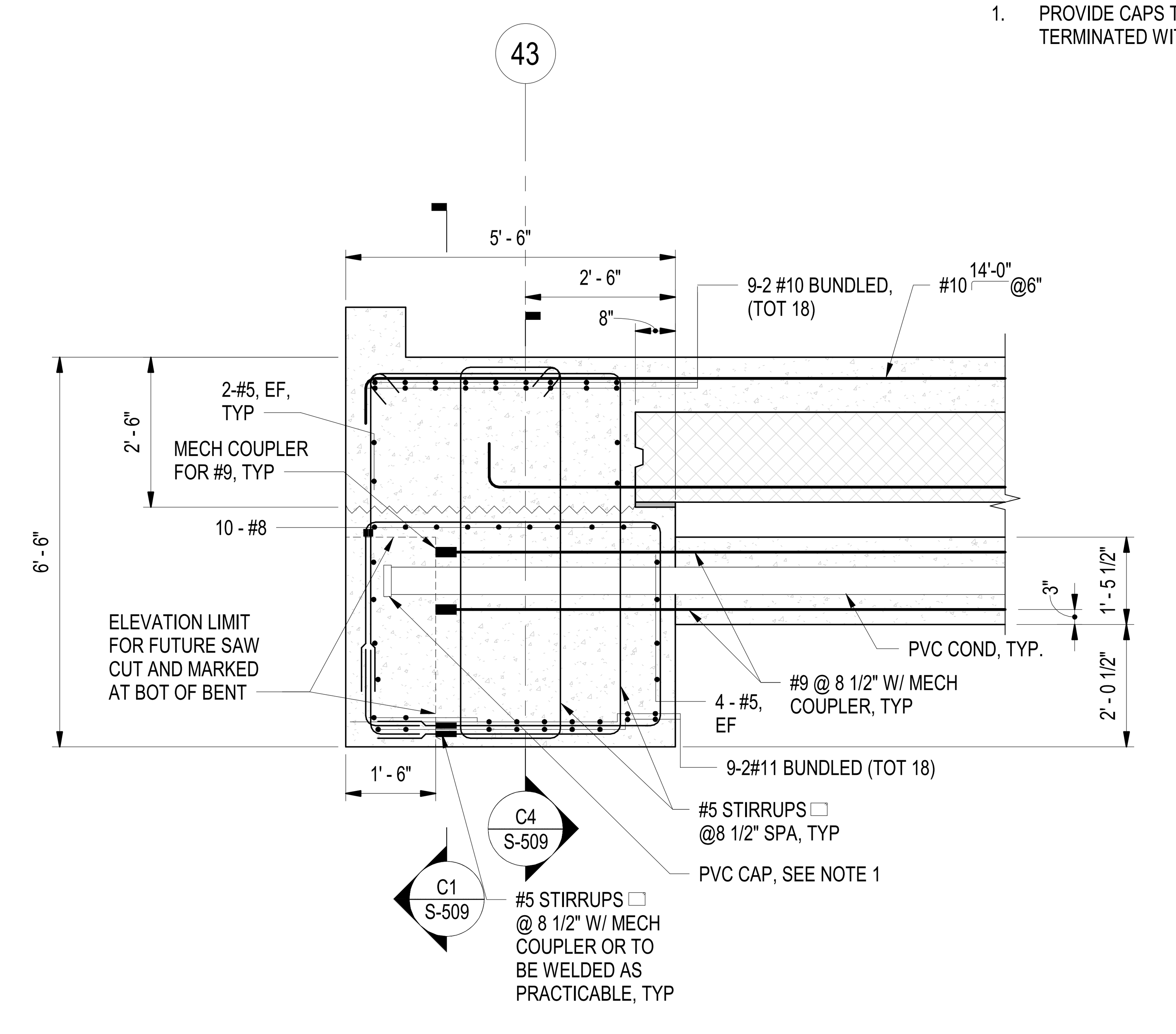
GENERAL SHEET NOTES:

1. PROVIDE CAPS TO SEAL OFF ALL PVC CONDUITS TERMINATED WITHIN BENT LINE 43, TYPICAL.



A1 REINFORCEMENT PLAN DETAILS AT BLOCK OUT

SCALE: 1/2" = 1'-0" (S-110)



A4 REINFORCEMENT ELEVATION DETAILS AT DUCTBANK BLOCKOUT

SCALE: 1/2" = 1'-0" (S-506)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES	JZ	DRW	AFC	CHK	MJF
PM/DM					MRI/CWJ
BRANCH MANAGER					NEK
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

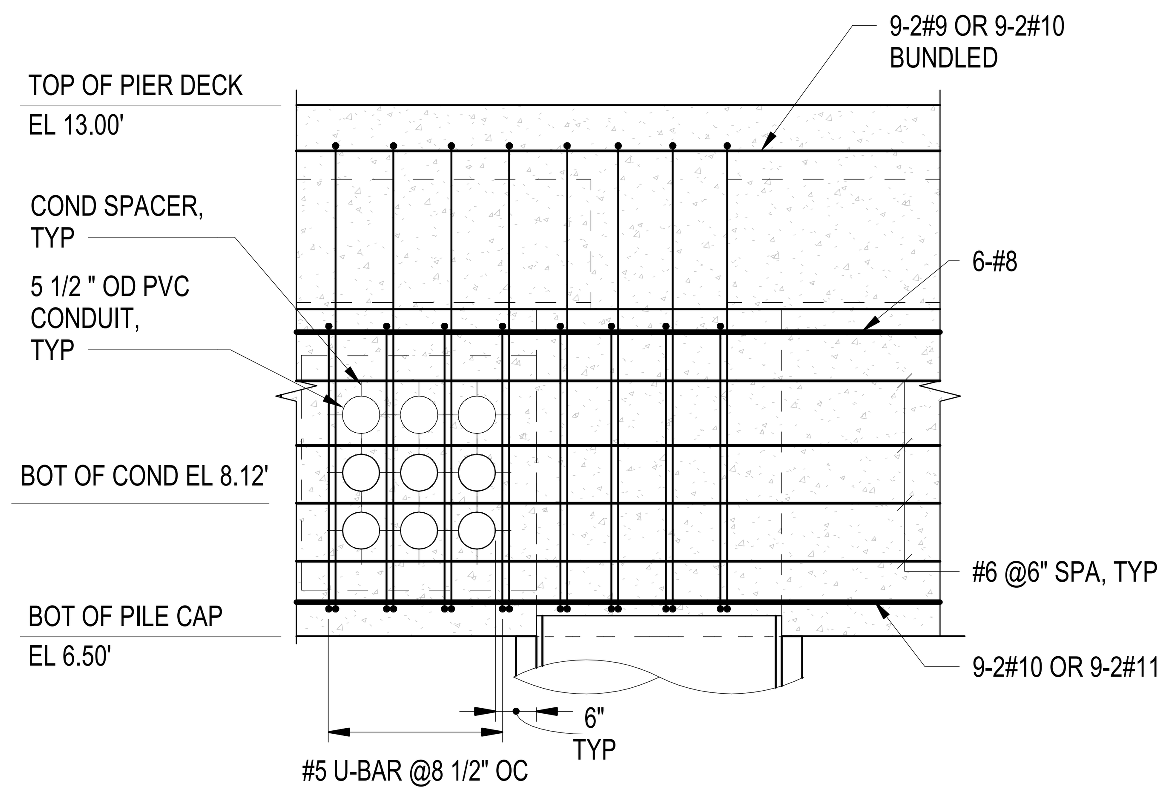
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
REINFORCEMENT DETAILS FOR CONCRETE BLOCKOUT AT BENT 43 - SHEET 2 OF 2

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936706
SHEET 137 OF 247
S-509

DRAWFORM REVISION: 25 AUGUST 2020

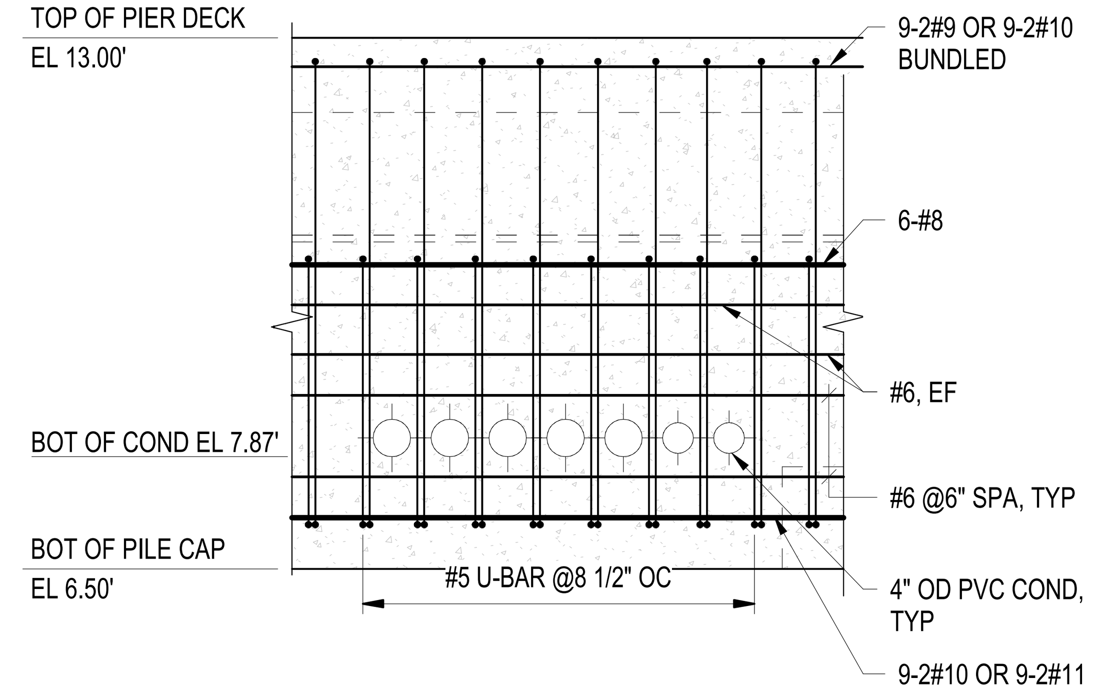
GENERAL SHEET NOTES:

1. ALL CONDUITS IN PILE CAP HAVE MINIMUM 3" CLEARANCE IN BETWEEN.
2. PROVIDE MINIMUM 3" CLEARANCE BETWEEN REINFORCEMENT (#6 @6" SPACING WITHIN PILE CAP AS TYPICAL) AND THE BOTTOM AND TOP ROWS OF CONDUITS.
3. SEE DRAWING S-504 FOR ADDITIONAL REINFORCEMENT.



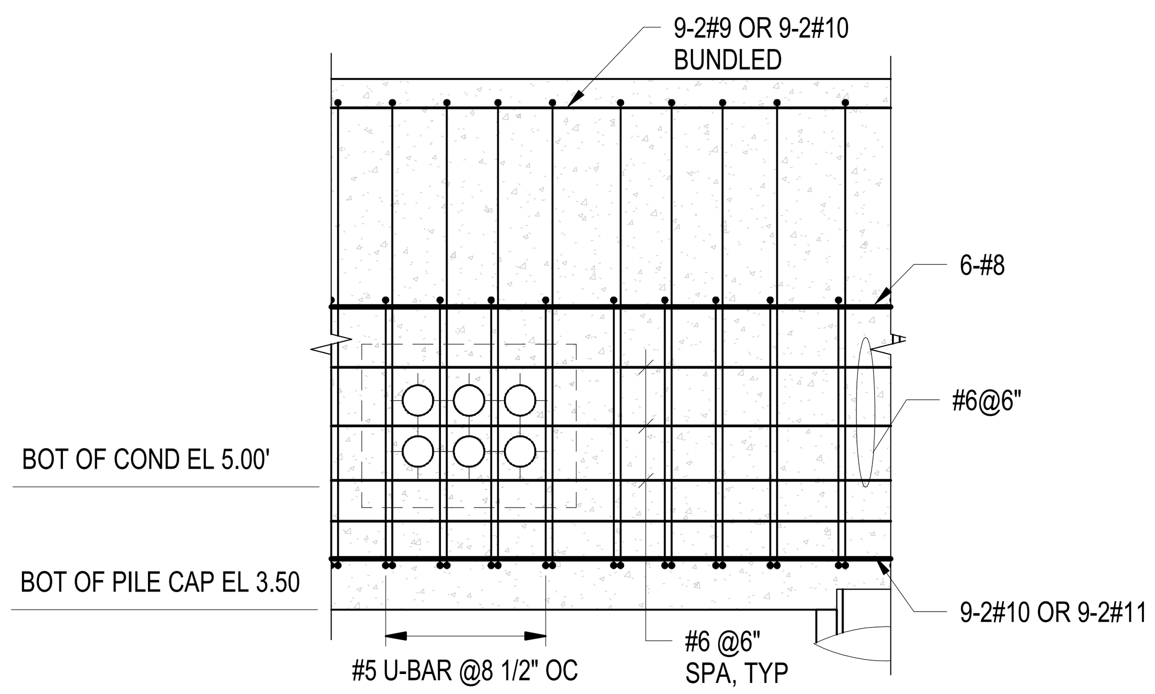
C1 TYPICAL REINFORCEMENT DETAILS FOR PILE CAP 3x3 CONDUITS

SCALE: 1/2" = 1'-0" (S-110, S-111, S-112, S-113, S-114)



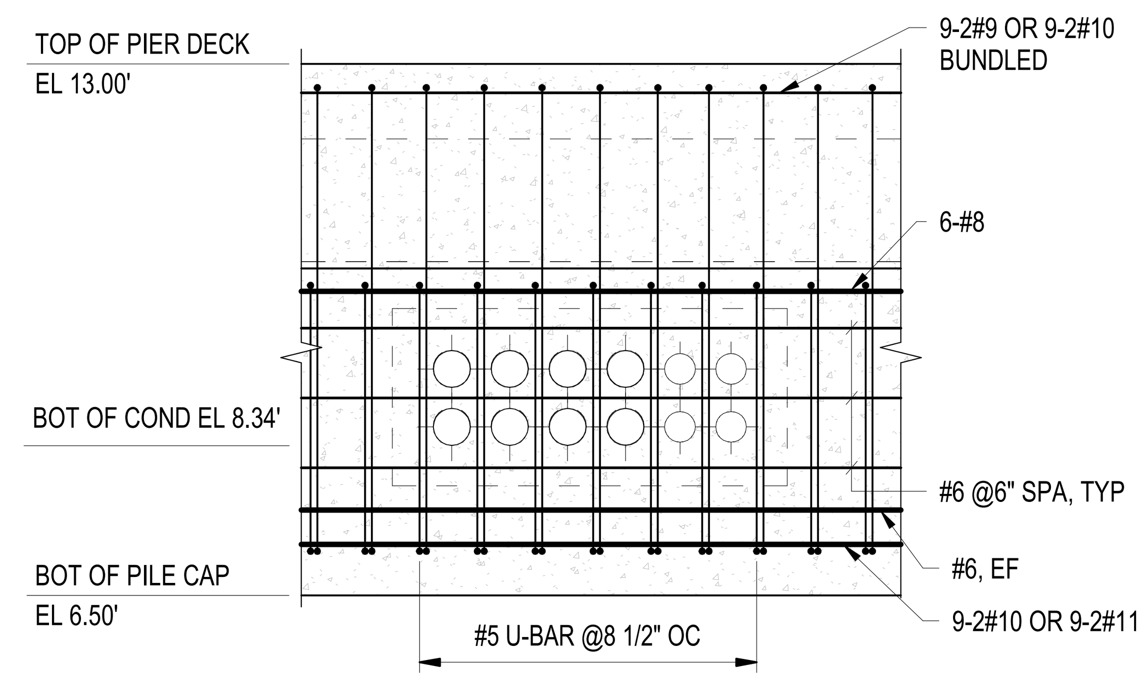
C2 TYPICAL REINFORCEMENT DETAILS FOR STRINGER 1x7 CONDUITS

SCALE: 1/2" = 1'-0" (S-110)



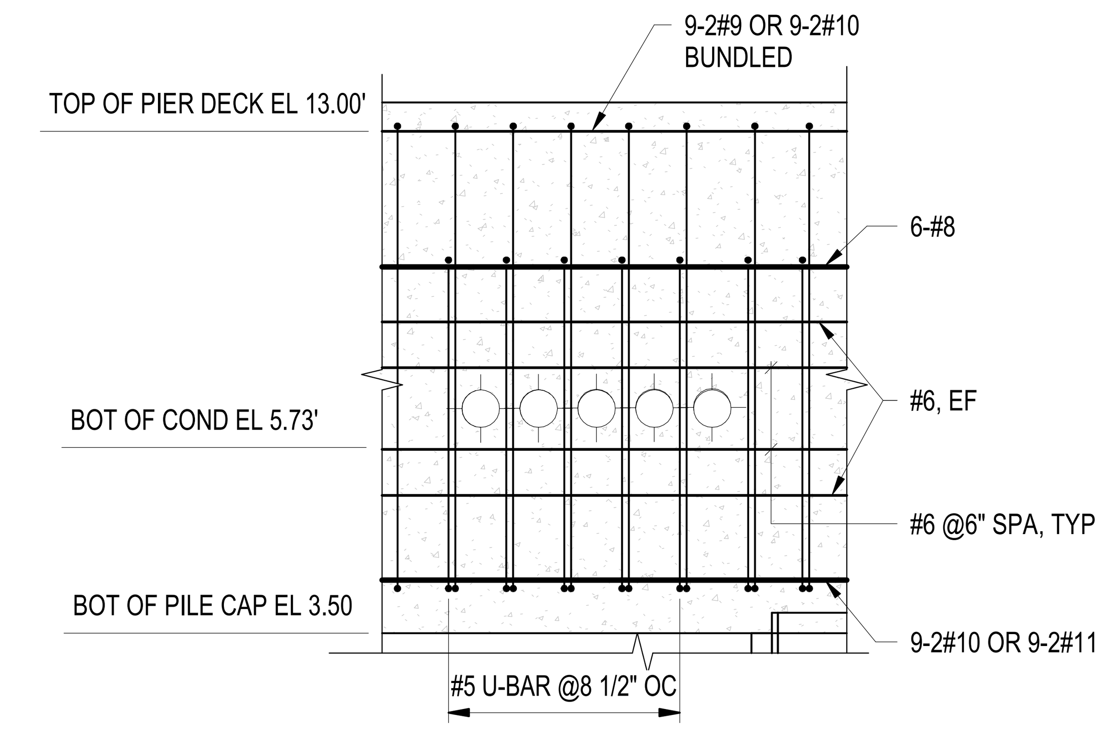
B1 TYPICAL REINFORCEMENT DETAILS FOR PILE CAP 2x3 CONDUITS

SCALE: 1/2" = 1'-0" (S-110, S-111, S-112, S-113, S-114, S-115, S-116)



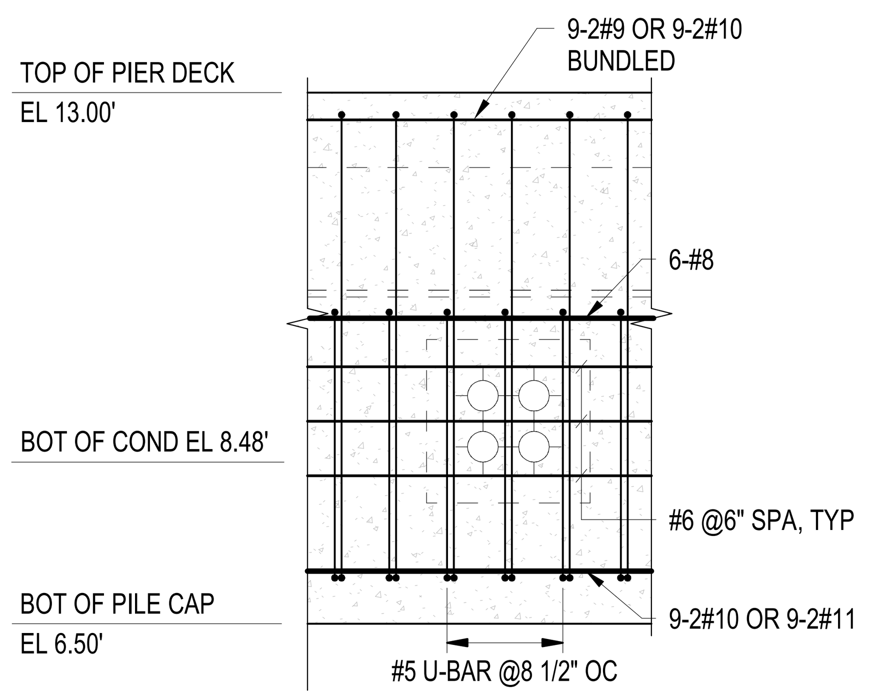
B2 TYPICAL REINFORCEMENT DETAILS FOR PILE CAP 2x6 CONDUITS

SCALE: 1/2" = 1'-0" (S-110, S-111, S-114, S-115)



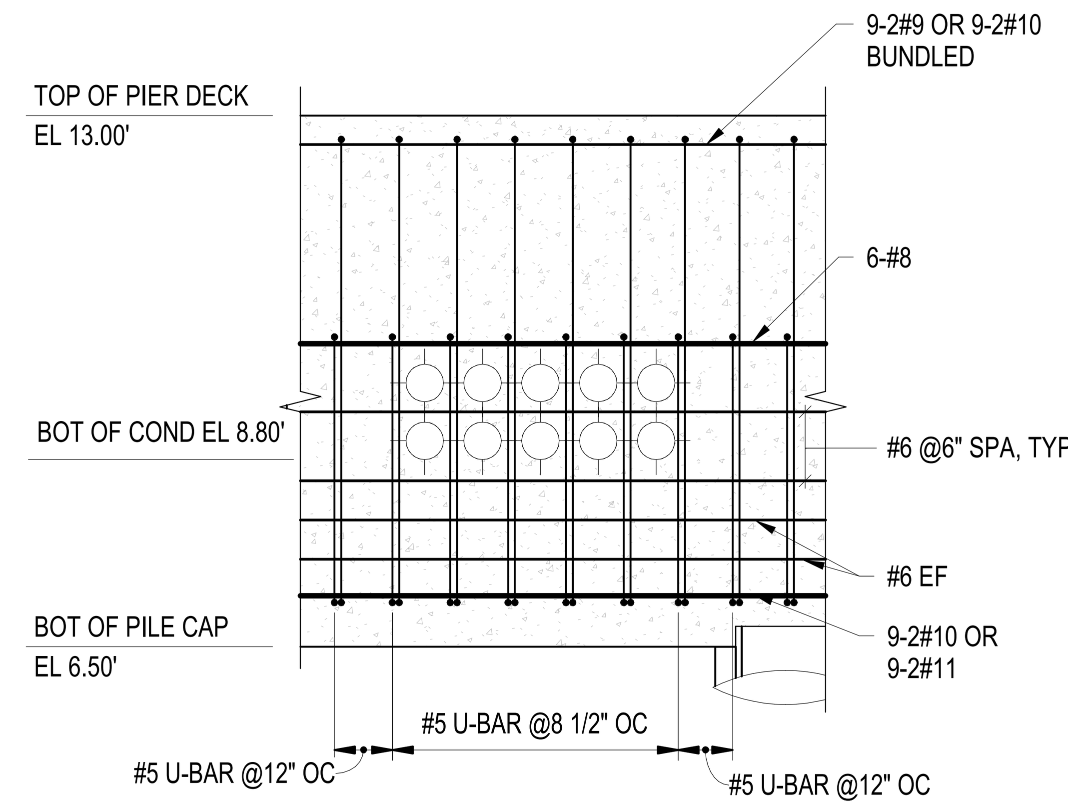
B4 TYPICAL REINFORCEMENT DETAILS FOR PILE CAP 1x5 CONDUITS

SCALE: 1/2" = 1'-0" (S-116)



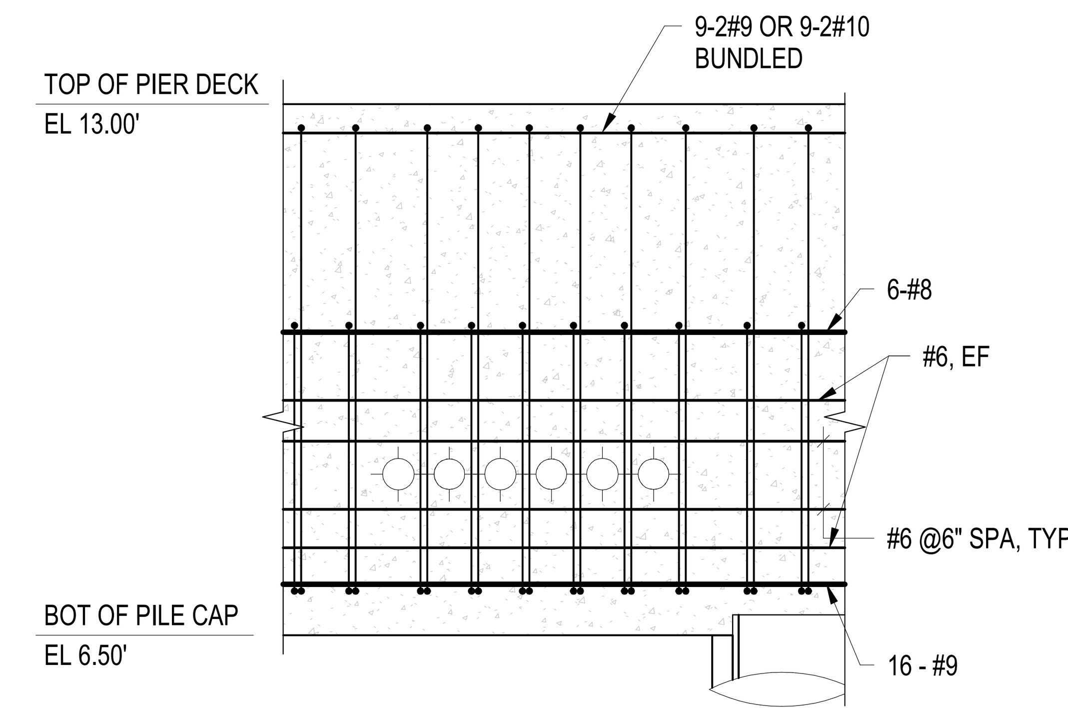
A1 TYPICAL REINFORCEMENT DETAILS FOR PILE CAP 2x2 CONDUITS

SCALE: 1/2" = 1'-0" (S-110)



A2 TYPICAL REINFORCEMENT DETAILS FOR PILE CAP 2x5 CONDUITS

SCALE: 1/2" = 1'-0" (S-110, S-115, S-116)



A4 TYPICAL REINFORCEMENT DETAILS FOR PILE CAP 1x4 CONDUITS

SCALE: 1/2" = 1'-0" (S-110, S-115, S-116)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

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A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW AFC CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
CONCRETE DUCTBANK REINFORCEMENT DETAILS

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936707

SHEET 138 OF 247

S-510

DRAWFORM REVISION: 25 AUGUST 2020

S-510
3/2/2026 7:36:43 PM
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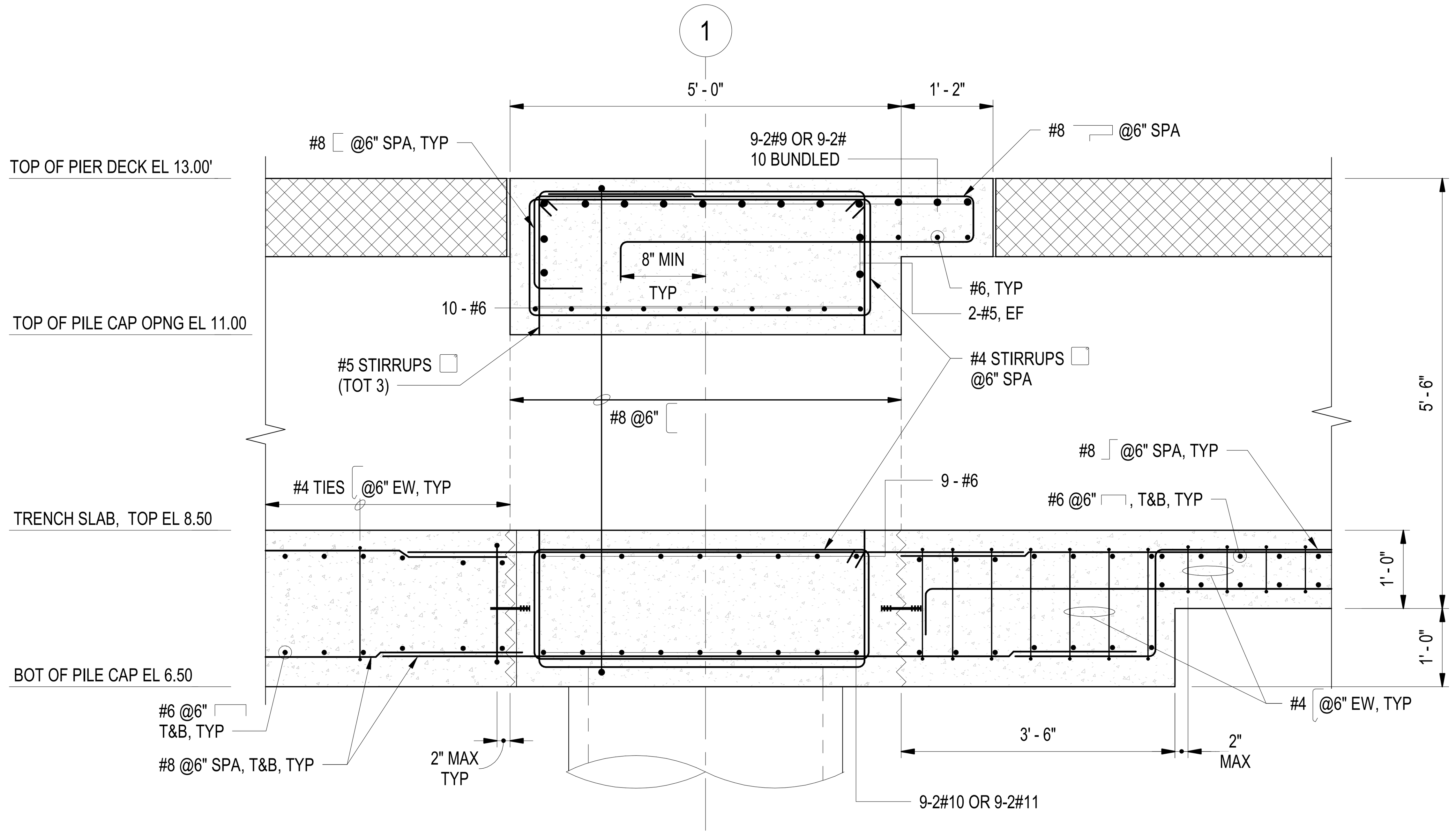
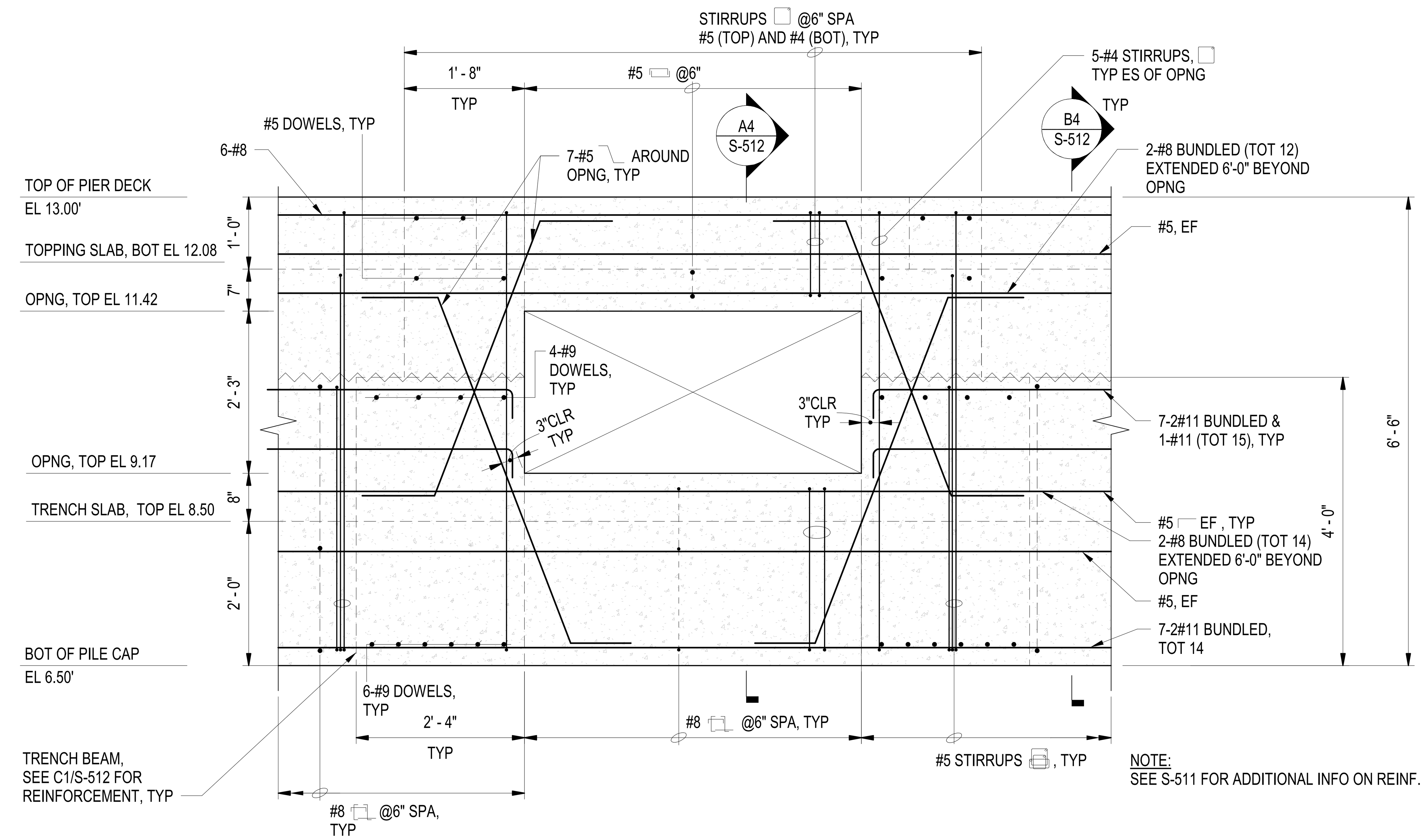


C1 REINFORCEMENT DETAILS AT BENT 1

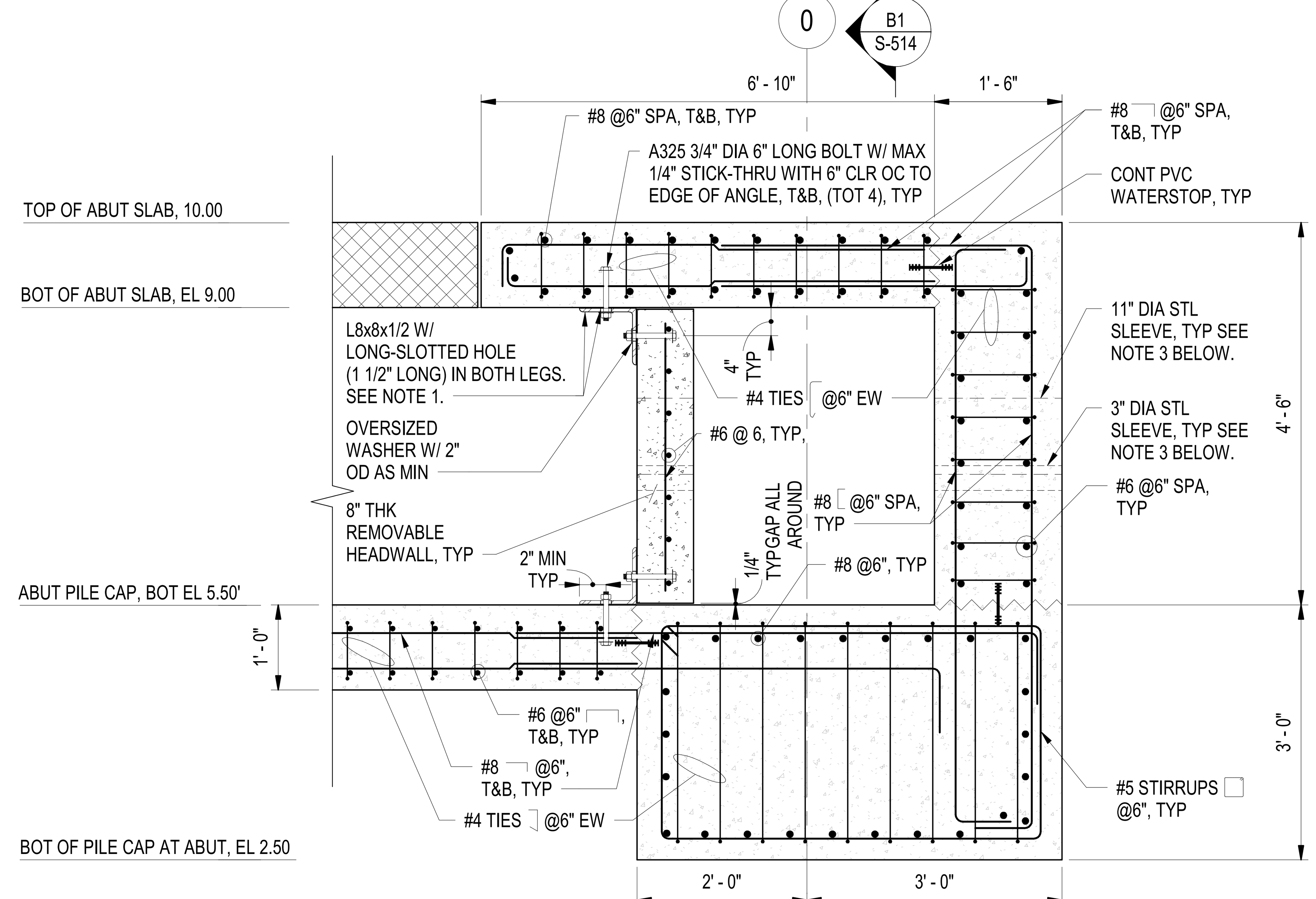
SCALE: 3/4" = 1'-0" (S-305)



NOTE:
SEE S-511 FOR ADDITIONAL INFO ON REINF.

C4 REINFORCEMENT DETAILS AROUND OPENING IN BEAM AT LATERAL TRENCH

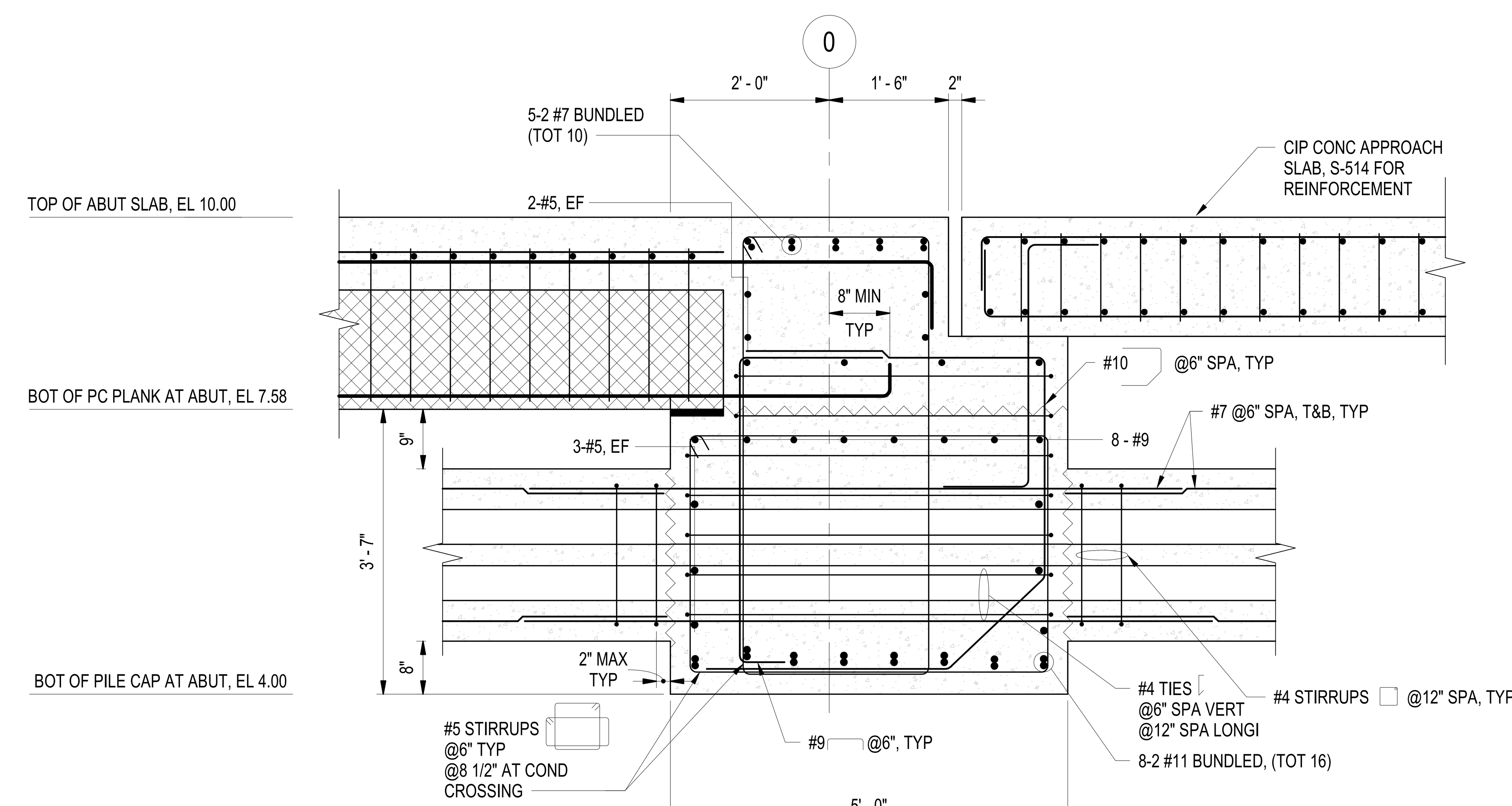
SCALE: 3/4" = 1'-0" (S-512)



NOTES:
1. REPLACE L8x8 BY STEEL PLATE (16" HEIGHT 1/2" THICK) WITH THE SAME LENGTH AT BENT LINES 10, 20 & 30 AT THE TOP OF CONCRETE HEADWALL.
2. ALL STEEL ANGLES AND PLATES ARE LOCATED AT CENTER OF TRENCH AND ALL BOLTS ARE LOCATED AT CENTER OF STEEL ANGLES OR PLATES WITH MINIMUM 4" TO THE NEAREST FACE OF CONCRETE
3. PROVIDE STEEL SLEEVES FOR UTILITY PIPES TO GO THROUGH ALL WALLS AT AND WITH UTILITY TRENCHES, TYPICAL. SEE DRAWING S-517 FOR ADDITIONAL REINFORCEMENT TO BE PROVIDED FOR SLEEVE OPENING LARGER THAN 8" DIAMETER.
4. REINFORCEMENT DETAILS IN CONCRETE HEAD WALL ARE ALSO APPLICABLE TO THE PARTITION WALL AS SHOWN IN DRAWINGS S-112, S-113 AND S-115 AND DOWEL ALL #6 REBAR INTO ADJACENT BEAMS AND SLABS AND MAINTAIN REBAR CONTINUITY.

A1 REINFORCEMENT DETAILS AT UTILITY TRENCH SLAB NEAR LANDSIDE

SCALE: 3/4" = 1'-0" (S-304, S-110, S-111, S-112, S-113, S-114, S-115, S-116, S-305, S-507, S-514)



A4 TYPICAL PILE CAP SECTION AT ABUTMENT WITH DUCTBANK CROSSING

SCALE: 3/4" = 1'-0" (S-116, S-506)

3/4" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

SYM | DESCRIPTION | DATE | APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA

Stantec

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES	JZ	DRW	AFC	CHK	MJF
PM/DM					MRI/CWJ
BRANCH MANAGER					NEK
CHIEF ENGR/ARCH					JWJ
FIRE PROTECTION					HPN

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CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
REINFORCEMENT DETAILS

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936708
SHEET 139 OF 247
S-511

DRAWFORM REVISION: 25 AUGUST 2020

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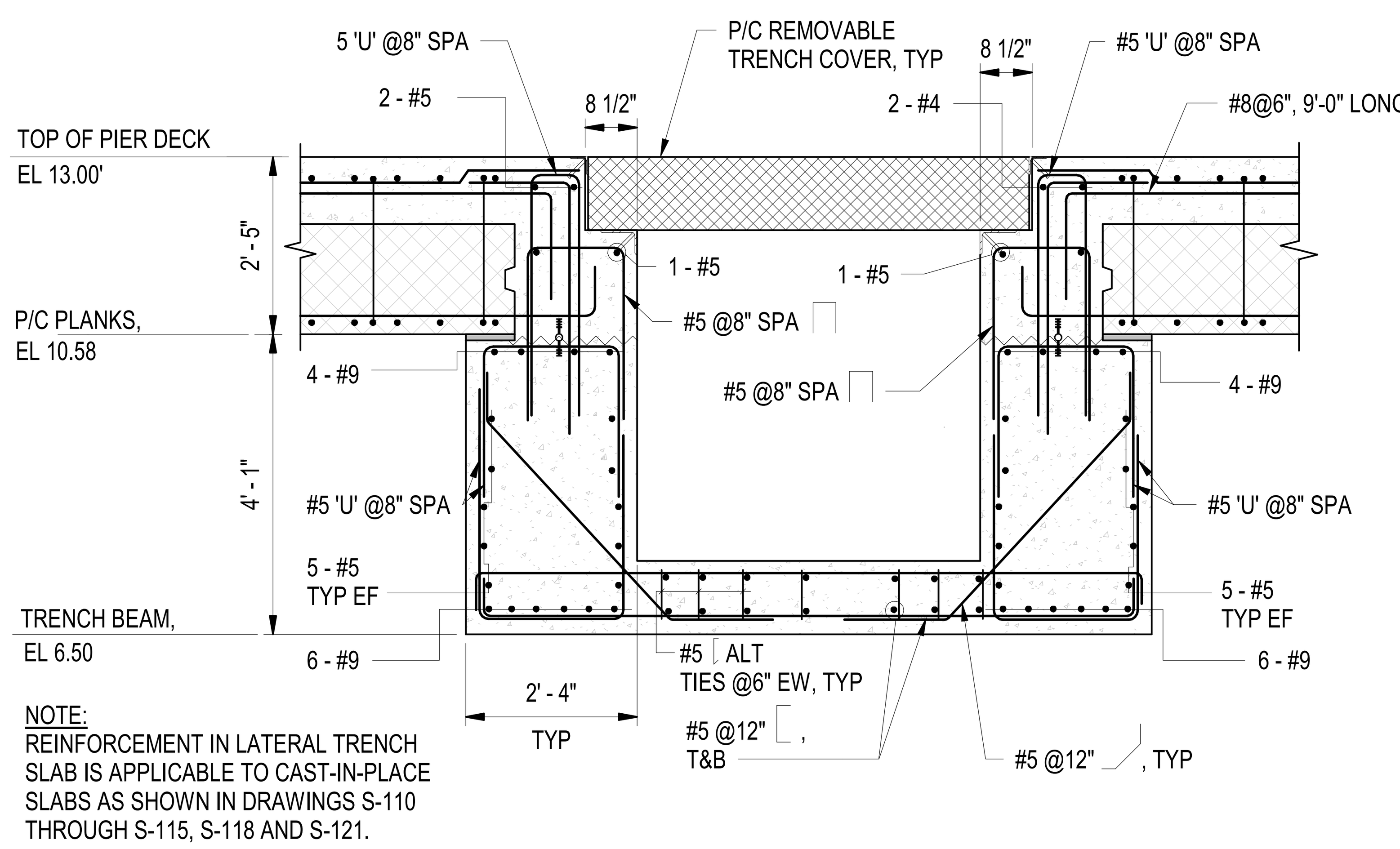
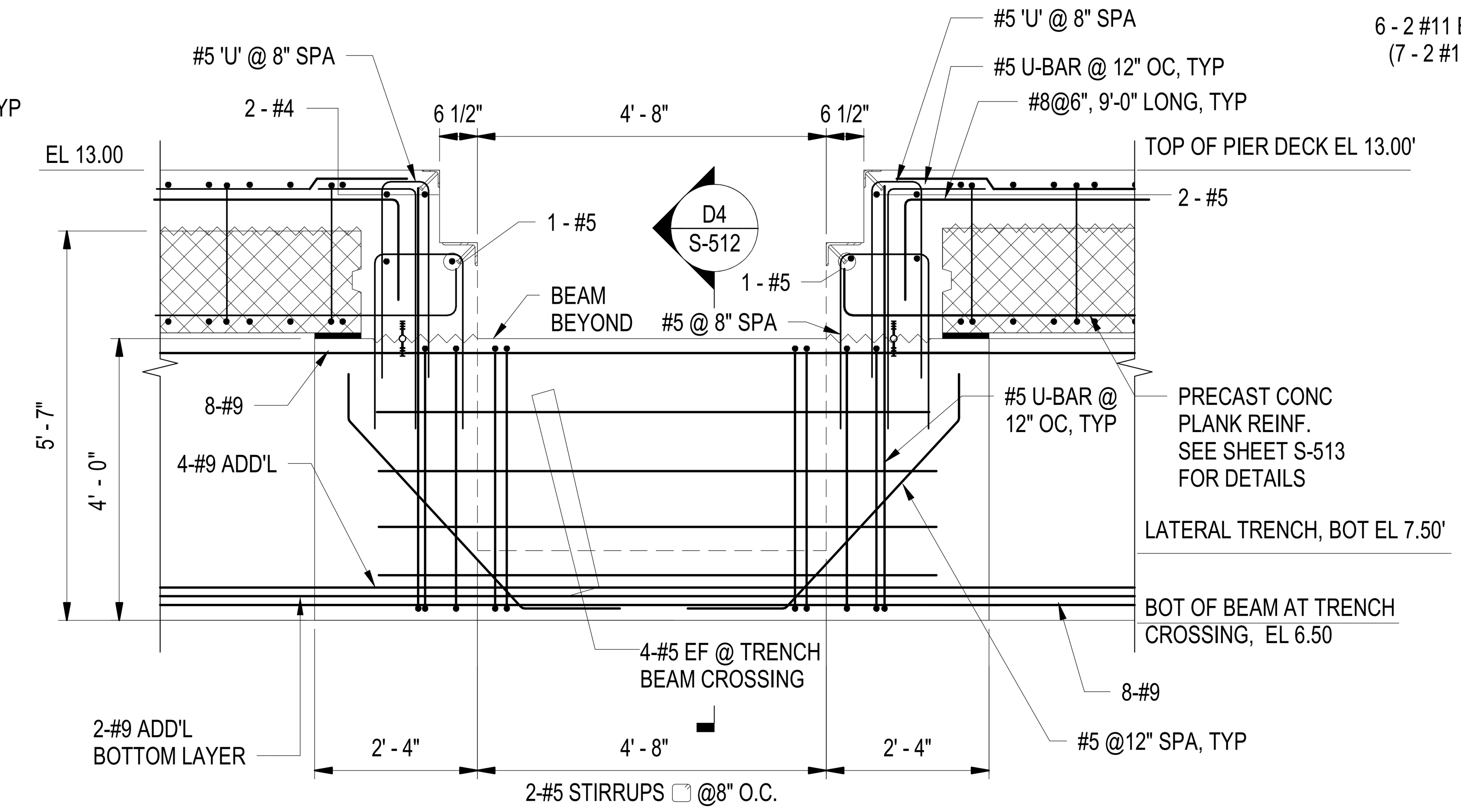
S-511



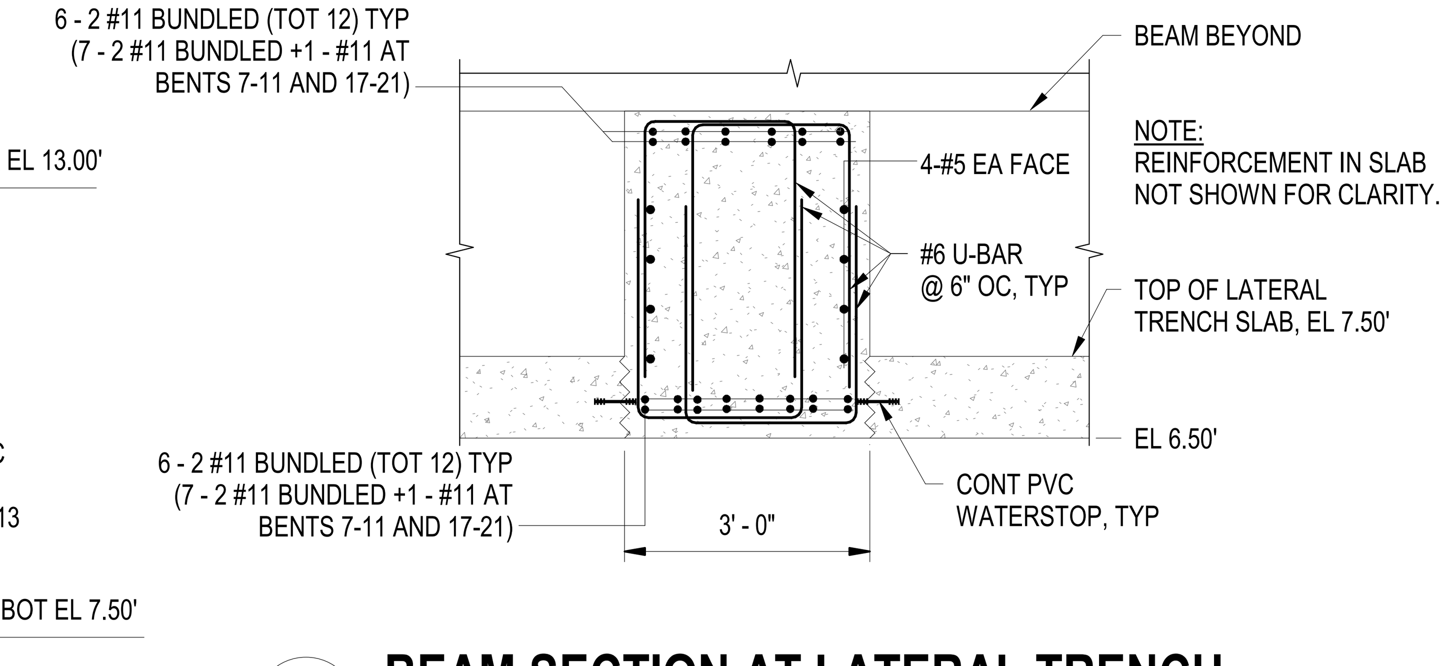
C1 TYPICAL LATERAL TRENCH SECTION

SCALE: 1/2" = 1'-0" (S-110, S-111, S-112, S-113, S-114, S-115, S-118, S-121)



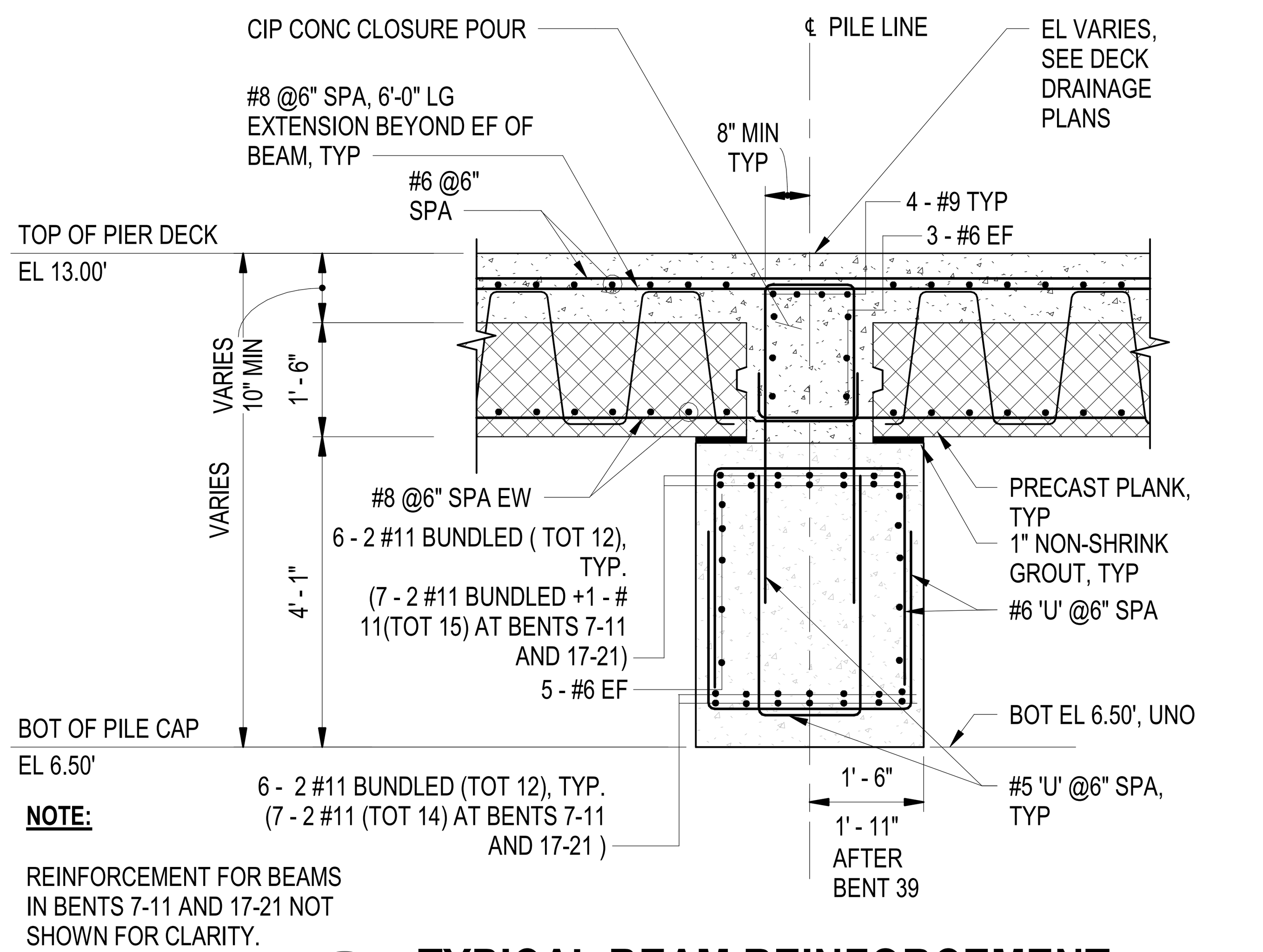
C3 TYPICAL LATERAL TRENCH SECTION AT BEAM CROSSING

SCALE: 1/2" = 1'-0" (S-122)



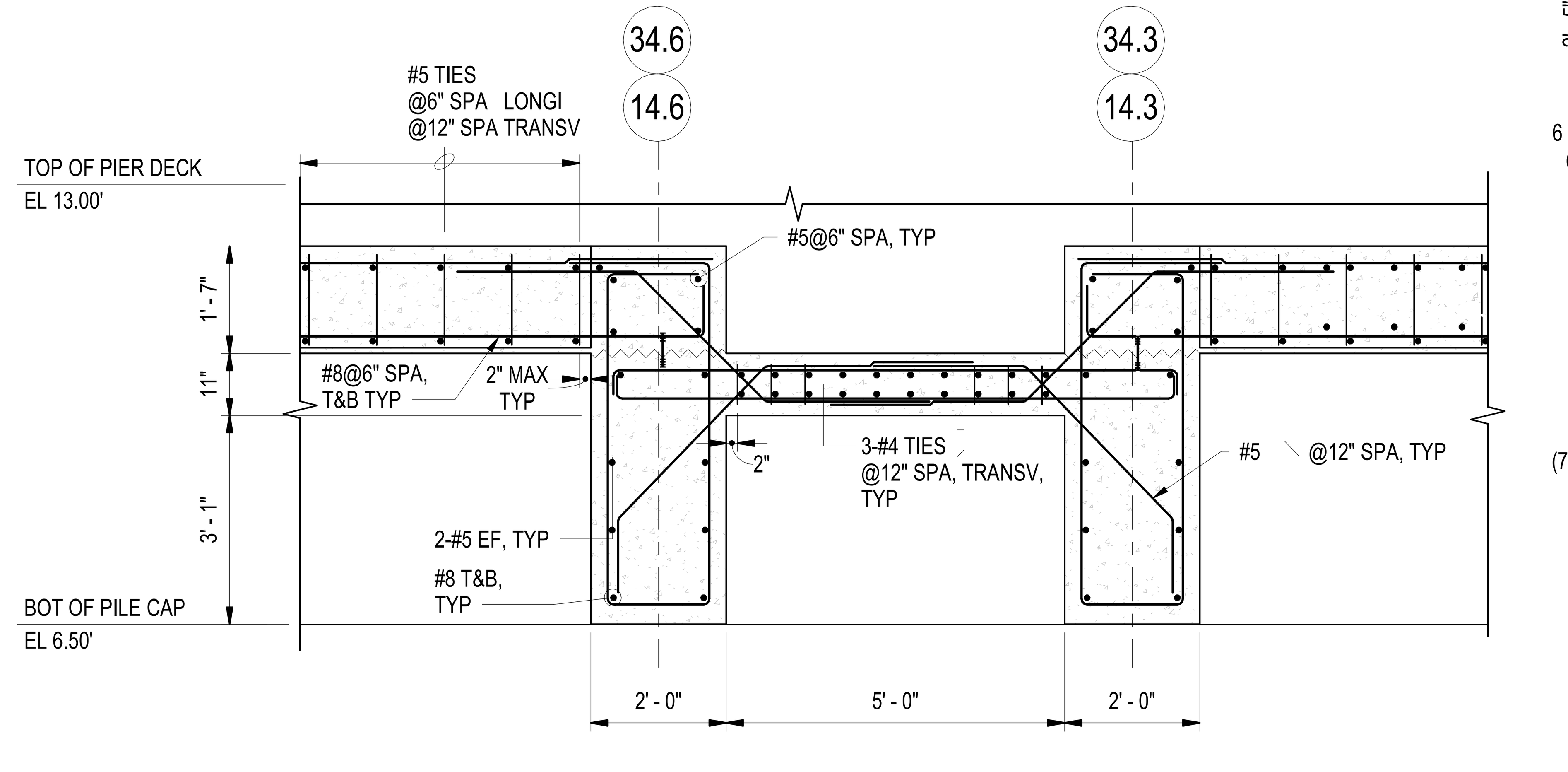
D4 BEAM SECTION AT LATERAL TRENCH

SCALE: 1/2" = 1'-0"



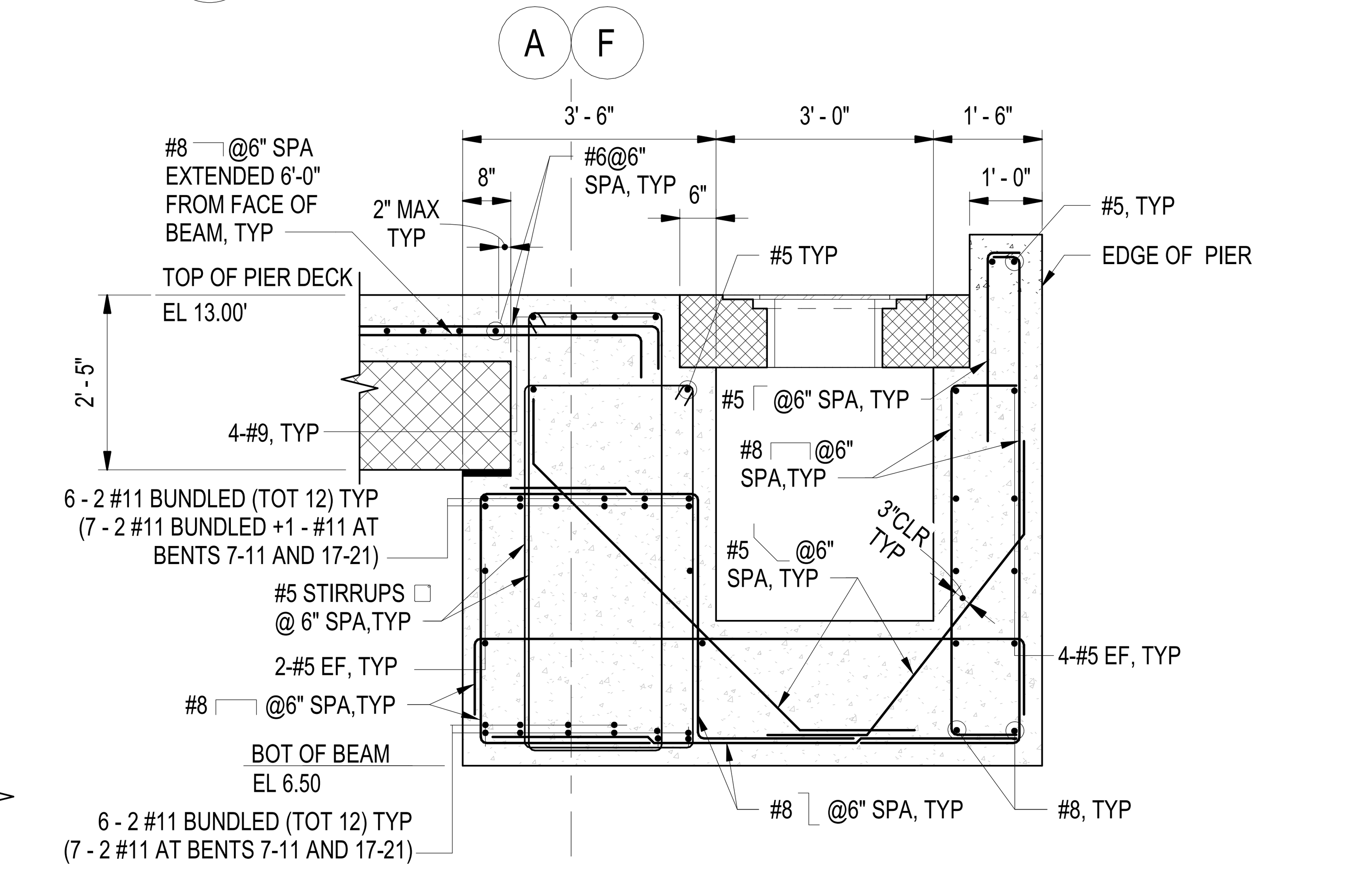
B1 TYPICAL BEAM REINFORCEMENT

SCALE: 1/2" = 1'-0" (S-110, S-111, S-112, S-113, S-114, S-115)



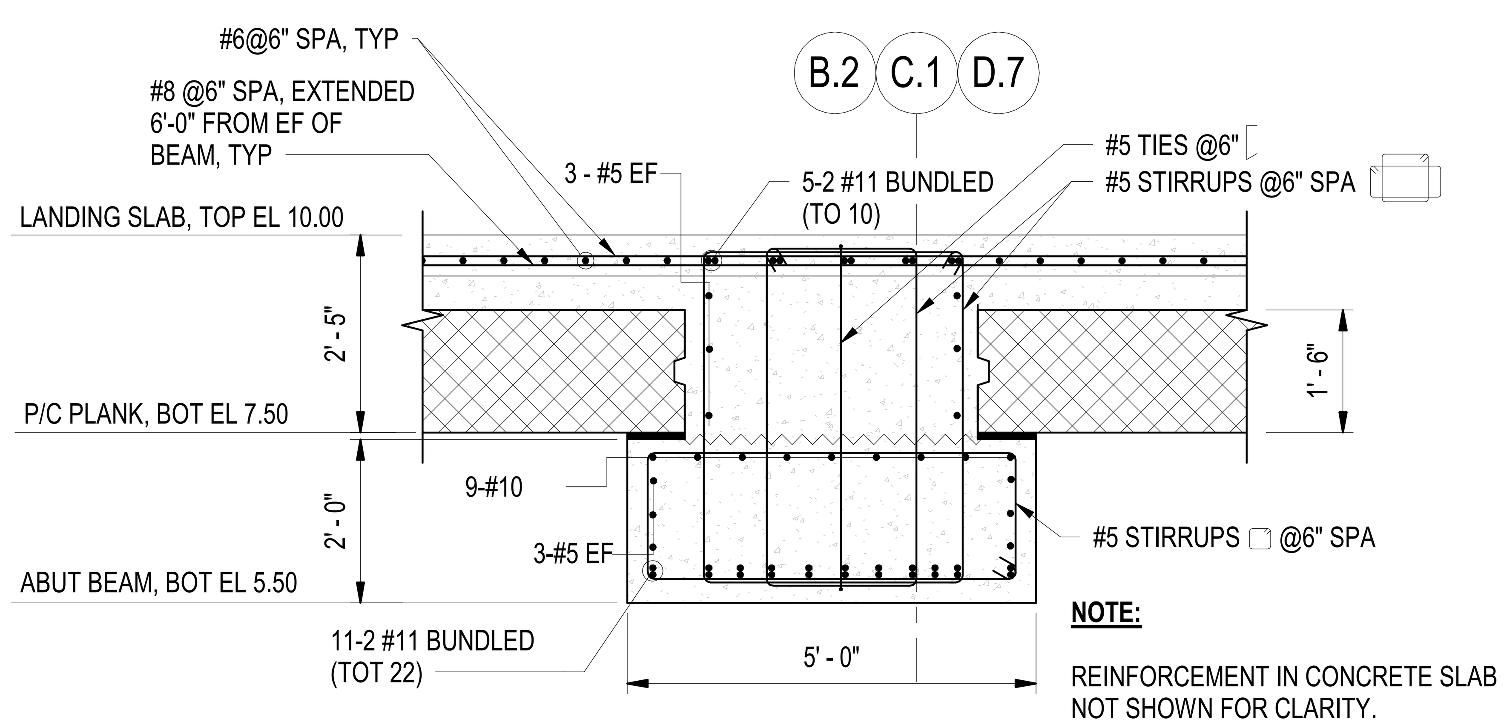
B3 TYPICAL REINFORCEMENT DETAILS FOR RAISED UTILITY TRENCH

SCALE: 1/2" = 1'-0" (S-110)(S-111, S-301)



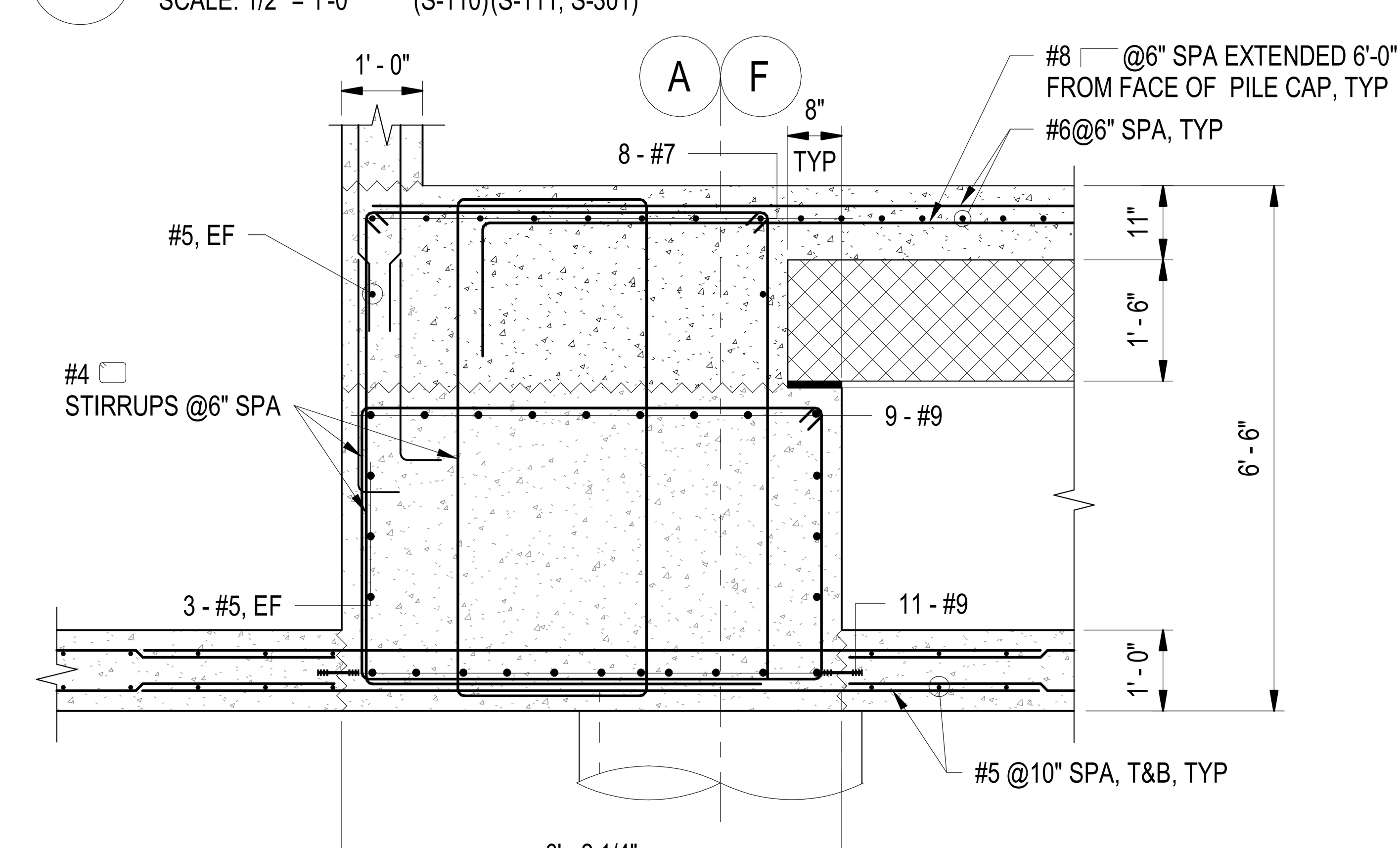
B4 TYPICAL REINFORCEMENT DETAILS FOR TRENCH BEAM

SCALE: 1/2" = 1'-0" (S-502, S-507, S-508, S-110, S-111, S-112, S-113, S-114, S-115, S-511)



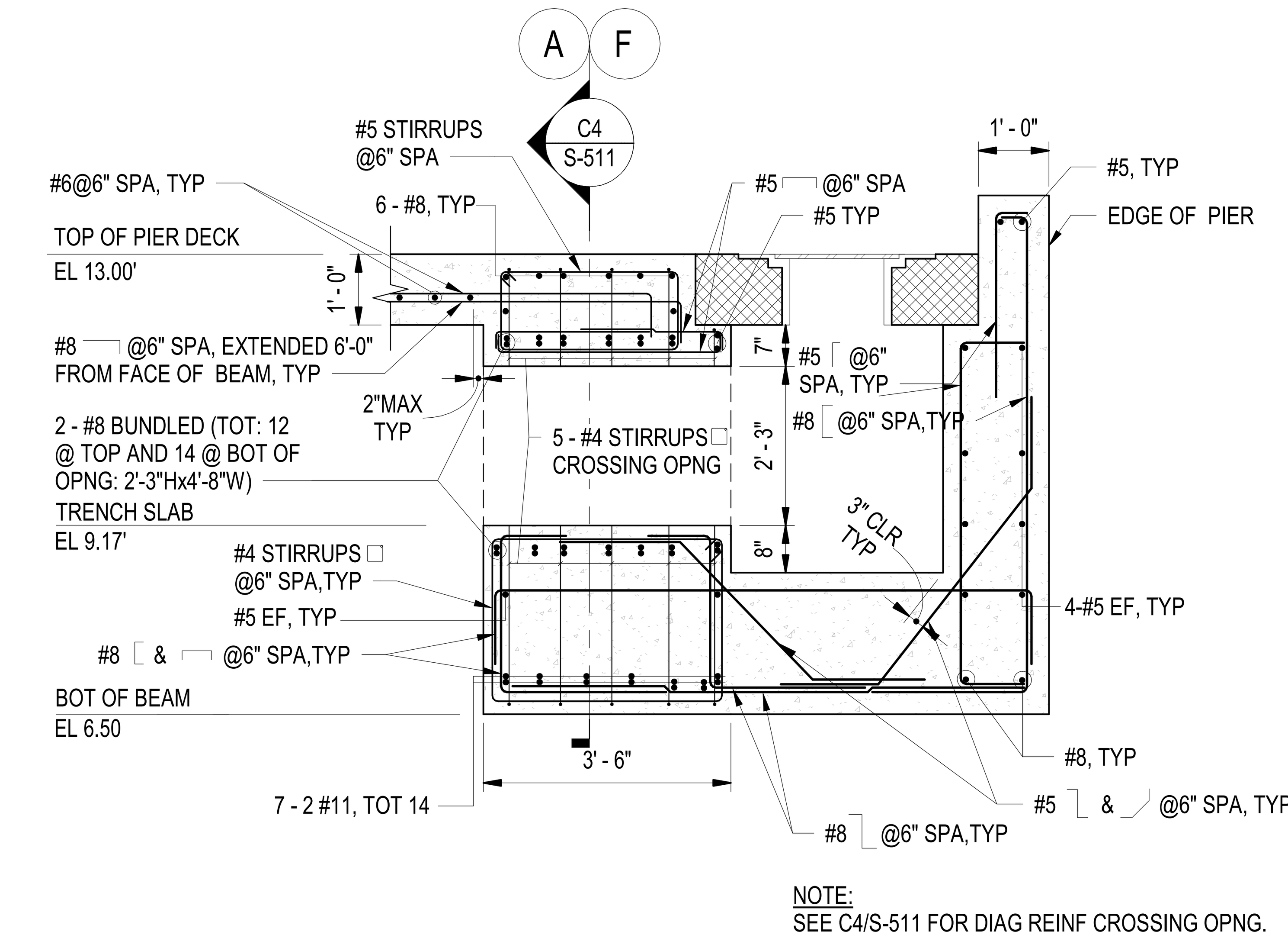
A1 TYPICAL ABUTMENT BEAM REINFORCEMENT

SCALE: 1/2" = 1'-0" (S-116, S-207, S-303)



A3 BEAM REINFORCEMENT DETAILS BELOW SUBSTATION WALL

SCALE: 1/2" = 1'-0" (S-110, S-402, S-403)



A4 REINFORCEMENT DETAILS FOR BEAM WITH OPENING

SCALE: 1/2" = 1'-0" (S-111, S-114, S-122, S-309, S-511)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

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PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

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NAVAL STATION NEWPORT NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

CONCRETE BEAM DETAILS

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936709

SHEET 140 OF 247

S-512

DRAWFORM REVISION: 26 AUGUST 2020

S-512
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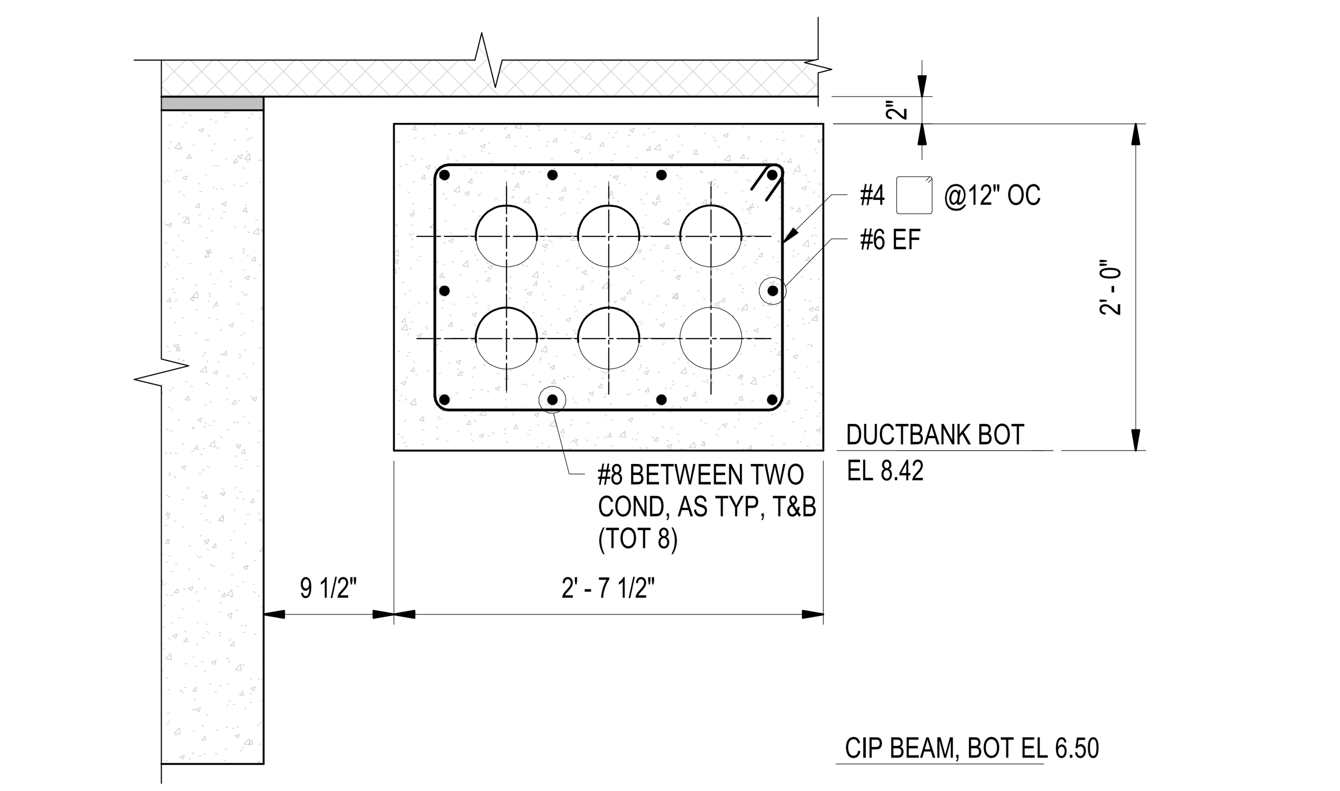
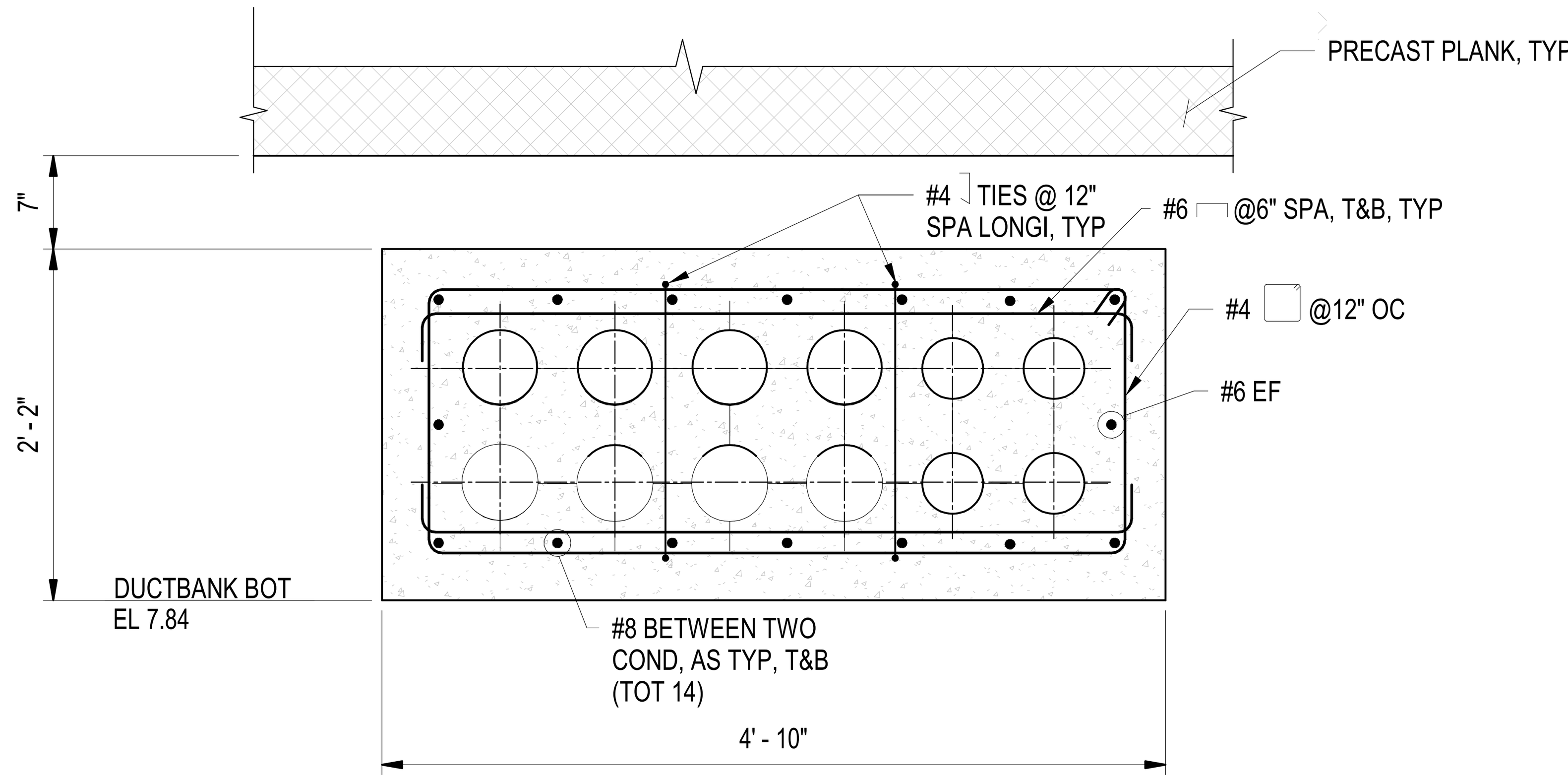


C1 **TYPICAL REINFORCEMENT DETAILS IN DUCTBANK 3.1 CONDUIT COLUMNS**

SCALE: 1" = 1'-0" (S-110, S-111, S-112, S-113, S-114, S-115, S-116)

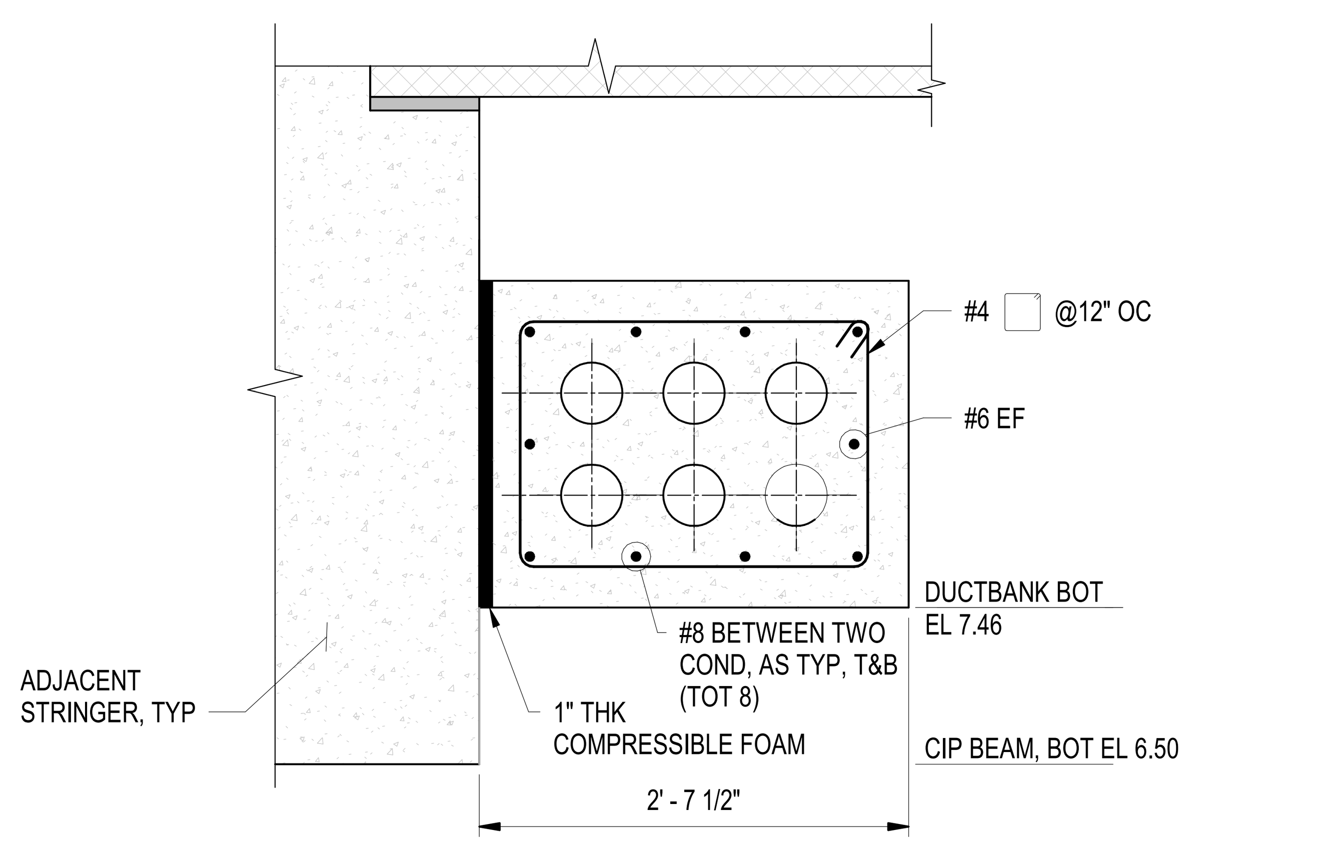


C2 **TYPICAL REINFORCEMENT DETAILS IN DUCTBANK 6 CONDUIT COLUMNS**

SCALE: 1" = 1'-0" (S-110) (S-114, S-115)

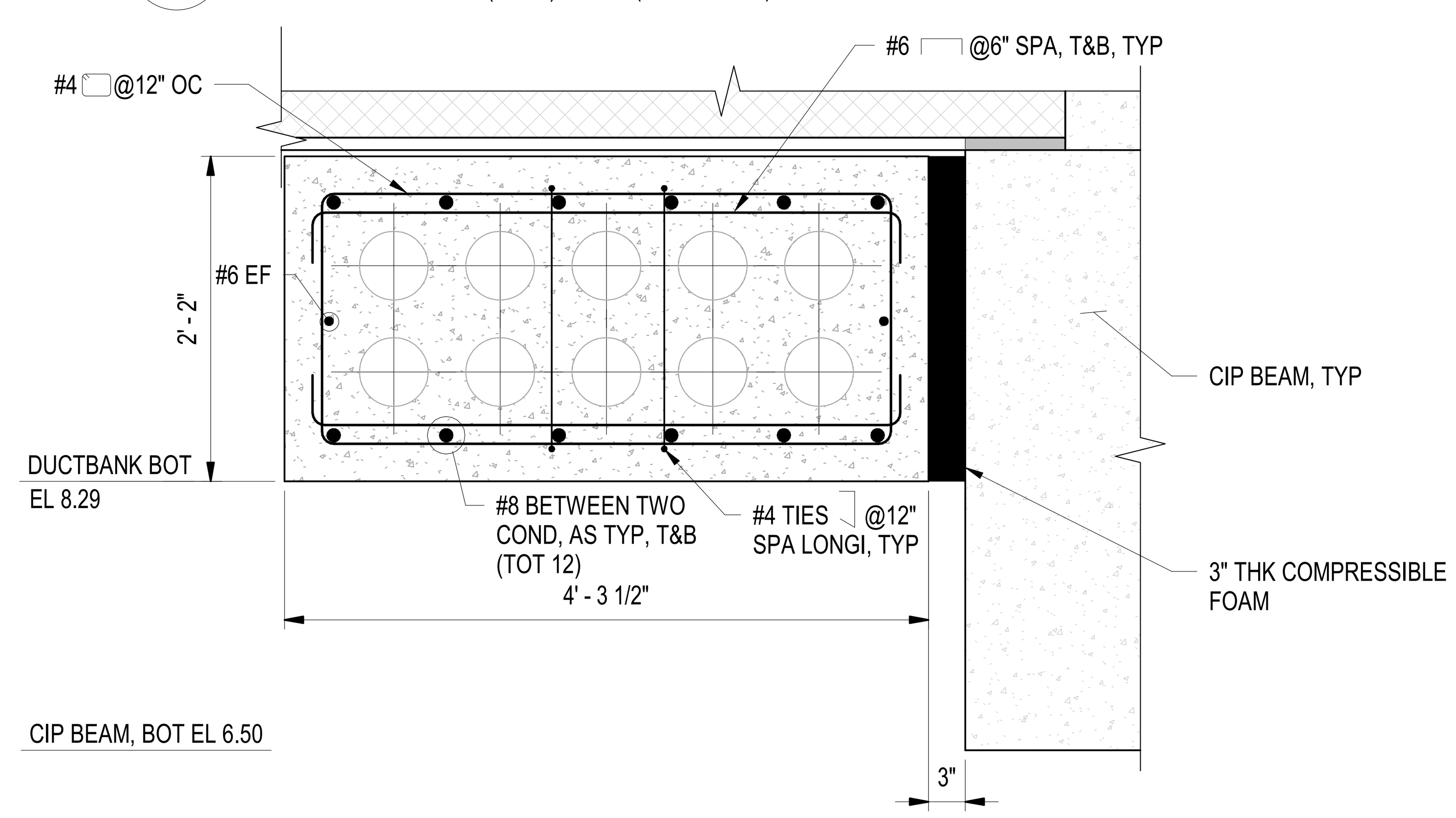
GENERAL SHEET NOTES:

1. ALL CONDUITS IN DUCTBANK HAVE MINIMUM 3" CLEARANCE IN BETWEEN
2. PROVIDE MINIMUM 3" CLEARANCE BETWEEN REINFORCEMENT (#6 @6" SPACING WITHIN PILE CAP AS TYPICAL)AND BOTTOM AND TOP ROWS OF CONDUITS.
3. USE 5,000 PSI MARINE CONCRETE FOR ALL CONDUIT DUCTBANKS.



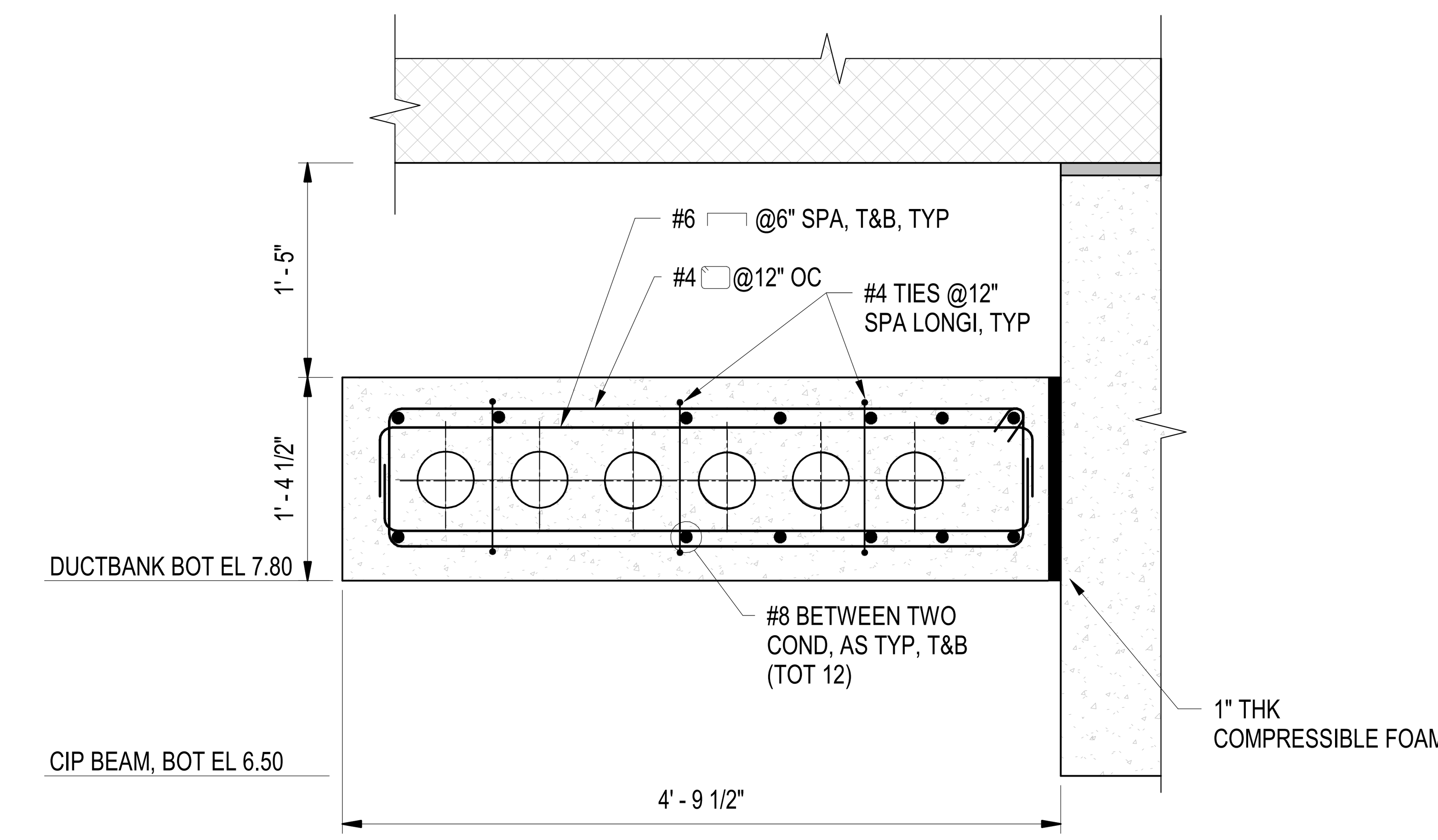
B1 **TYPICAL REINFORCEMENT DETAILS IN DUCTBANK 3 CONDUIT COLUMNS**

SCALE: 1" = 1'-0" (S-110, S-111, S-114, S-115, S-116)



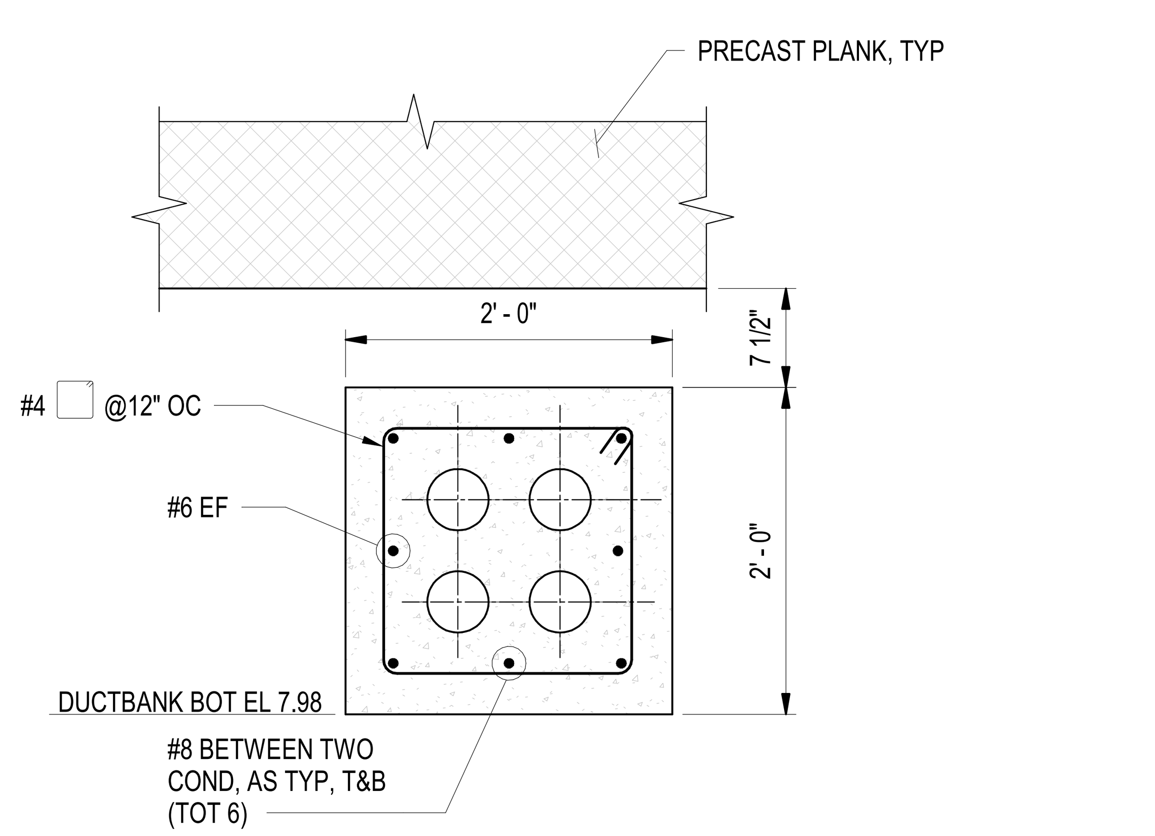
B2 **TYPICAL REINFORCEMENT DETAILS IN DUCTBANK 5 CONDUIT COLUMNS**

SCALE: 1" = 1'-0" (S-115)



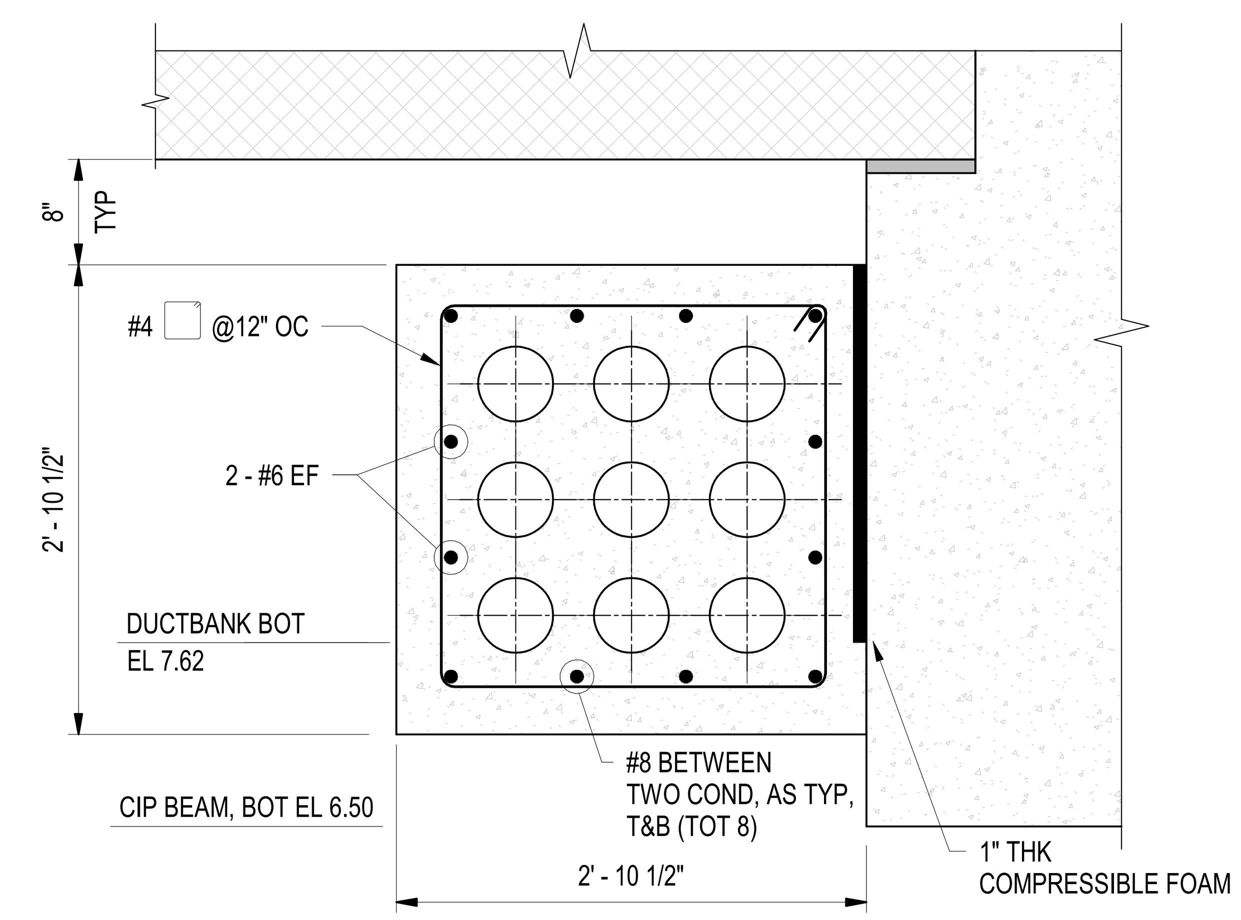
B4 **TYPICAL REINFORCEMENT DETAILS IN DUCTBANK 6 CONDUIT COLUMNS**

SCALE: 1" = 1'-0" (S-110)



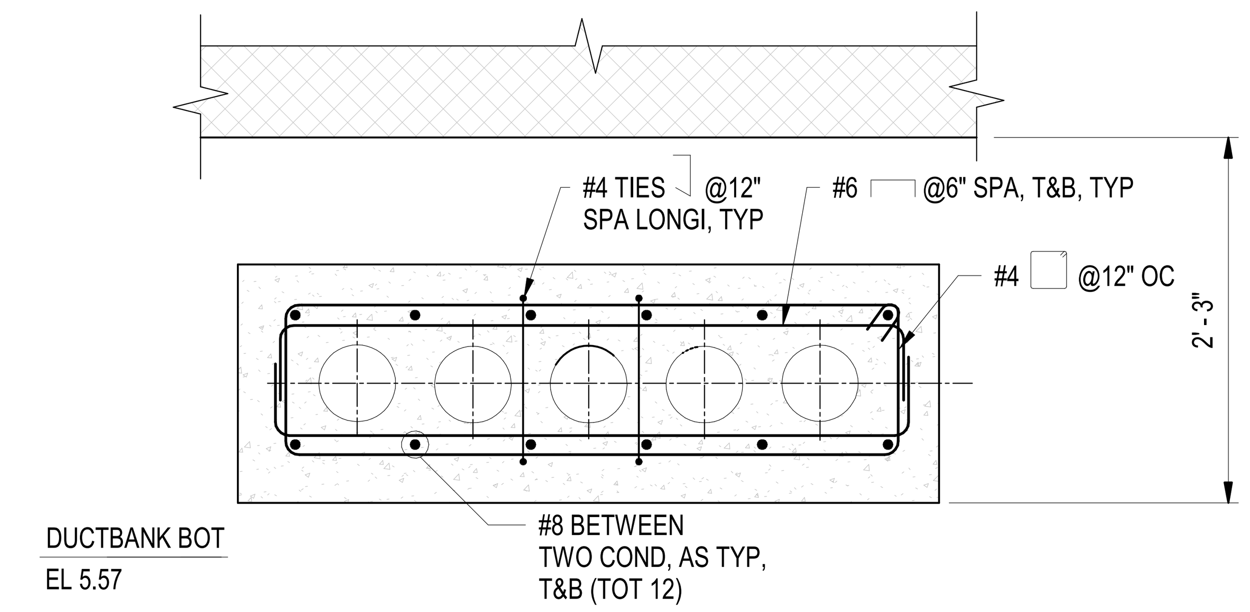
A1 **TYPICAL REINFORCEMENT DETAILS IN DUCTBANK 2 CONDUIT COLUMNS**

SCALE: 1" = 1'-0" (S-110, S-111, S-114, S-116)



A2 **TYPICAL REINFORCEMENT DETAILS IN DUCTBANK NEXT TO STRINGER 3 CONDUIT COLUMNS**

SCALE: 1" = 1'-0" (S-110, S-114)



A4 **TYPICAL REINFORCEMENT DETAILS IN DUCTBANK 5 CONDUIT COLUMNS 1 ROW**

SCALE: 1" = 1'-0" (S-116)

1" = 1' - 0" 0 6" 1' 2' 3'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

Stantec

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PM/DM MRI/CWJ
BRANCH MANAGER NEK
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
TYPICAL PIER DUCTBANK DETAILS

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO.
SHEET 141 OF 247
S-513

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:36:51 PM S-513
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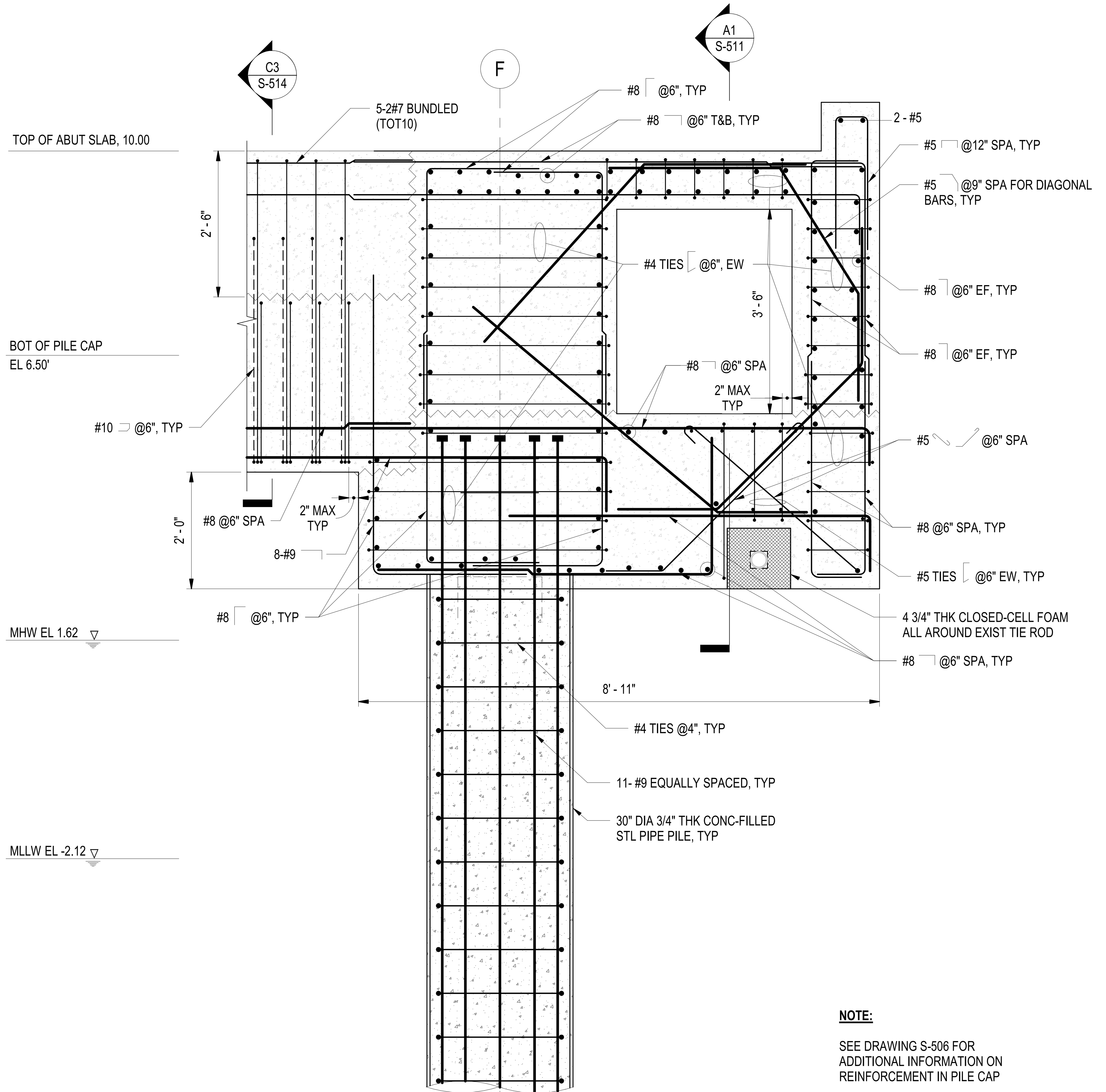


NOTE:

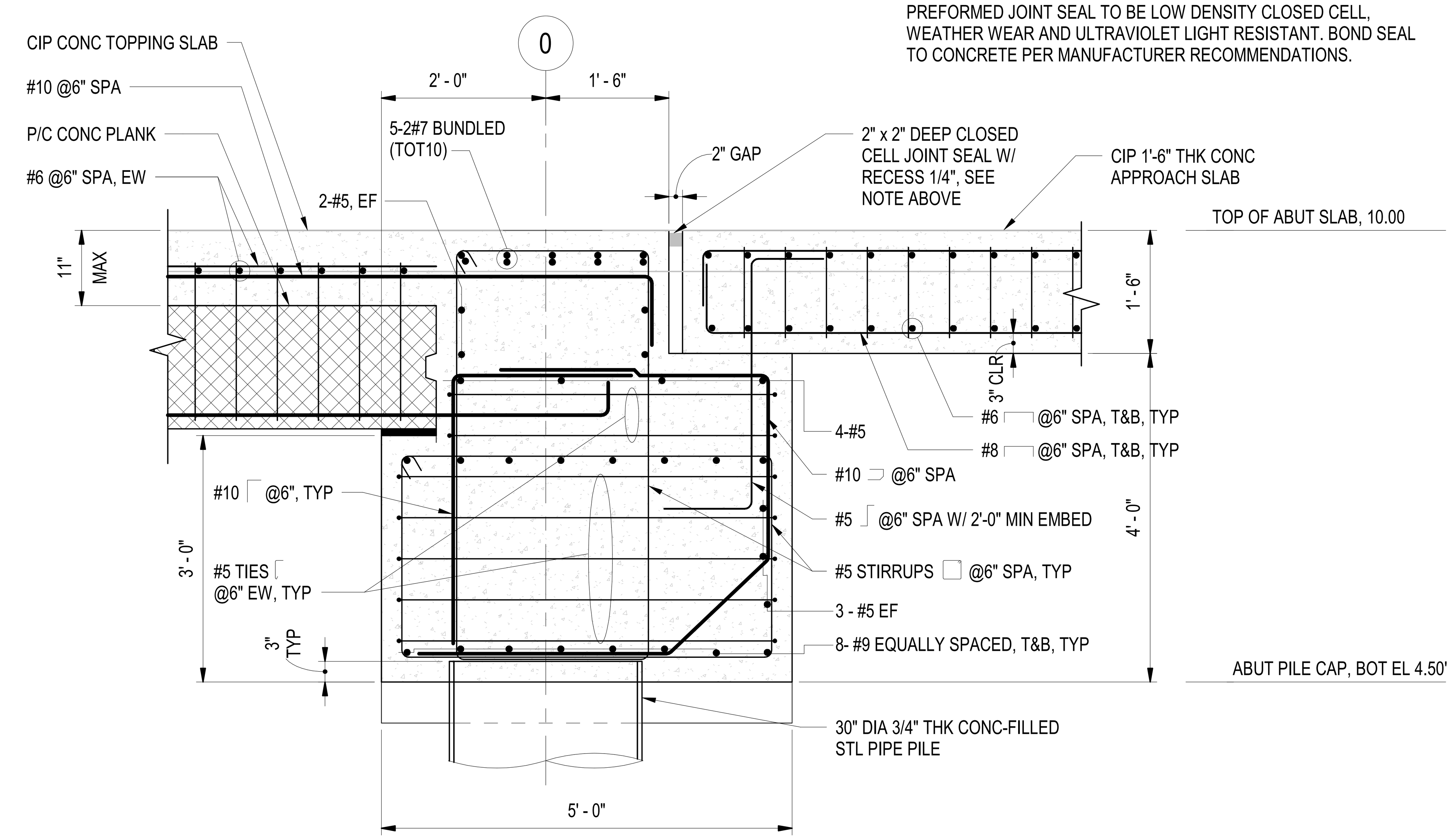
SEE DRAWING S-506 FOR ADDITIONAL INFORMATION ON REINFORCEMENT IN PILE CAP

B1 UTILITY TRENCH REINFORCED PILE CAP DETAIL

SCALE: 3/4" = 1'-0"   (S-503, S-511)

NOTE:

PREFORMED JOINT SEAL TO BE LOW DENSITY CLOSED CELL, WEATHER WEAR AND ULTRAVIOLET LIGHT RESISTANT. BOND SEAL TO CONCRETE PER MANUFACTURER RECOMMENDATIONS.



C3 TYPICAL PILE CAP REINFORCEMENT & APPROACH SLAB CONNECTION DETAIL

SCALE: 3/4" = 1'-0"   (S-303, S-508)

3/4" = 1' - 0"   0 6" 1' 2' 3' 4' 5'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA
SEAL

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PM/DM   MRI/CWJ

BRANCH MANAGER   NEK

CHIEF ENG/ARCH   JWJ

FIRE PROTECTION   HPN

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LANDSIDE PIPE PILES CONNECTION DETAILS

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

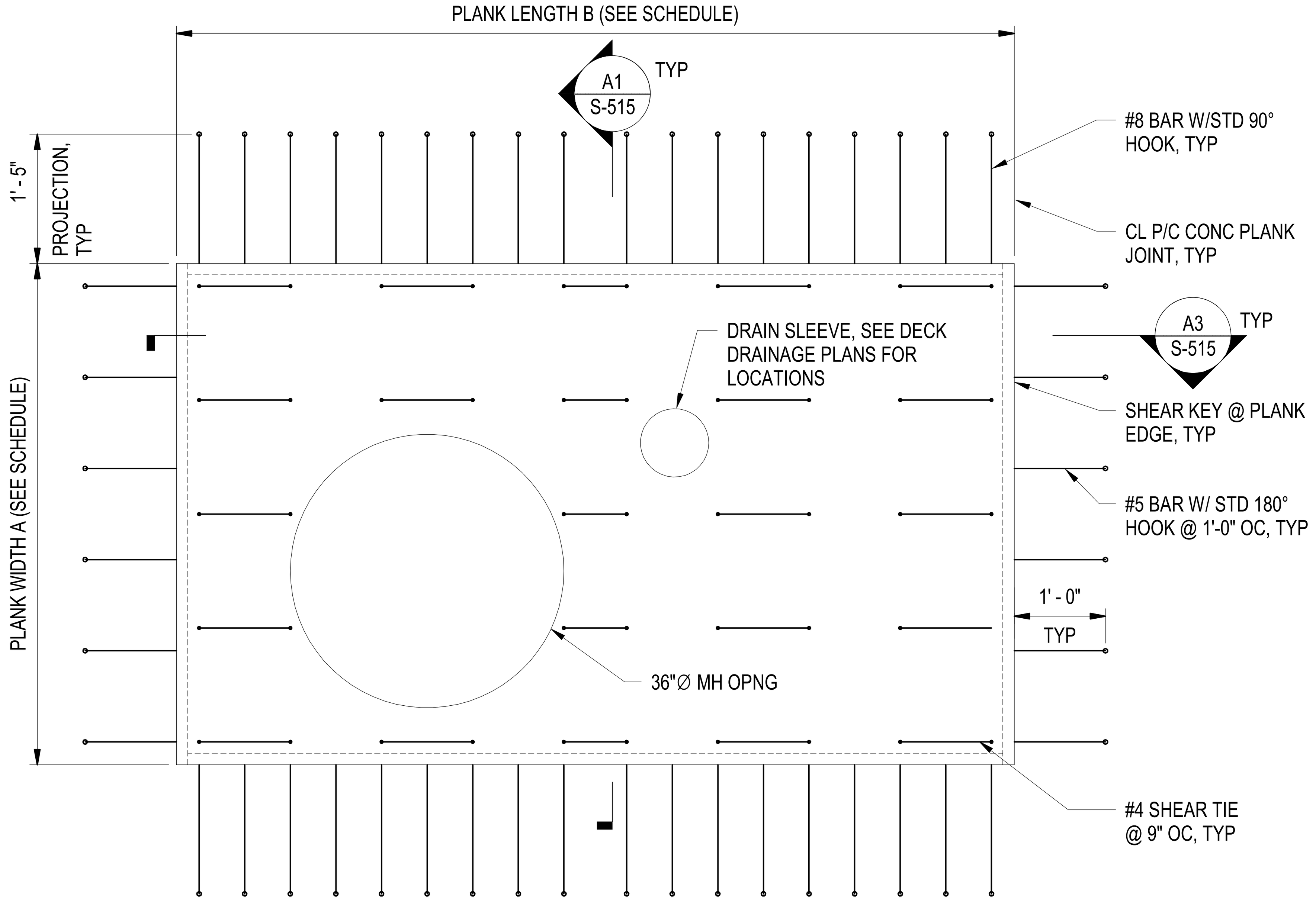
NAVFAC DRAWING NO. 12936711

SHEET 142 OF 247

S-514

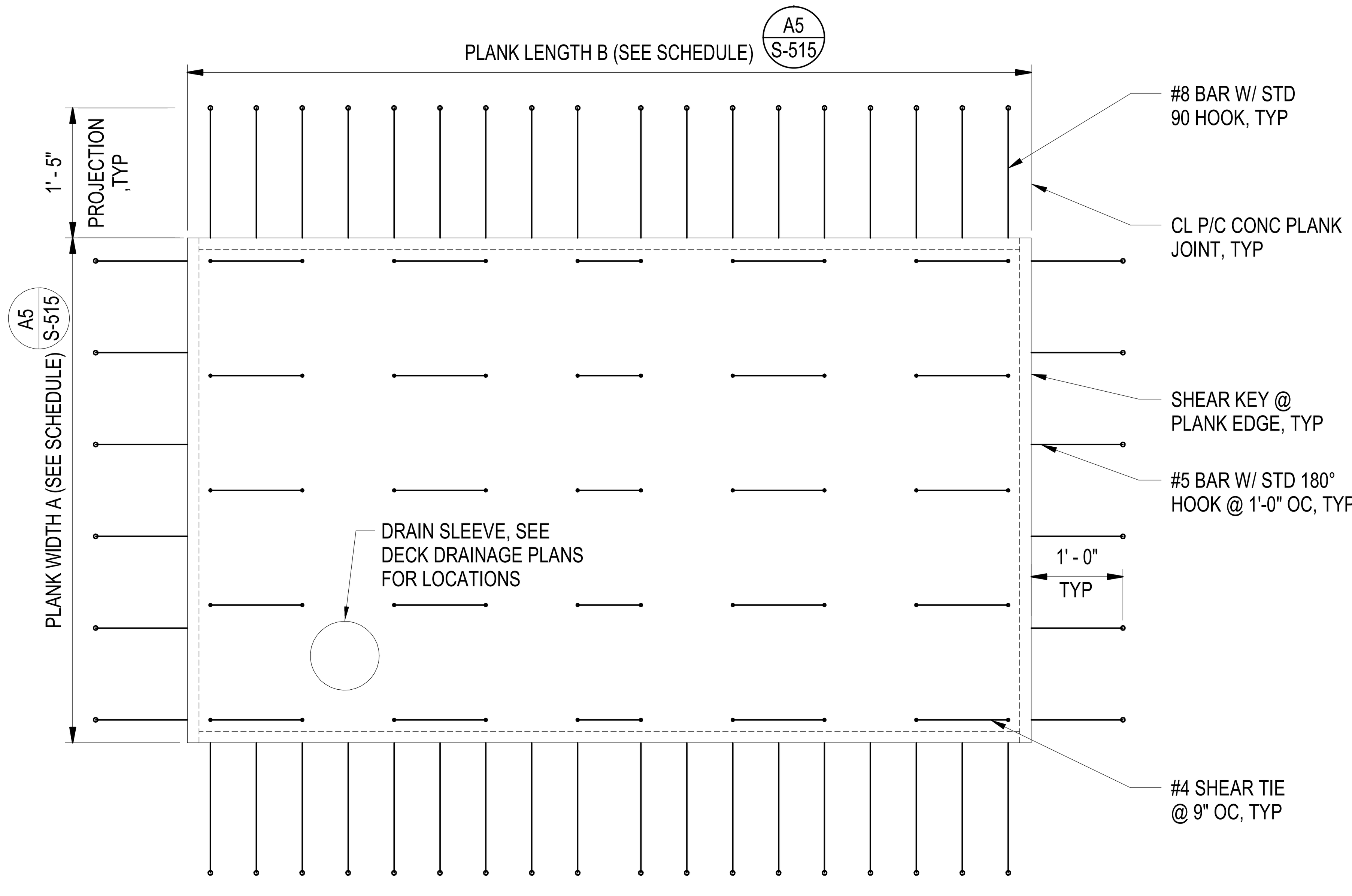
DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:36:51 PM   S-514
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt



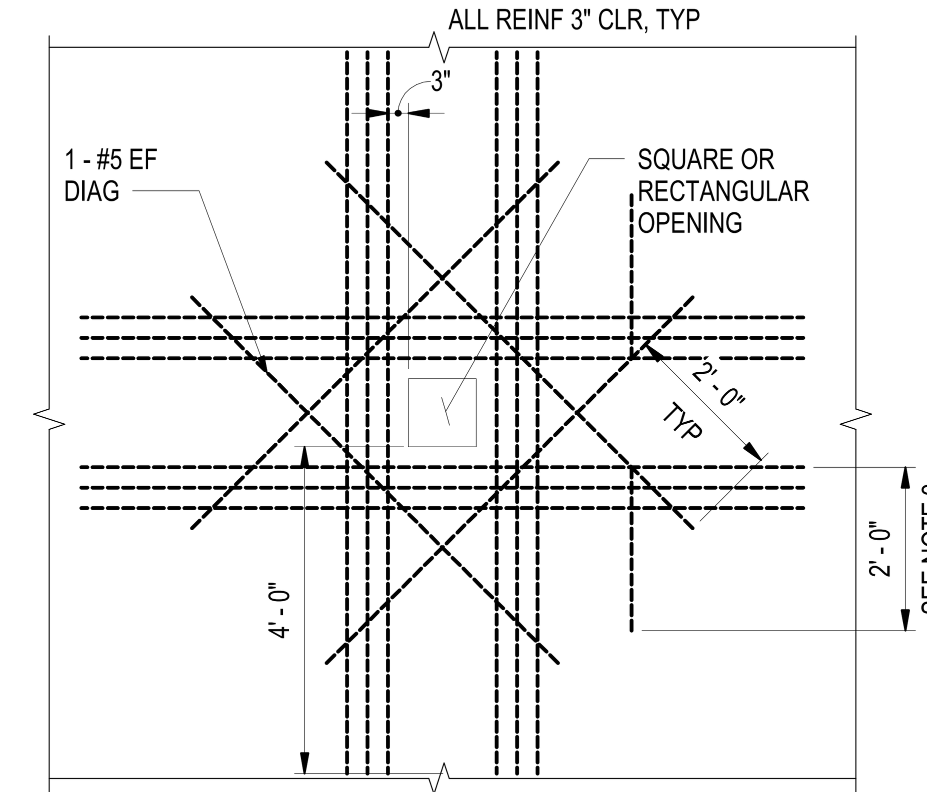
C1 TYPE 2 PRECAST PLANKS WITH MANHOLE OPENING

SCALE: 3/4" = 1'-0"



C3 PRECAST PLANKS WITH SLEEVE OPENING

SCALE: 3/4" = 1'-0"



NOTES:

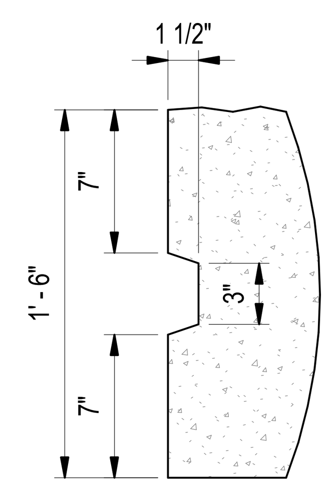
1. PROVIDE ADDITIONAL REINFORCING, (MINIMUM OF ONE-HALF THE NUMBER OF PRINCIPLE REINFORCING BARS INTERRUPTED BY THE OPENING) ON EACH SIDE AND EACH FACE OF THE OPENING.
2. FOR OPENINGS THAT ARE 12" PER SIDE SMALLER, ADDITIONAL REINFORCEMENT IS NOT REQUIRED.
3. FOR DIAGONAL REINFORCEMENT NEAR EDGES OF PRECAST PLANKS PROVIDE HOOKED TAILS AS SHOWN ABOVE.
4. SEE DRAWING S-517 FOR ADDITIONAL REINFORCEMENT AROUND CIRCULAR OPENINGS.

C4 ADDITIONAL REINFORCING AROUND OPENING

SCALE: 1/2" = 1'-0" S-517

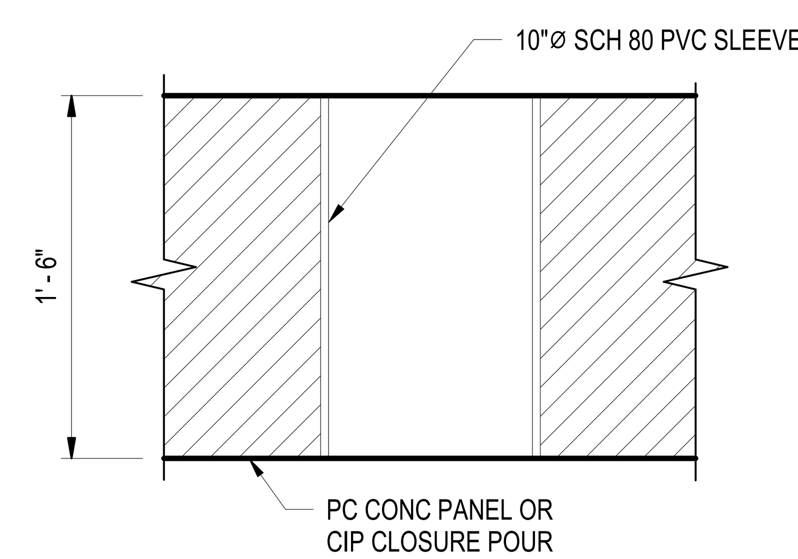
GENERAL SHEET NOTES:

1. FOR MANHOLE OPENING LOCATIONS, SEE S-117 THROUGH S-123.
2. FOR DRAIN SLEEVE LOCATIONS, SEE S-124 THROUGH S-130.
3. FOR DETAILS OF CAST-IN-PLACE CONCRETE TOPPING SLAB, SEE DRAWING S-512..
4. SEE DRAWING S-517 FOR ADDITIONAL REINFORCEMENT AROUND CIRCULAR OPENING.



B4 SHEAR KEY DETAIL

SCALE: 1 1/2" = 1'-0"



NOTE:

FOR DRAIN DETAIL, SEE SHEET S-522.

B3 SLEEVE DETAIL

SCALE: 1 1/2" = 1'-0"

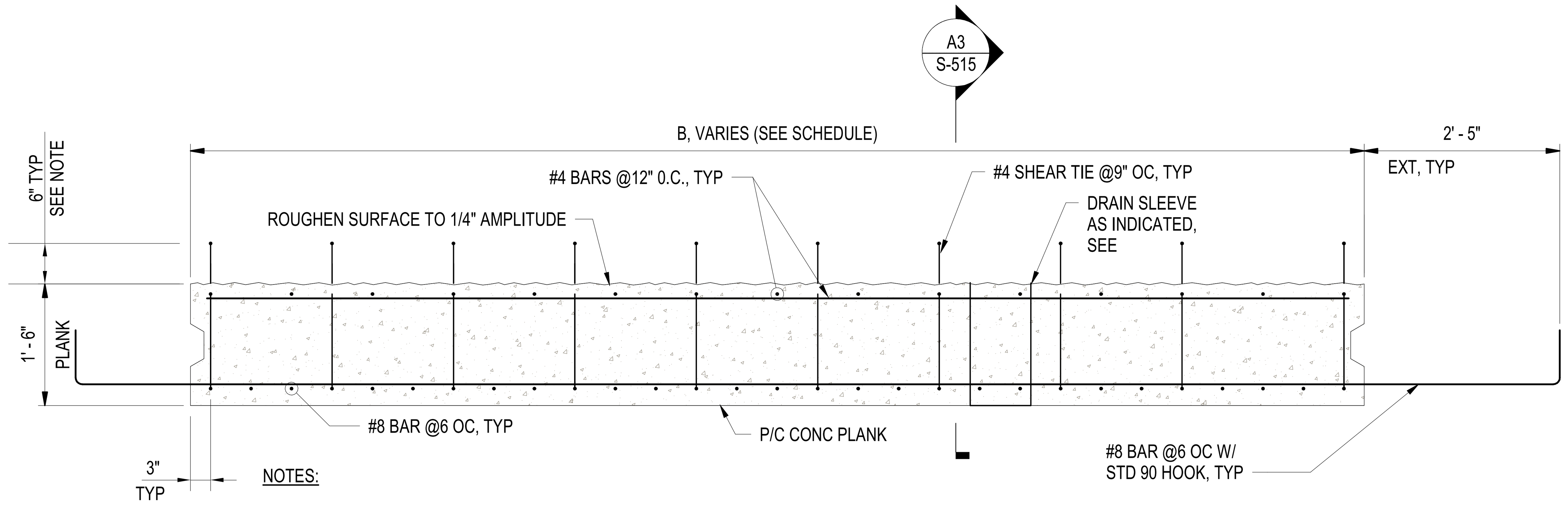
P/C CONC PLANK SCHEDULE				
MARK[1]	QUANTITY	PLANK THICKNESS	A (WIDTH)	B (LENGTH)
1A	56	1' - 6"	10' - 1"	21' - 4"
1B	51	1' - 6"	12' - 10"	21' - 4"
1C	5	1' - 6"	10' - 1"	17' - 10"
1D	4	1' - 6"	12' - 10"	17' - 10"
1E	6	1' - 6"	10' - 1"	5' - 11"
1F	9	1' - 6"	12' - 10"	5' - 11"
1G	4	1' - 6"	10' - 1"	6' - 8"
1H	6	1' - 6"	12' - 10"	6' - 8"
1I	2	1' - 6"	10' - 1"	11' - 10"
1J	3	1' - 6"	12' - 10"	11' - 10"
1K	1	1' - 6"	10' - 1"	14' - 10"
1L	1	1' - 6"	10' - 5 1/2"	14' - 10"
1M	1	1' - 6"	11' - 9"	14' - 10"
2B	29	1' - 6"	12' - 10"	21' - 4"
2D	2	1' - 6"	12' - 10"	17' - 10"
2N	1	1' - 6"	22' - 1"	14' - 10"
3A	1	1' - 6"	10' - 1"	21' - 4"
3B	5	1' - 6"	12' - 10"	21' - 4"
3D	2	1' - 6"	12' - 10"	17' - 10"
4B	7	1' - 6"	12' - 10"	21' - 4"
5B	3	1' - 6"	12' - 10"	21' - 4"

NOTE:

1. THE FOLLOWING CONVENTION IS APPLICABLE FOR THE NUMBER AT EACH PLANK MARK:
   1: NO OPENING
   2: 8" DIA OPENING
   3: 36" DIA OPENING
   4: 8" DIA AND 36" DIA OPENINGS
   5: 2X36" DIA OPENINGS

A5 PRECAST PLANK SCHEDULE

NOT TO SCALE (S-515)

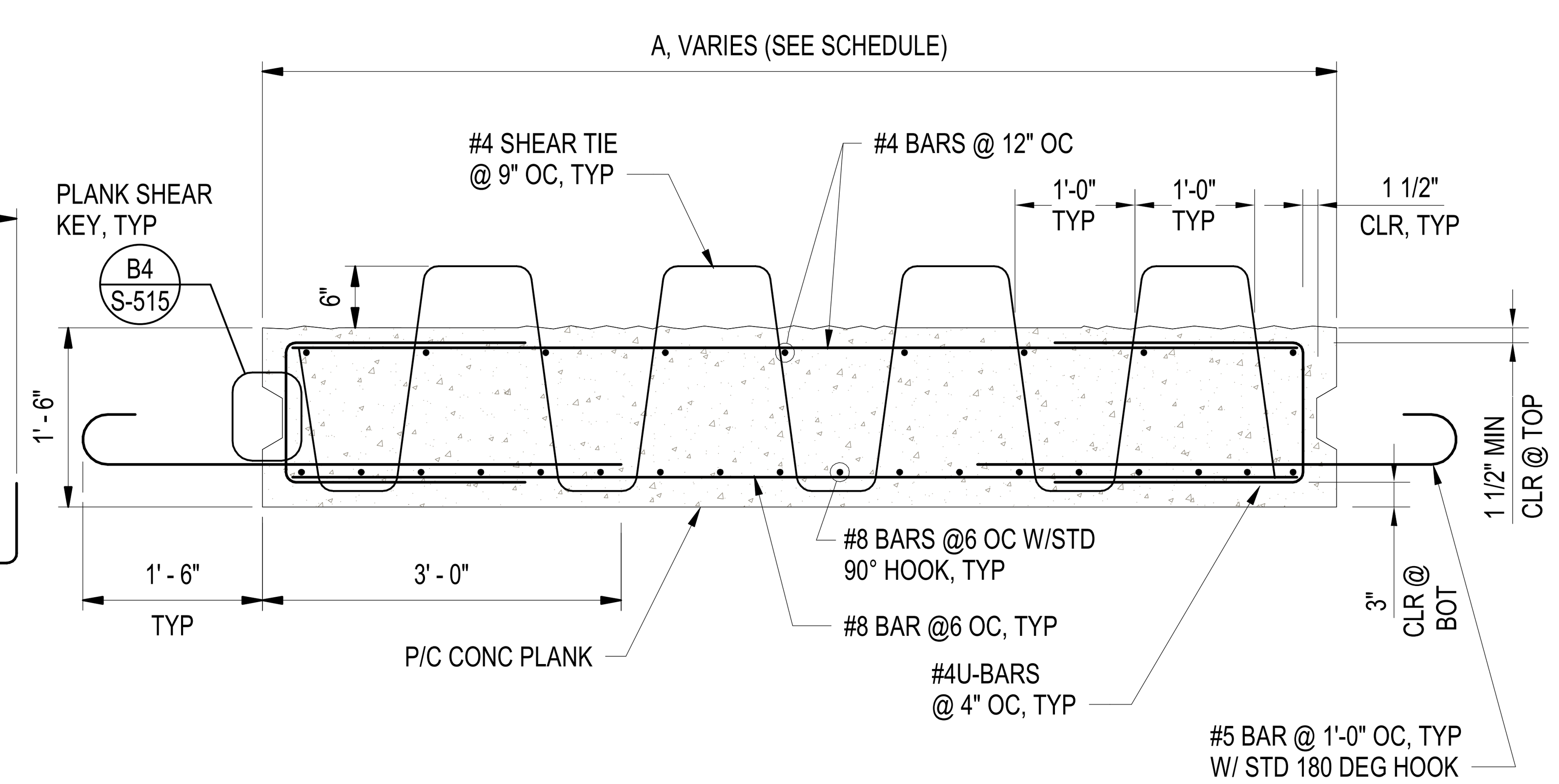


NOTES:

1. THE CONTRACTOR MUST ENSURE THAT 6" DIMENSION SHOWN IS MAINTAINED SO THAT THE TOPPING SLAB REINFORCEMENT IS PROPERLY LOCATED.

A1 PRECAST PLANKS LONGITUDINAL SECTION

SCALE: 3/4" = 1'-0" (S-504)



A3 PRECAST PLANKS TRANSVERSE SECTION

SCALE: 3/4" = 1'-0"

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

1 1/2" = 1' - 0" 0 3" 6" 9" 1' 2'

3/4" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

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FIRE PROTECTION		HPN

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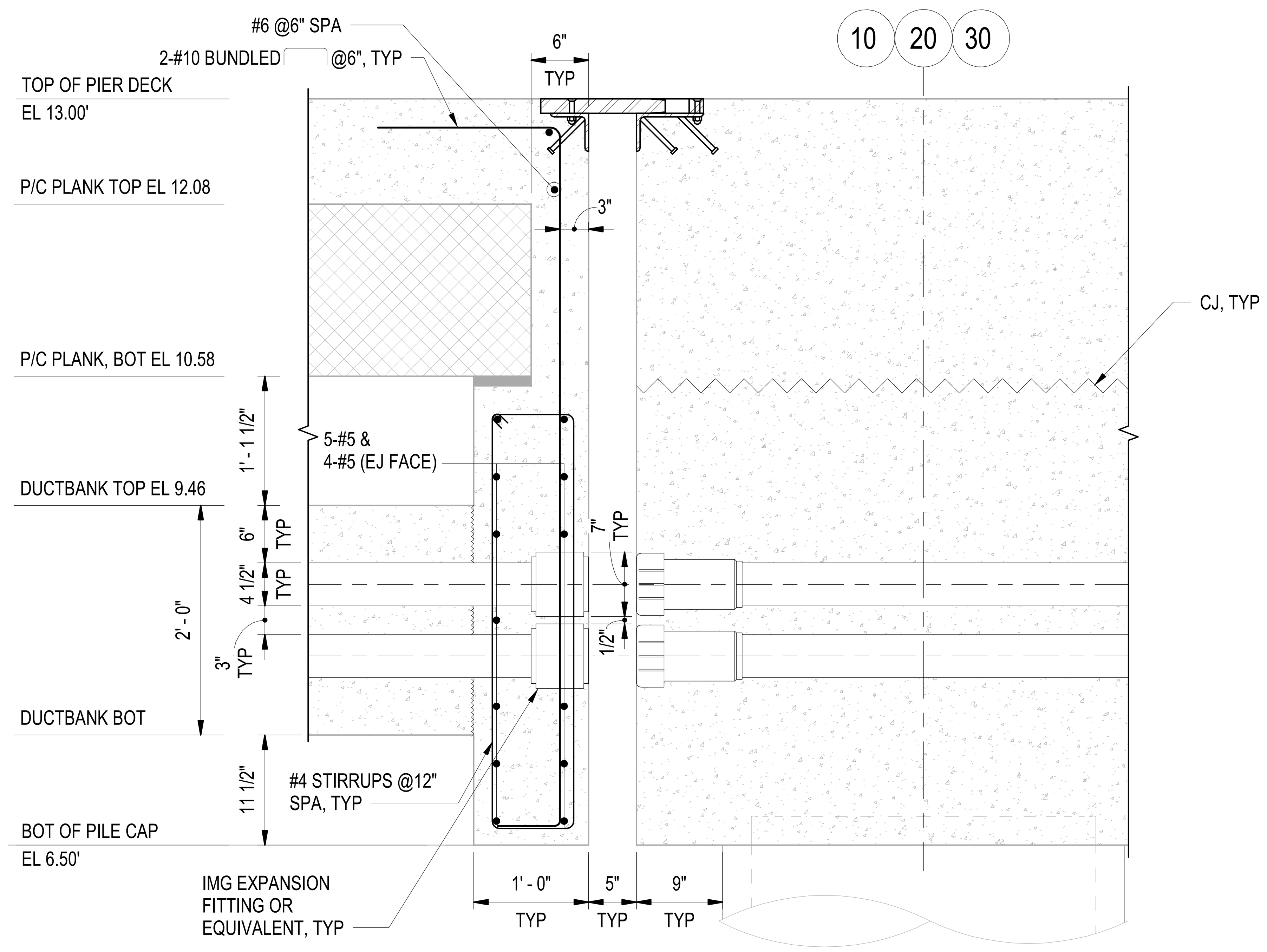
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PRECAST CONCRETE PLANK DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936712
SHEET 143 OF 247
S-515

DRAWFORM REVISION: 25 AUGUST 2020

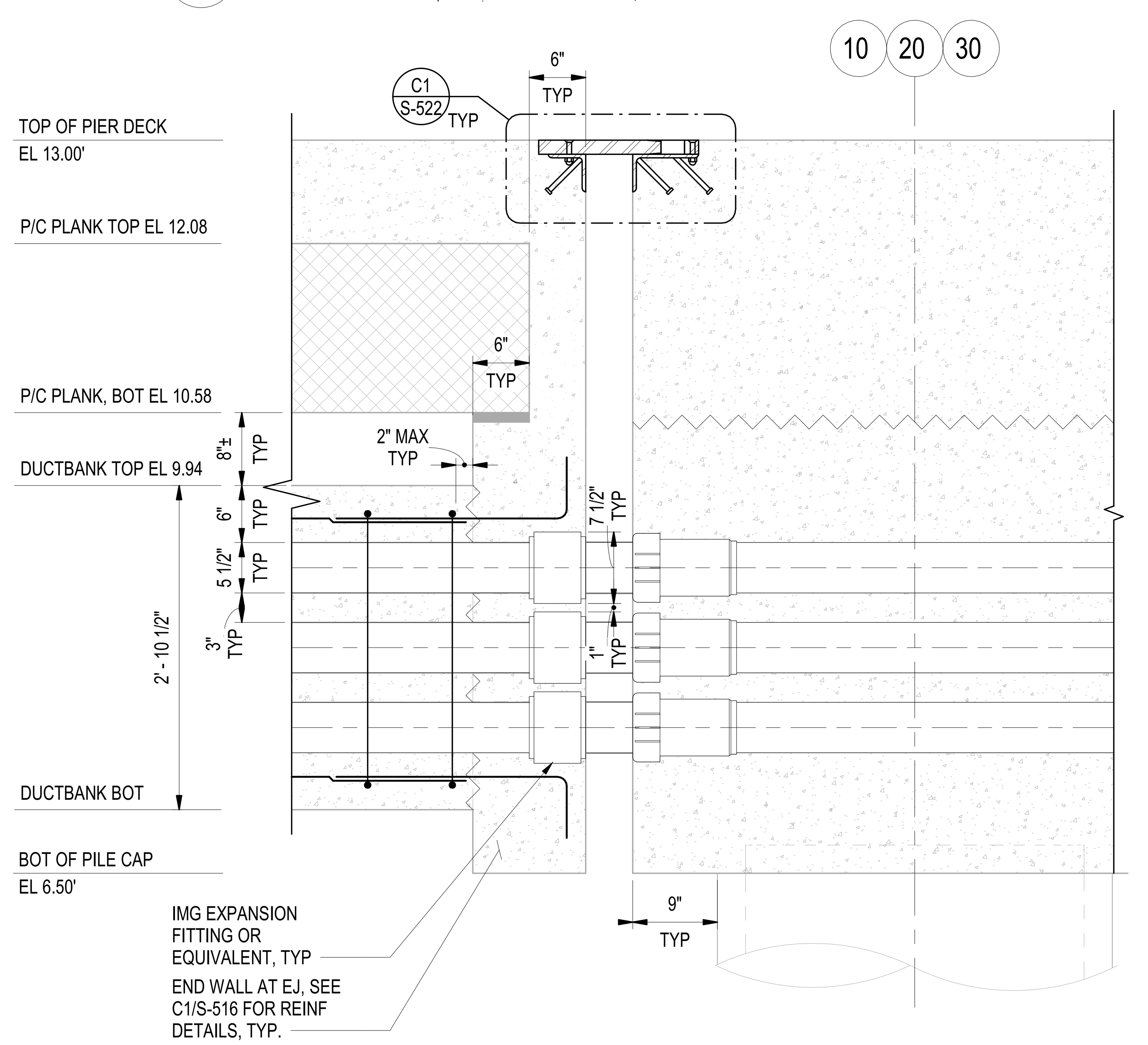
3/2/2026 7:36:53 PM S-515
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-173829-S.rvt

1 2 3 4 5



TYPICAL DETAILS FOR EXPANSION JOINT WITH CONDUIT CROSSING-TWO ROWS

C1
SCALE: 1" = 1'-0" (S-113, S-114, S-504, S-516)



TYPICAL DETAILS FOR EXPANSION JOINT WITH CONDUIT CROSSING-THREE ROWS

A1
SCALE: 1" = 1'-0" (S-111)

GENERAL SHEET NOTES:

1. ADJUST REINFORCEMENT AS REQUIRED AT CONDUIT AND PIPE.
2. REINFORCEMENT AT DUCTBANK, PRECAST PLANKS, PILE CAP AND TOPPING SLAB ARE NOT SHOWN FOR CLARITY.
3. ELEVATIONS FOR DUCT BANKS CROSSING ALL EXPANSION JOINTS MAY VARY.
4. SEE DRAWING S-513 FOR REINFORCEMENT DETAILS FOR DUCTBANK, TYPICAL.
5. SEE ELECTRICAL DRAWINGS FOR DETAILS ON CONDUIT FITTINGS.

1" = 1'-0" 0 6" 1' 2' 3'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA
SEAL

Stantec

A/E INFO

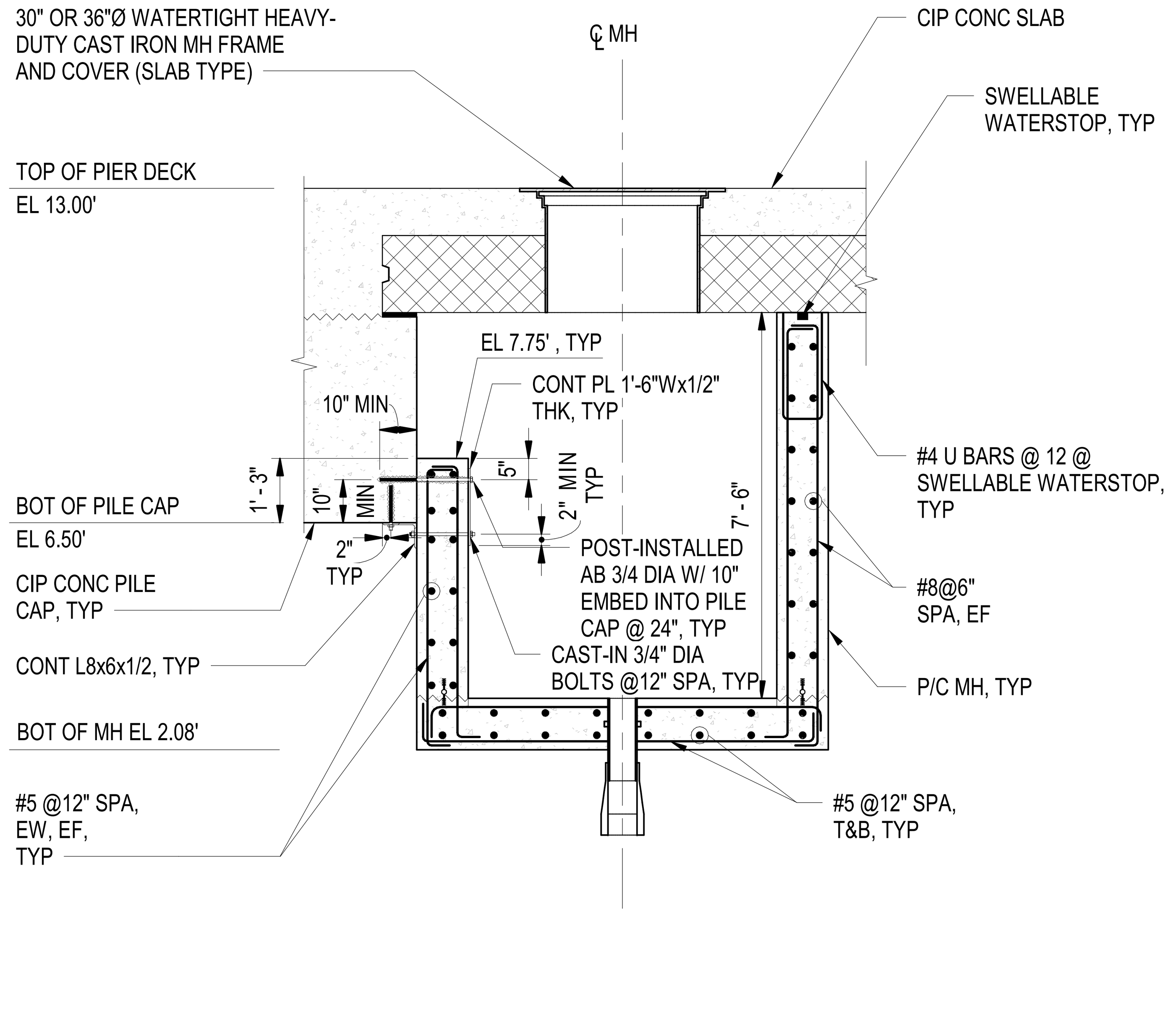
APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES	JZ	DRW	AFC	CHK	MJF
PM/DM					MRI/CWJ
BRANCH MANAGER					NEK
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
CONDUITS AND FITTINGS AT EXPANSION JOINT

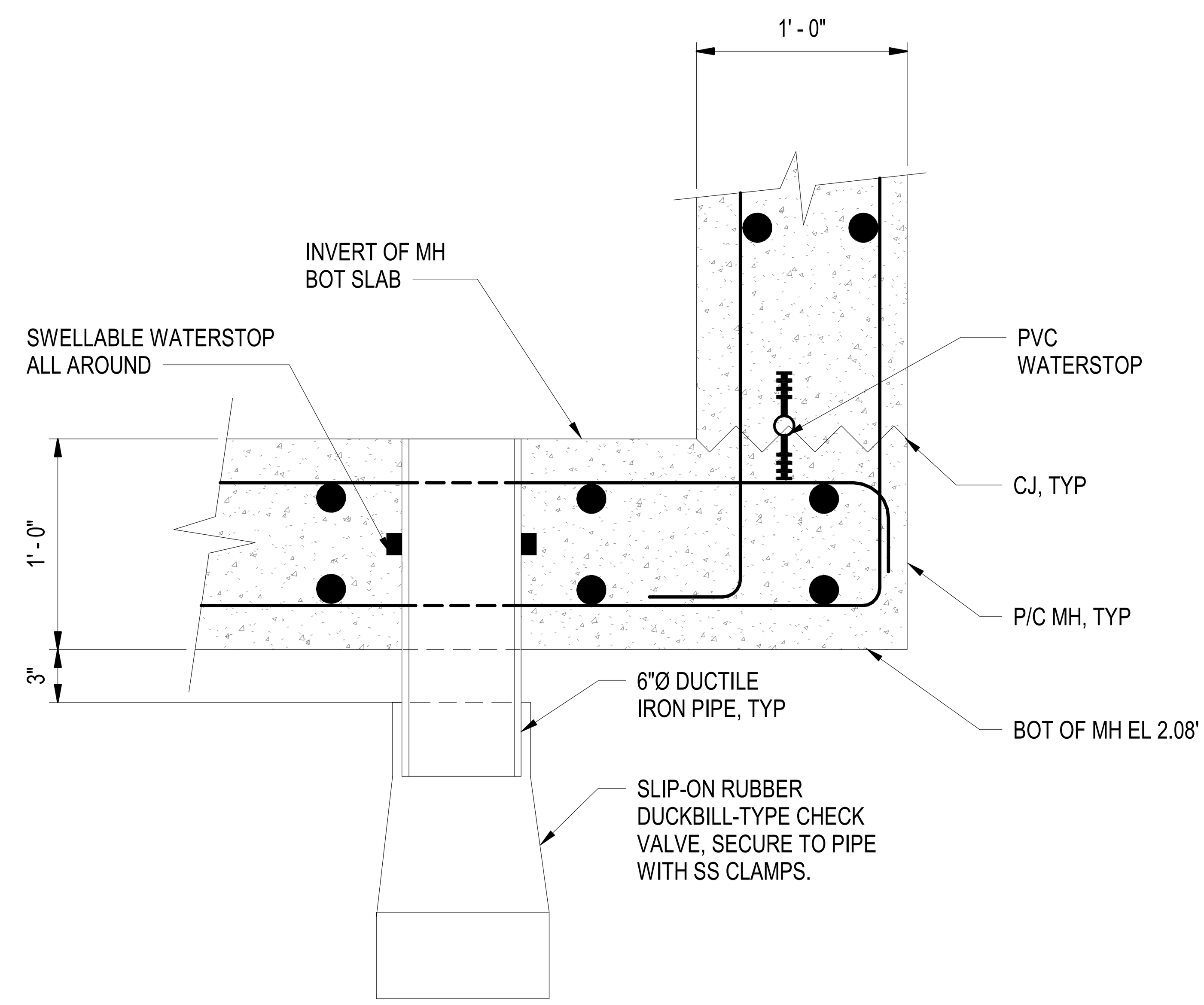
SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936713
SHEET 144 OF 247
S-516
DRAWFORM REVISION: 25 AUGUST 2020

S-516
3/2/2026 7:36:54 PM
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt



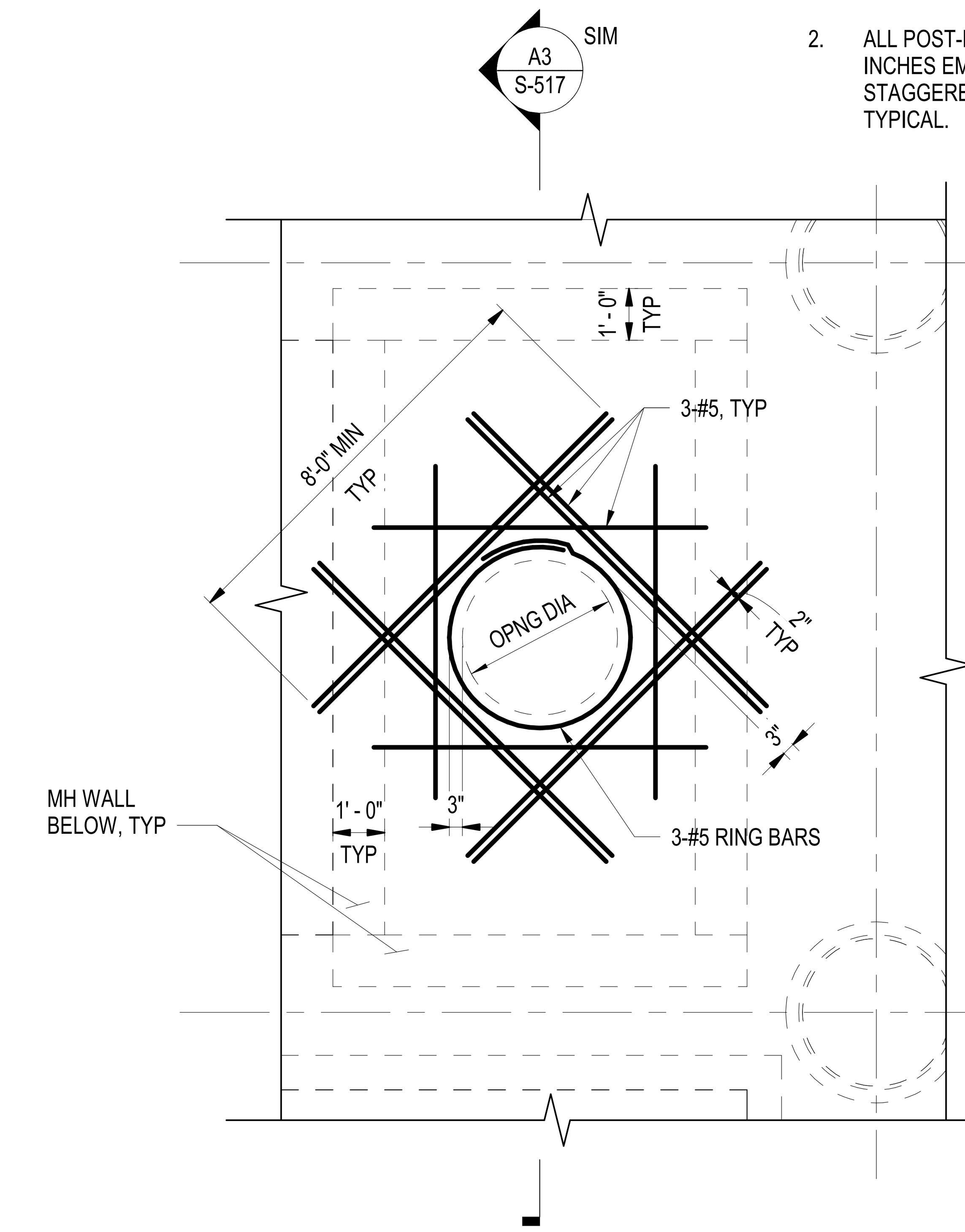
C1 LATERAL SECTION ALONG PRECAST MANHOLE

SCALE: 3/8" = 1'-0"



C3 TYPICAL MANHOLE DRAIN

SCALE: 1 1/2" = 1'-0"



NOTES:

1. PLACE REINFORCEMENT AROUND OPENING AT 3 TYPICAL LOCATIONS:

1.1 AT MANHOLE OPENING ON PIER DECK: ABOVE BOTTOM REBAR CURTAIN, BELOW TOP REBAR CURTAIN OF PLANK, AND BELOW TOPPING SLAB REBAR CURTAIN. BEND REBAR AS REQUIRED BY DRAWING S-515.

1.2. AT OTHER OPENINGS:
1.2.1 EQUALLY SPACED THROUGH THICKNESS W/ MIN 2" CLR COVER FOR WALL/SLAB WITH 8" THICK;
1.2.2 AT OTHER LOCATIONS: ABOVE BOTTOM REBAR CURTAIN, BELOW TOP REBAR CURTAIN, AND AT THE CENTER OF THICKNESS FOR OTHER THICKNESS.

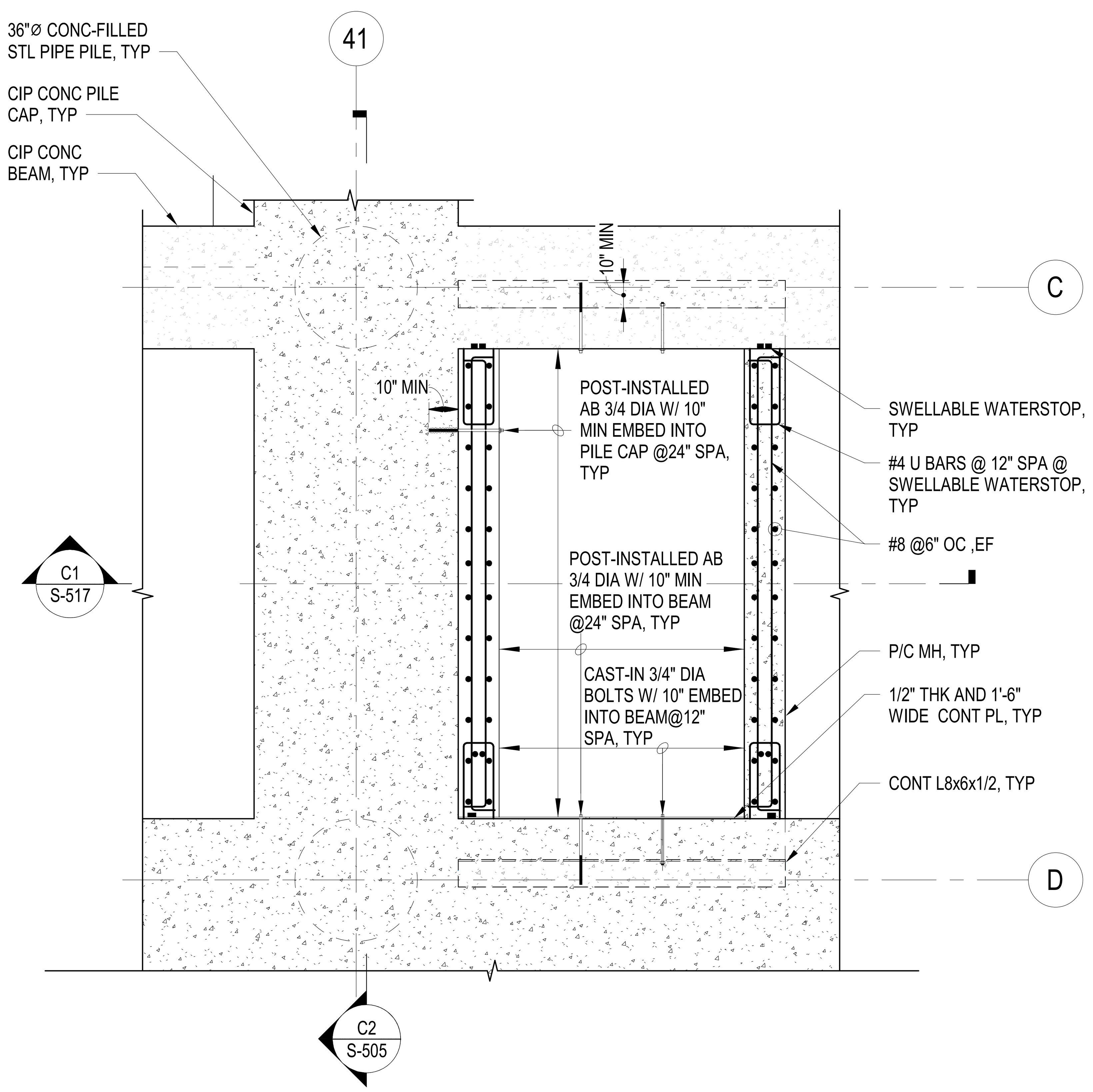
2. SEE DRAWING S-515 FOR SQUARE OR RECTANGULAR OPENINGS.

B4 ADDITIONAL REINFORCEMENT AROUND MANHOLE (SHOWN) AND OTHER CIRCULAR OPENINGS

SCALE: 3/8" = 1'-0" (S-128) (S-128, S-402, S-508, S-511, S-515, S-534)

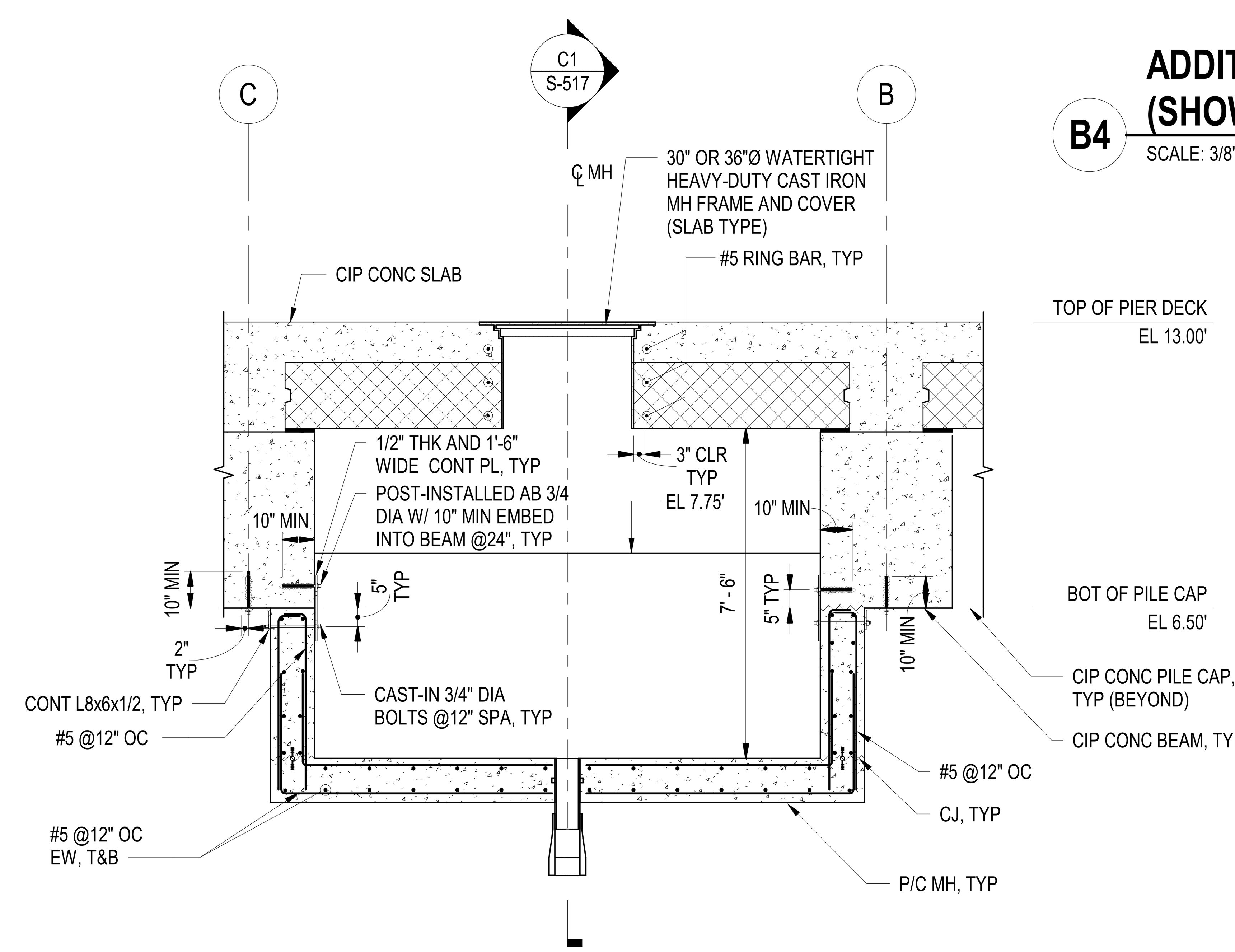
GENERAL SHEET NOTES:

1. ALL STEEL ANGLES, PLATES, BOLTS, NUTS AND WASHERS MUST BE A316/A316L STAINLESS STEEL.
2. ALL POST-INSTALLED ANCHORS MUST HAVE MINIMUM 10 INCHES EMBEDMENT INTO PILE CAP OR BEAMS AND BE STAGGERED OUT 6" AS MINIMUM TO AVOID CONFLICTS, TYPICAL.



A1 TYPICAL PRECAST MANHOLE PLAN

SCALE: 3/8" = 1'-0"



A3 LONGITUDINAL SECTION ALONG PRECAST MANHOLE

SCALE: 3/8" = 1'-0" (S-307, S-517)

1 1/2" = 1' - 0" 0 3" 6" 9" 1' 2'

3/8" = 1' - 0" 0 1' 2' 3' 4' 5' 10'

APPR	DATE	DESCRIPTION	SYM

NAVFAC

JINQUAN ZHONG C 82242 CIVIL REGISTERED PROFESSIONAL ENGINEER STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW AFC CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PRECAST MANHOLE DETAILS

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936714

SHEET 145 OF 247

S-517

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:36:56 PM S-517
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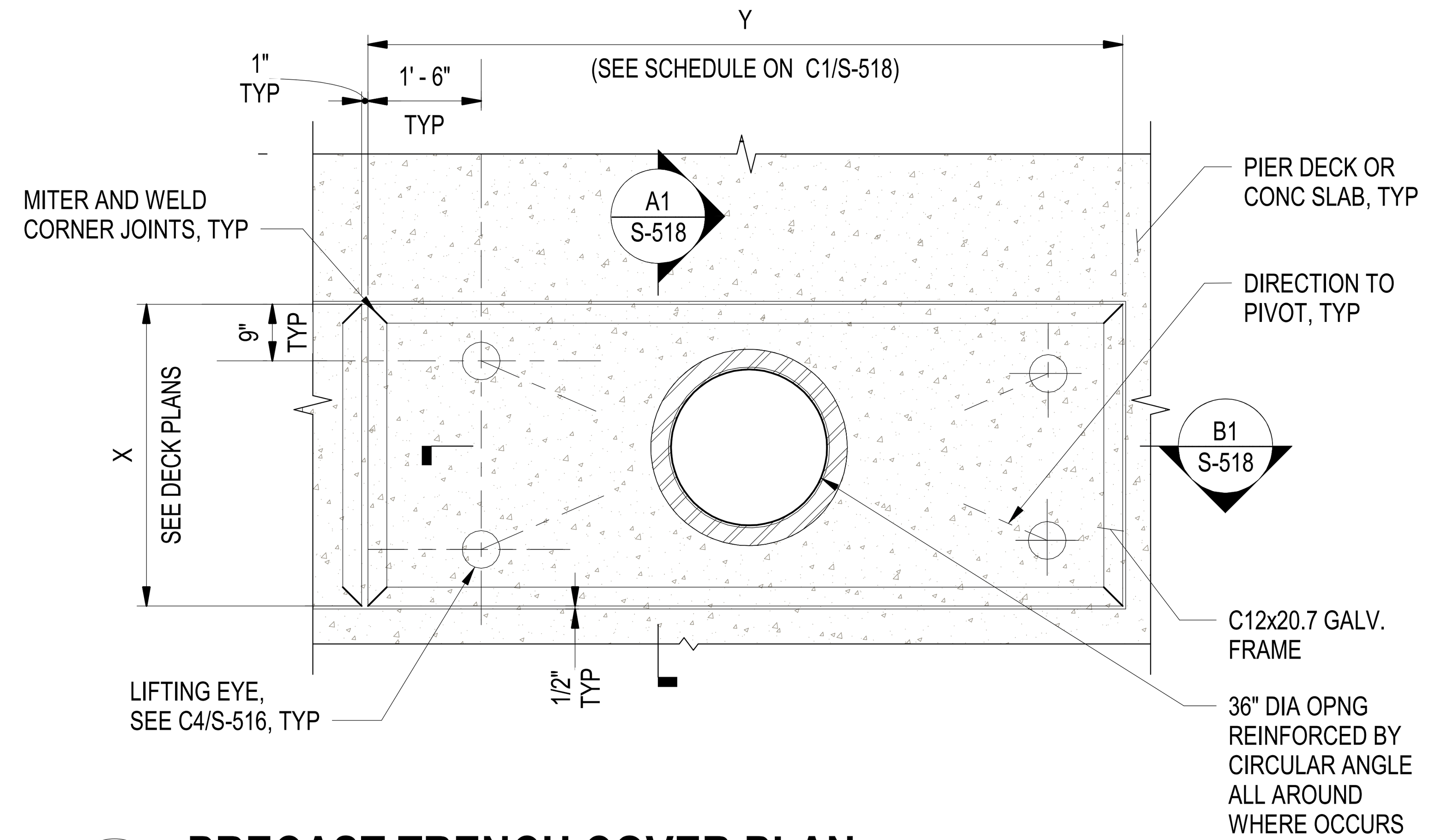
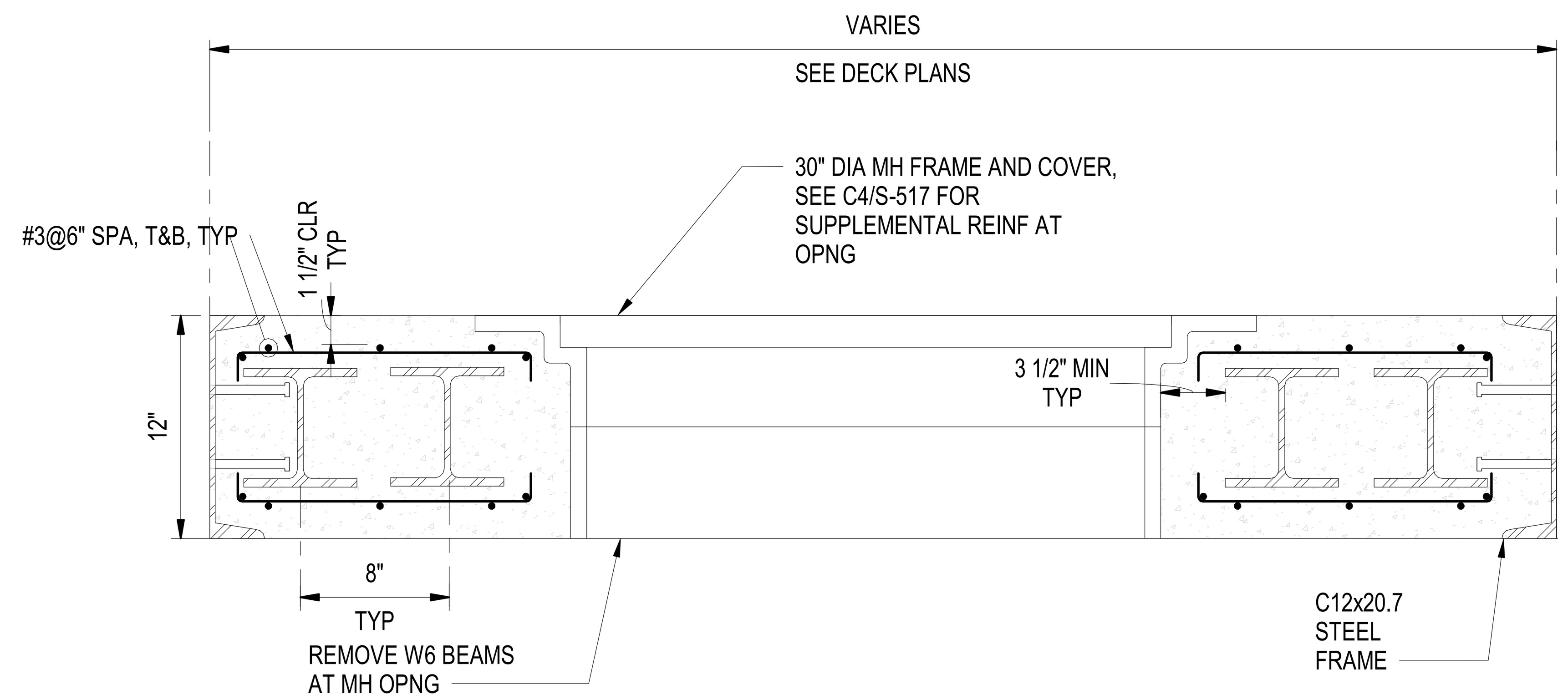
P/C TRENCH COVER SCHEDULE				
MARK	QUANTITY	THICKNESS	X (WIDTH)	Y (LENGTH)
A	132	1' - 0"	4' - 0"	9' - 11 3/4"
C	12	1' - 0"	4' - 0"	8' - 2 3/4"
D	8	1' - 0"	4' - 0"	8' - 0"
E	3	1' - 0"	4' - 0"	10' - 6"
F	4	1' - 0"	6' - 0"	9' - 11"
G	1	1' - 0"	6' - 0"	11' - 9"



C1 PRECAST TRENCH COVER PLAN

SCALE: 1/2" = 1'-0"

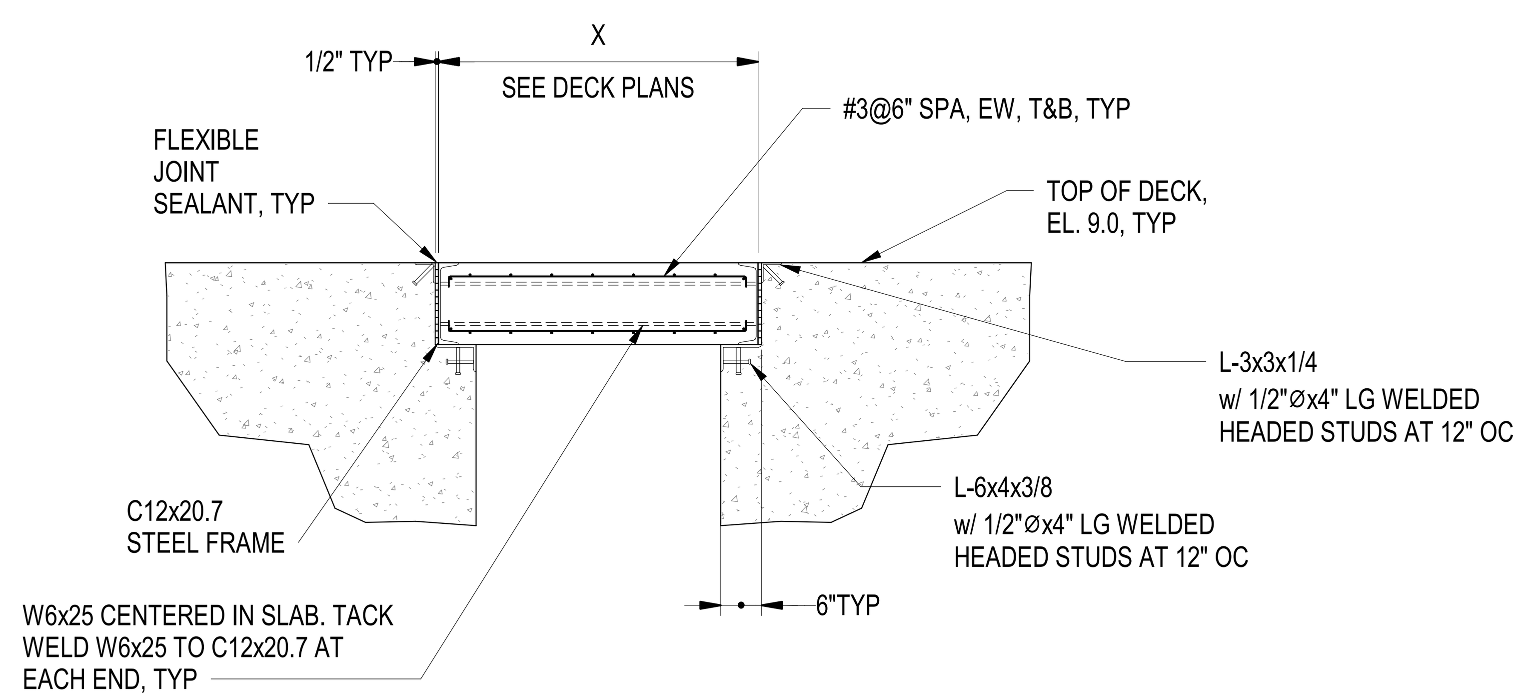


NOTE:

1. ADJUST W6s AT MH OPNGS. MAINTAIN 8" OC, SPACING BETWEEN BEAMS.
2. OUTRIGGERS ARE PROHIBITED FROM BEING PLACED ON TRENCH COVERS WITH MANHOLE OPNGS.

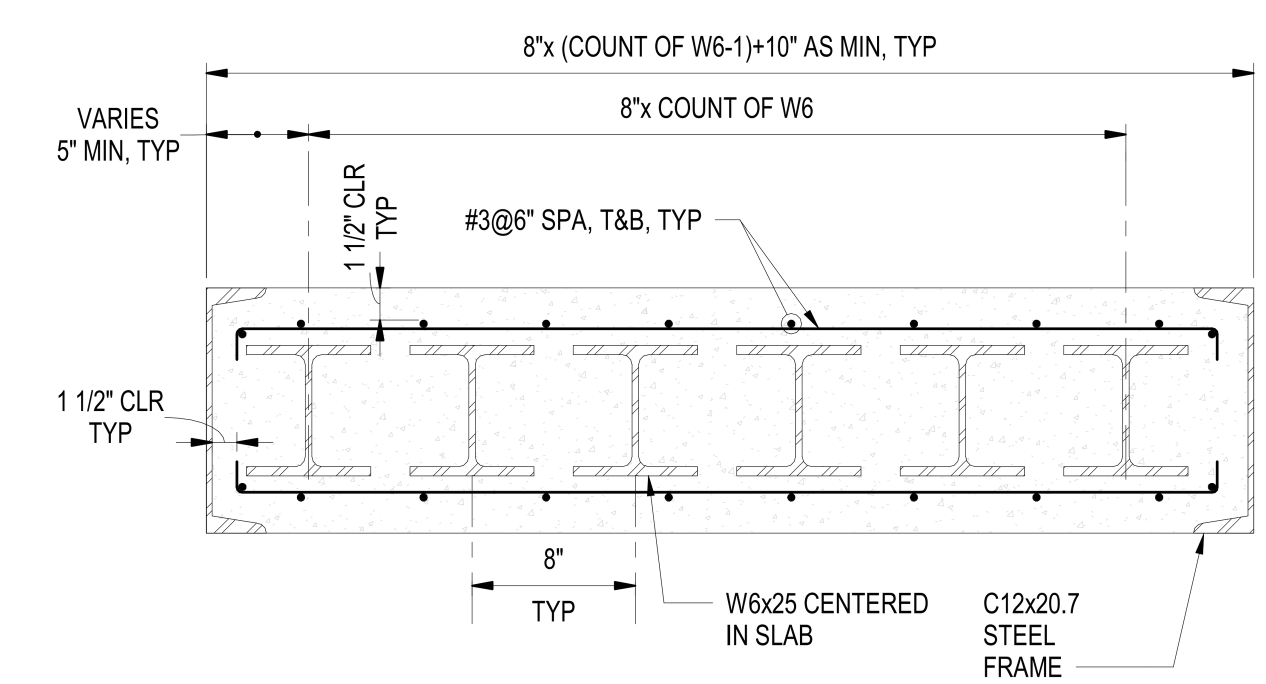
B1 PRECAST TRENCH COVER LONGITUDINAL SECTION AT MANHOLE

SCALE: 1 1/2" = 1'-0"



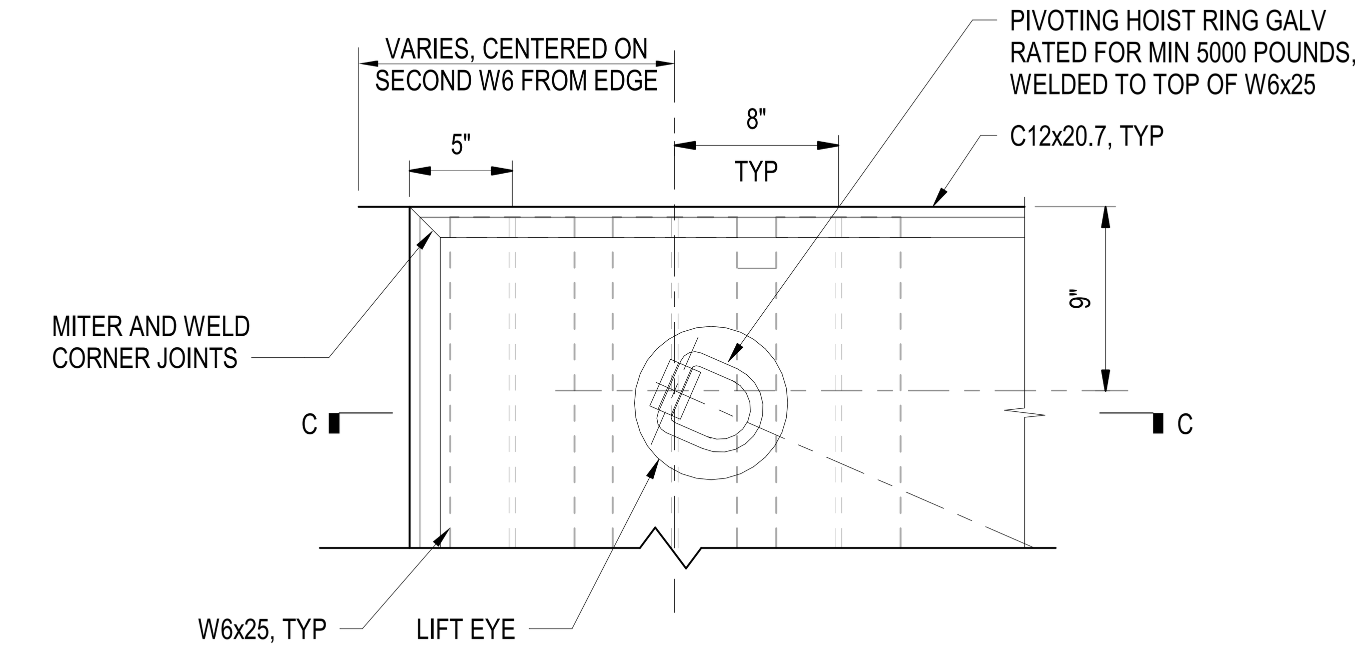
A1 PRECAST TRENCH COVER TRANSVERSE SECTION

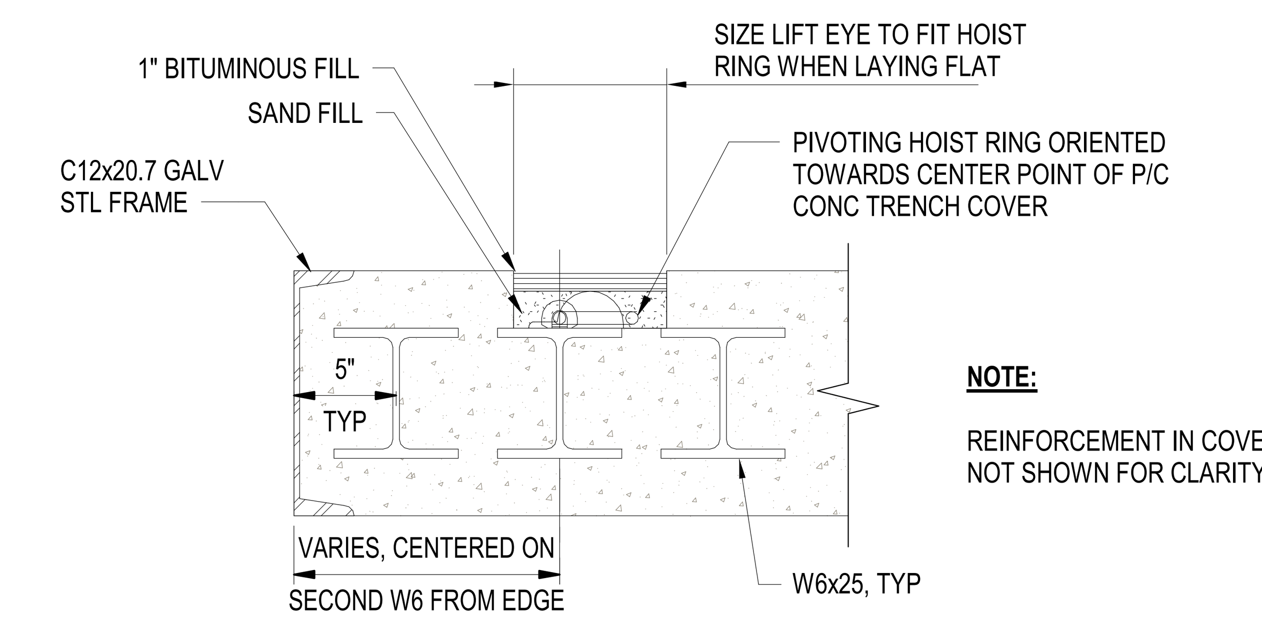
SCALE: 1/2" = 1'-0"



A3 PRECAST TRENCH COVER LONGITUDINAL SECTION

SCALE: 1 1/2" = 1'-0"





NOTE:

REINFORCEMENT IN COVER NOT SHOWN FOR CLARITY.

SECTION C-C

C4 LIFTING EYE

SCALE: 1 1/2" = 1'-0"

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

1 1/2" = 1' - 0" 0 3" 6" 9" 1' 2'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW AFC CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER

UTILITY TRENCH COVER DETAILS

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936715

SHEET 146 OF 247

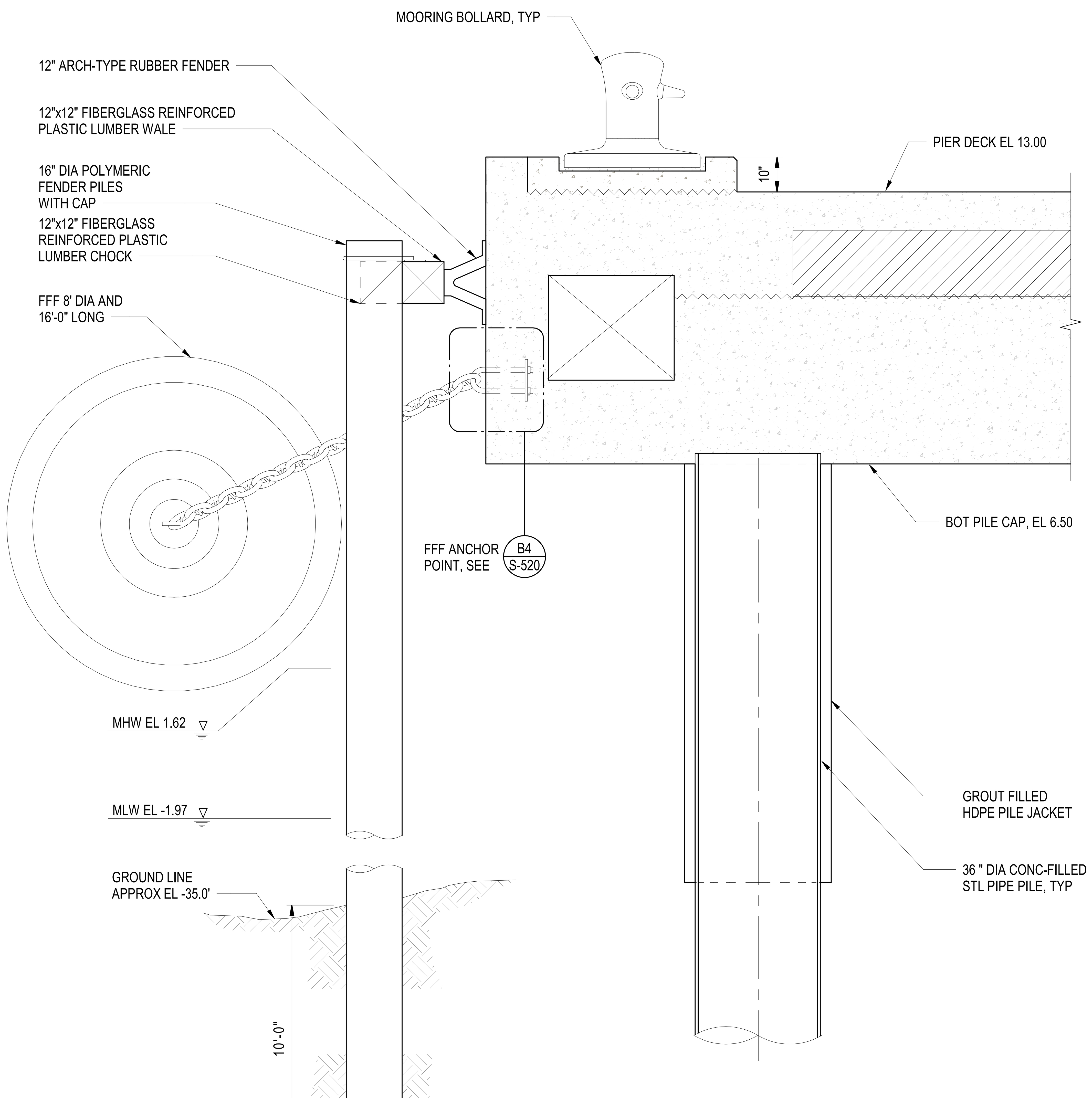
S-518

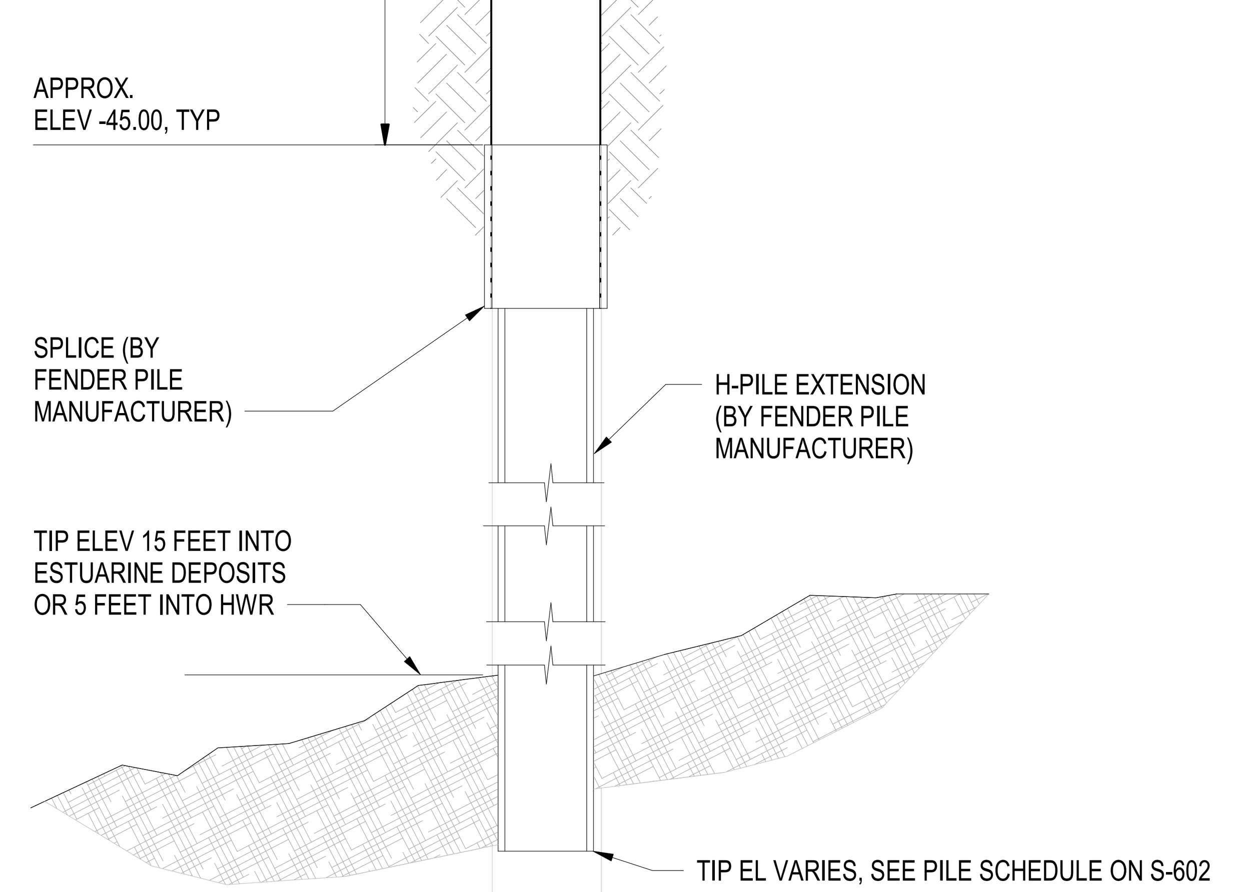
DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:36:57 PM S-518
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

GENERAL SHEET NOTES:

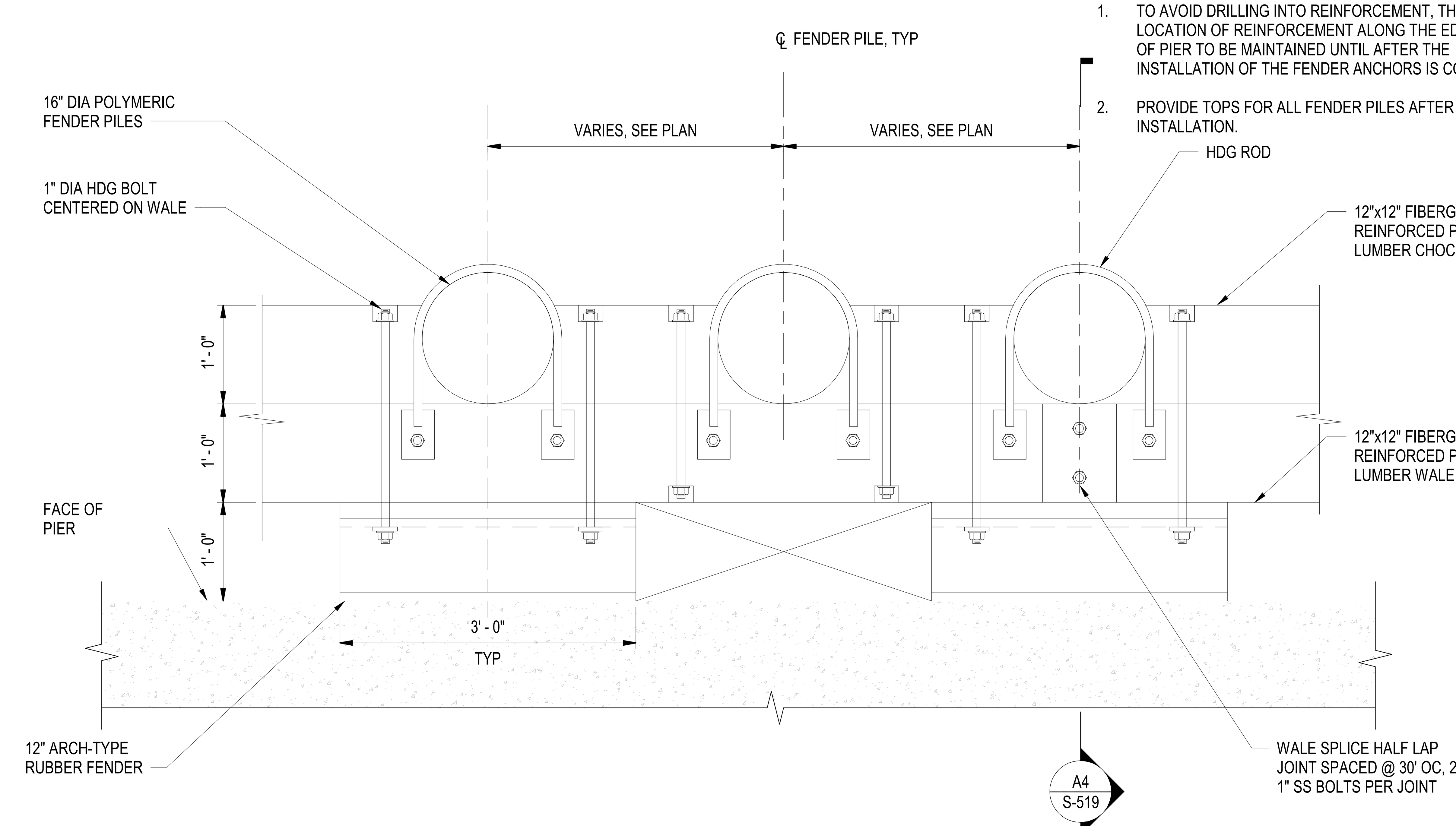
1. TO AVOID DRILLING INTO REINFORCEMENT, THE LOCATION OF REINFORCEMENT ALONG THE EDGE OF PIER TO BE MAINTAINED UNTIL AFTER THE INSTALLATION OF THE FENDER ANCHORS IS COMPLETE.
2. PROVIDE TOPS FOR ALL FENDER PILES AFTER INSTALLATION.





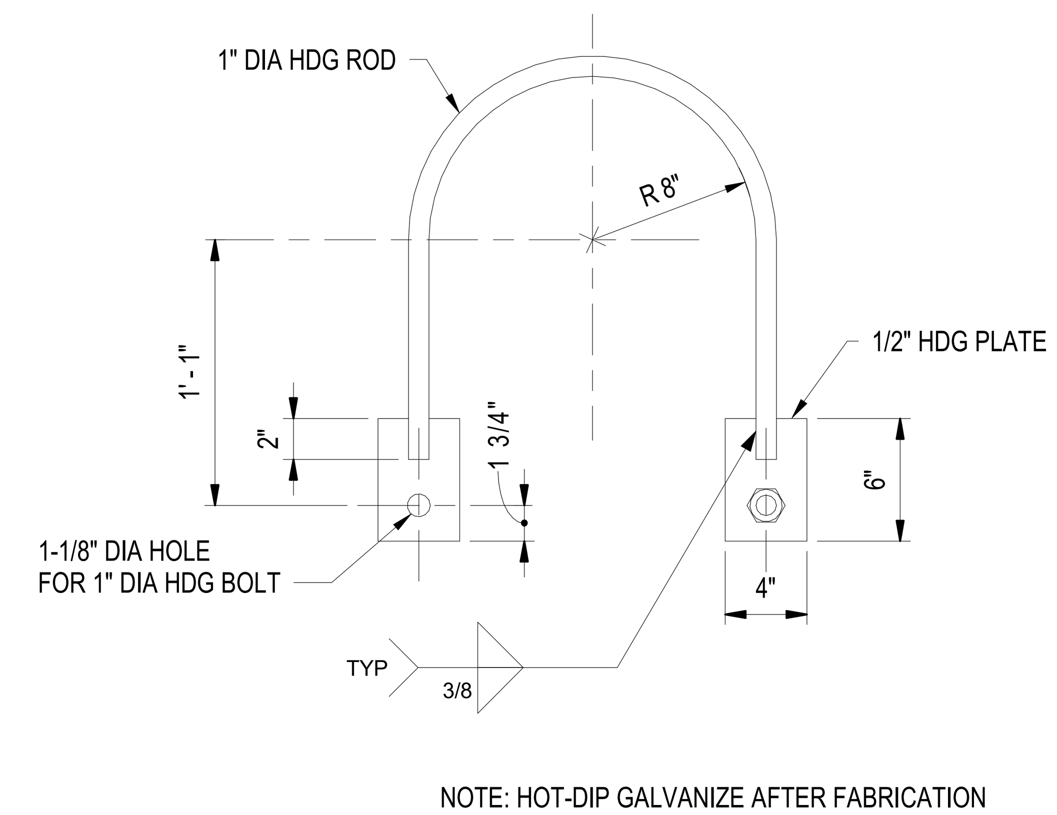
A1 FENDER SECTION

SCALE: 1/2" = 1'-0" (S-103, S-131)



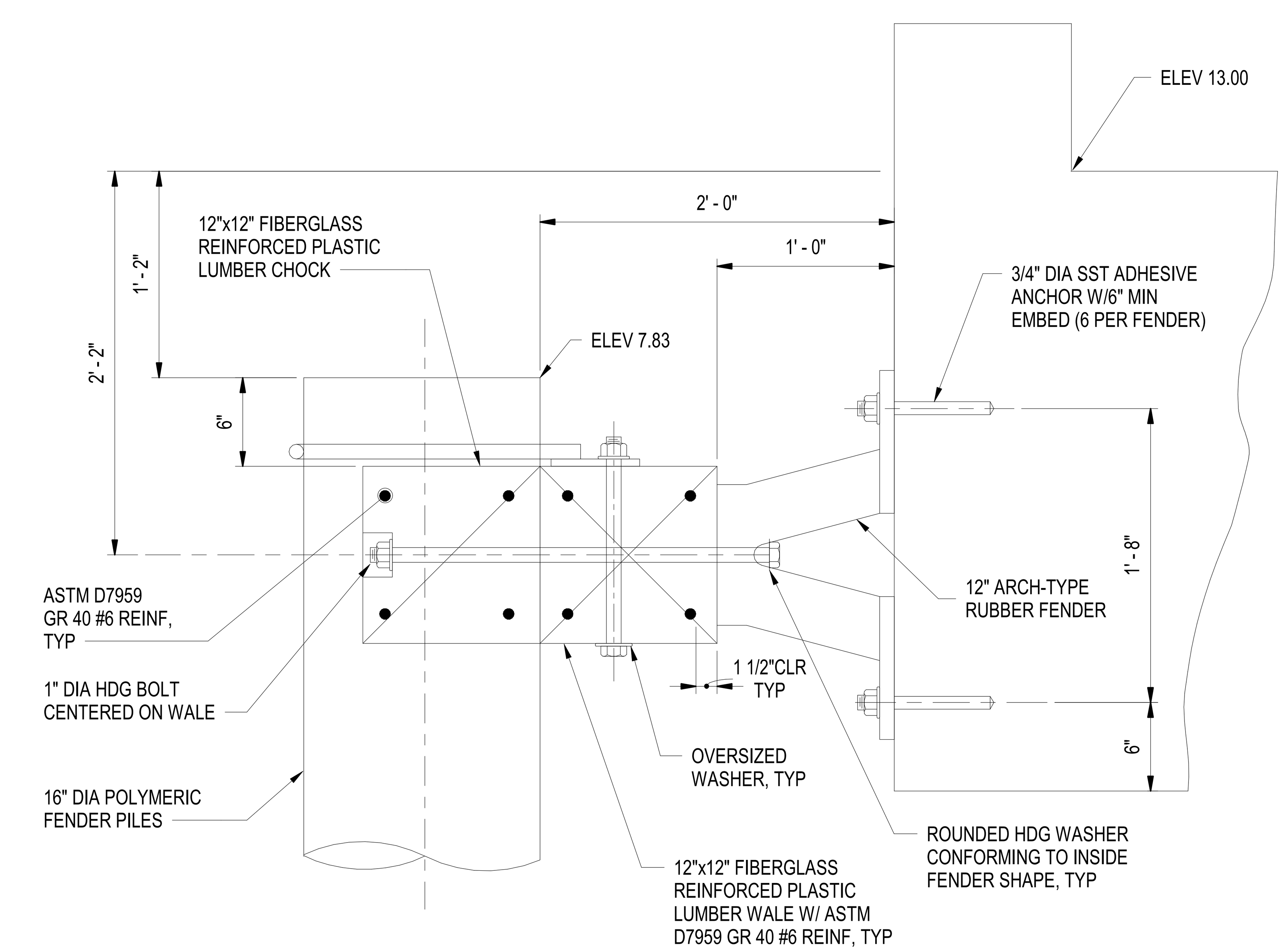
C4 FENDER LAYOUT - POLYMERIC PILES

SCALE: 1" = 1'-0" (S-131)



A2 FENDER PILE CONNECTION DETAIL

SCALE: 1 1/2" = 1'-0" (S-520)



A4 FENDER CONNECTION DETAIL

SCALE: 1 1/2" = 1'-0"

1/2" = 1'-0" 0 6" 1' 2' 3' 4' 5'

1" = 1'-0" 0 6" 1' 2' 3'

1 1/2" = 1'-0" 0 3" 6" 9" 1' 2'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

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FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

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SATISFACTORY TO DATE 06 FEB 2026

DES JZ	DRW AFC	CHK MJF	
PM/DM			MRI/CWJ
BRANCH MANAGER			NEK
CHIEF ENG/ARCH			JWJ
FIRE PROTECTION			HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
FENDER SYSTEM DETAILS - SHEET 1 OF 2

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936716
SHEET 147 OF 247

S-519

DRAWFORM REVISION: 26 AUGUST 2020

3/2/2026 7:36:58 PM S-519
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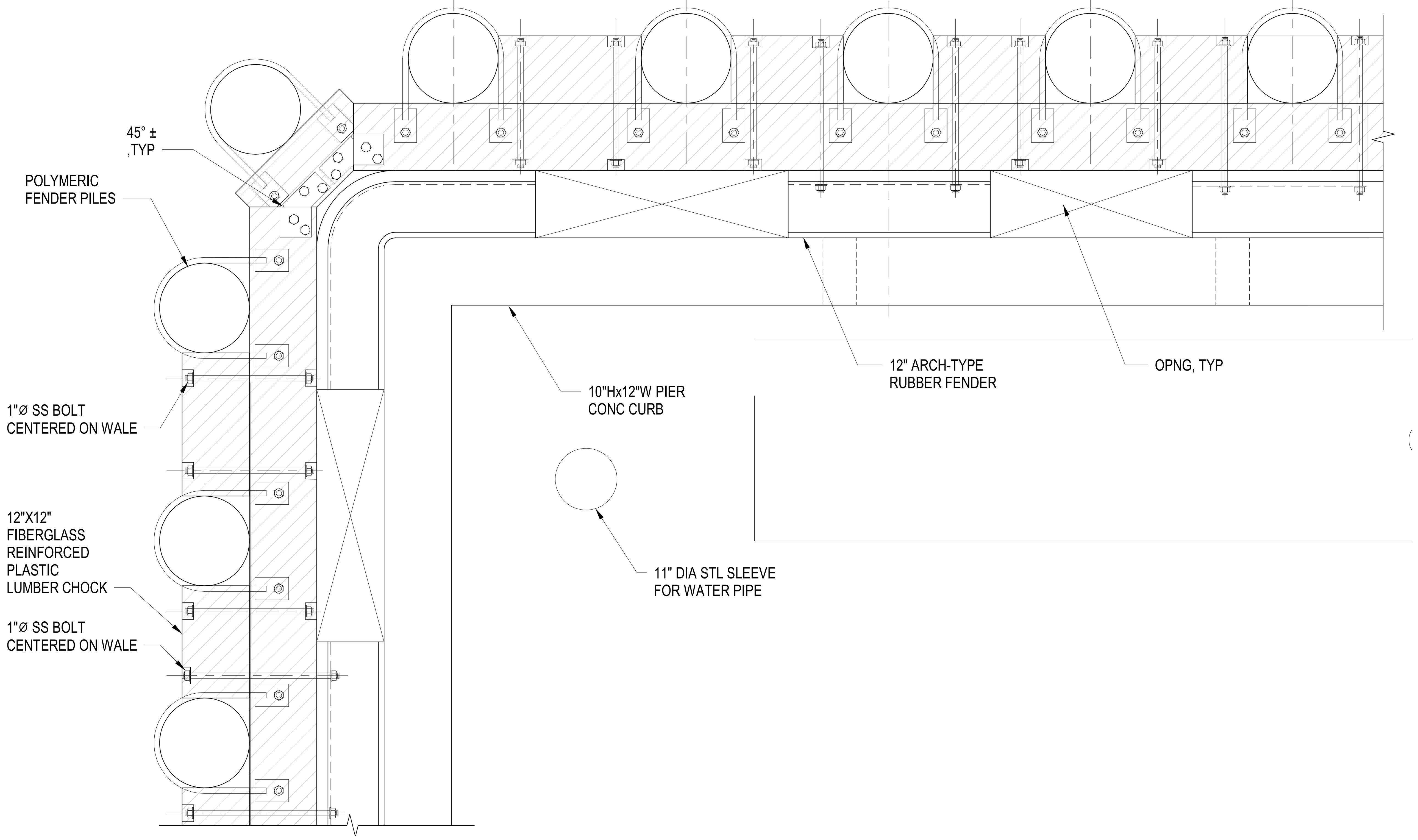
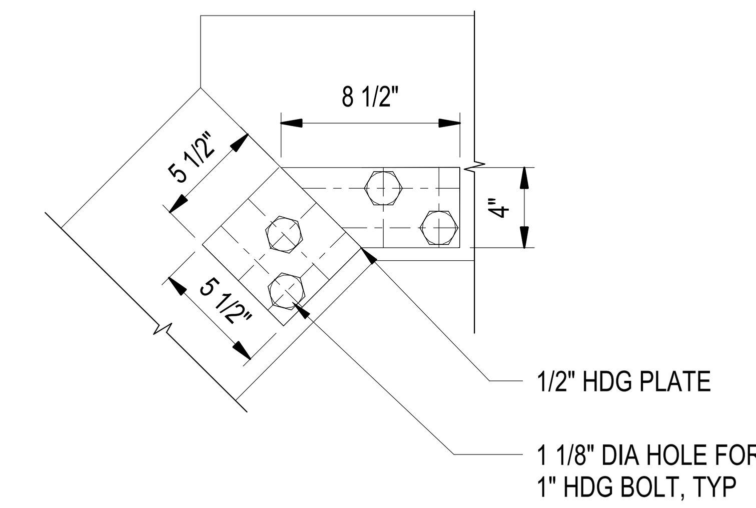
GENERAL SHEET NOTES:

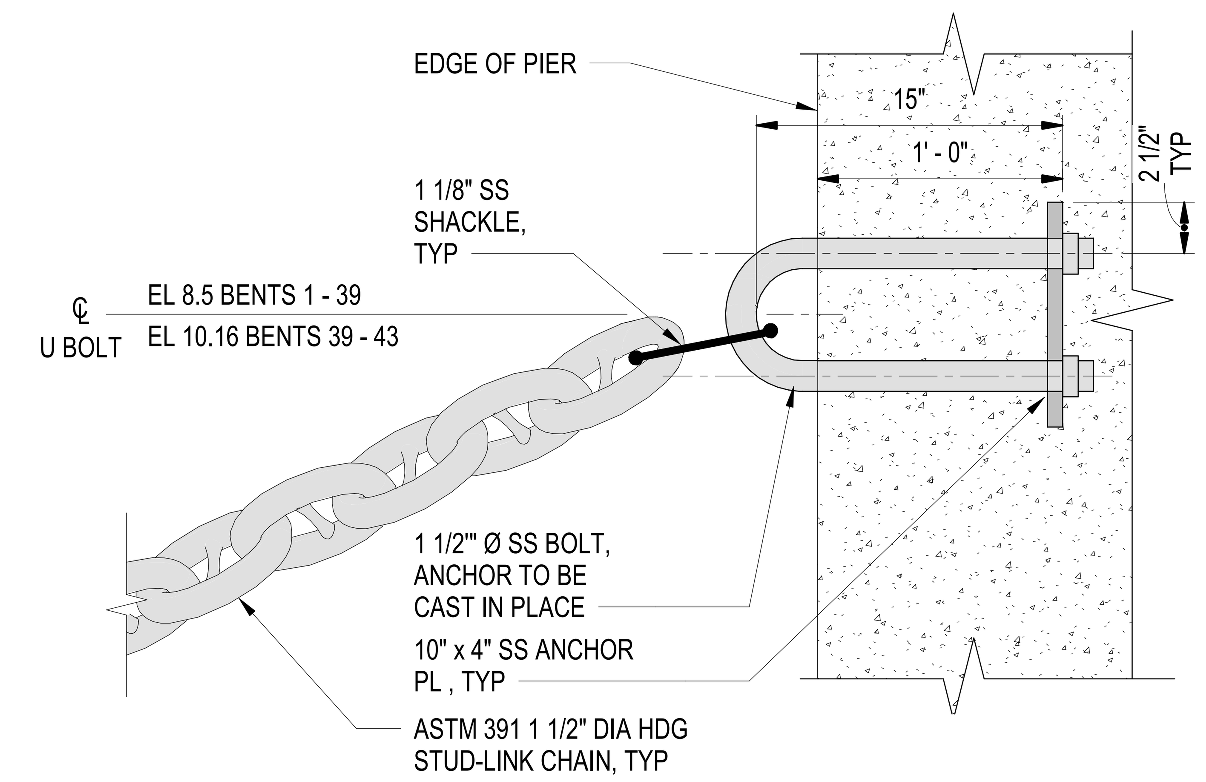
1. ALL STEEL ANGLES, PLATES, AND BOLTS MUST BE 316 STAINLESS STEEL.



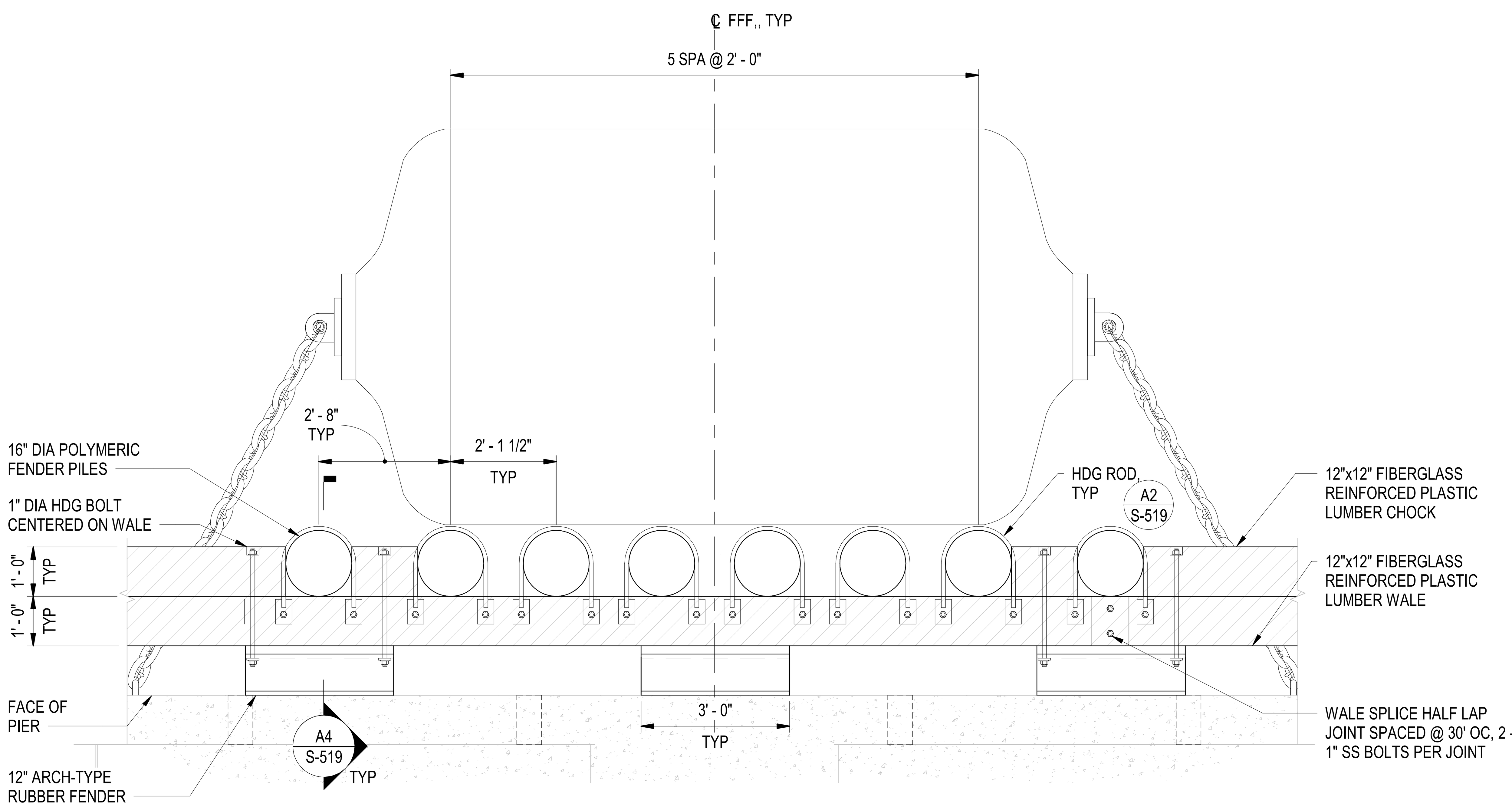
B1 **TYPICAL CORNER FENDER ARRANGEMENT**
SCALE: 3/4" = 1'-0" (S-131)



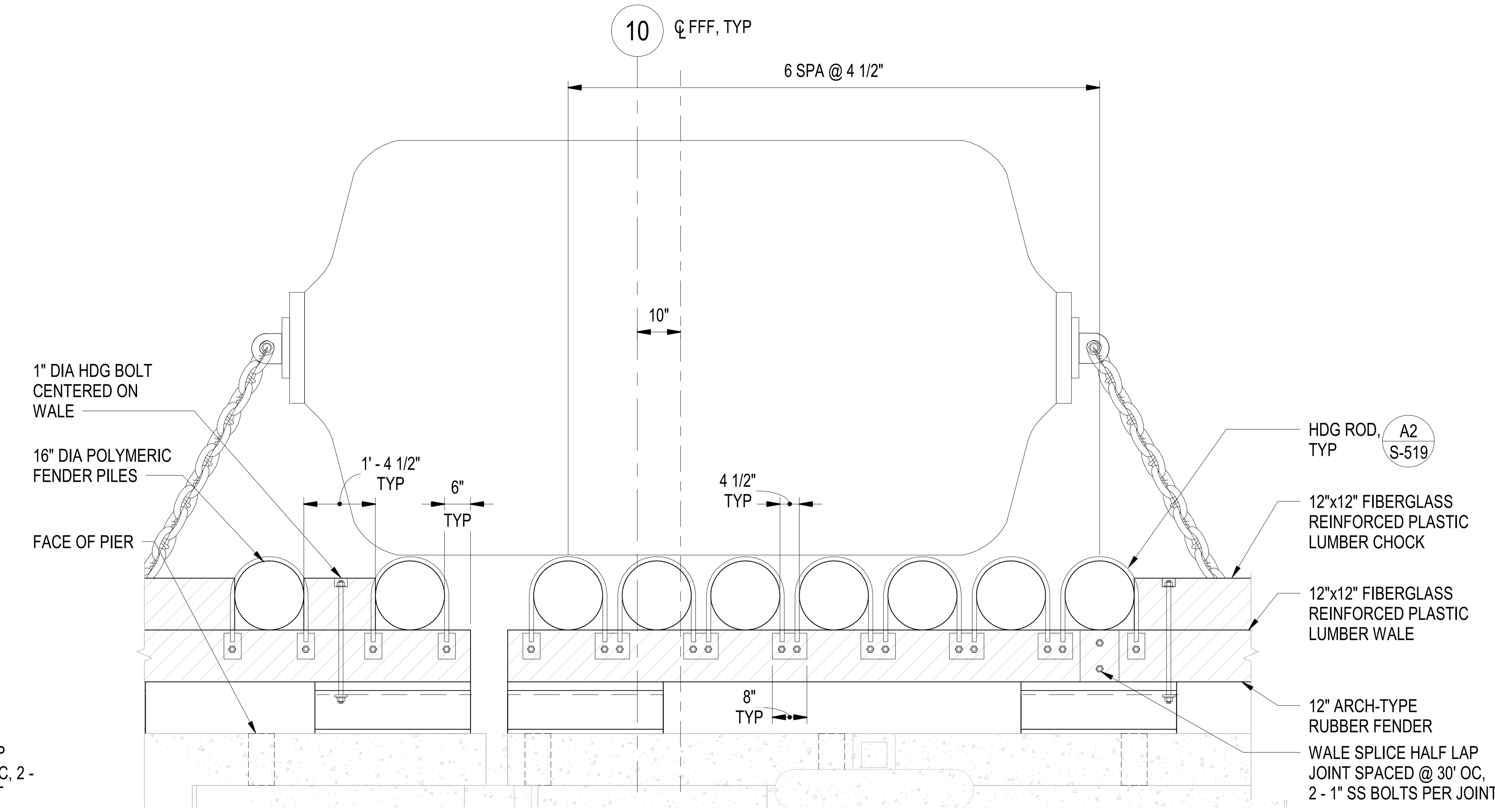
D4 **CORNER FENDER SPLICE PLATE DETAIL**
SCALE: 1 1/2" = 1'-0"



B4 **FLOATING FOAM FILLED FENDER ANCHOR DETAIL**
SCALE: 1 1/2" = 1'-0" (S-519)



A1 **TYPICAL FENDER PILE DETAILS AT FOAM-FILLED FLOATING FENDER**
SCALE: 1/2" = 1'-0" (S-131)



A4 **TYPICAL FENDER PILE DETAILS NEAR BENT 10**
SCALE: 1/2" = 1'-0" (S-135)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'
3/4" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA
SEAL

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PM/DM					MRI/CWJ
BRANCH MANAGER					NEK
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
FENDER SYSTEM DETAILS - SHEET 2 OF 2

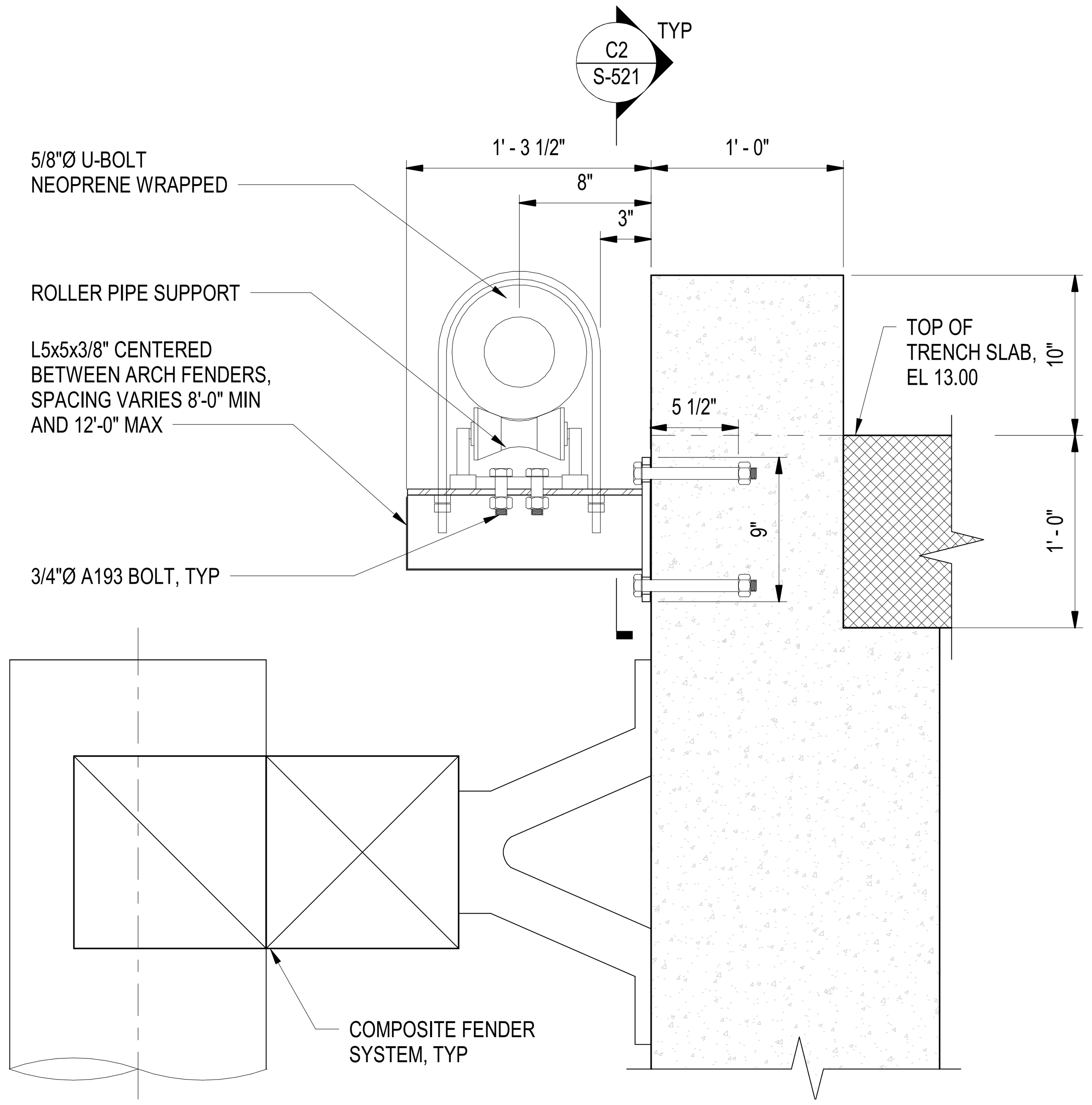
SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936717
SHEET 148 OF 247
S-520

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:37:01 PM S-520
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

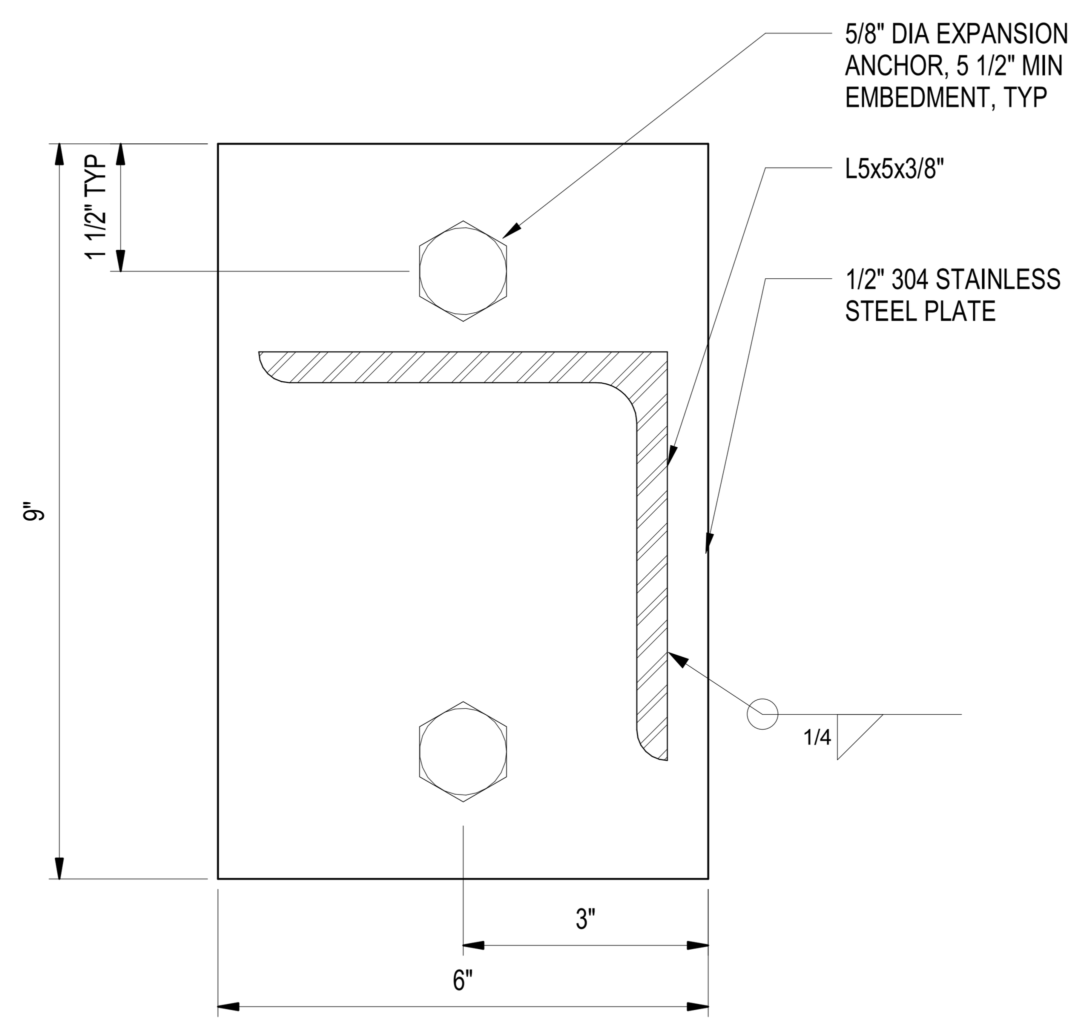
GENERAL SHEET NOTES:

1. ALL STEEL ANGLES, PLATES, AND BOLTS MUST BE 316 STAINLESS STEEL.



C1 **PIPE SUPPORT AT PIER EDGE DETAIL**

SCALE: 1 1/2" = 1'-0"

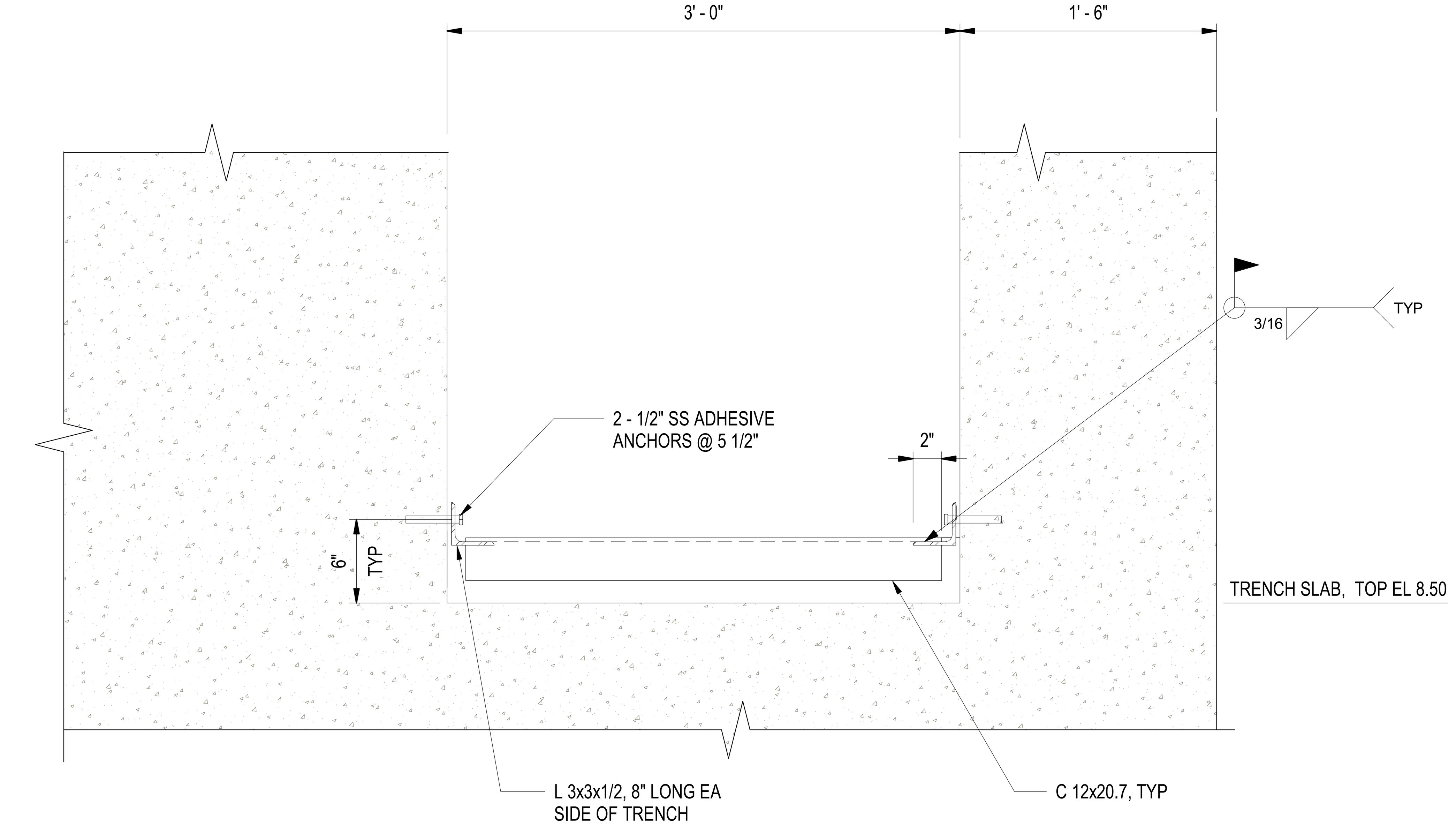


NOTES:

FOR CONNECTION AT LADDERS ANGLES ARE L3x3x1/2" AND L4x4x1/2". ALL DIMENSIONS REMAIN UNCHANGED.

C2 **PIPE SUPPORT CONNECTION DETAIL**

SCALE: 6" = 1'-0"

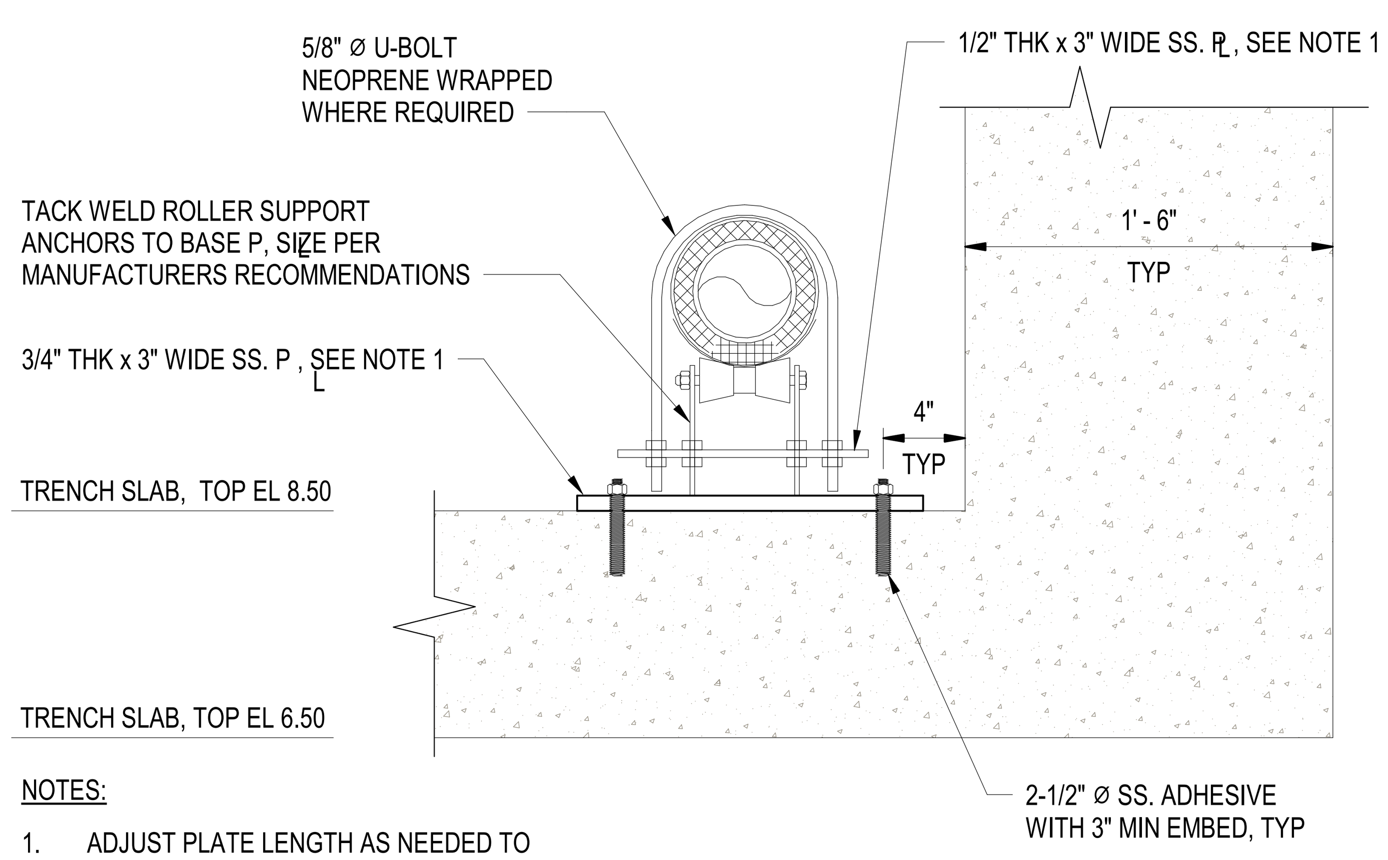


NOTES:

SEE MECHANICAL DETAIL DRAWINGS FOR ADDITIONAL INFORMATION.

C4 **PIPE MAIN TRENCH SUPPORT TYPICAL DETAIL**

SCALE: 1 1/2" = 1'-0"

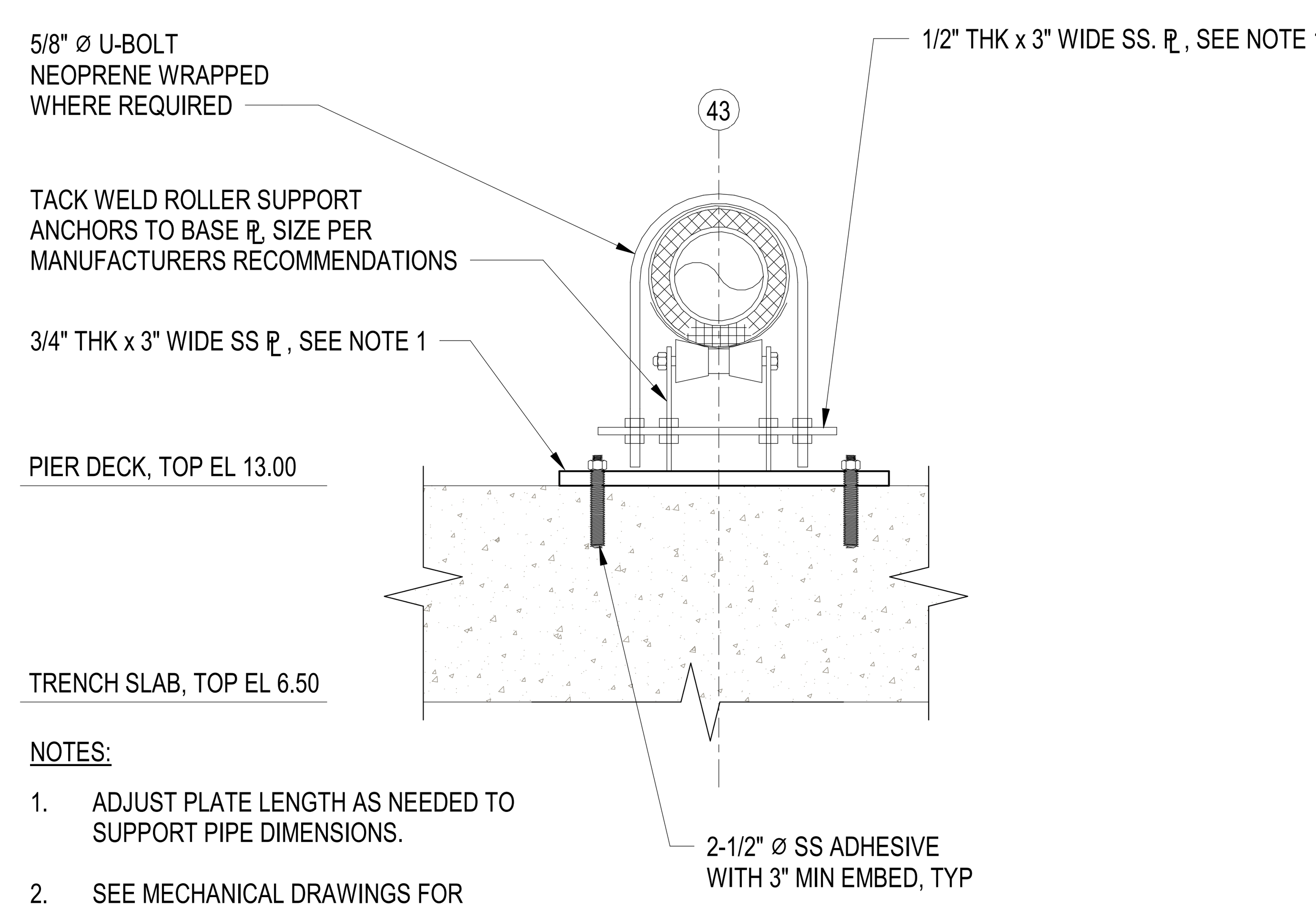


NOTES:

1. ADJUST PLATE LENGTH AS NEEDED TO SUPPORT PIPE DIMENSIONS.
2. SEE MECHANICAL DRAWINGS FOR ADDITIONAL INFORMATION.

A1 **LATERAL UTILITY TRENCH PIPE SUPPORTS**

SCALE: 1 1/2" = 1'-0"



NOTES:

1. ADJUST PLATE LENGTH AS NEEDED TO SUPPORT PIPE DIMENSIONS.
2. SEE MECHANICAL DRAWINGS FOR ADDITIONAL INFORMATION.

A2 **SUPPORT DETAILS FOR PIPE EXPOSED ON PIER DECK**

SCALE: 1 1/2" = 1'-0"

6" = 1' - 0" 0 1" 2" 4" 6"

1 1/2" = 1' - 0" 0 3" 6" 9" 1' 2'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

Stantec

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BRANCH MANAGER			NEK
CHIEF ENG/ARCH			JWJ
FIRE PROTECTION			HPN

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CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
PIPE SUPPORT DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936718
SHEET 149 OF 247

S-521

DRAWFORM REVISION: 25 AUGUST 2020

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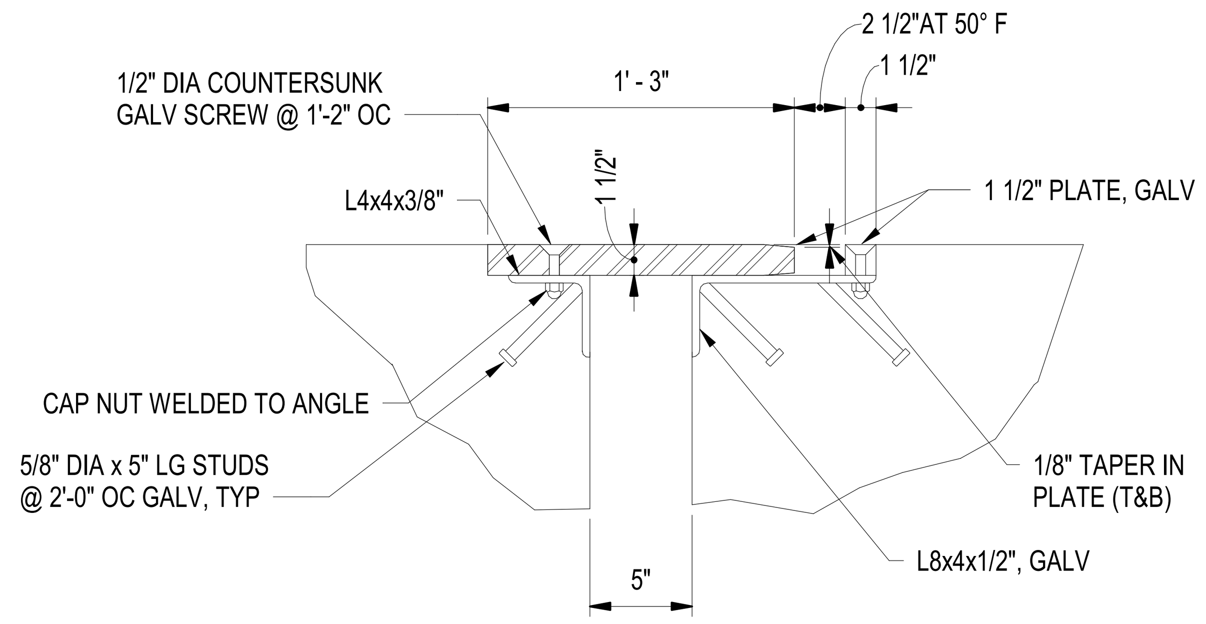
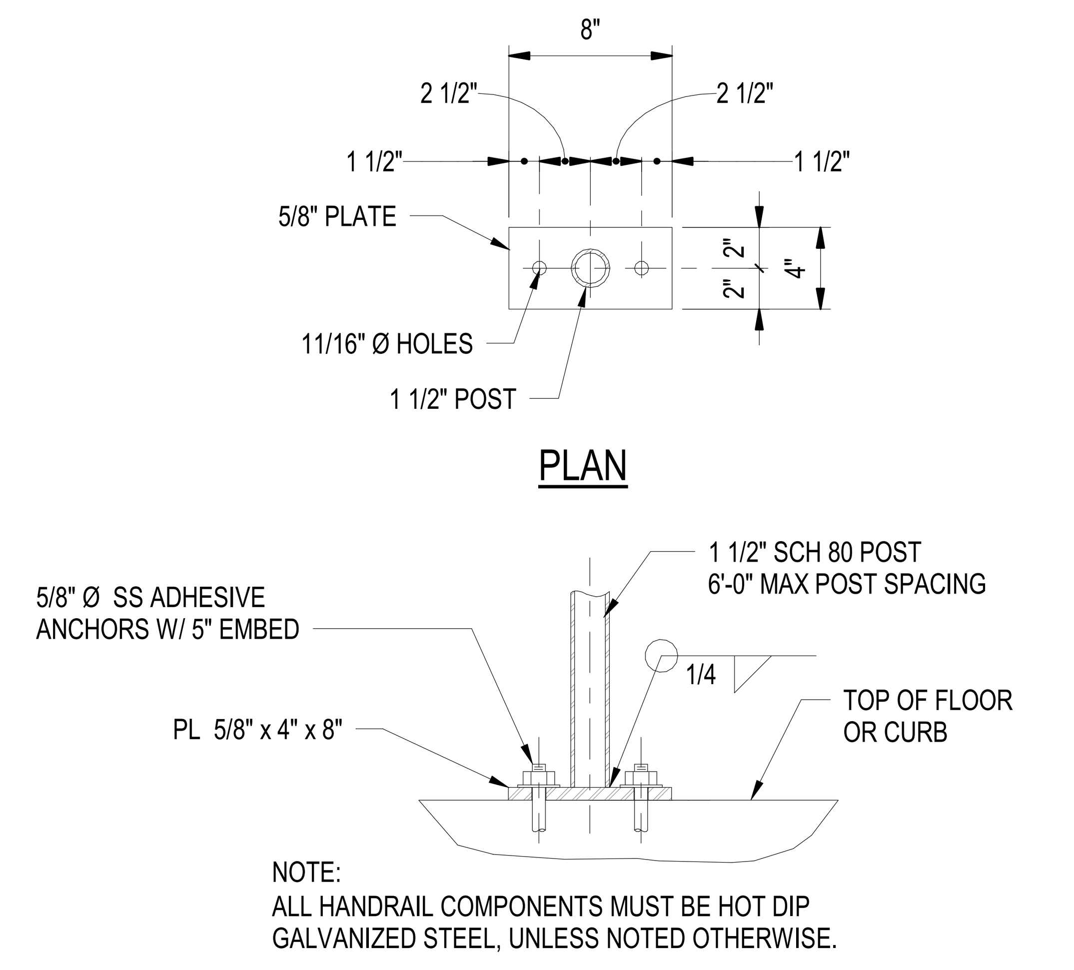


D1 TYPICAL EXPANSION JOINT COVER (5")

SCALE: 1 1/2" = 1'-0" (S-519, S-504, S-516)

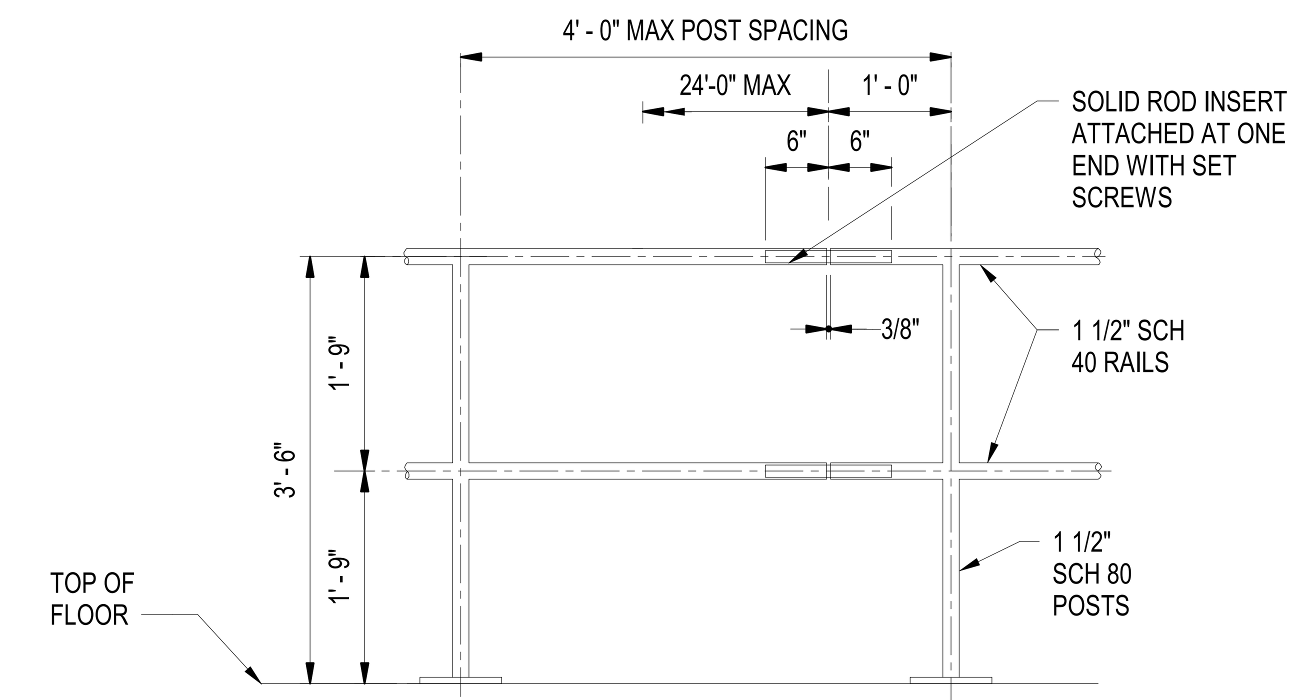


NOTE:
ALL HANDRAIL COMPONENTS MUST BE HOT DIP GALVANIZED STEEL, UNLESS NOTED OTHERWISE.

SECTION

C3 POST CONNECTION TO CONCRETE - STEEL

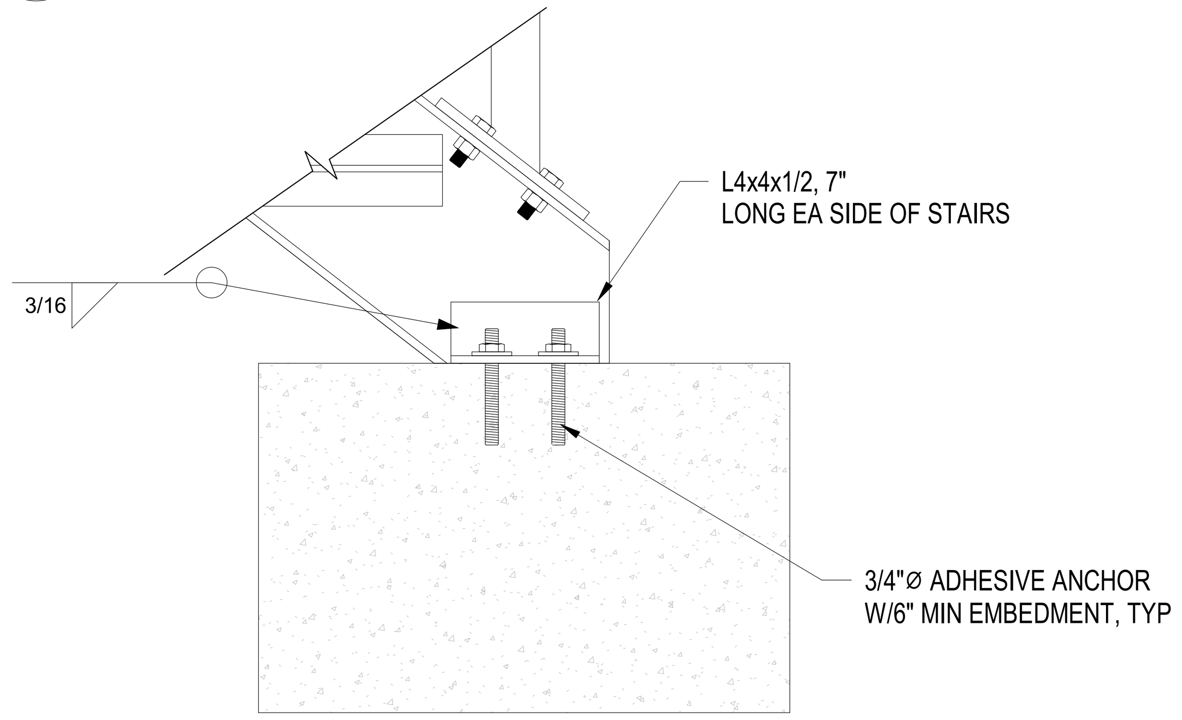
SCALE: 1 1/2" = 1'-0"



NOTE:
ALL HANDRAIL COMPONENTS MUST BE HOT DIP GALVANIZED STEEL, UNLESS NOTED OTHERWISE.

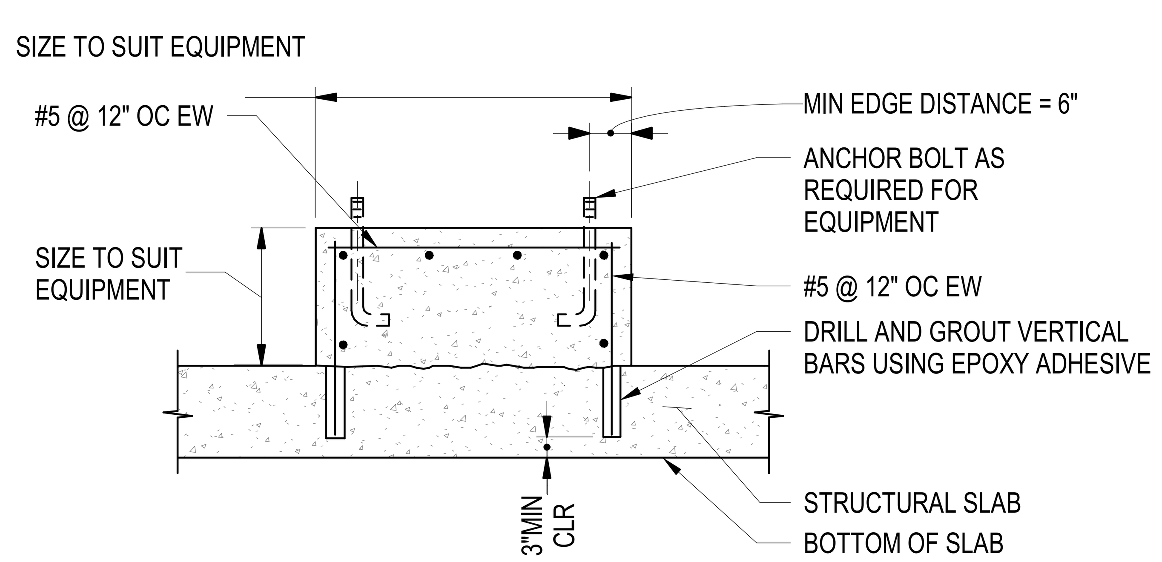
D4 TYPICAL HANDRAIL ELEVATION - STEEL

SCALE: 3/4" = 1'-0" (S-402, S-403, S-534)



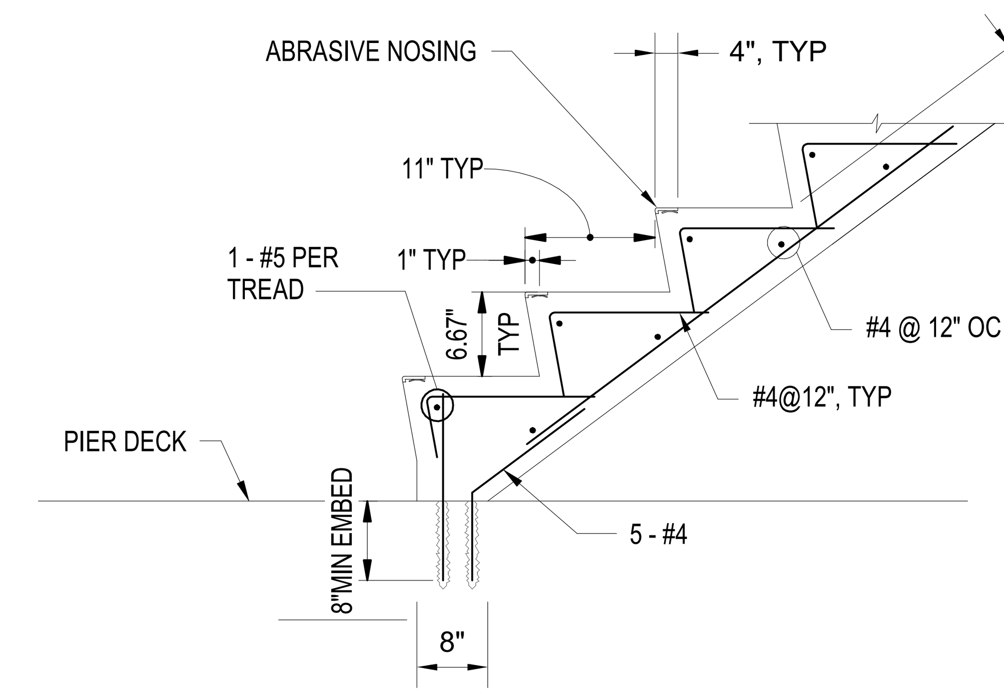
C1 TYPICAL CONNECTION DETAILS FOR STAIR LANDING

SCALE: 1 1/2" = 1'-0" (S-516)



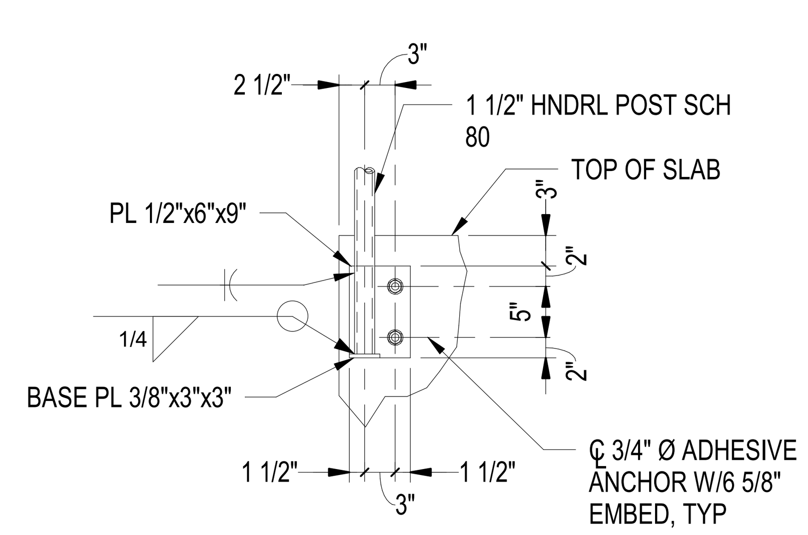
B1 TYPICAL EQUIPMENT SUPPORT

SCALE: 1/2" = 1'-0" S-307



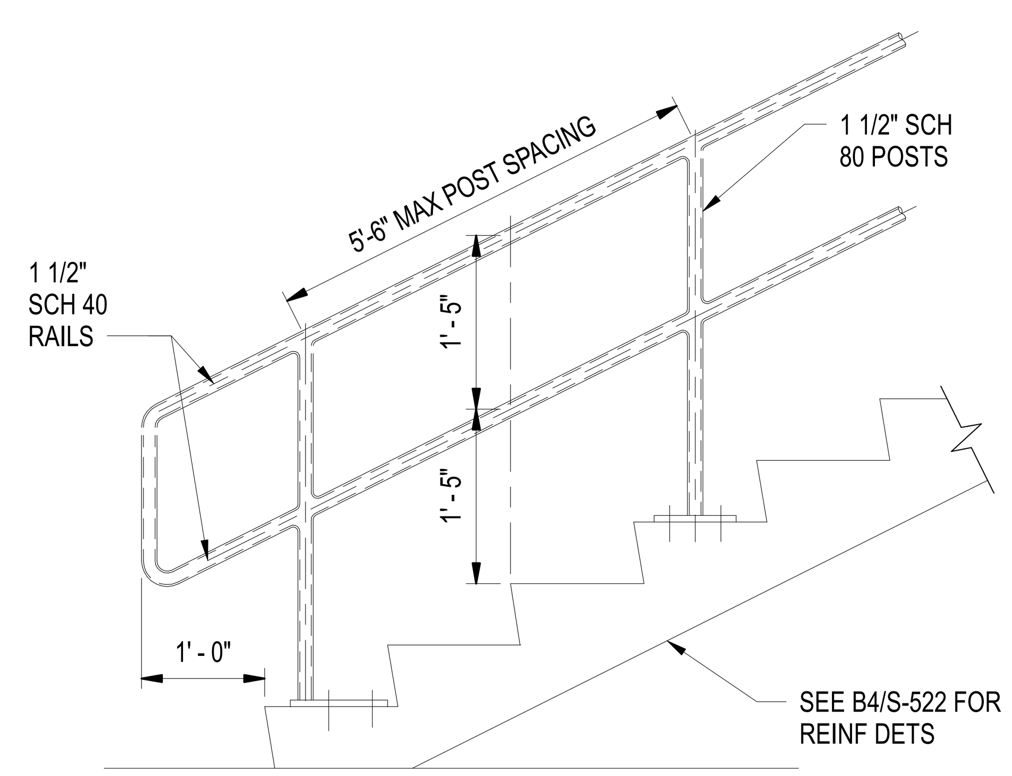
B4 TYPICAL CONCRETE STAIR

SCALE: 3/4" = 1'-0" S-522



A1 SIDE MOUNTED HANDRAIL DETAIL

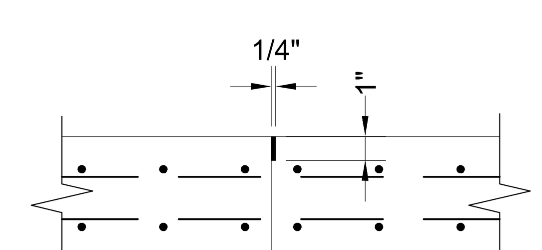
SCALE: 3/4" = 1'-0" (S-402, S-403)



NOTE:
HANDRAIL MATERIAL MUST BE AS SHOWN ON TYPICAL HANDRAIL ELEVATION

A2 HANDRAIL ELEVATION AT CONCRETE STAIRS

SCALE: 3/4" = 1'-0" (S-402, S-403)

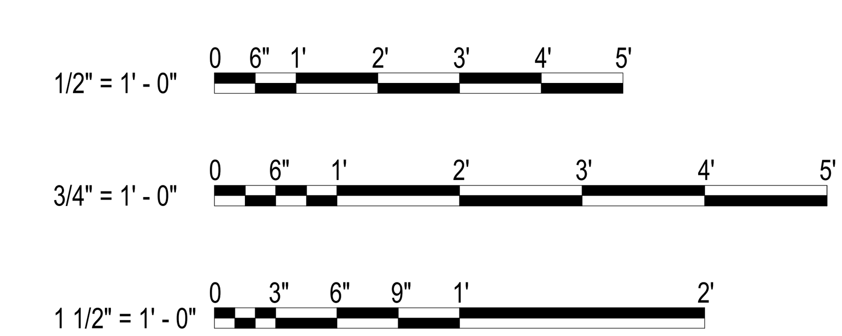


SECTION

NOTE:
FORM 1/4" x 1" SLOT IN SLAB. CLEAN SLOT AND FILL WITH NON-SHRINK GROUT 28 DAYS AFTER PLACING CONCRETE.

A3 TYPICAL CONSTRUCTION JOINT IN SLAB

SCALE: 3/4" = 1'-0"



SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES	JZ	DRW	AFC	CHK	MJF

PM/DM	MRI/CWJ
BRANCH MANAGER	NEK
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT
NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

MISCELLANEOUS SECTIONS & DETAILS - SHEET 1 OF 2

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936719
SHEET 150 OF 247

S-522

DRAWFORM REVISION: 25 AUGUST 2020

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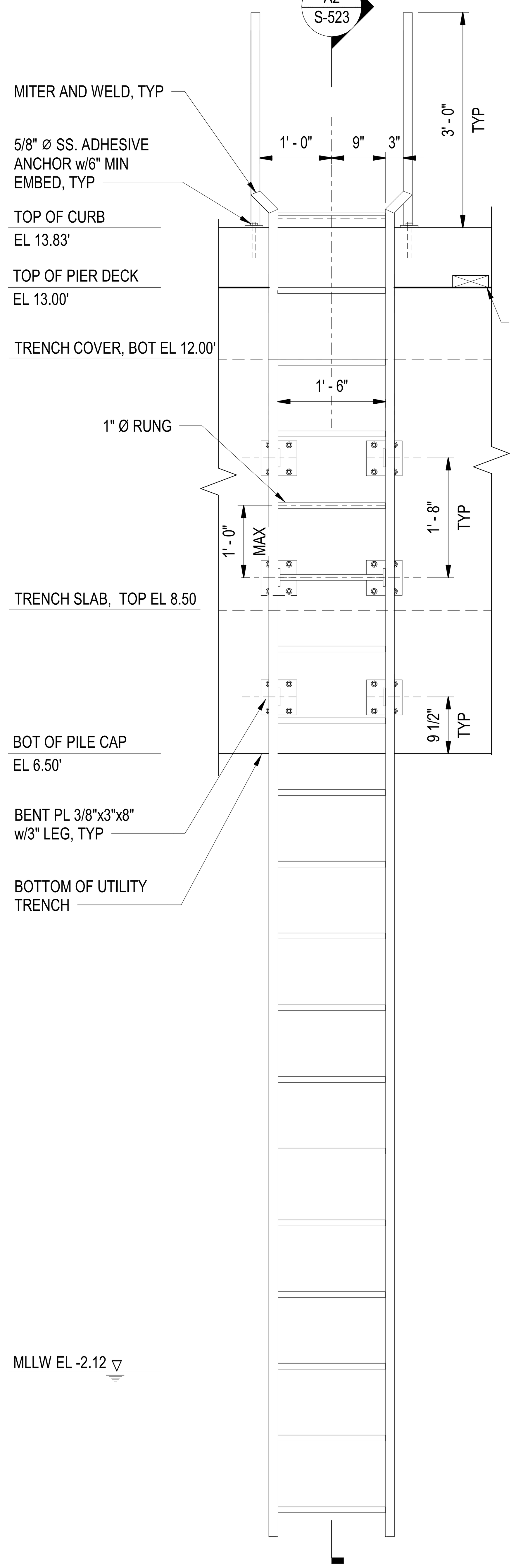
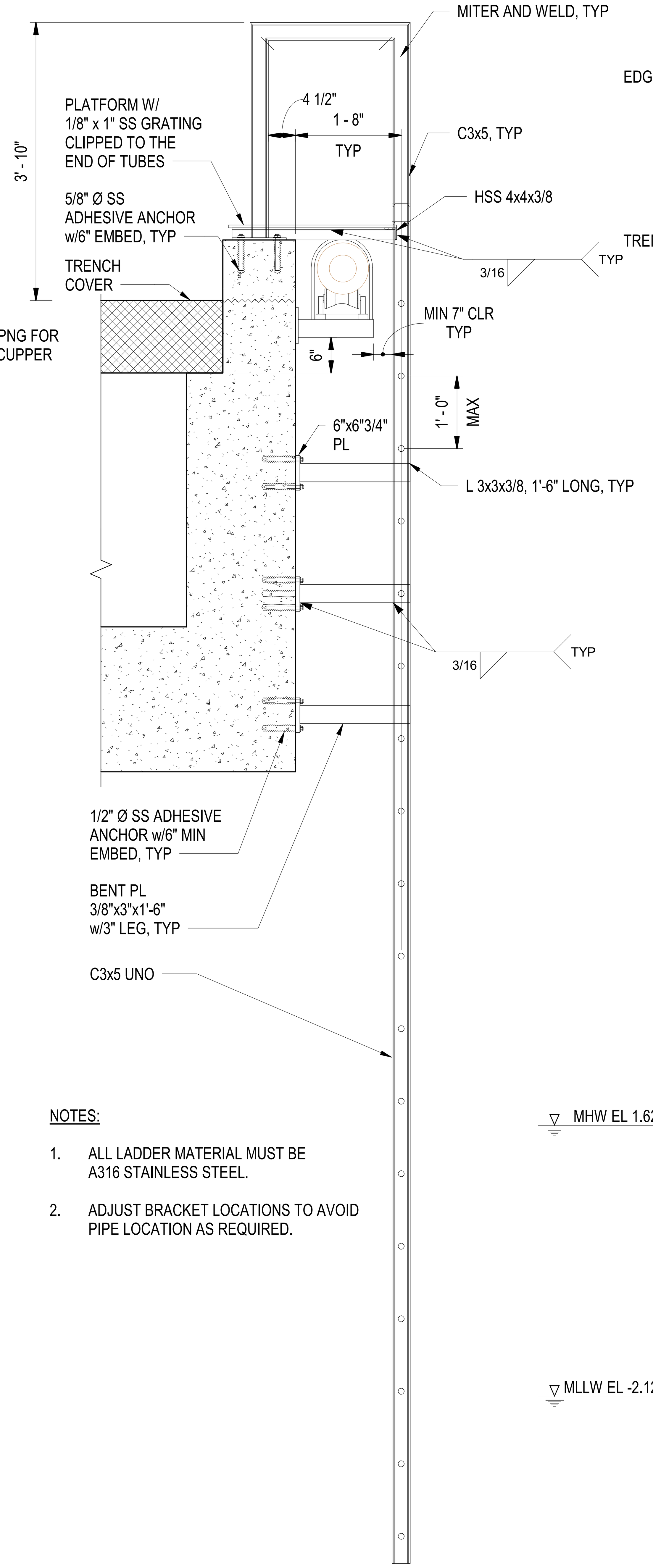
S-522



A1 EMERGENCY LADDER ELEVATION

SCALE: 3/4" = 1'-0"

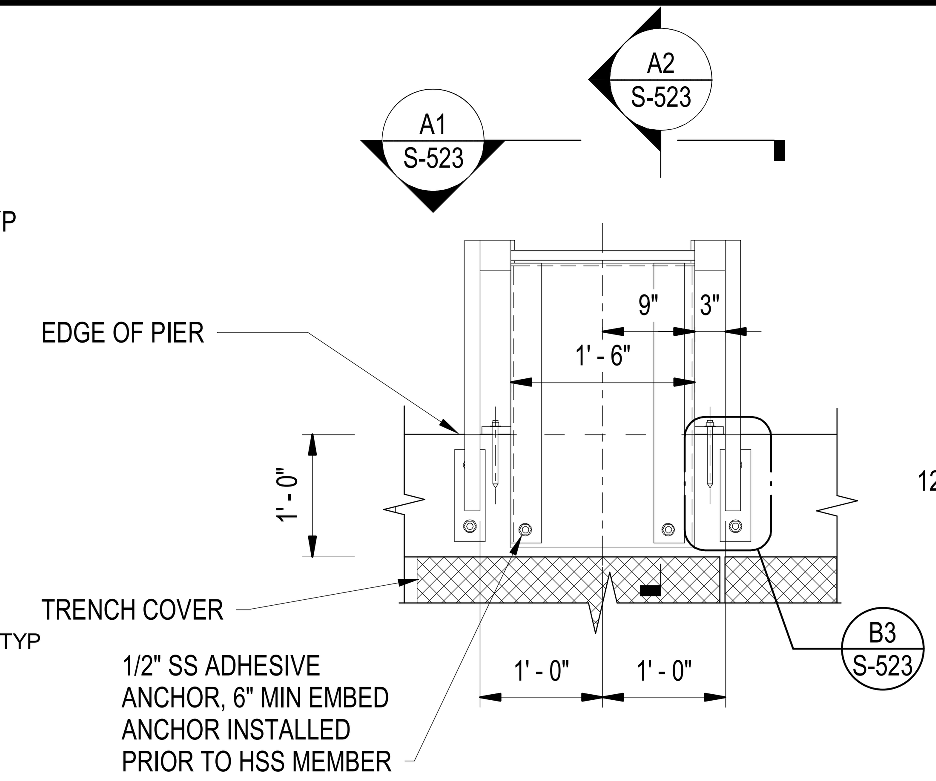


NOTES:

1. ALL LADDER MATERIAL MUST BE A316 STAINLESS STEEL.
2. ADJUST BRACKET LOCATIONS TO AVOID PIPE LOCATION AS REQUIRED.

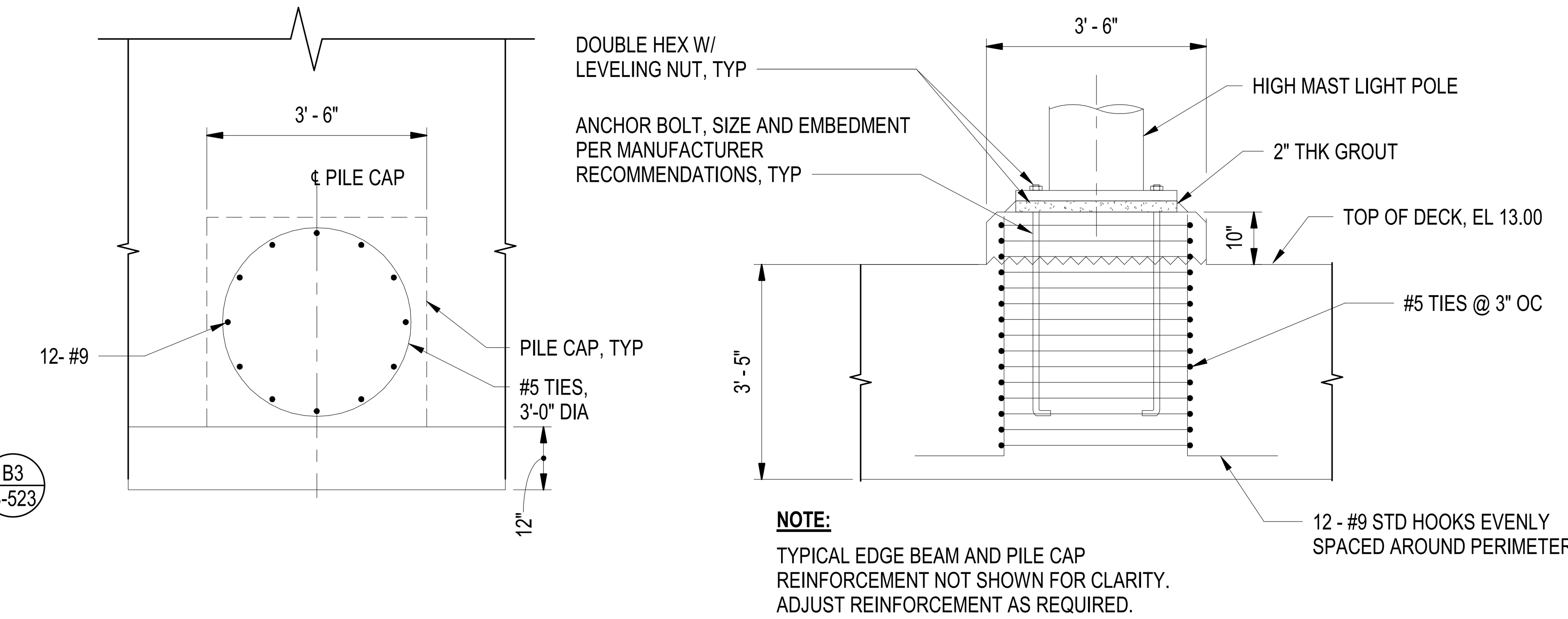
A2 EMERGENCY LADDER TYPICAL SECTION

SCALE: 3/4" = 1'-0"



D3 EMERGENCY LADDER PLAN

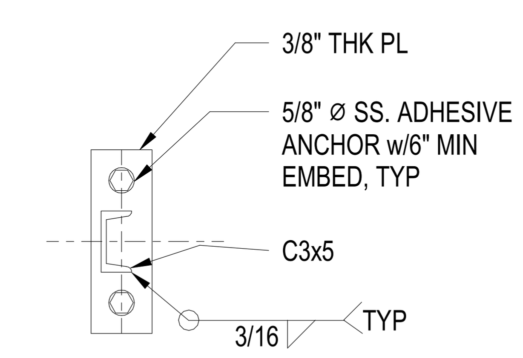
SCALE: 3/4" = 1'-0" (S-128, S-523)



NOTE:
TYPICAL EDGE BEAM AND PILE CAP REINFORCEMENT NOT SHOWN FOR CLARITY. ADJUST REINFORCEMENT AS REQUIRED.

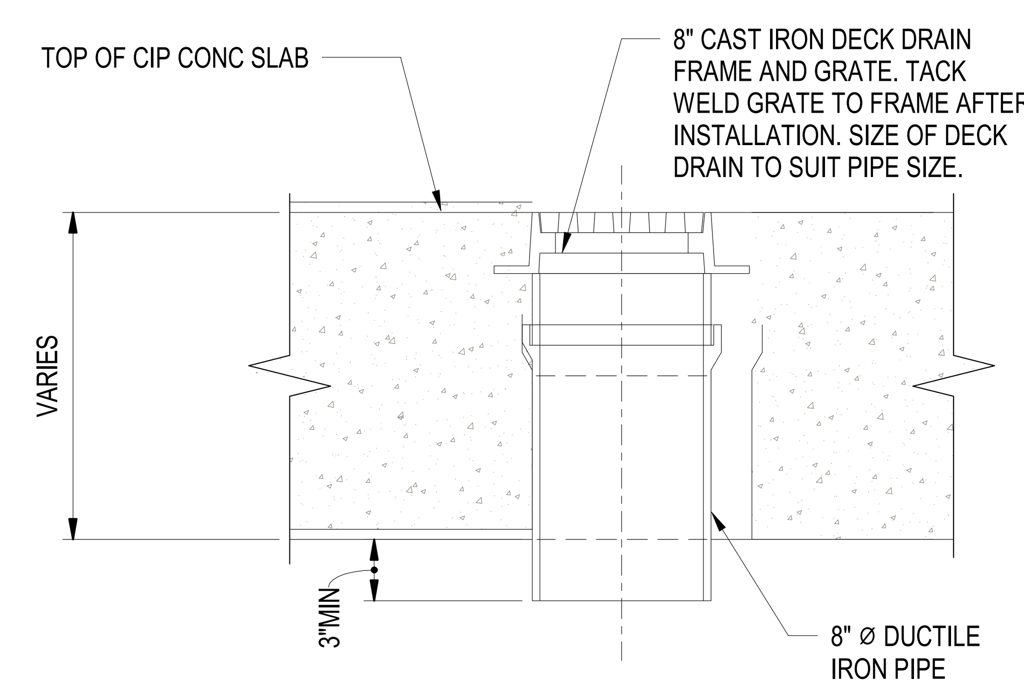
D4 LIGHT POLE BASE

SCALE: 1/2" = 1'-0"



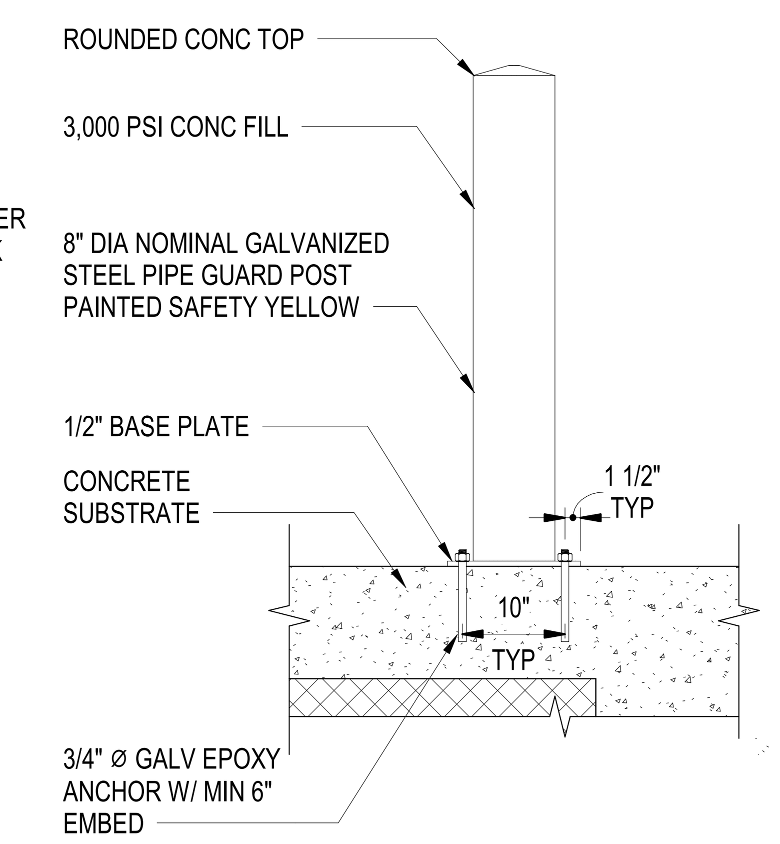
B3 EMERGENCY LADDER DETAIL

SCALE: 1 1/2" = 1'-0"



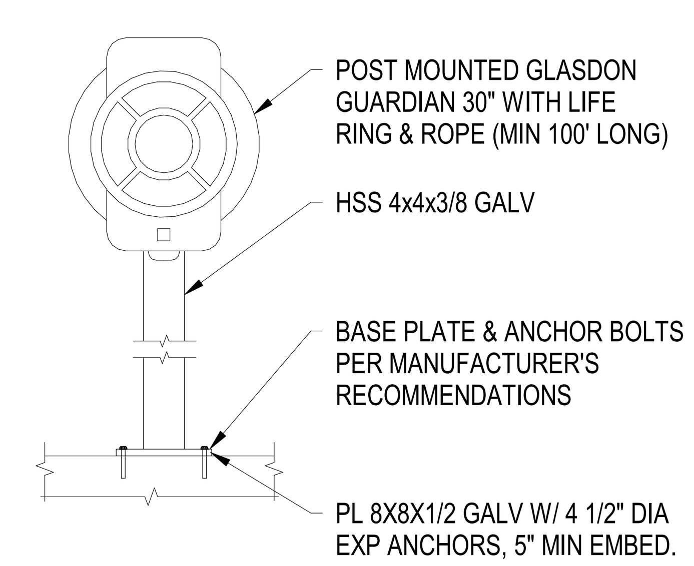
B4 TYPICAL DECK DRAIN

SCALE: 1 1/2" = 1'-0"



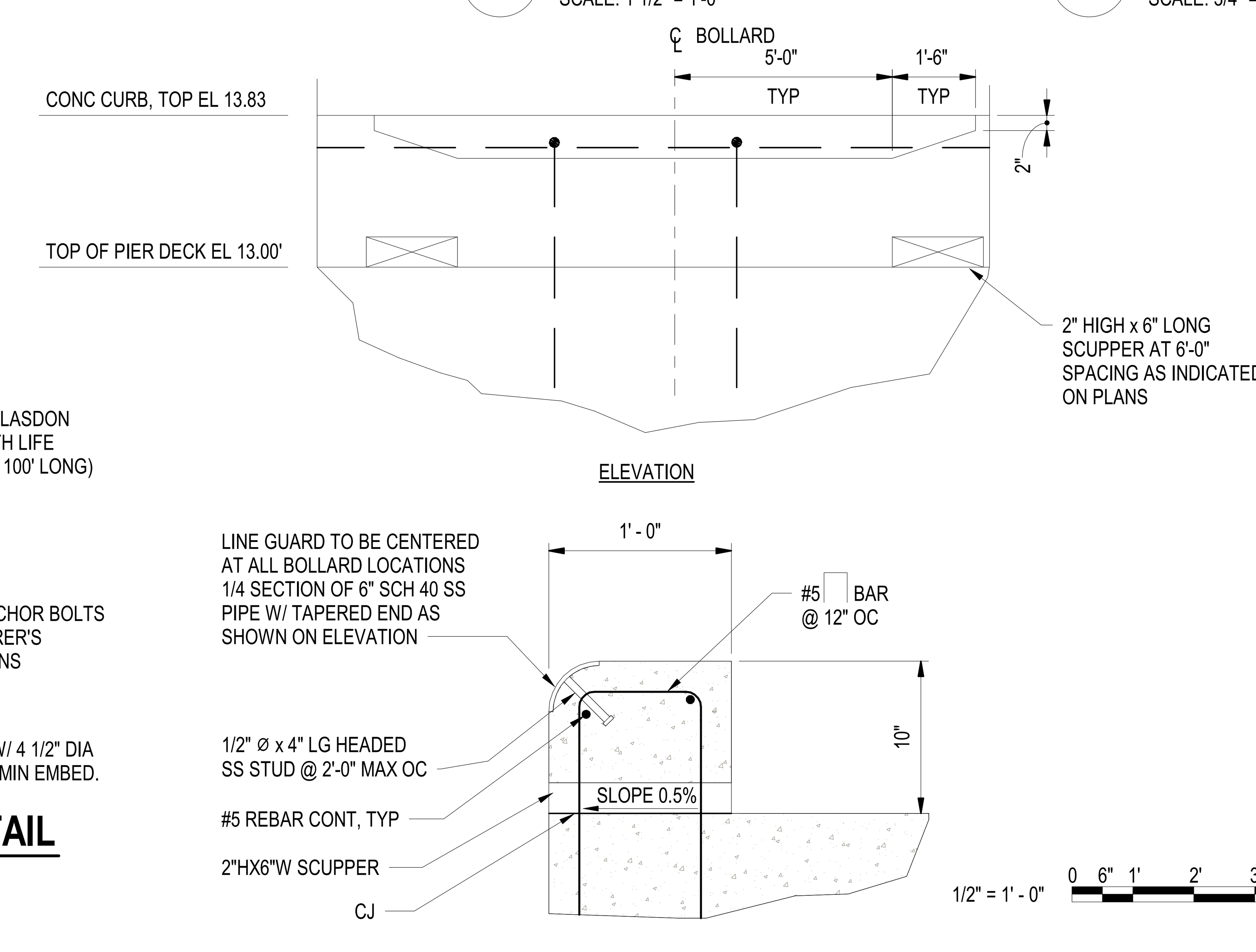
B5 TYPICAL GUARD POST

SCALE: 3/4" = 1'-0"



A3 LIFE RING DETAIL

SCALE: 1 1/2" = 1'-0"



A4 CURB DETAIL

SCALE: 1 1/2" = 1'-0" (S-124)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

1 1/2" = 1' - 0" 0 3" 6" 9" 1' 2'

3/4" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

(S-102, S-121)

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S-523

SYM | DESCRIPTION | DATE | APPR

NAVFAC

JINQUAN ZHONG C 82242 CIVIL REGISTERED PROFESSIONAL ENGINEER STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES JZ	DRW AFC	CHK MJF	
PM/DM			MRI/CWJ
BRANCH MANAGER			NEK
CHIEF ENG/ARCH			JWJ
FIRE PROTECTION			HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
MISCELLANEOUS SECTIONS & DETAILS - SHEET 2 OF 2

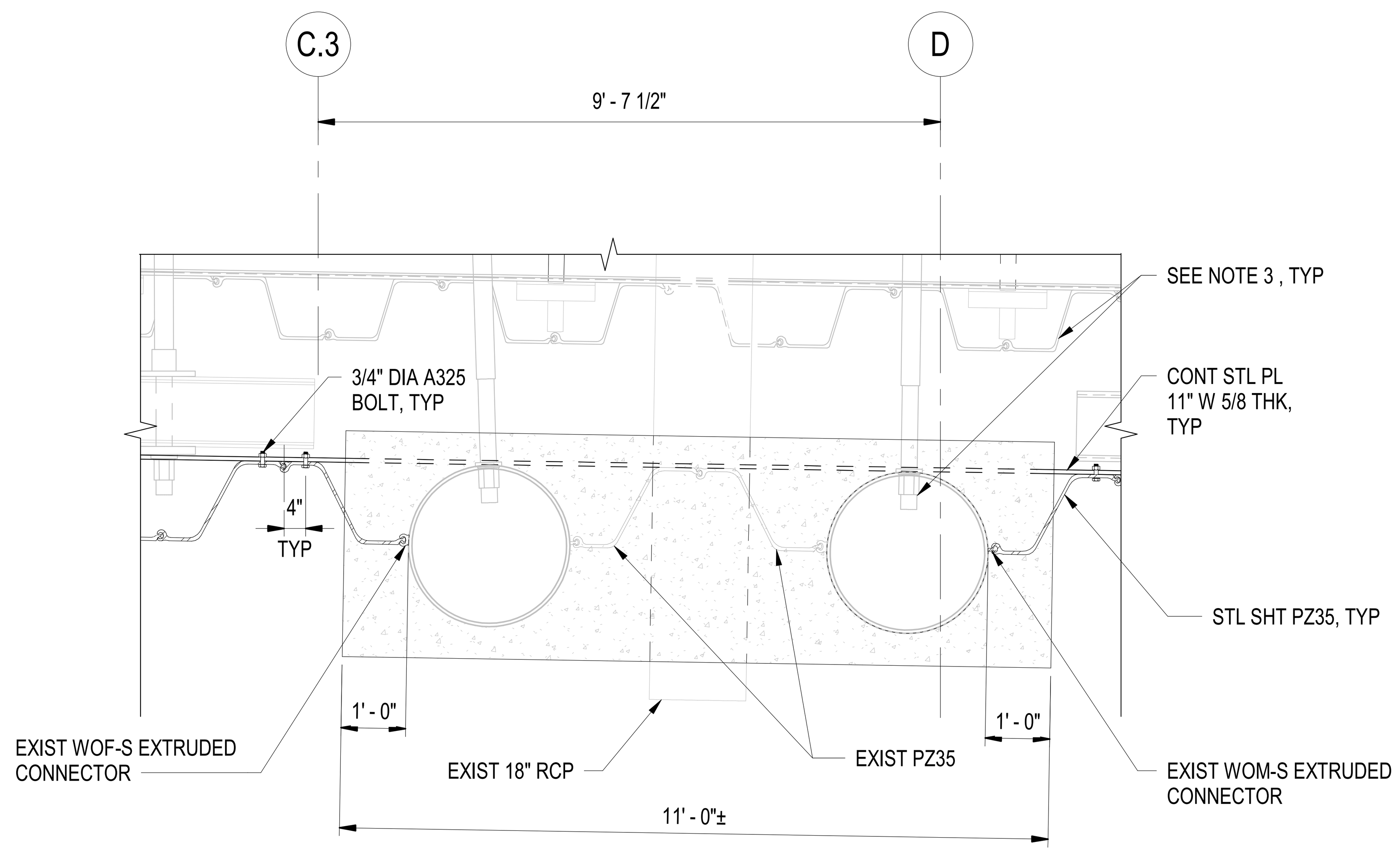
SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936720
SHEET 151 OF 247

S-523

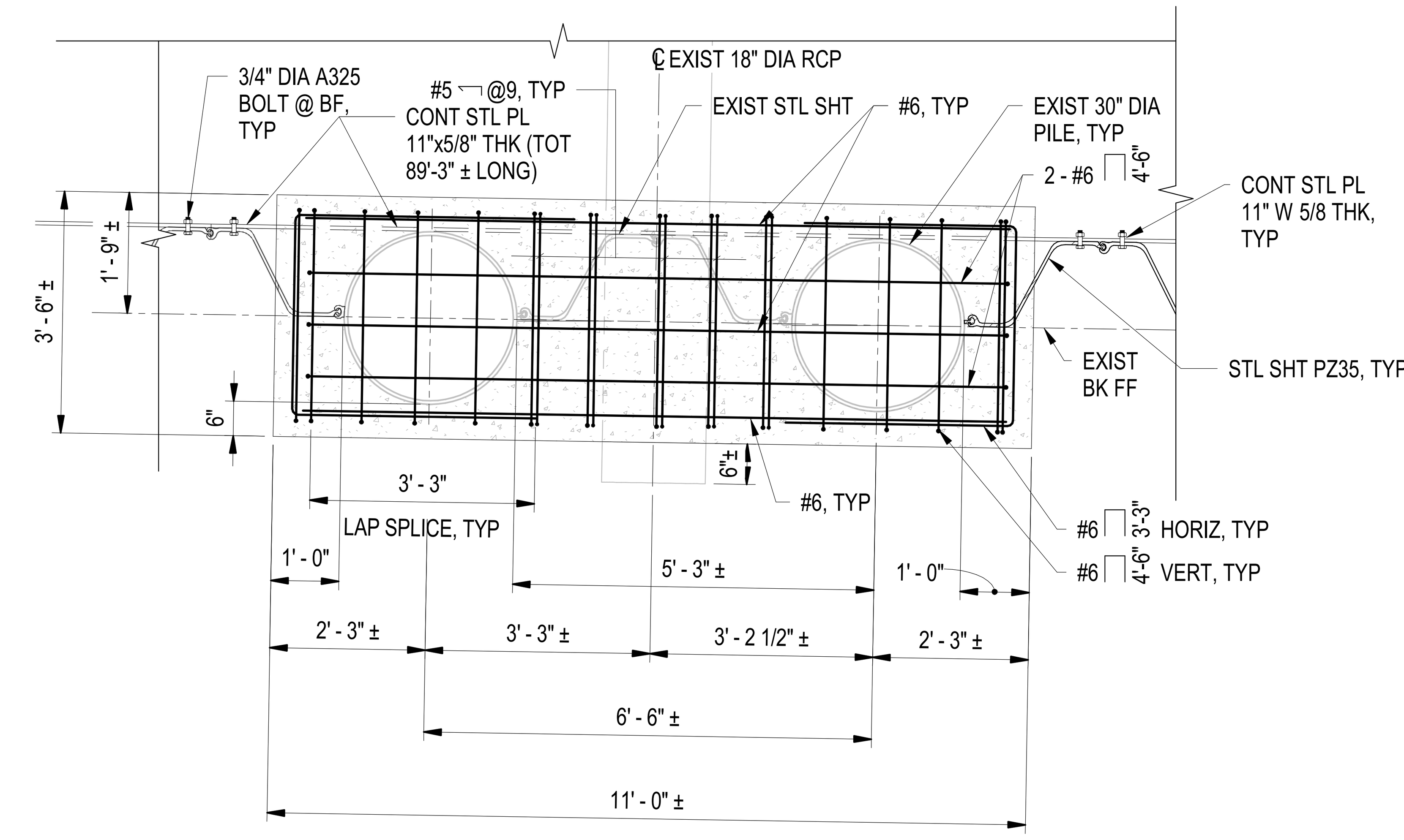
DRAWFORM REVISION: 25 AUGUST 2020

GENERAL SHEET NOTES:

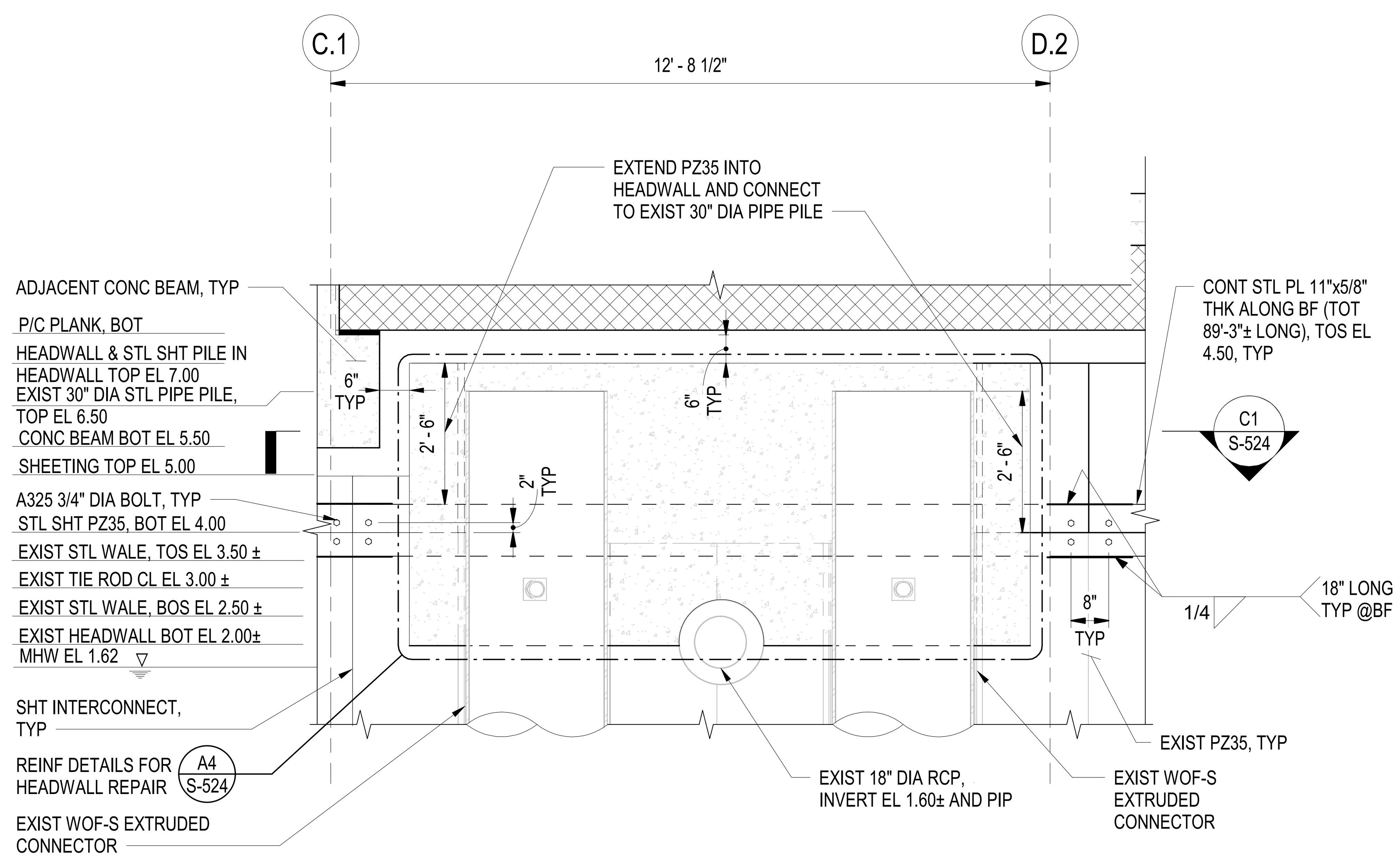
1. FOR LIMITS REQUIRED FOR SHEETING REPAIR, SEE DRAWING S-207.
2. VERIFY IN THE FIELD FOR EXISTING EXTRUDED CONNECTOR AT EXISTING STEEL PIPE PILES FOR SHEETING CONNECTION.
3. SEE ATTACHMENT A FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.
4. SEE DRAWING S-525 FOR ADDITIONAL STEEL DETAILS.
5. DRILL SHEETING TO PULL REINFORCEMENT THROUGH. DO NOT TORCH.



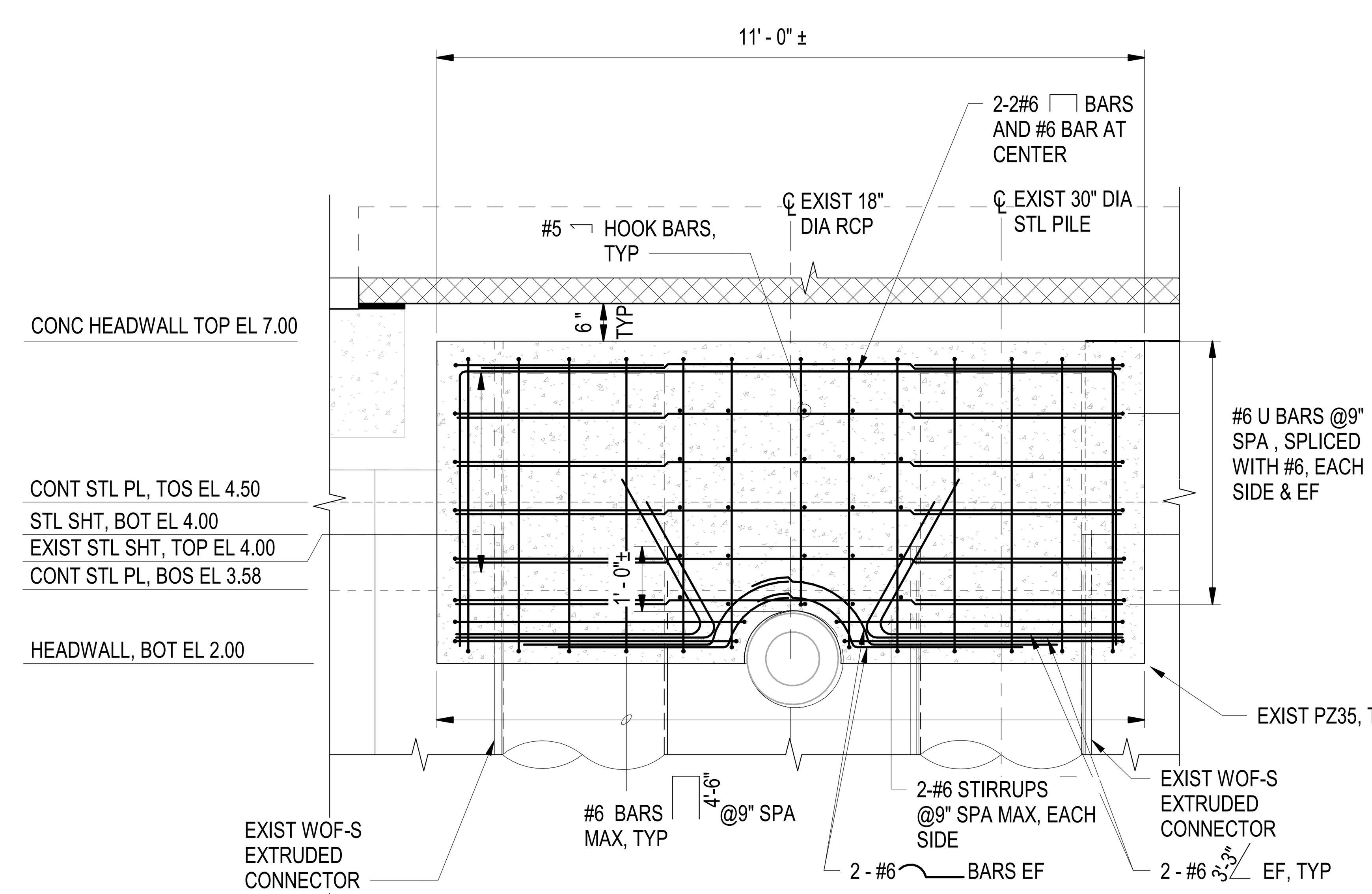
C1 **SHEETING CONNECTION DETAILS AT CONCRETE HEADWALL**
SCALE: 1/2" = 1'-0" S-524



C4 **HEAD WALL REINFORCEMENT - PLAN VIEW**
SCALE: 1/2" = 1'-0"



A1 **TYPICAL SHEETING REPAIR AT CONCRETE HEADWALL**
SCALE: 1/2" = 1'-0" (S-207, S-302)



NOTE:
SEE A1/S-524 FOR DETAILS ON BOLTS AND WELDS.

A4 **REINFORCEMENT DETAILS FOR HEADWALL REPAIR**
SCALE: 1/2" = 1'-0" (S-303)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA
SEAL

Stantec
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026

DES JZ	DRW AFC	CHK MJF
PM/DM		MRI/CWJ
BRANCH MANAGER		NEK
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
S45N BULKHEAD MODIFICATION DETAILS - SHEET 1 OF 2

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936721
SHEET 152 OF 247
S-524

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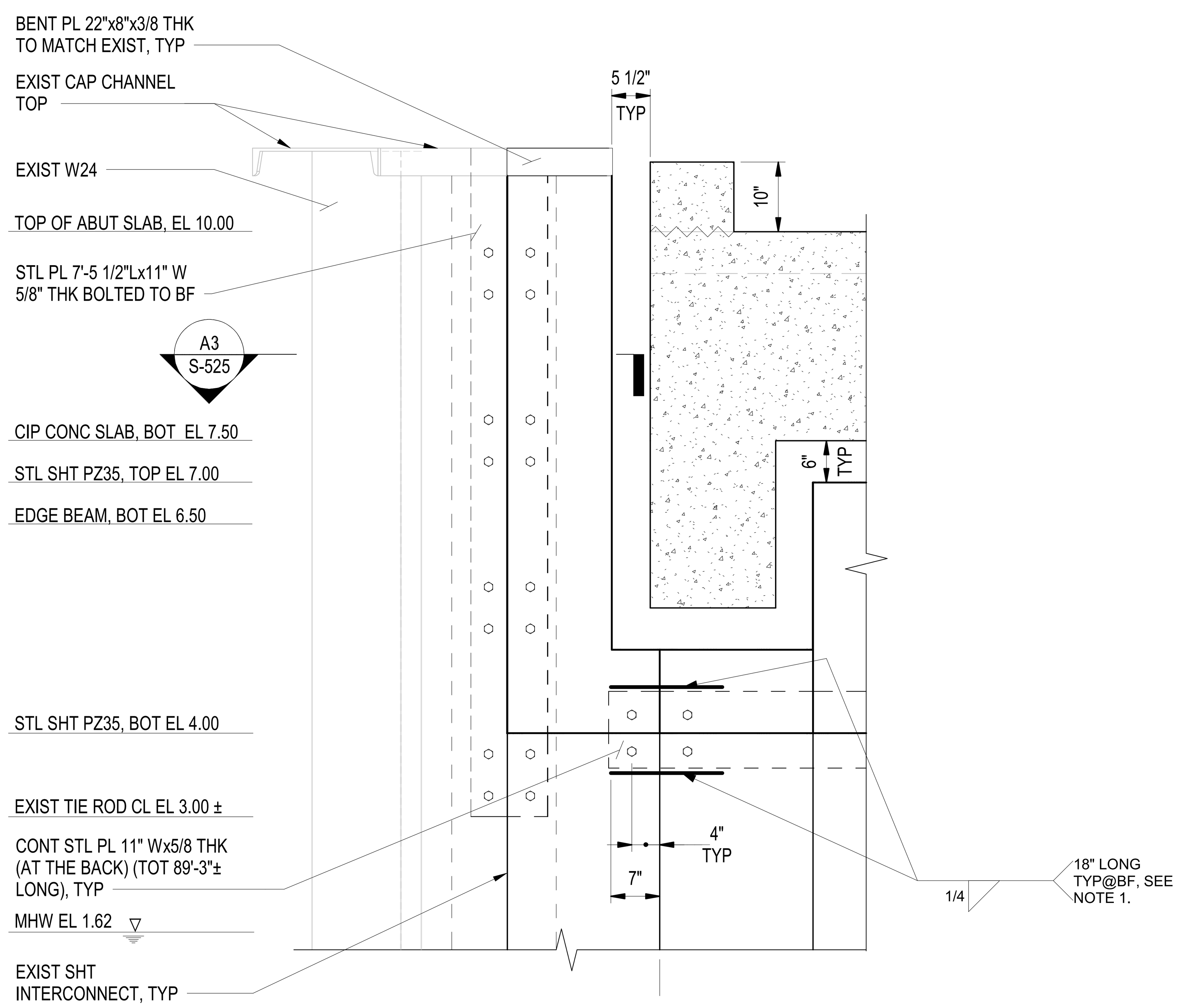
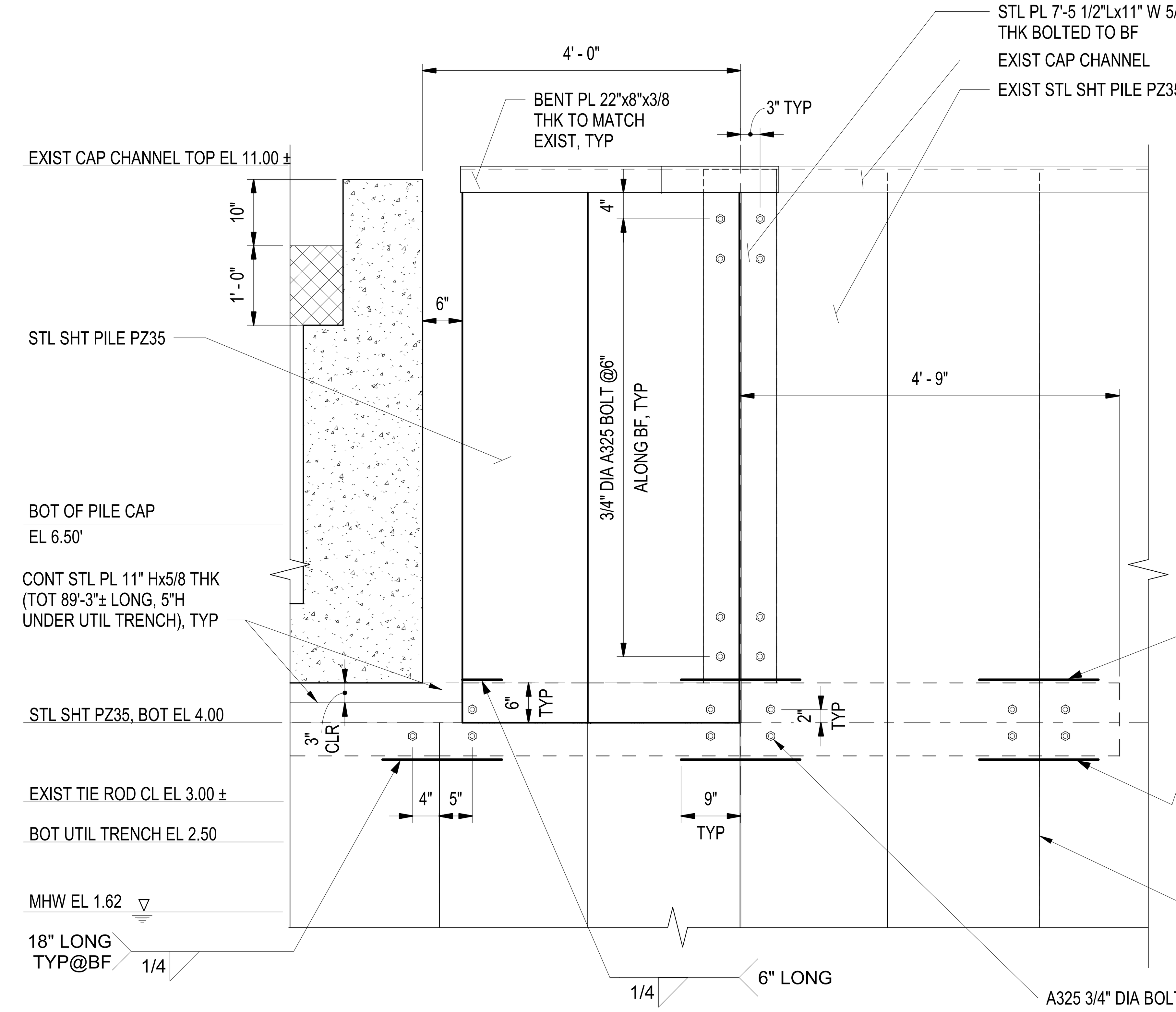


B1 SHEETING REPAIR ELEVATION AT NORTH END OF PIER

SCALE: 3/4" = 1'-0" (S-207)

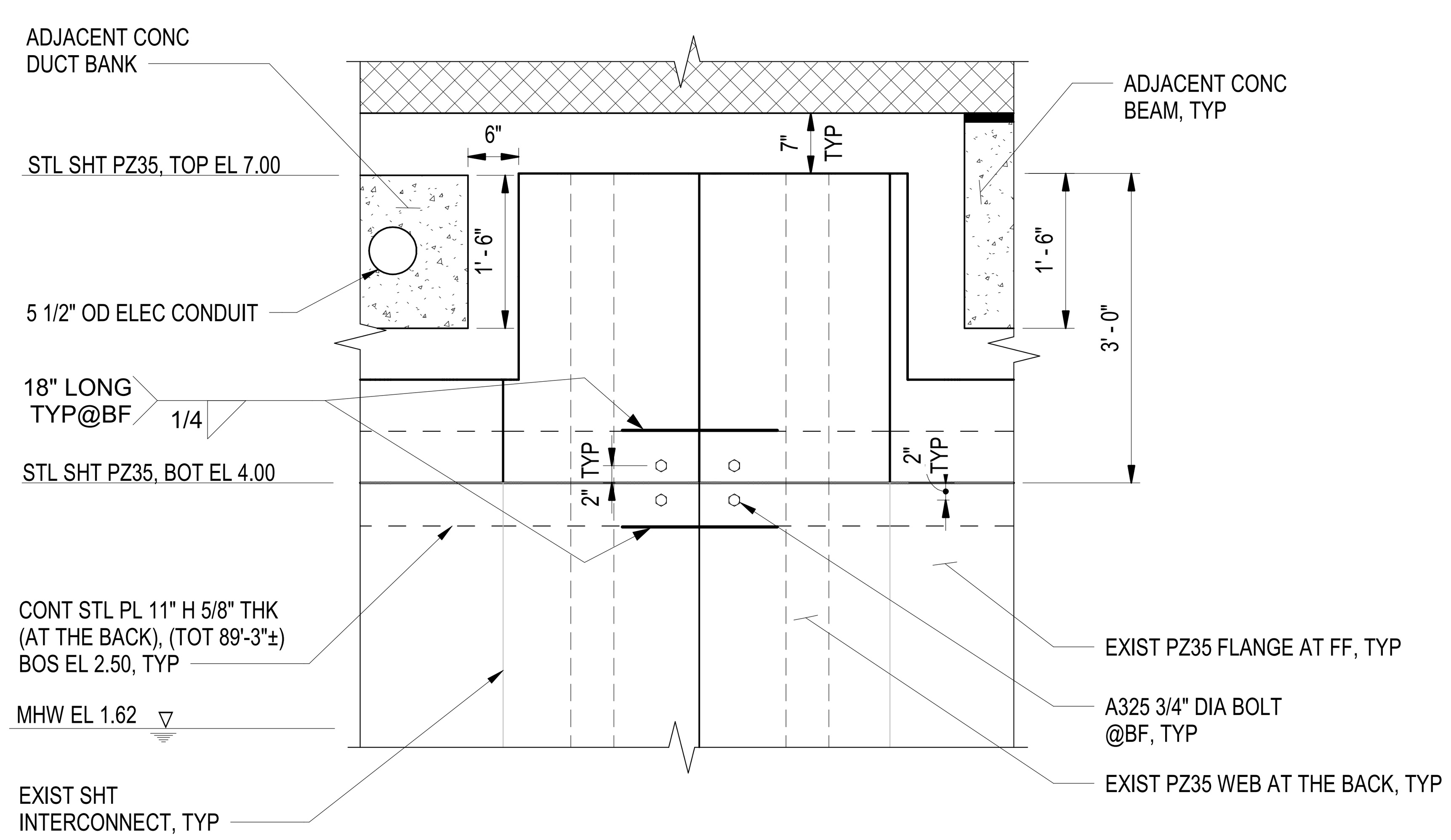


B3 SHEETING REPAIR ELEVATION AT SOUTH END OF PIER

SCALE: 3/4" = 1'-0" (S-207)

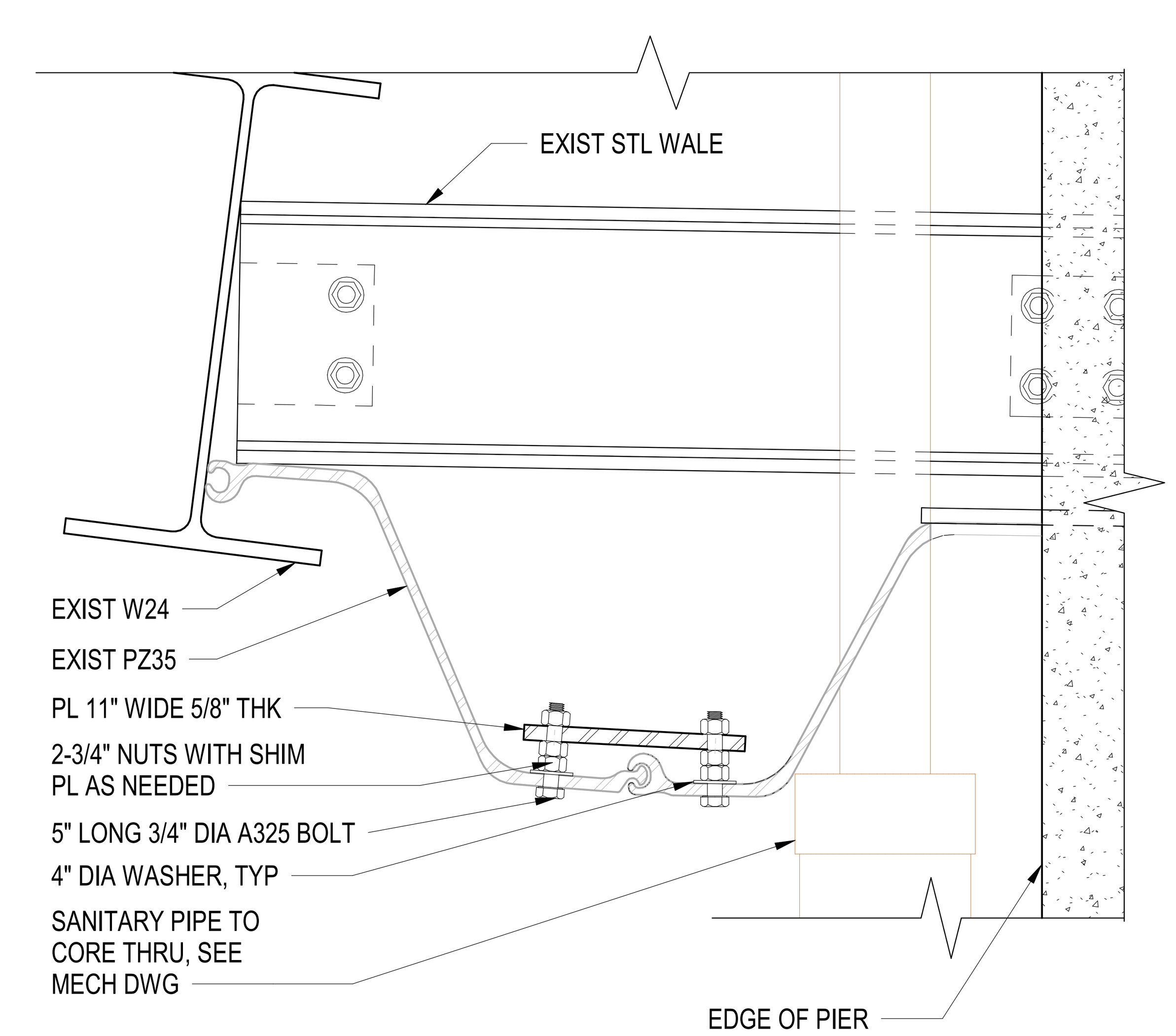
NOTES:

1. FOR LIMITS REQUIRED FOR SHEETING REPAIR, SEE DRAWING S-207.
2. SEE ATTACHMENT A IN THE SPECIFICATIONS FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.



A1 TYPICAL SHEETING REPAIR ELEVATION AT PIER CROSSING

SCALE: 3/4" = 1'-0" (S-207)



A3 S45N REPAIRED BULKHEAD AT NORTH END OF PIER

SCALE: 1 1/2" = 1'-0"

1 1/2" = 1' - 0" 0 3" 6" 9" 1' 2'

3/4" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW AFC CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
S45N BULKHEAD MODIFICATION DETAILS - SHEET 2 OF 2

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936722

SHEET 153 OF 247

S-525

DRAWFORM REVISION: 25 AUGUST 2020

S-525
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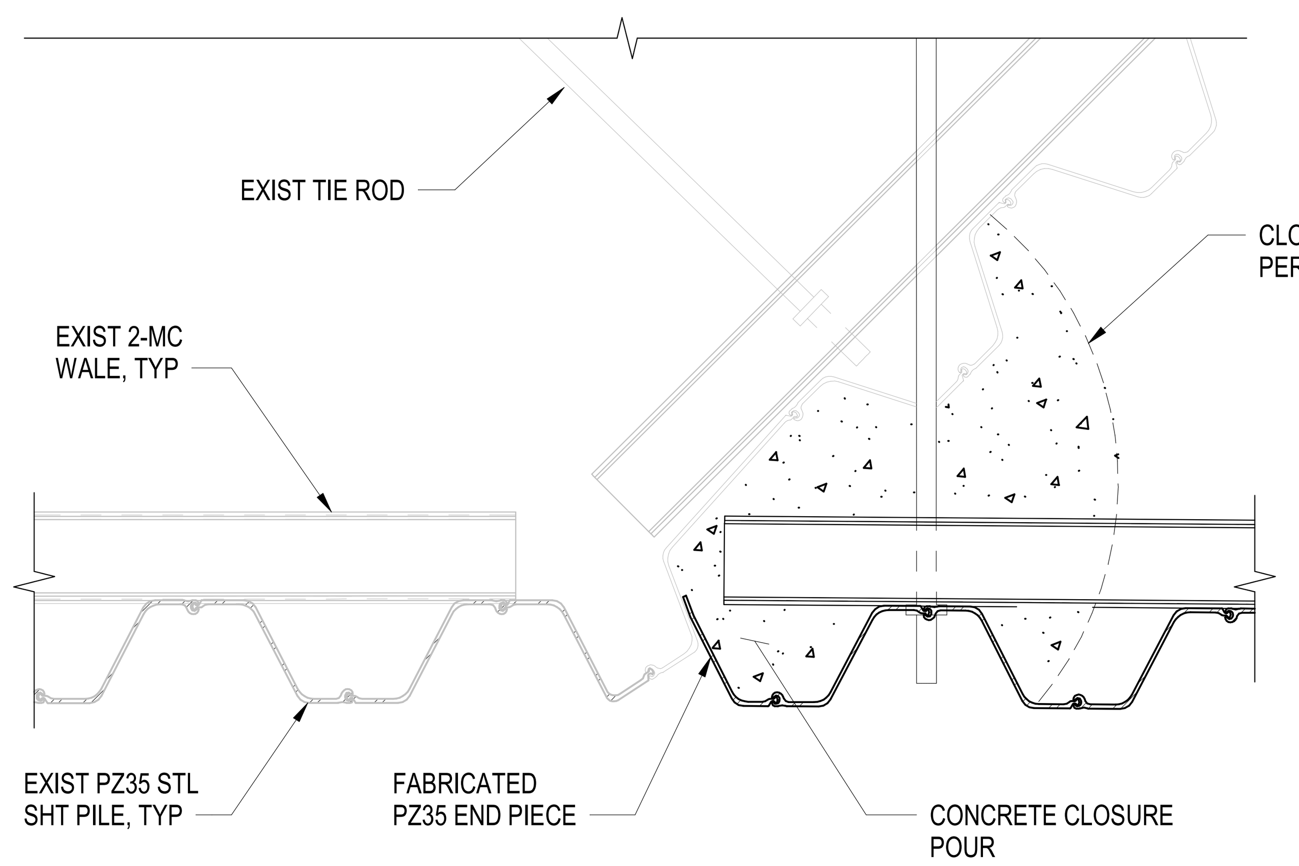
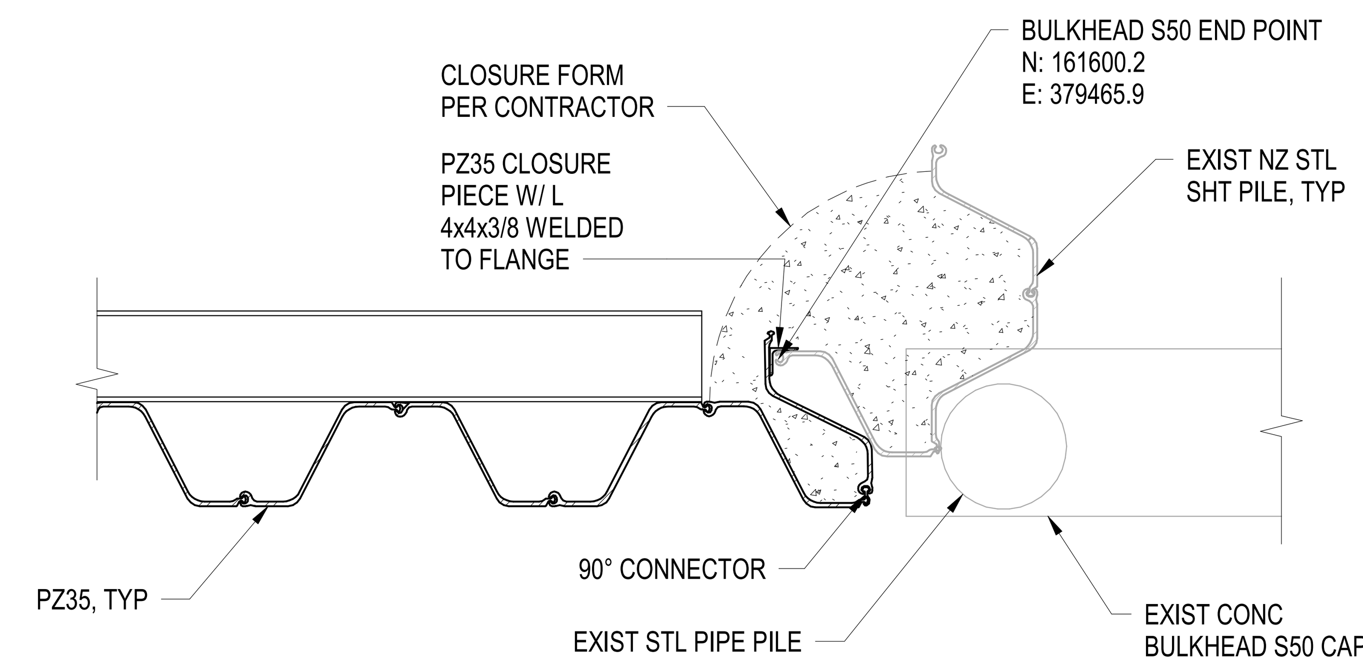


C1 BULKHEAD S45S TO S45N CONNECTION DETAIL

SCALE: 1/2" = 1'-0" (S-404)



C3 BULKHEAD S45S TO S50 CONNECTION DETAIL

SCALE: 1/2" = 1'-0" (S-404)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS
VICTOR M. GOTONI
CIVIL
No. 41734
REGISTERED
PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES MJF DRW AFC CHK JZ

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA

NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER

S45S BULKHEAD DETAILS - SHEET 1 OF 3 (BASE BID)

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936723

SHEET 154 OF 247

S-526

DRAWFORM REVISION: 25 AUGUST 2020

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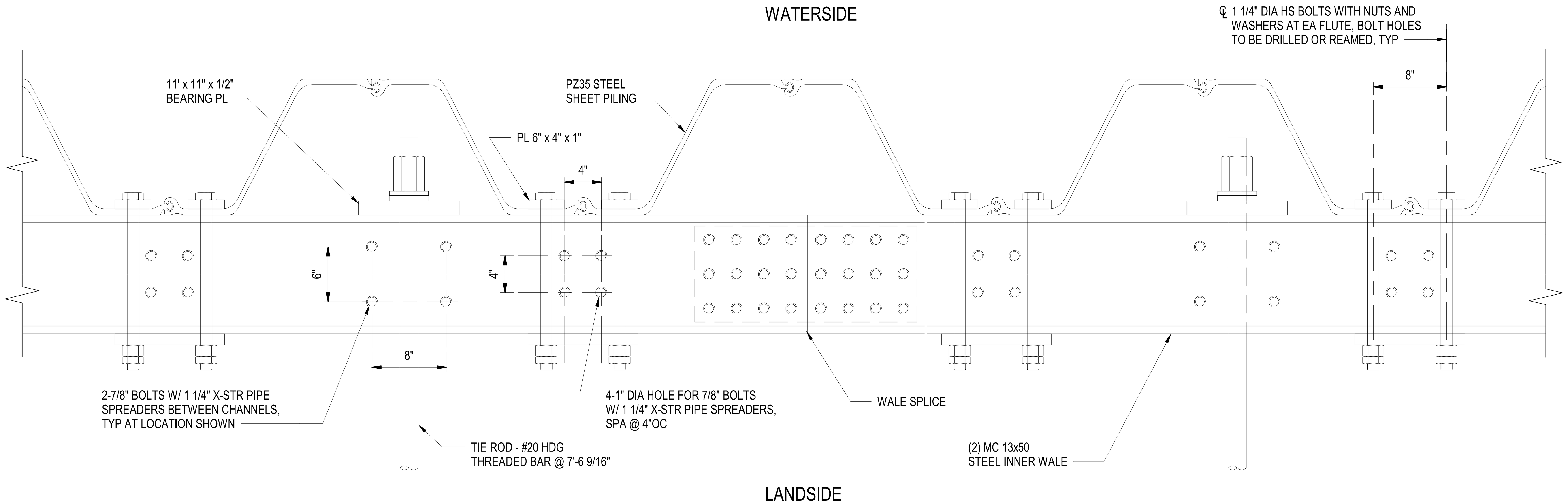
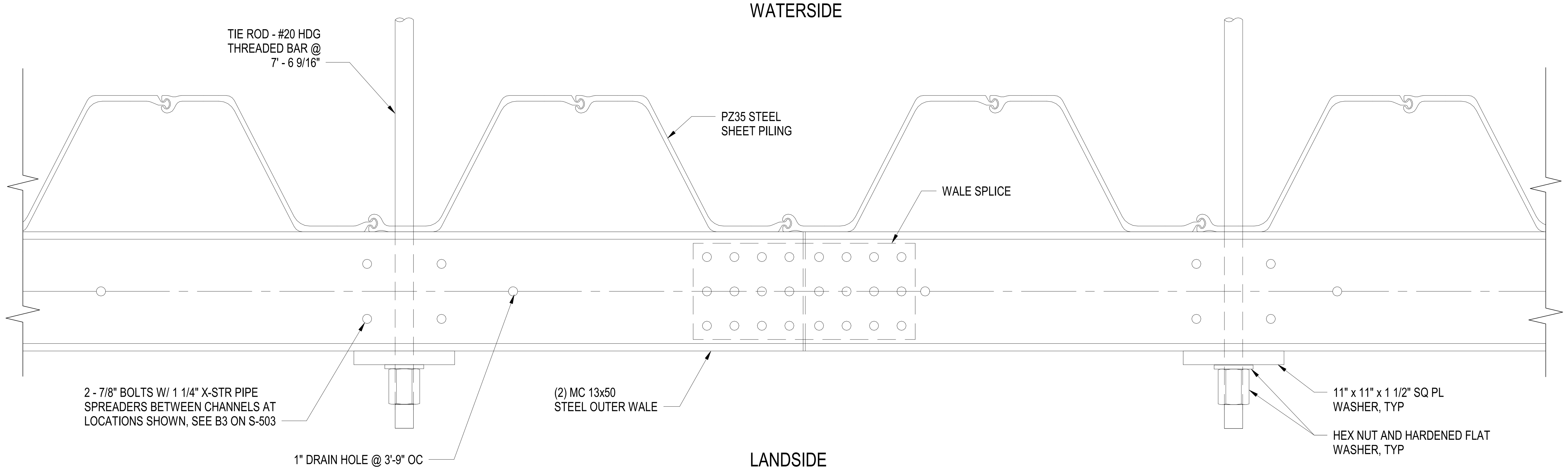
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt



C2 DETAIL - TYPICAL BULKHEAD WALE AND TIE ROD LAYOUT

SCALE: 1 1/2" = 1'-0"



A2 DETAIL - TYPICAL DEADMAN WALE AND TIE ROD LAYOUT

SCALE: 1 1/2" = 1'-0"

1 1/2" = 1' - 0" 0 3" 6" 9" 1' 2'

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF MASSACHUSETTS VICTOR M. GOTONI CIVIL No. 41734 REGISTERED PROFESSIONAL ENGINEER

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES MJF DRW AFC CHK JZ

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
S45S BULKHEAD DETAILS - SHEET 2 OF 3 (BASE BID)

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936724

SHEET 155 OF 247

S-527

DRAWFORM REVISION: 25 AUGUST 2020

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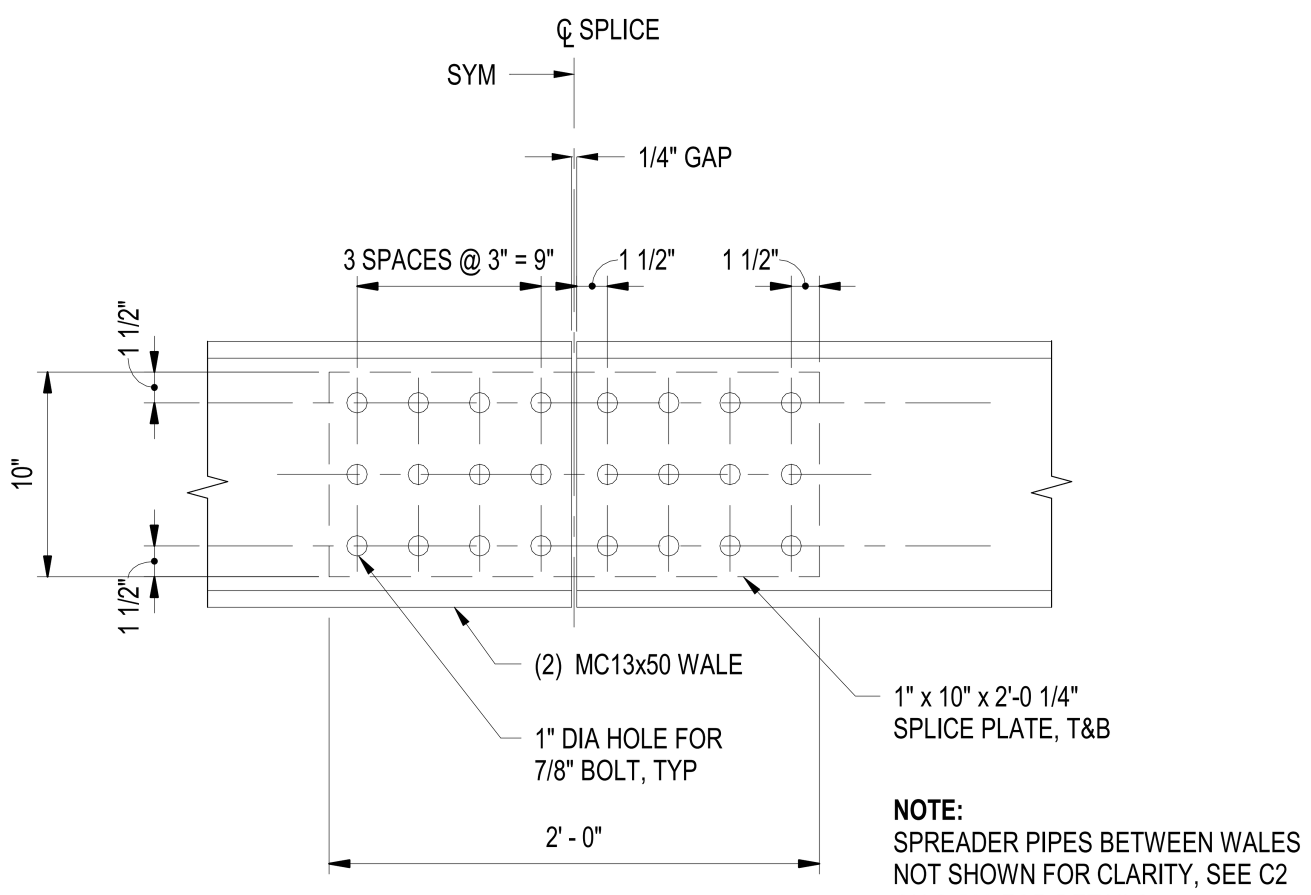
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

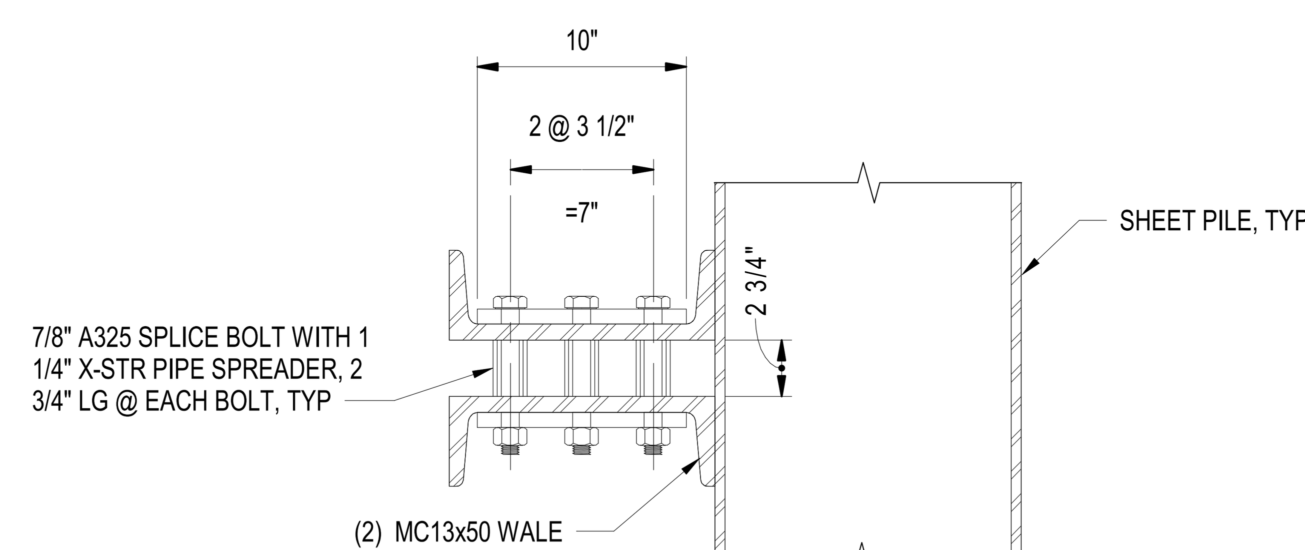
GENERAL SHEET NOTES:

1. COATING AT ALL INFIELD WELDS MUST BE TOUCHED UP AFTER WELDING.

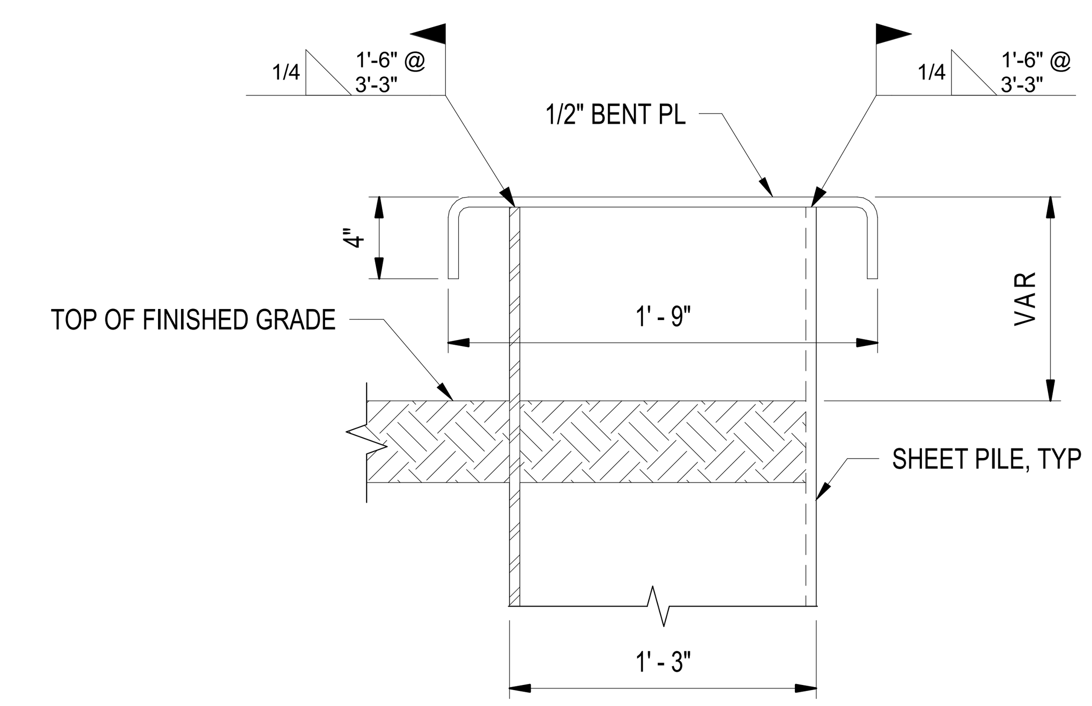


NOTE:
SPREADER PIPES BETWEEN WALES NOT SHOWN FOR CLARITY, SEE C2 ON S-503 FOR DETAILS.

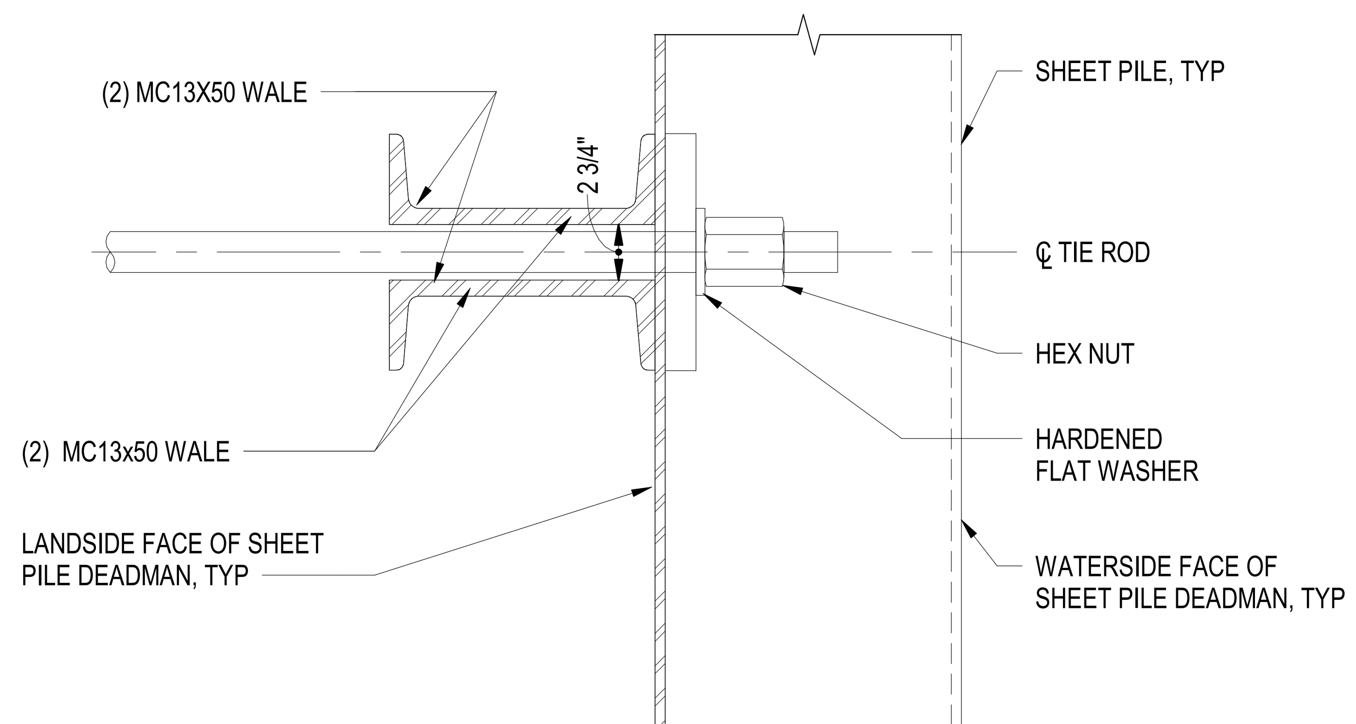
C1 DETAIL - WALE SPLICE
SCALE: 1 1/2" = 1'-0"



C2 SECTION - WALE SPLICE CONNECTION
SCALE: 1 1/2" = 1'-0"

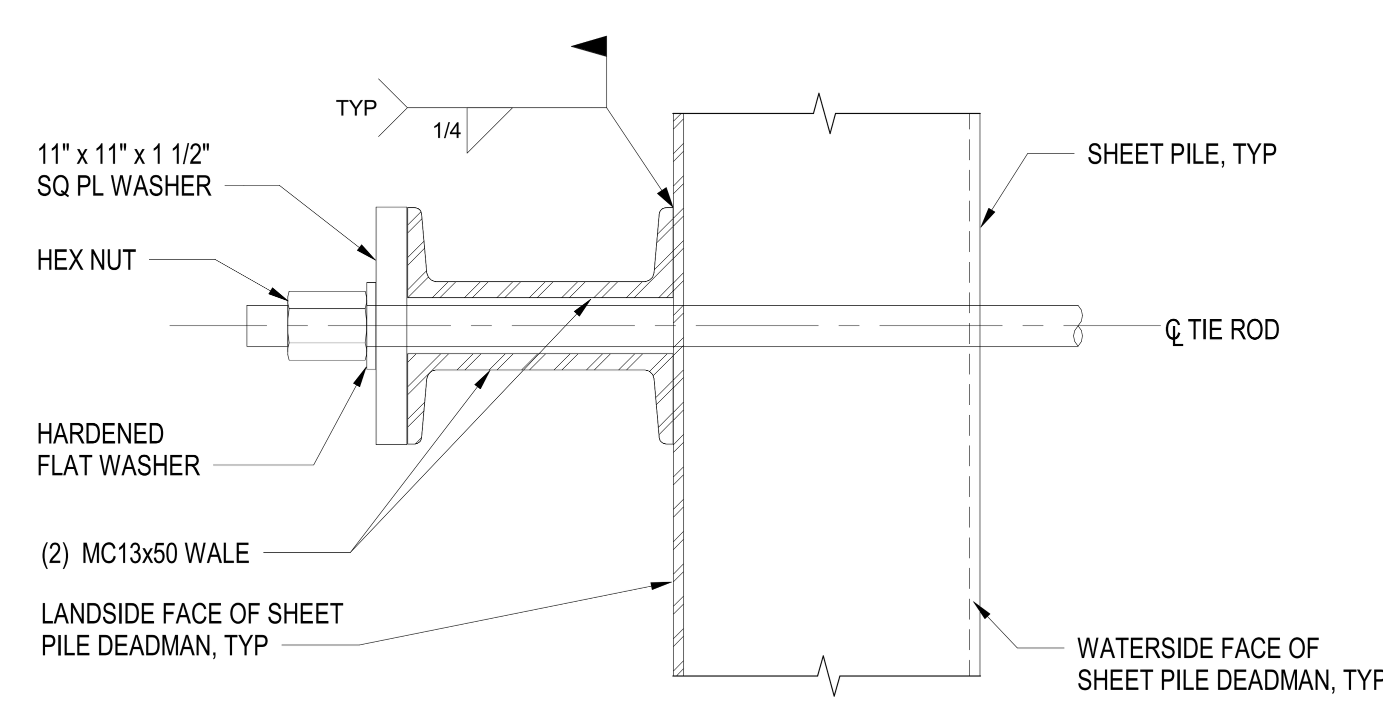


C4 DETAIL - BULKHEAD CAP
SCALE: 1 1/2" = 1'-0"



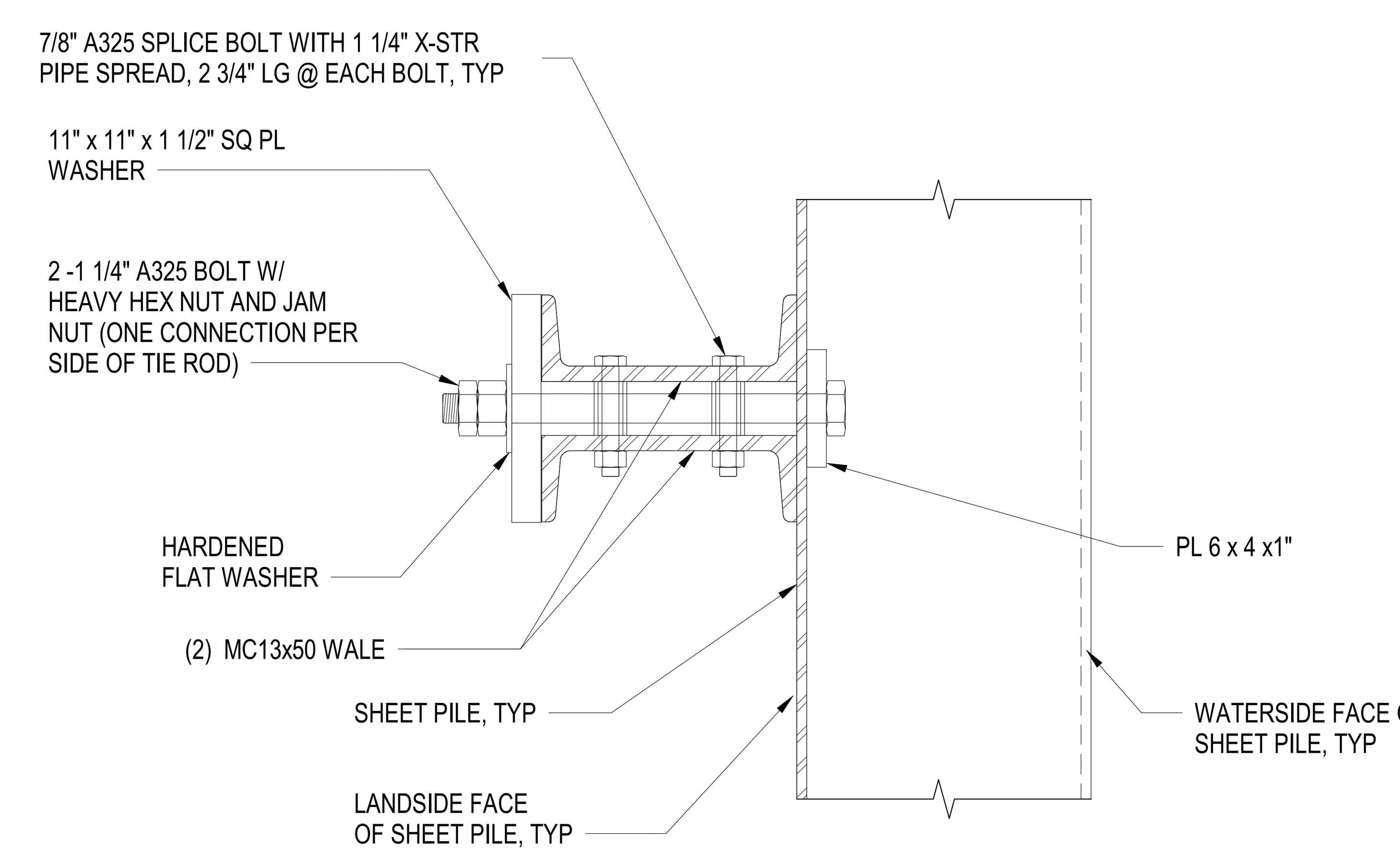
NOTE:
SPREADER PIPES BETWEEN THE (2) MC 13x50 ARE NOT SHOWN FOR CLARITY.

A1 SECTION - TYPICAL BULKHEAD WALE CONNECTION WITH TIE ROD
SCALE: 1 1/2" = 1'-0"



NOTE:
SPREADER PIPES BETWEEN THE (2) MC 13x50 ARE NOT SHOWN FOR CLARITY

A2 SECTION - TYPICAL BULKHEAD WALE CONNECTION AT TIE ROD
SCALE: 1 1/2" = 1'-0"



A4 SECTION - TYPICAL BULKHEAD WALE CONNECTION AT OTHER LOCATIONS
SCALE: 1 1/2" = 1'-0"

1 1/2" = 1' - 0" 0 3" 6" 9" 1' 2'

SYM | DESCRIPTION | DATE | APPR

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COMMONWEALTH OF MASSACHUSETTS VICTOR M. GOTONI CIVIL No. 41734 REGISTERED PROFESSIONAL ENGINEER
SEAL

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FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES MJF | DRW AFC | CHK JZ
PM/DM MRI/CWJ
BRANCH MANAGER NEK
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
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NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
S45S BULKHEAD DETAILS - SHEET 3 OF 3 (BASE BID)

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936725
SHEET 156 OF 247
S-528
DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:37:14 PM S-528
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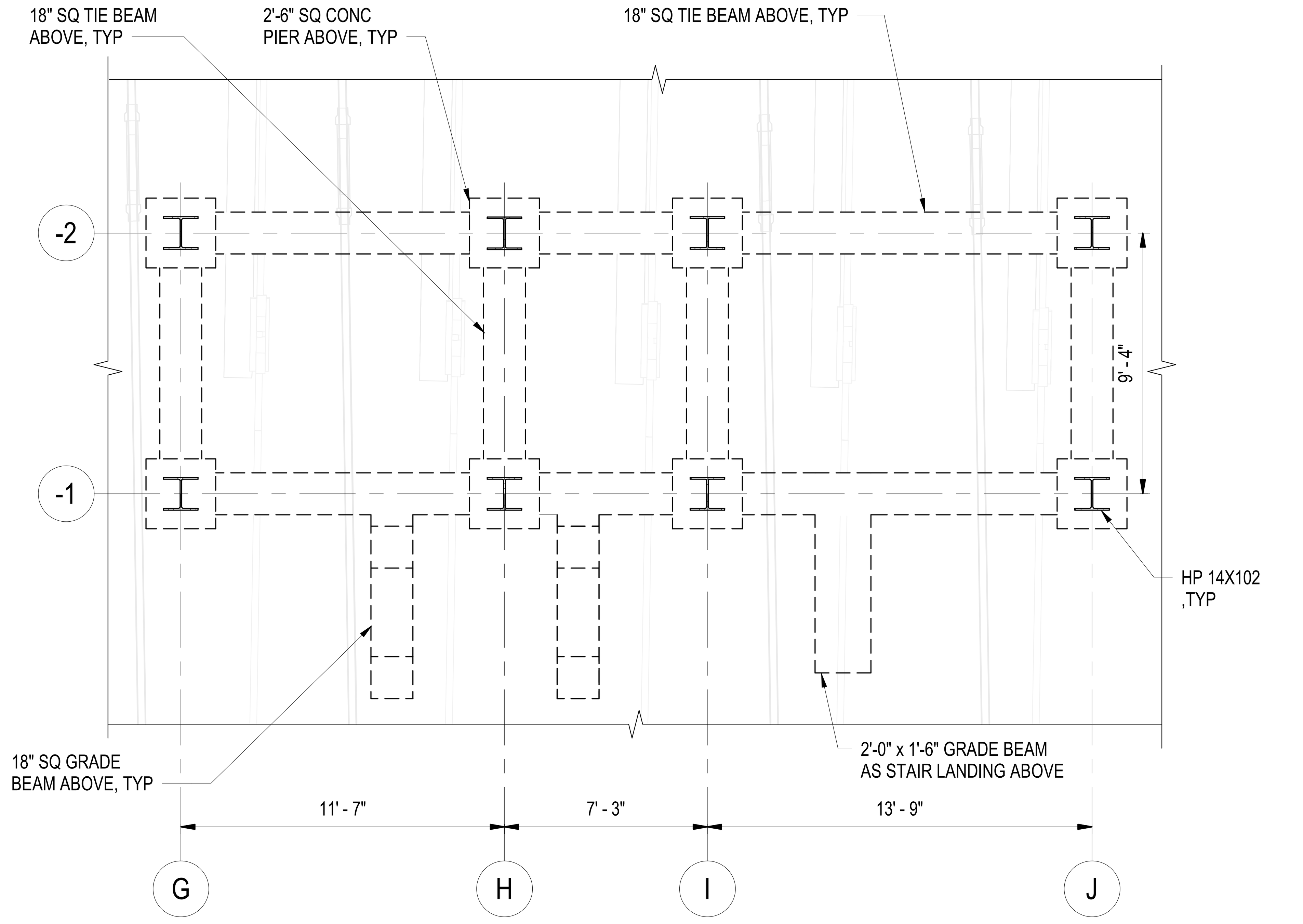
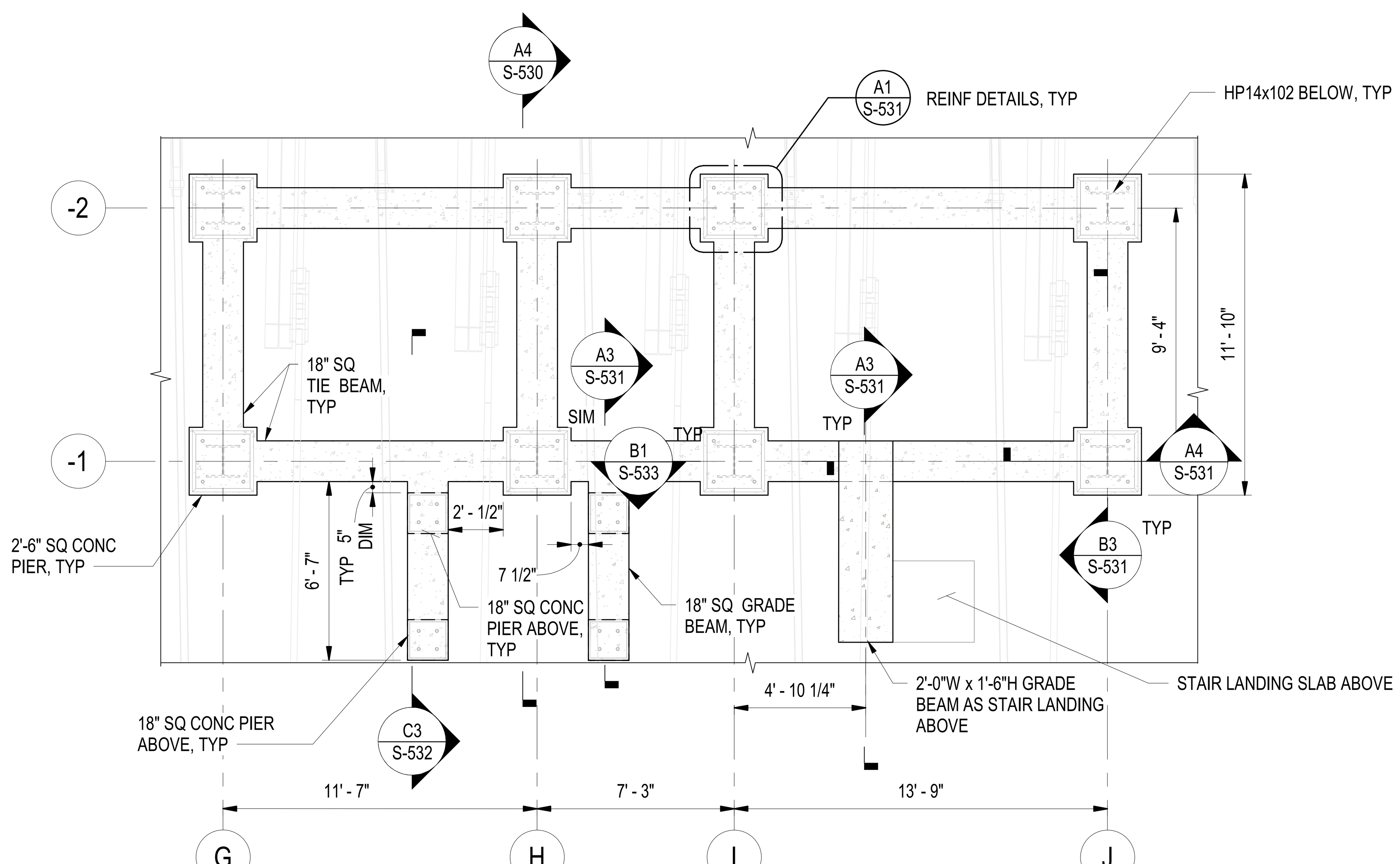


C1 PILE PLAN FOR AIR COMPRESSOR ENCLOSURE

SCALE: 1/4" = 1'-0"



A1 FOUNDATION PLAN FOR COMPRESSOR ENCLOSURE

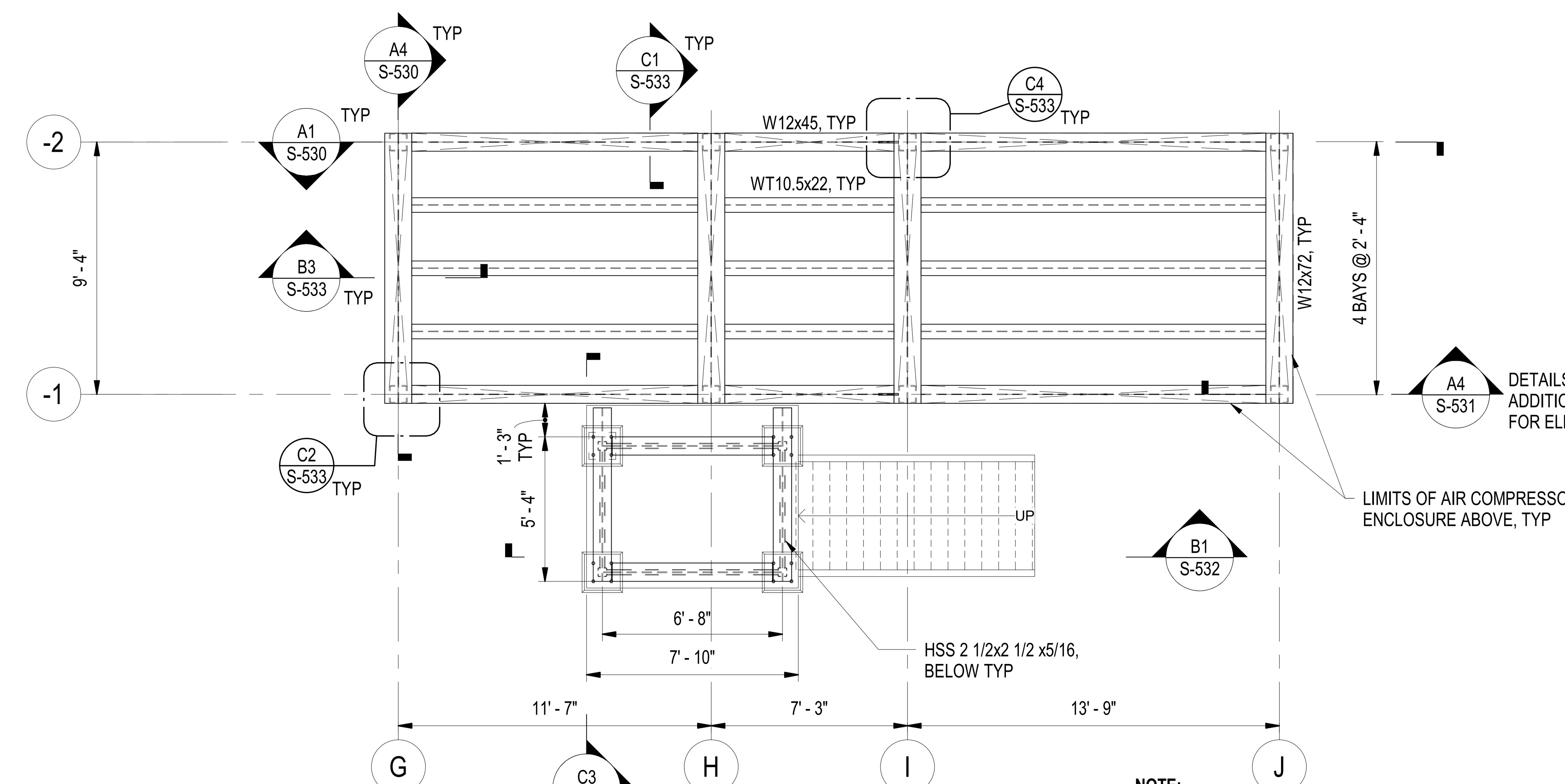
SCALE: 1/4" = 1'-0" (S-116, S-129, S-401)

GENERAL SHEET NOTES:

1. FOR DIMENSIONS WITH ±, FIELD VERIFY AND COORDINATE WITH MECHANICAL EQUIPMENT SPECIFICATIONS.
2. SPACING AND SIZE OF FRAMING ELEMENTS ARE BASED ON MANUFACTURER ASSEMBLY DRAWINGS. CONTRACTOR TO VERIFY ALL DIMENSION AND MEMBERS WITH PRODUCT TO BE INSTALLED PRIOR TO FABRICATION OR INSTALLING ANY FOUNDATION ELEMENTS.
3. LOCATE STAIRS TO ACCESS AIR COMPRESSOR ENCLOSURE, STAIRS TO ALIGN WITH AIR COMPRESSOR DOORWAY. SEE S-534 THROUGH S-537 FOR DETAILS.
4. ROUTING OF UTILITIES AND DUCT OPENING IN BASE TO BE COORDINATED PRIOR TO FABRICATING ANY ELEMENTS.
5. CROSSING BRACING TO BE PROVIDED IN LONGITUDINAL AND TRANSVERSE DIRECTIONS BETWEEN ALL HSS COLUMNS. SEE S-534 THROUGH S-537 FOR BRACING DETAILS.
6. VERIFY LOCATIONS OF EXISTING TIE RODS, STEEL BATTER PILES AND CONCRETE DEADMAN IN THE FIELD TO PROTECT EXISTING TIE RODS AND AVOID PILE OBSTRUCTION.
7. SEE ATTACHMENT A FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.

LEGEND:

CROSS BRACING



NOTE:

FRAMING BASED ON KAESER AC ENCLOSURE USED AS THE BASIS OF DESIGN (SEE MECHANICAL DWGS). ADJUST FRAMING AS REQUIRED TO SUIT UNIT SELECTED. DESIGN UNIT WEIGHT = 34,000 LBS

A4 STEEL FRAMING PLAN

SCALE: 1/4" = 1'-0"

S-540

1/4" = 1' - 0" 0 1' 5' 10' 15'

SYM	DESCRIPTION	DATE	APPR

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REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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SATISFACTORY TO DATE 06 FEB 2026

DES	JZ	DRW	AFC	CHK	MJF
PM/DM					MRI/CWJ
BRANCH MANAGER					NEK
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
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CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
AIR COMPRESSOR FOUNDATION & STEEL FRAME PART PLAN

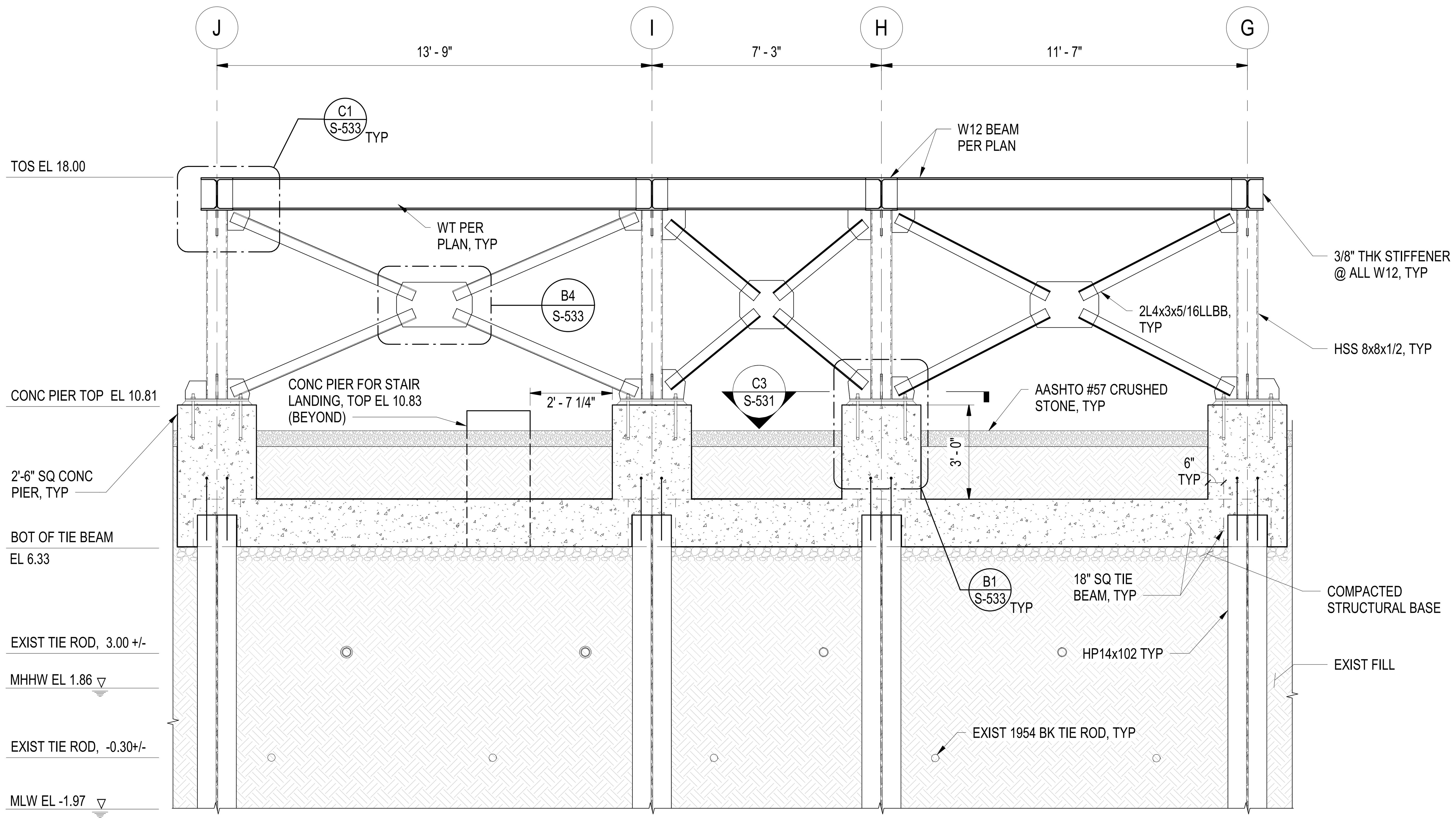
SCALE:	AS NOTED
EPROJECT NO.	1826479
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12936730
SHEET 157	OF 247

S-529

DRAWFORM REVISION: 25 AUGUST 2020

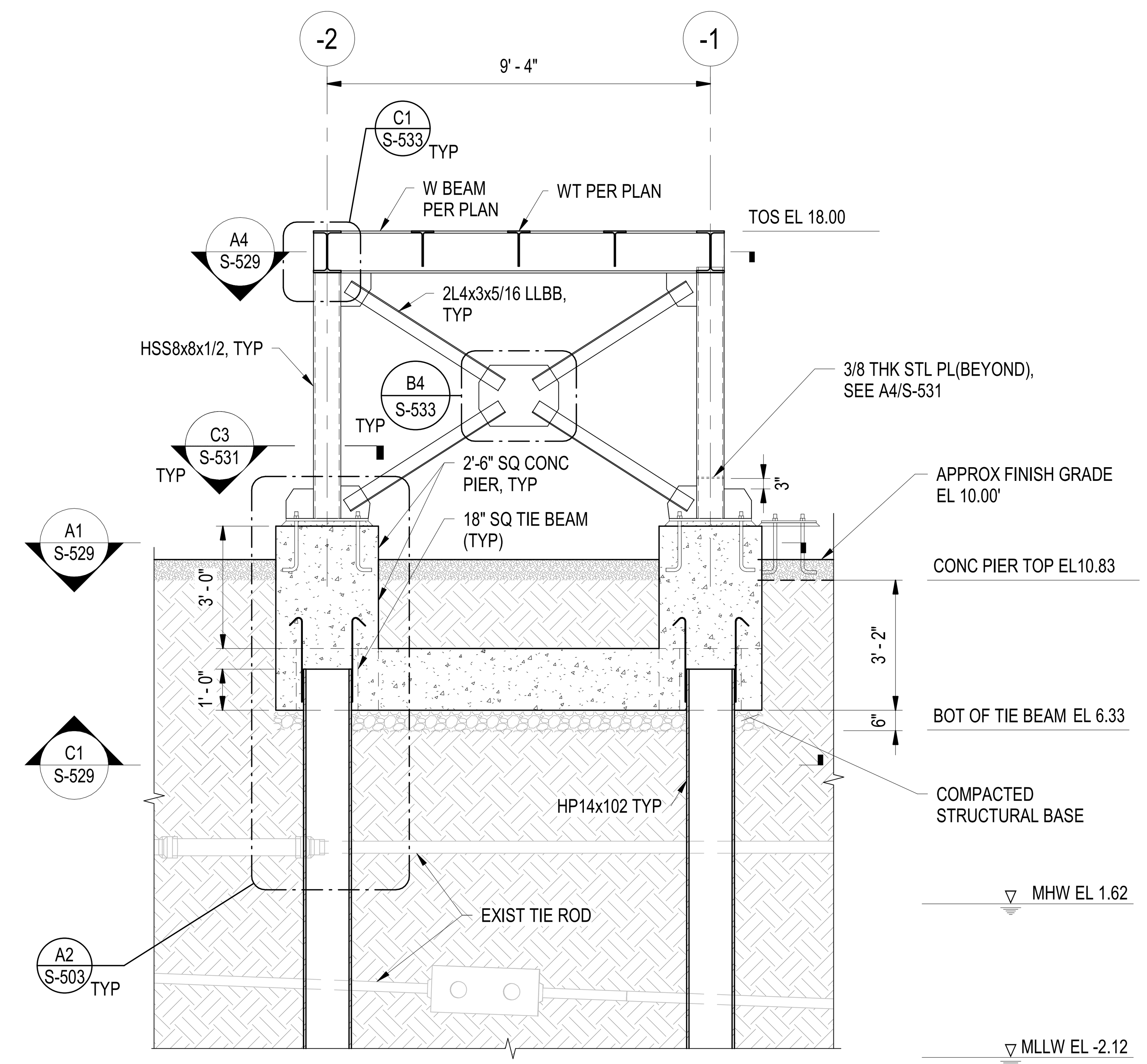
GENERAL SHEET NOTES:

1. SUBSURFACE PROFILES SHOWN ARE APPROXIMATE BASED ON LIMITED GEOTECHNICAL DATA.
2. SEE ATTACHMENT A IN THE SPECIFICATIONS FOR ADDITIONAL INFORMATION FOR EXISTING CONDITIONS.
3. BASE STEEL FRAMING FOR AC ENCLOSURE NOT SHOWN FOR CLARITY.
4. SEE NOTE 5 ON S-109 FOR PILE CONSTRUCTION WITHIN EXISTING RIPRAP.
5. SEE DRAINAGE DRAWINGS S-137 AND S-138 FOR LONGITUDINAL SLOPE, TYPICAL.



A1 **LONGITUDINAL SECTION AT AC ENCLOSURE**

SCALE: 3/8" = 1'-0" (S-529)



A4 **CONCRETE FOUNDATION ELEVATIONS AND REINFORCEMENT DETAILS**

SCALE: 3/8" = 1'-0" (S-529)

3/8" = 1' - 0" 0 1' 2' 3' 4' 5' 10'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

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BRANCH MANAGER					NEK
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

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NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

AIR COMPRESSOR DETAILS

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO.
SHEET 158 OF 247

S-530

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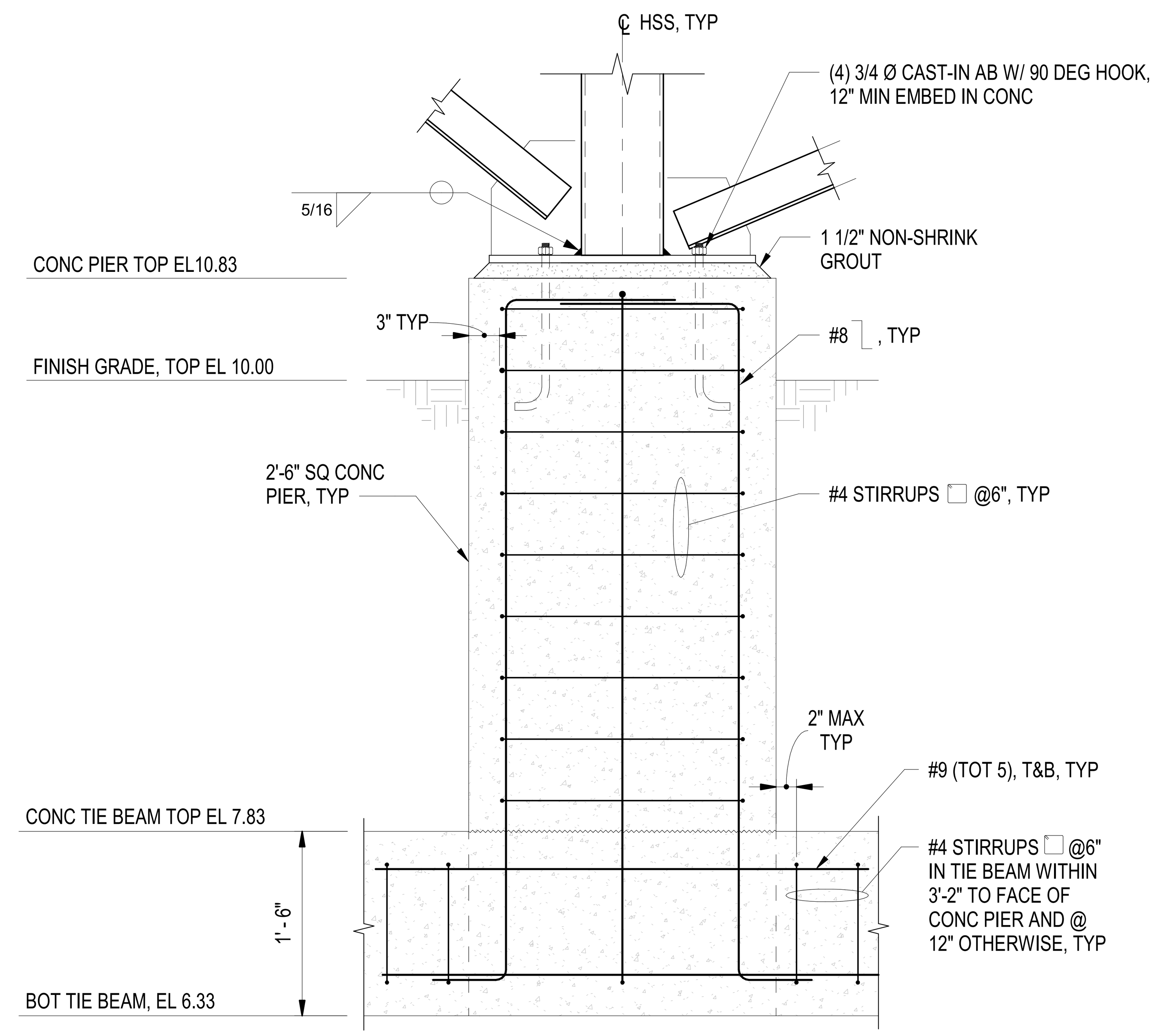
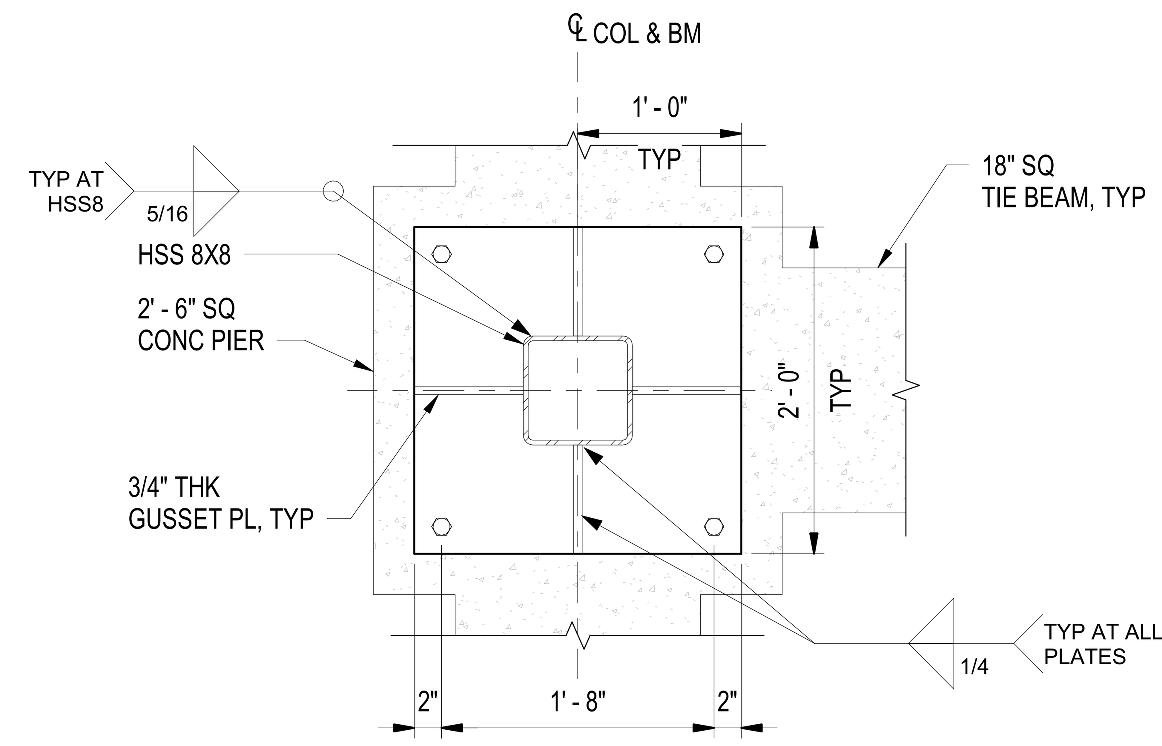


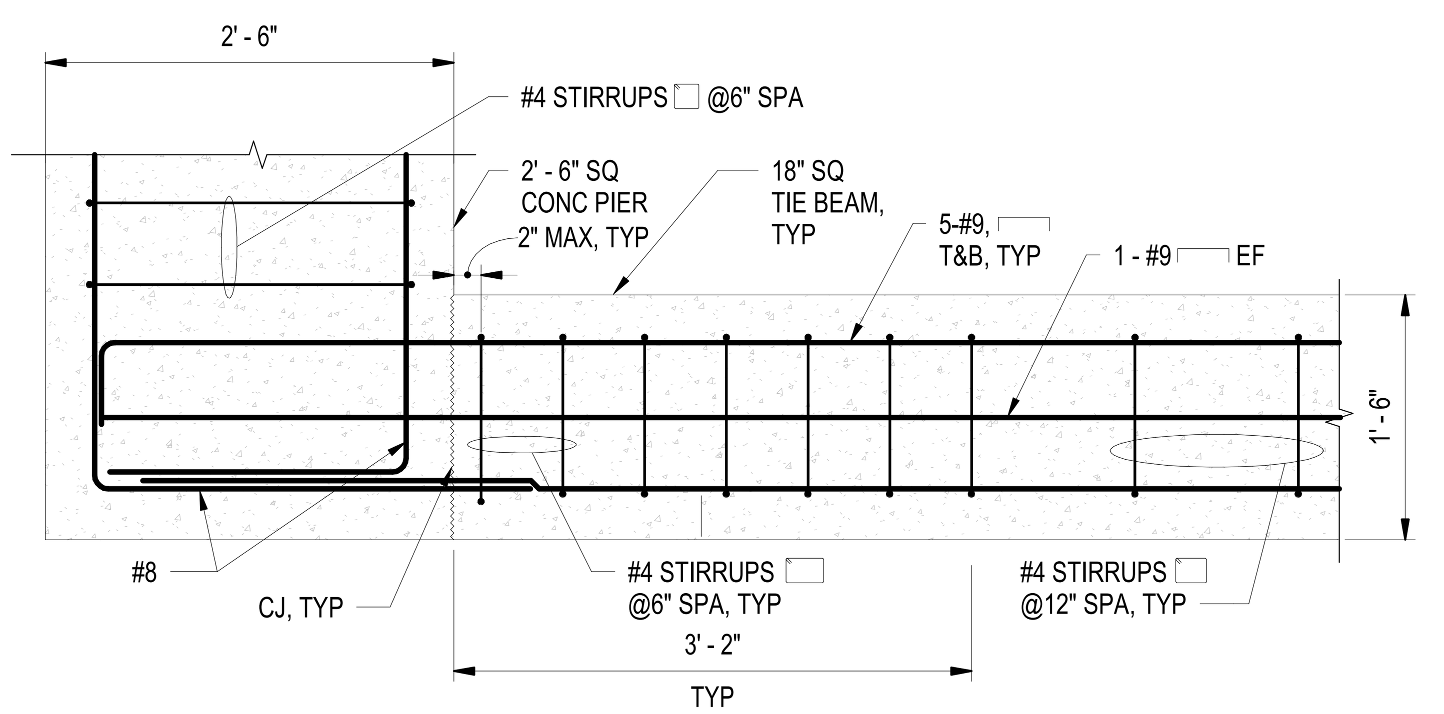
B1 TYPICAL DETAILS FOR CONCRETE PIER CONNECTION

SCALE: 1" = 1'-0" (S-531)



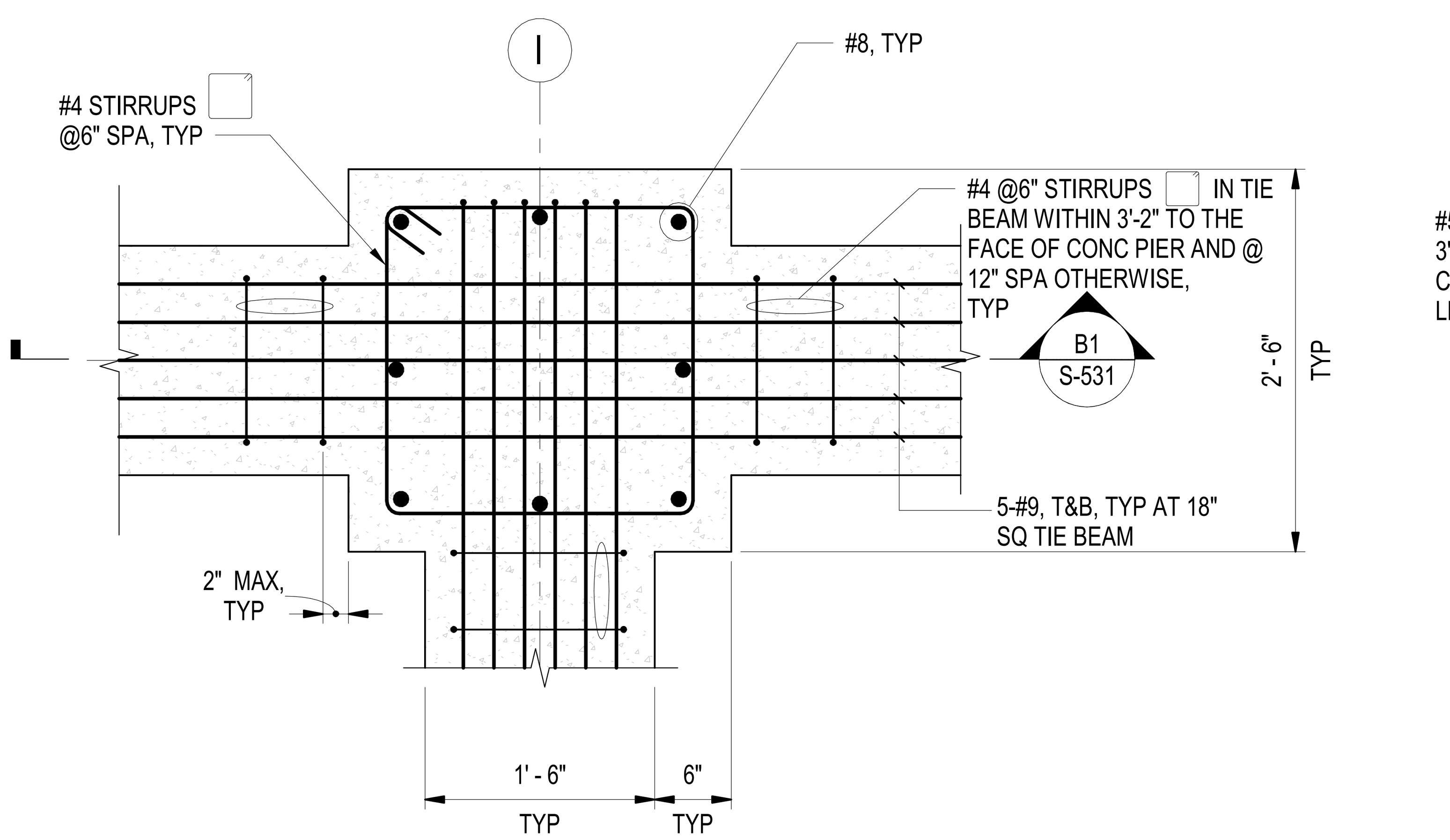
C3 TYPICAL DETAILS FOR COLUMN BASE

SCALE: 1" = 1'-0" (S-530)



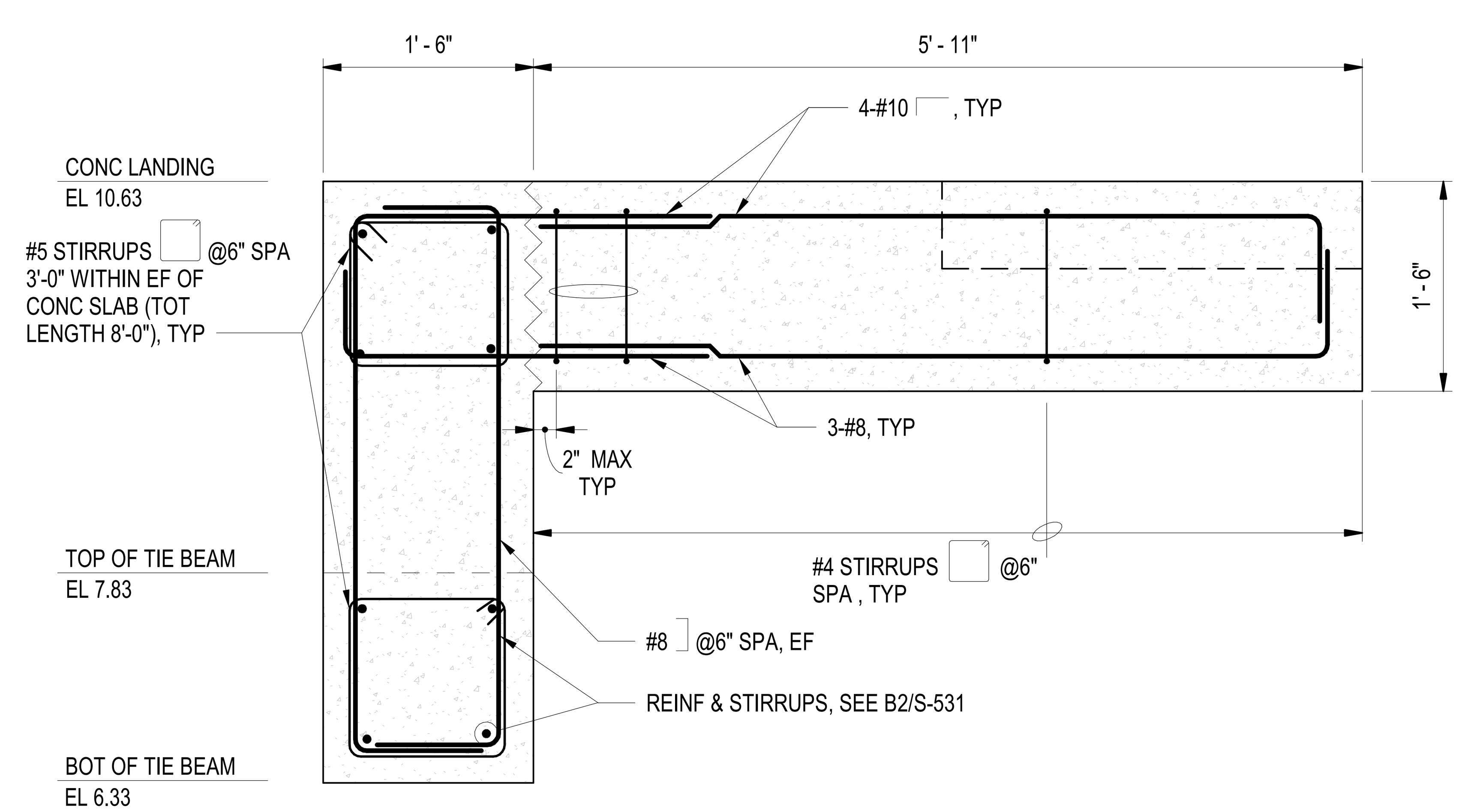
B3 TYPICAL REBAR DETAILS FOR TIE BEAM TO CONCRETE PIER

SCALE: 1" = 1'-0" (S-529)



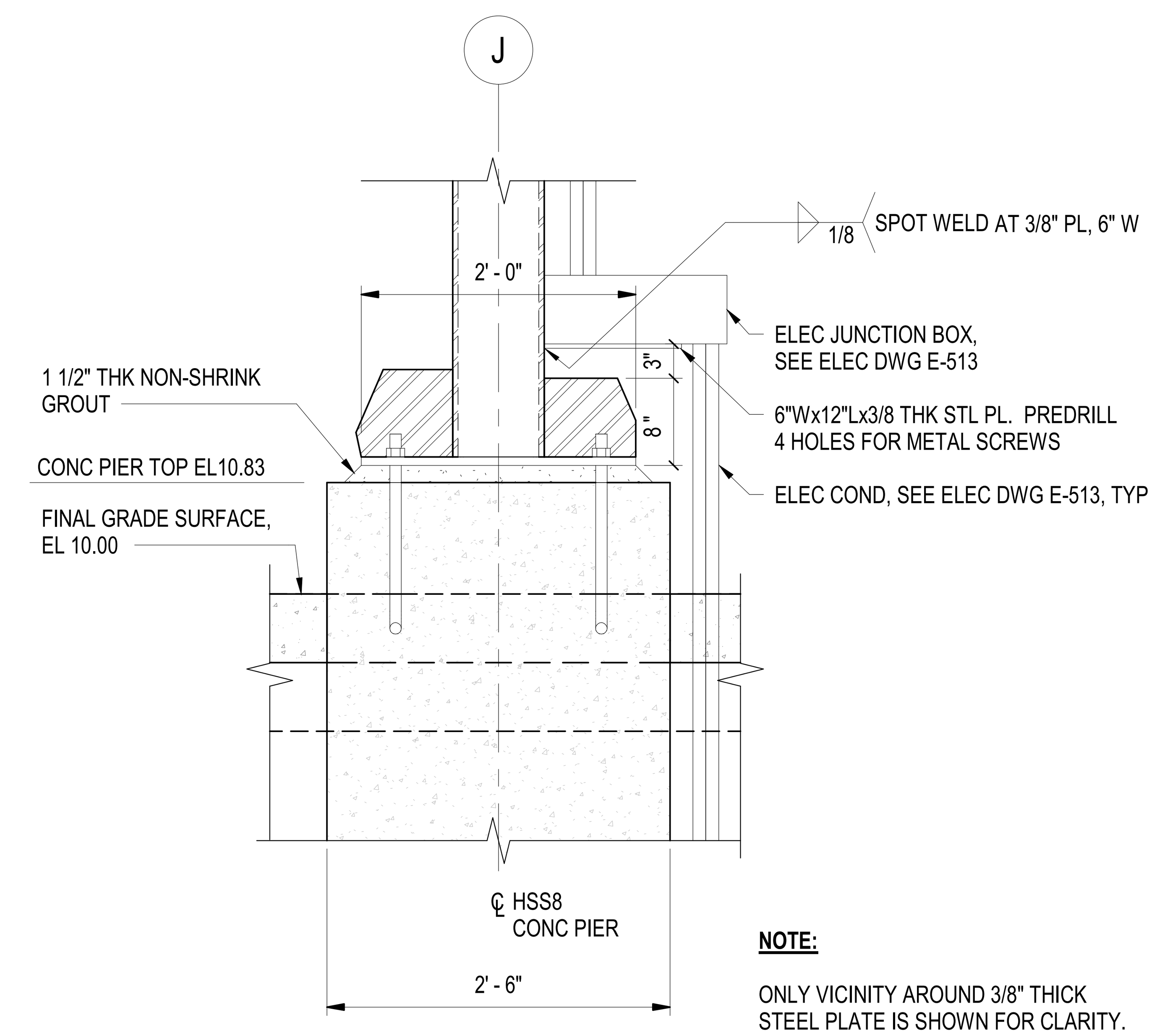
A1 TYPICAL REINFORCEMENT DETAILS AT BEAM & CONCRETE PIER INTERSECTION

SCALE: 1" = 1'-0" (S-529)



A3 REINFORCEMENT DETAILS FOR GRADE BEAM

SCALE: 1" = 1'-0" (S-529)



NOTE:

ONLY VICINITY AROUND 3/8" THICK STEEL PLATE IS SHOWN FOR CLARITY.

A4 ADDITIONAL SUPPORT FOR ELECTRICAL ROUTING AT ONE HSS COLUMN

SCALE: 1" = 1'-0" (S-529)

1" = 1' - 0" 0 6" 1' 2' 3'

SYM	DESCRIPTION	DATE	APPR

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JINQUAN ZHONG C 82242 CIVIL REGISTERED PROFESSIONAL ENGINEER STATE OF CALIFORNIA

SEAL

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FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
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NAVAL STATION NEWPORT NEWPORT, RI

USCG OCR HOMEPORT INITIATIVE OPC PIER

AIR COMPRESSOR FOUNDATION DETAILS

SCALE: AS NOTED

EPROJECT NO.: 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936728

SHEET 159 OF 247

S-531

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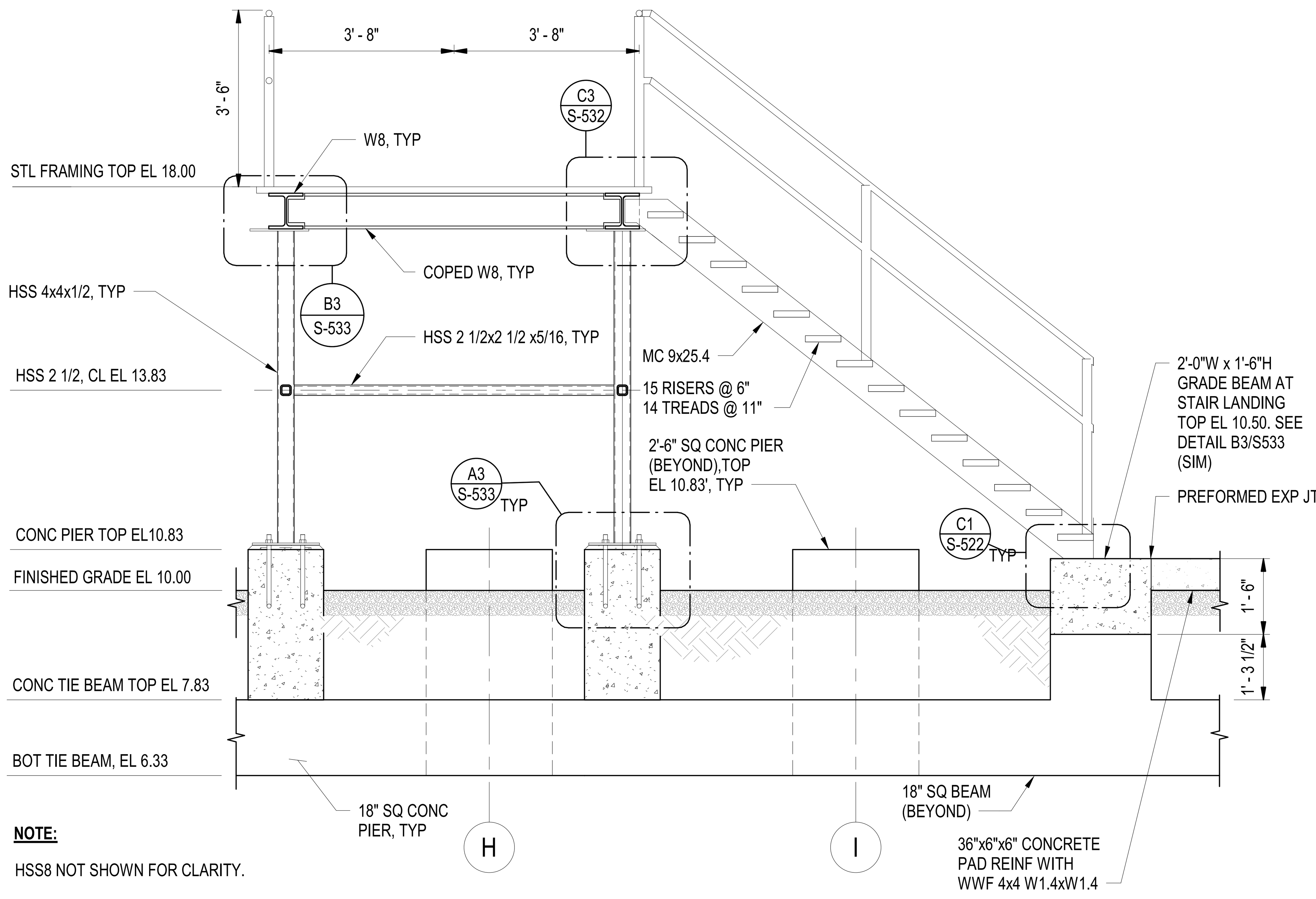
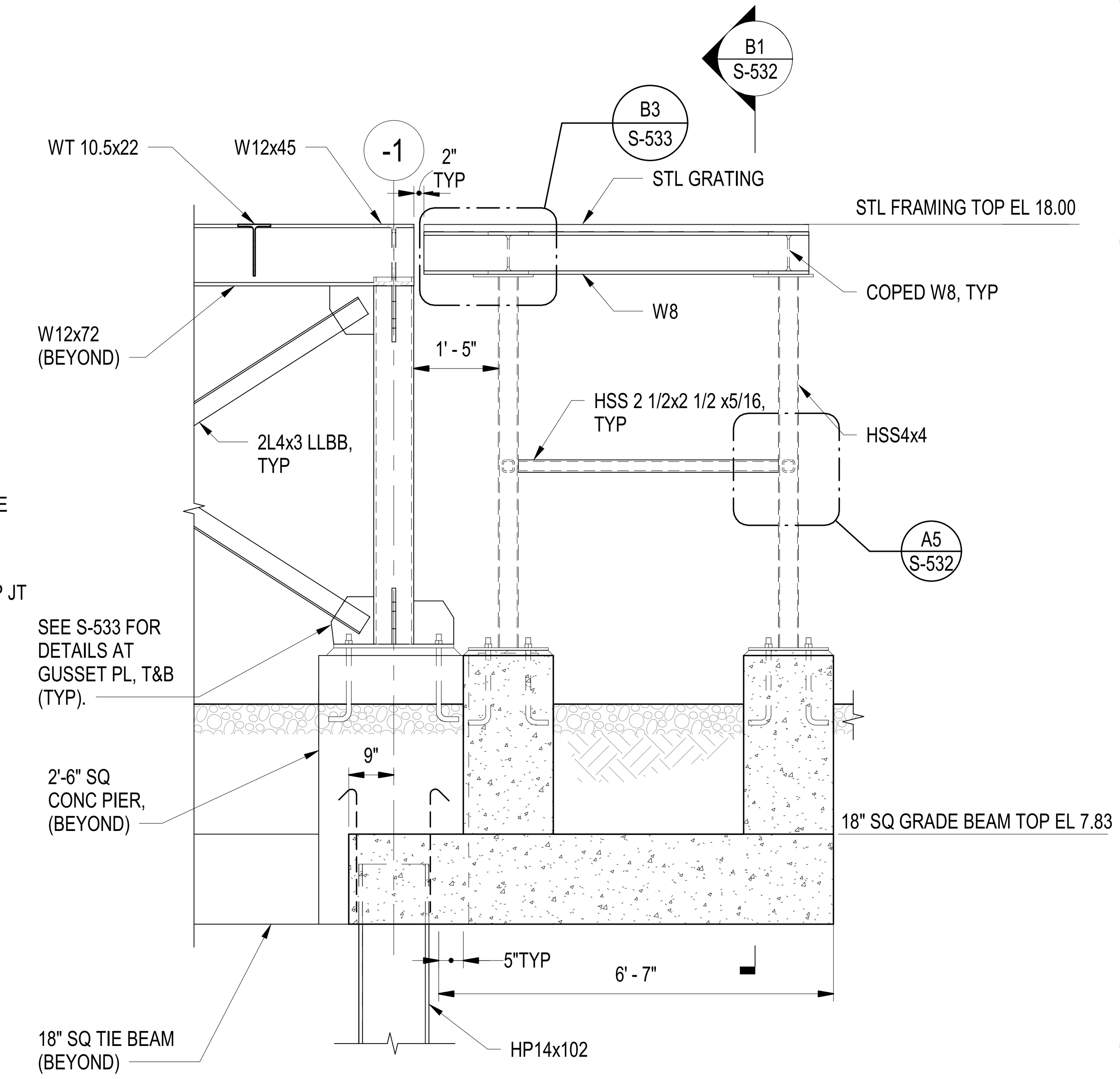


B1 **ELEVATION - STAIR PLATFORM FOR AIR COMPRESSOR ENCLOSURE**
SCALE: 1/2" = 1'-0" (S-529)



C3 **STAIR CONCRETE FOUNDATION SECTION**
SCALE: 1/2" = 1'-0" (S-529)

DESIGN CRITERIA:

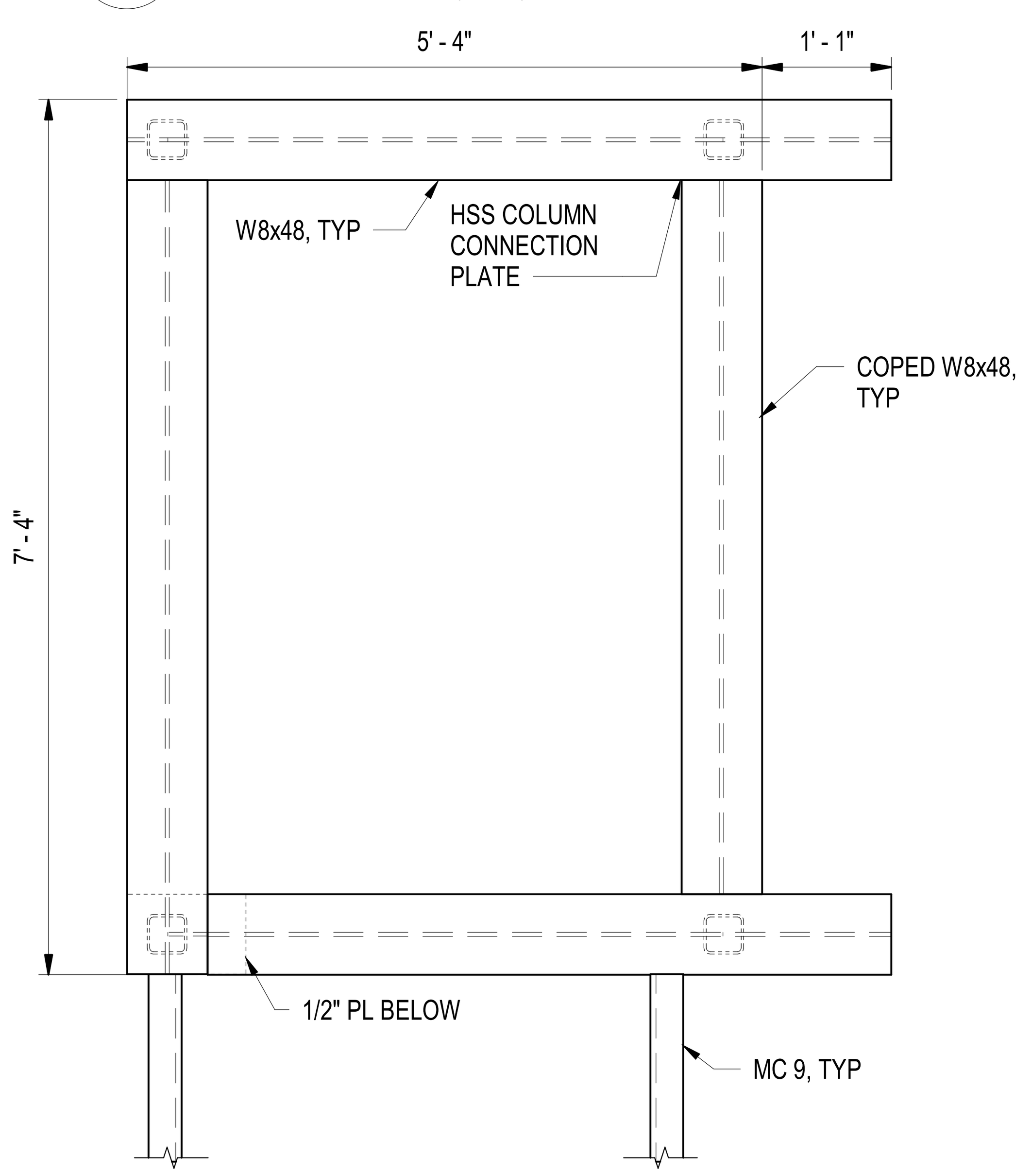
1. PLATFORM LIVE LOADING: 50 psf
2. STAIR TREADS: 250 PSF (5 TIMES ANTICIPATED LIVE LOAD) OR 1000 POUND CONCENTRATED LIVE LOAD.
3. HANDRAIL: 50PLF OR 200LB CONCENTRATED LIVE LOAD.

STRUCTURAL STEEL NOTES:

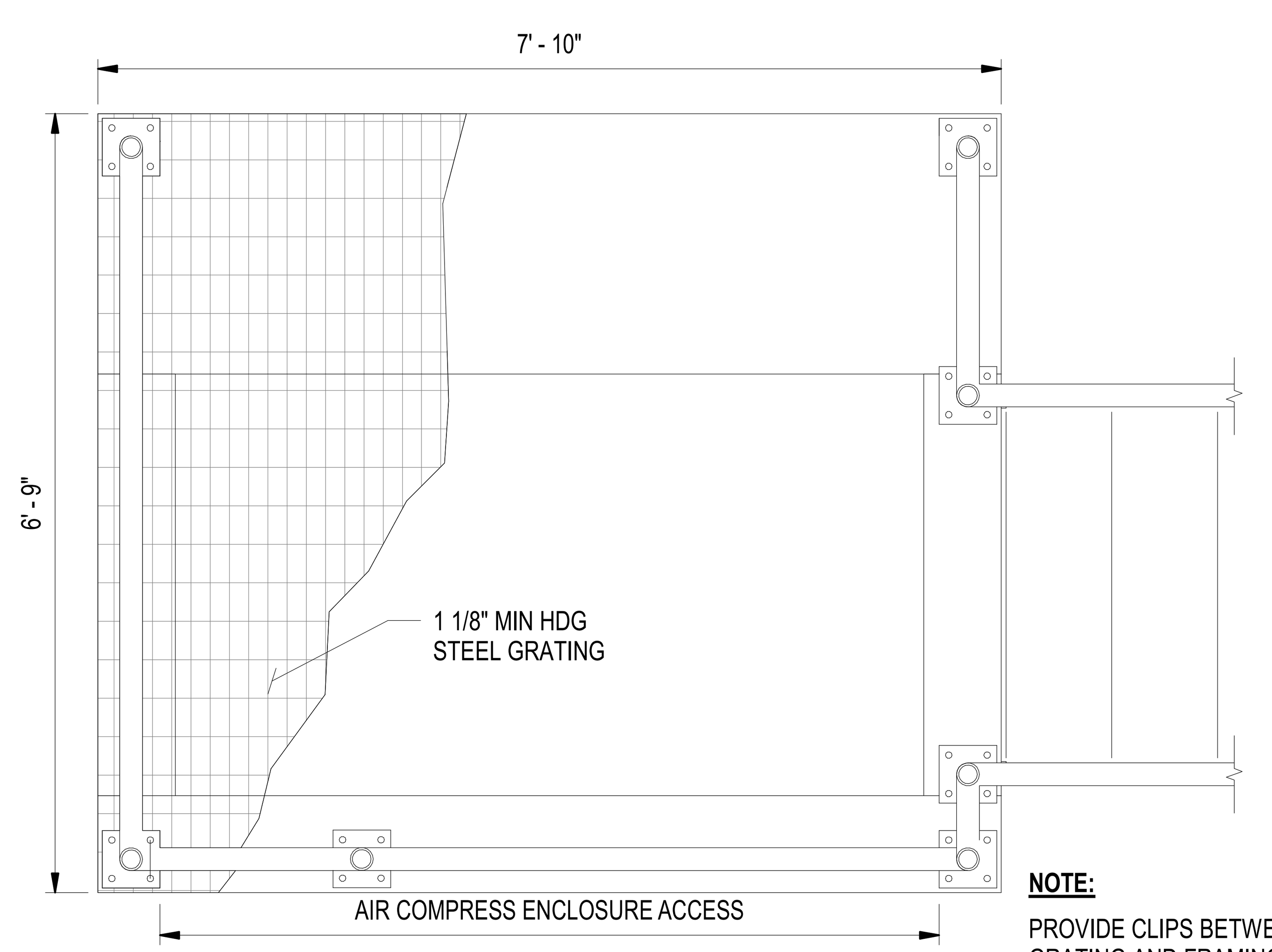
1. STAIRS AND RAIL MUST BE DESIGNED IN COMPLIANCE WITH OSHA REQUIREMENTS.
2. FABRICATE AND ERECT STRUCTURAL STEEL CONFORMING TO THE REQUIREMENTS OF AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC), "STEEL CONSTRUCTION MANUAL".
3. SUBMIT ERECTION PLANS AND SHOP DETAILS AND RECEIVE APPROVAL FROM THE CONTRACTING OFFICER BEFORE PROCEEDING WITH FABRICATION.
4. PROVIDE STRUCTURAL STEEL WIDE-FLANGE SHAPES CONFORMING TO ASTM A992; HSS MEMBERS CONFORMING TO ASTM A500; CHANNELS, PLATES, AND ANGLES CONFORMING TO ASTM A36; AND THREADED RODS CONFORMING TO ASTM A 36.
5. PROVIDE HOT DIP GALVANIZED ANCHOR BOLTS CONFORMING TO ASTM F1554.
6. PROVIDE BOLTS CONFORMING TO ASTM A306, NUTS CONFORMING TO ASTM A563, AND WASHERS CONFORMING TO ASTM F436.
7. WELD IN COMPLIANCE WITH AMERICAN WELDING SOCIETY AWS D1.1, "STRUCTURAL WELDING CODE".
8. WELD ALL SHOP CONNECTIONS WITH CLASS E70XX ELECTRODES. PROVIDE FIELD CONNECTIONS WITH HIGH STRENGTH BOLTED CONNECTIONS UNLESS NOTED OTHERWISE.
9. FRAMING MEMBERS AND CONNECTIONS MUST BE DESIGNED BY FABRICATOR AND RECEIVE APPROVAL FROM THE CONTRACTING OFFICER.
10. MILL BOTTOM OF ALL COLUMNS AND FINISH TOP OF ALL BASE PLATES IN ACCORDANCE WITH AISC SPECIFICATIONS. WELD BASE PLATES TO BOTTOM OF COLUMNS.
11. ALL STRUCTURAL STEEL MUST BE HOT-DIP GALVANIZED, UNLESS NOTED OTHERWISE.

CONCRETE NOTES:

1. MARINE CONCRETE COMPRESSIVE STRENGTH: 5000 psi
2. DETAIL AND CONSTRUCT REINFORCED CONCRETE IN ACCORDANCE WITH THE AMERICAN CONCRETE INSTITUTE ACI SPEC-301-20, "SPECIFICATION FOR STRUCTURAL CONCRETE.
3. DETAIL REINFORCING STEEL IN ACCORDANCE WITH AMERICAN CONCRETE INSTITUTE ACI 315R-18, "GUIDE TO PRESENTING REINFORCING STEEL DESIGN DETAILS," AND ACI SP-66, "ACI DETAILING MANUAL".
4. PROVIDE REINFORCEMENT CONFORMING TO ASTM A615, GRADE 60. GALVANIZED REINFORCING STEEL MUST BE USED.
5. CHAMFER ALL EXPOSED CONCRETE EDGES 3/4" UNLESS NOTED OTHERWISE.

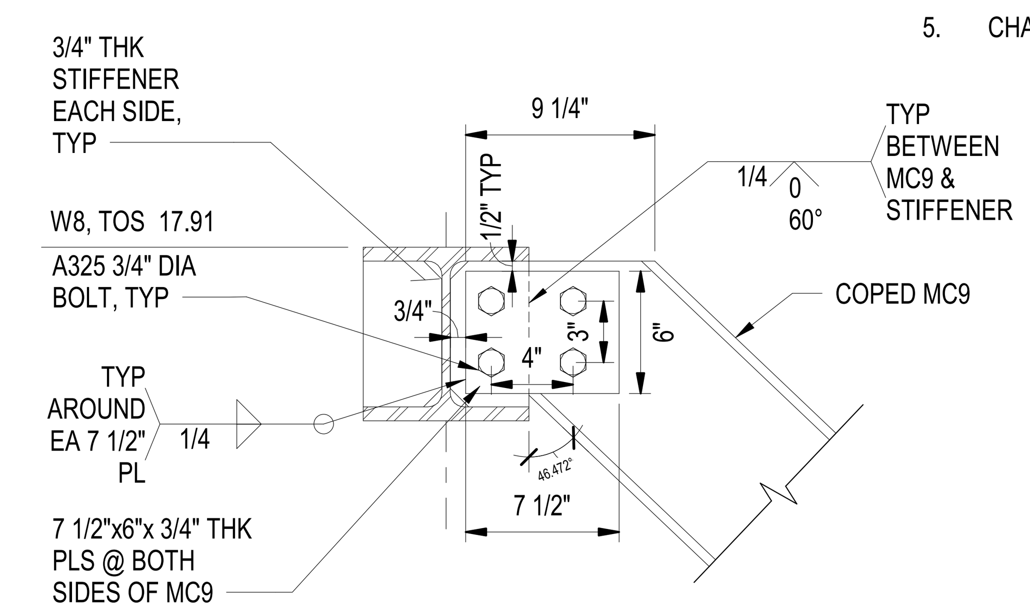


A1 **PLAN - PLATFORM FRAMING**
SCALE: 3/4" = 1'-0"

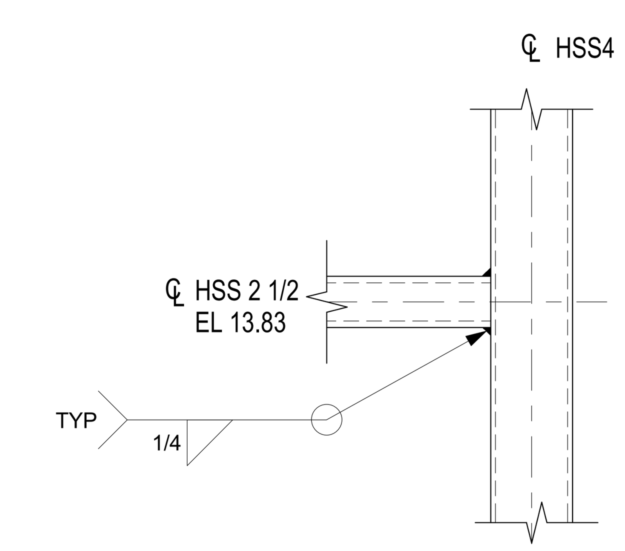


A2 **PLAN - PLATFORM DECK**
SCALE: 3/4" = 1'-0"

NOTE:
PROVIDE CLIPS BETWEEN GRATING AND FRAMING MEMBERS.



A3 **TYPICAL DETAILS FOR MC AND W8 CONNECTION**
SCALE: 1 1/2" = 1'-0"



A5 **TYPICAL CONNECTION FROM HSS2 1/2 TO HSS4**
SCALE: 1 1/2" = 1'-0" (S-532)

1/2" = 1' - 0" 0 6" 1' 2' 3' 4' 5'
1" = 1' - 0" 0 6" 1' 2' 3'
1 1/2" = 1' - 0" 0 3" 6" 9" 1' 2'

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA
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FIRE PROTECTION					HPN

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NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
AIR COMPRESSOR STAIRS ACCESS DETAILS - SHEET 1 OF 2

SCALE: AS NOTED
EPROJECT NO.: 1826479
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NAVFAC DRAWING NO. 12936729
SHEET 160 OF 247
S-532

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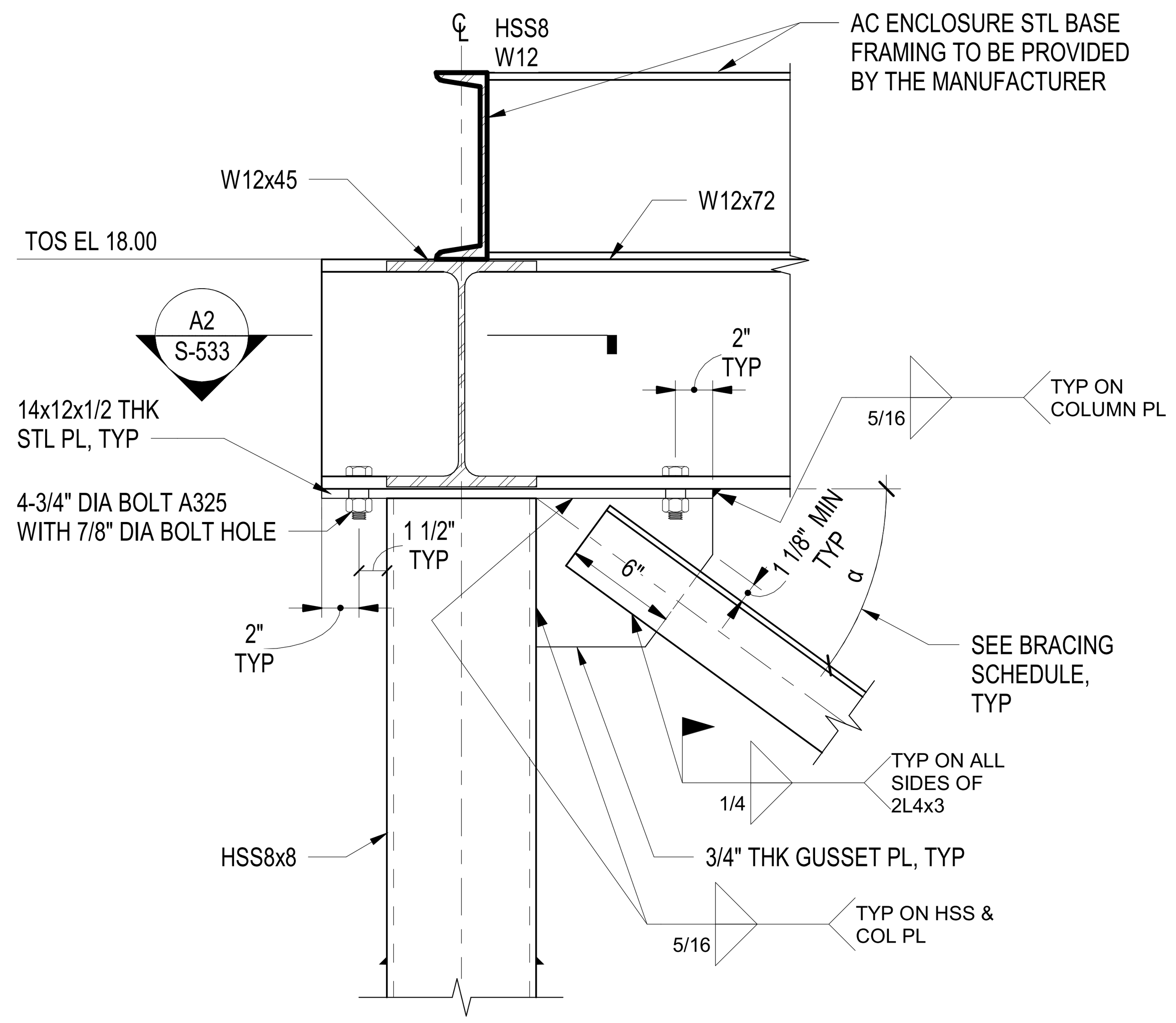
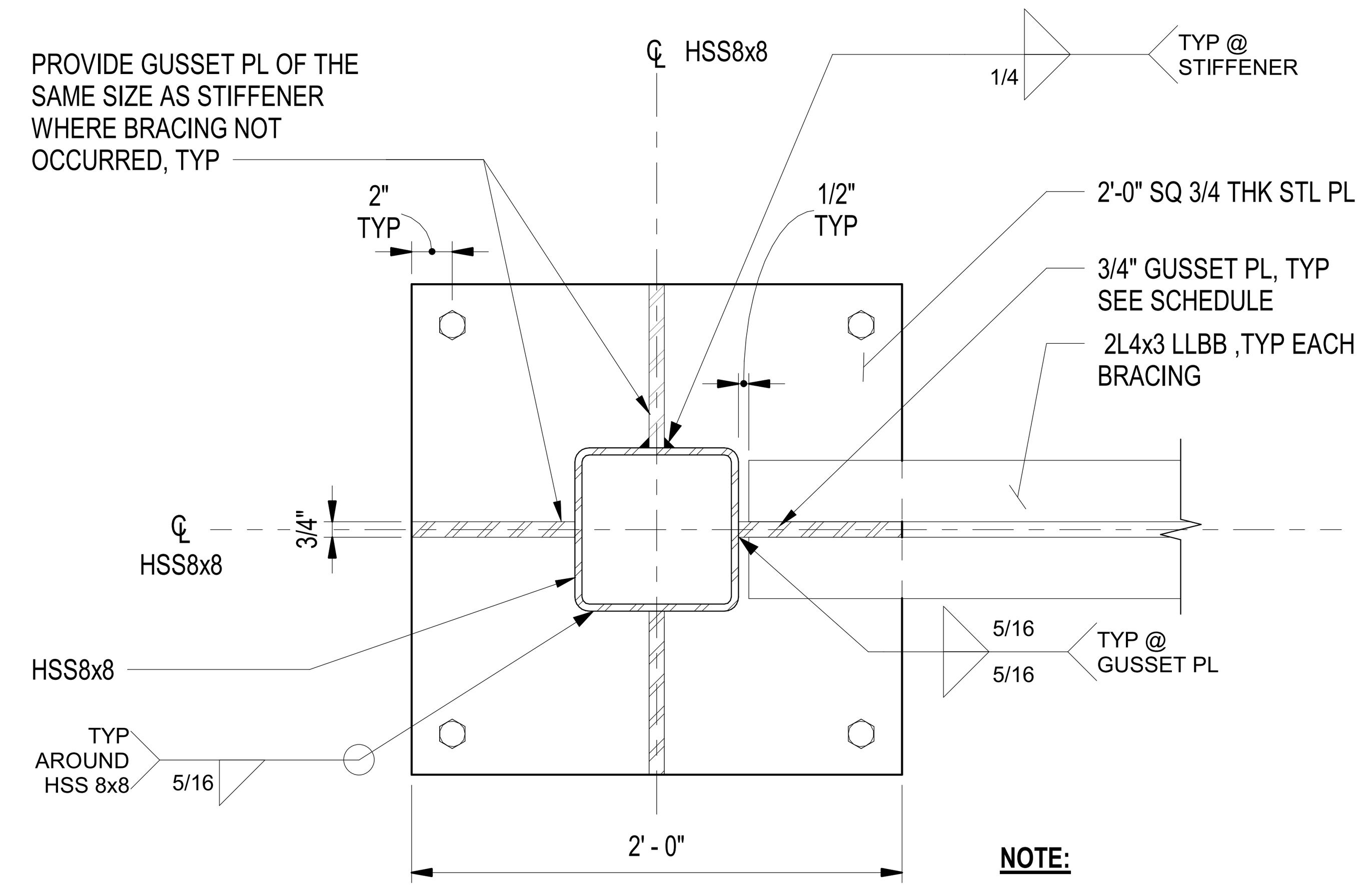


C1 TYPICAL GUSSET PLATE TO HSS TOP CONNECTION

SCALE: 1 1/2" = 1'-0" (S-529)

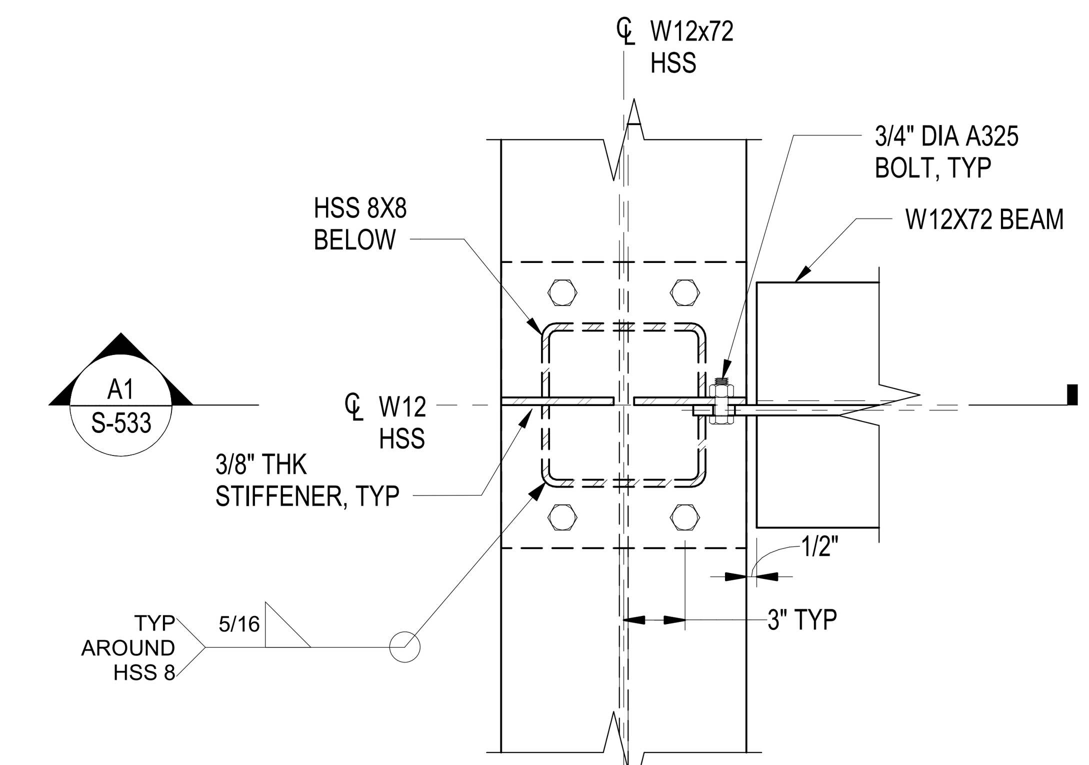


NOTE:

ONLY ONE SET OF 2L4x3 LLBB IS SHOWN FOR CLARITY

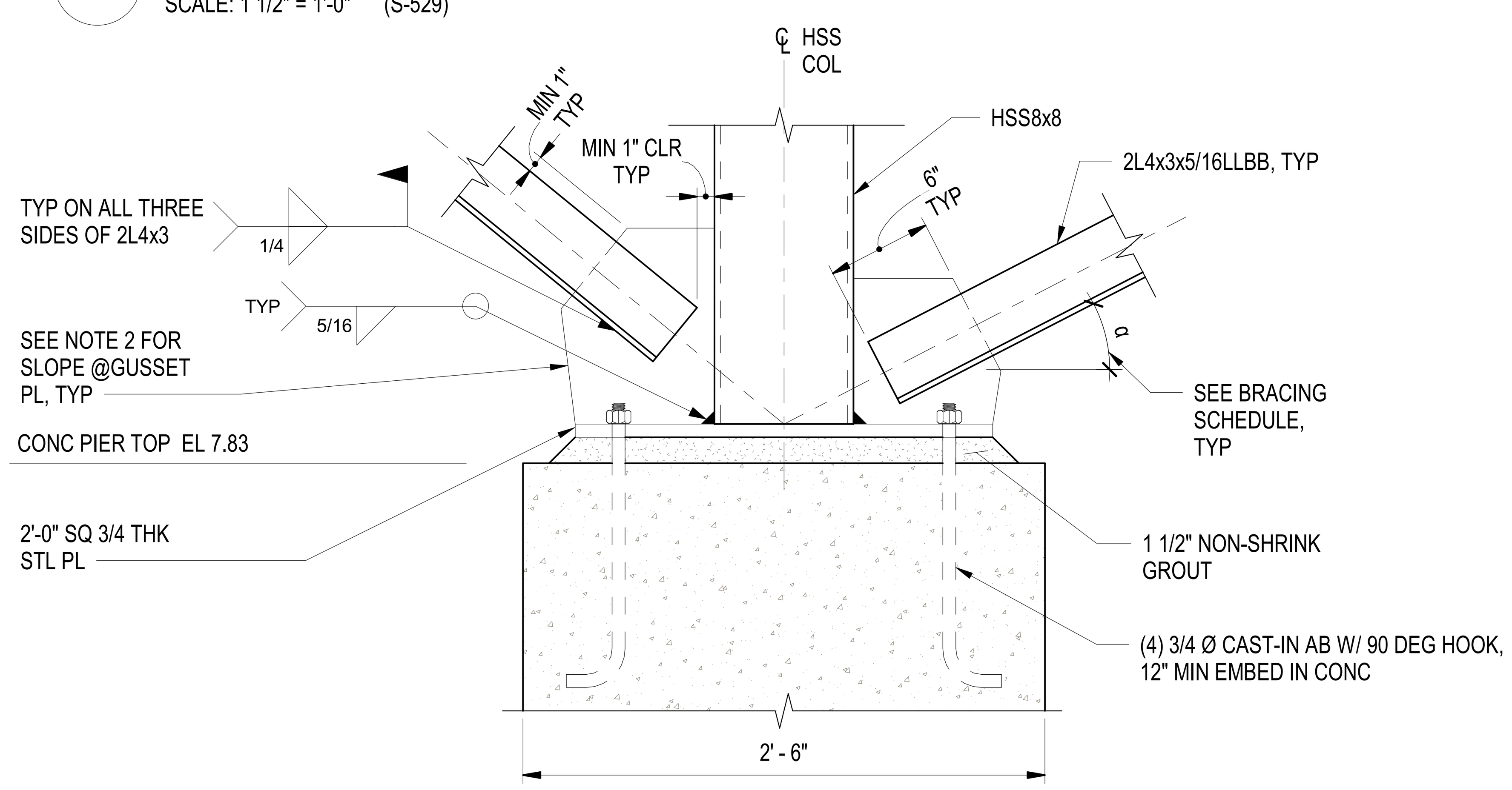
C2 TYPICAL GUSSET PLATE TO HSS BASE CONNECTION PLAN

SCALE: 1 1/2" = 1'-0" (S-529)



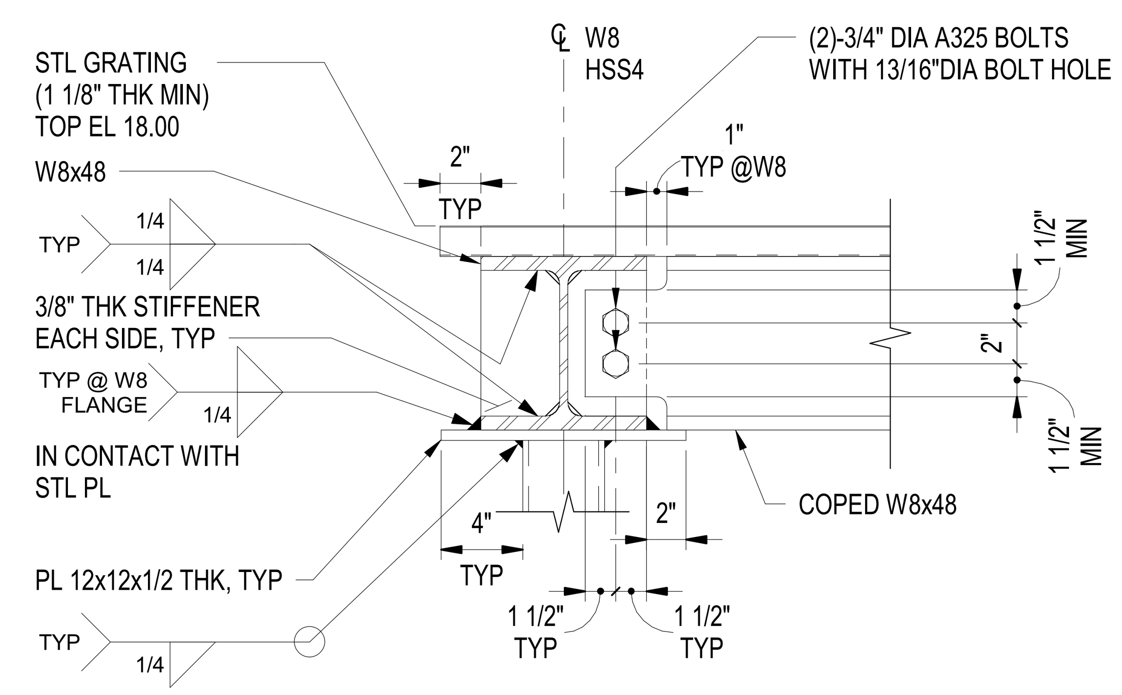
C4 TYPICAL DETAILS FOR GIRDER TO GIRDER CONNECTION

SCALE: 1 1/2" = 1'-0" (S-529)



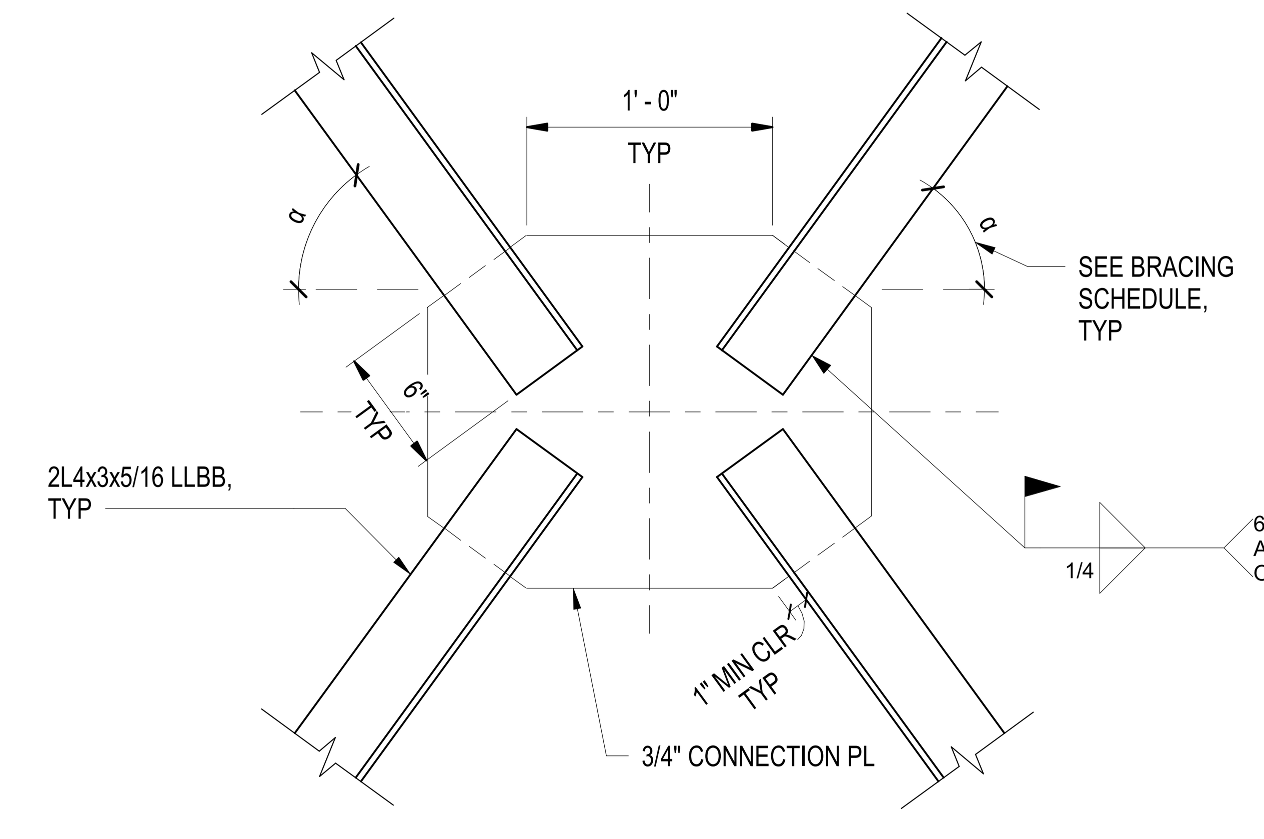
B1 TYPICAL GUSSET PLATE TO HSS BASE CONNECTION SECTION

SCALE: 1 1/2" = 1'-0" (S-503) (S-503, S-529, S-530)



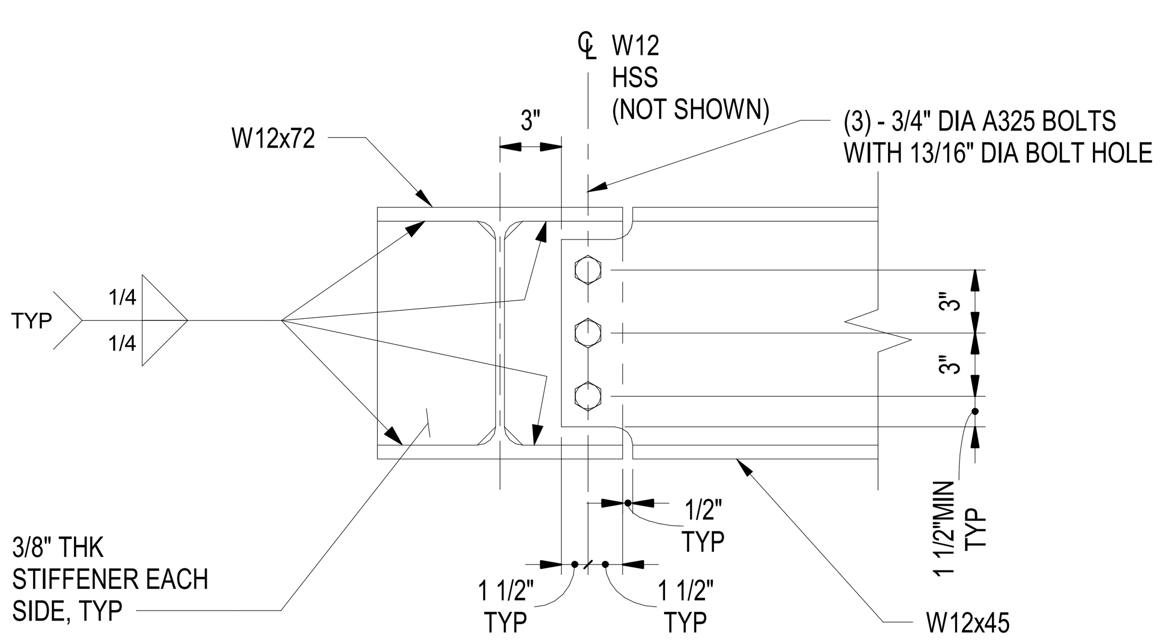
B3 TYPICAL DETAILS FOR W BEAM TO WT JOIST BEAM CONNECTION

SCALE: 1 1/2" = 1'-0" (S-529, S-532)



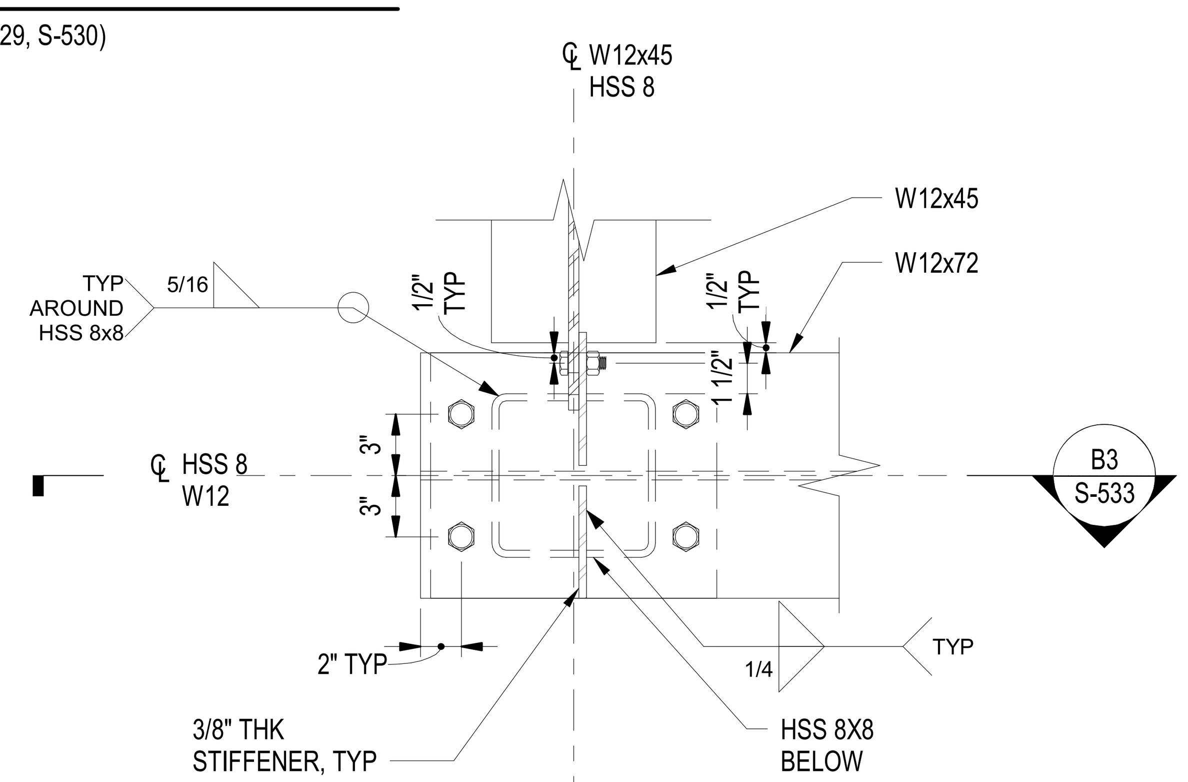
B4 TYPICAL CROSS BRACING CONNECTION DETAIL

SCALE: 1 1/2" = 1'-0" (S-530)



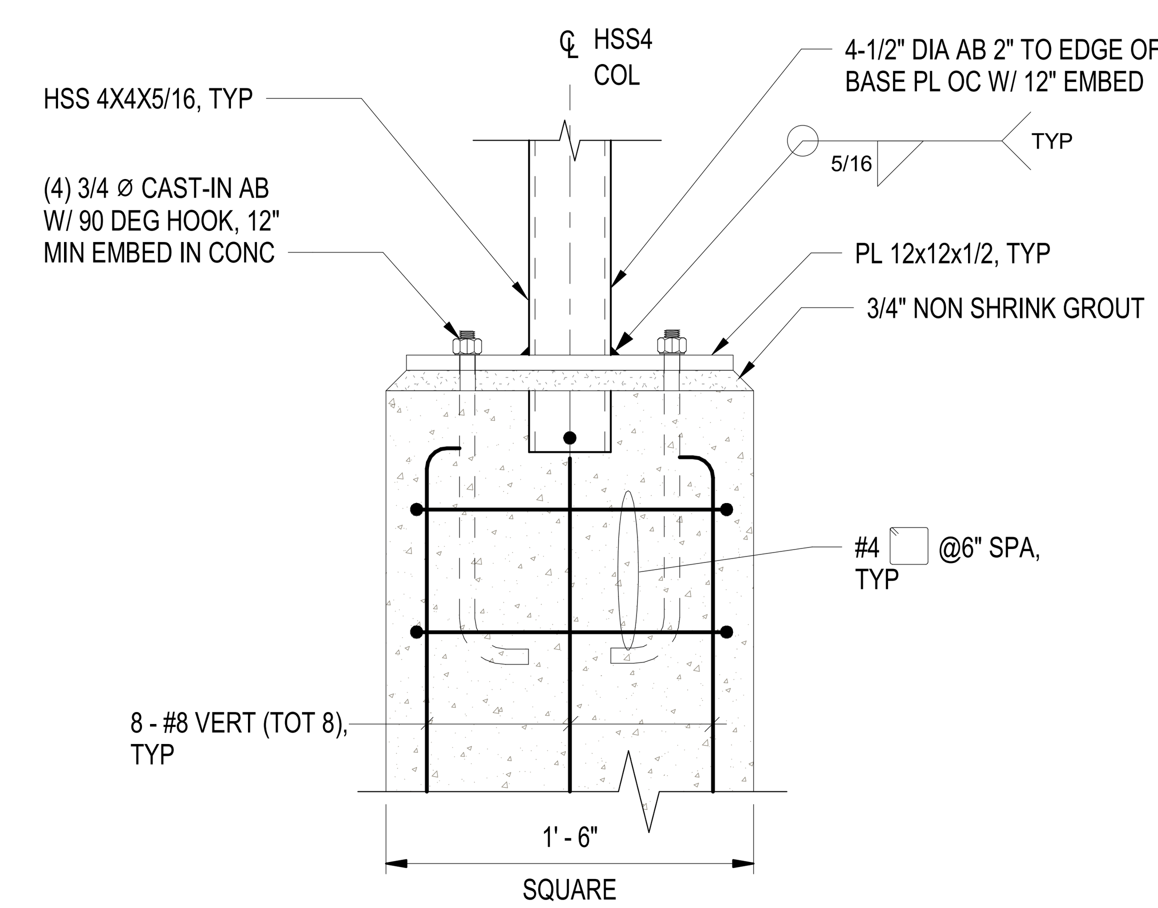
A1 TYPICAL W12 TO W12 CONNECTION

SCALE: 1 1/2" = 1'-0"



A2 TYPICAL GIRDER TO BEAM CONNECTION

SCALE: 1 1/2" = 1'-0"



A3 STAIR HSS COLUMN BASE

SCALE: 1 1/2" = 1'-0" (S-532)

CROSS BRACING SCHEDULE

GL	GL	α
H	I	27.29°
I	J	39.44°
H	G	23.47°
-1	-2	32.63°

NOTES:

1. SEE DETAILS FOR CHAMFER DIMENSIONS ON GUSSET PLATE, TYP
2. SLOPE BOTTOM FREE EDGE OF GUSSET PLATE TO PROVIDE MINIMUM 1" CLEARANCE FROM GUSSET PLATE CORNER TO 2L4x3.

1 1/2" = 1' - 0" 0 3" 6" 9" 1' 2'

SYM | DESCRIPTION | DATE | APPR

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JINOUAN ZHONG
REGISTERED PROFESSIONAL ENGINEER
C 82242
CIVIL
STATE OF CALIFORNIA
SEAL

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FIRE PROTECTION HPN

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AIR COMPRESSOR STAIRS ACCESS DETAILS - SHEET 2 OF 2

SCALE: AS NOTED

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SHEET 161 OF 247

S-533

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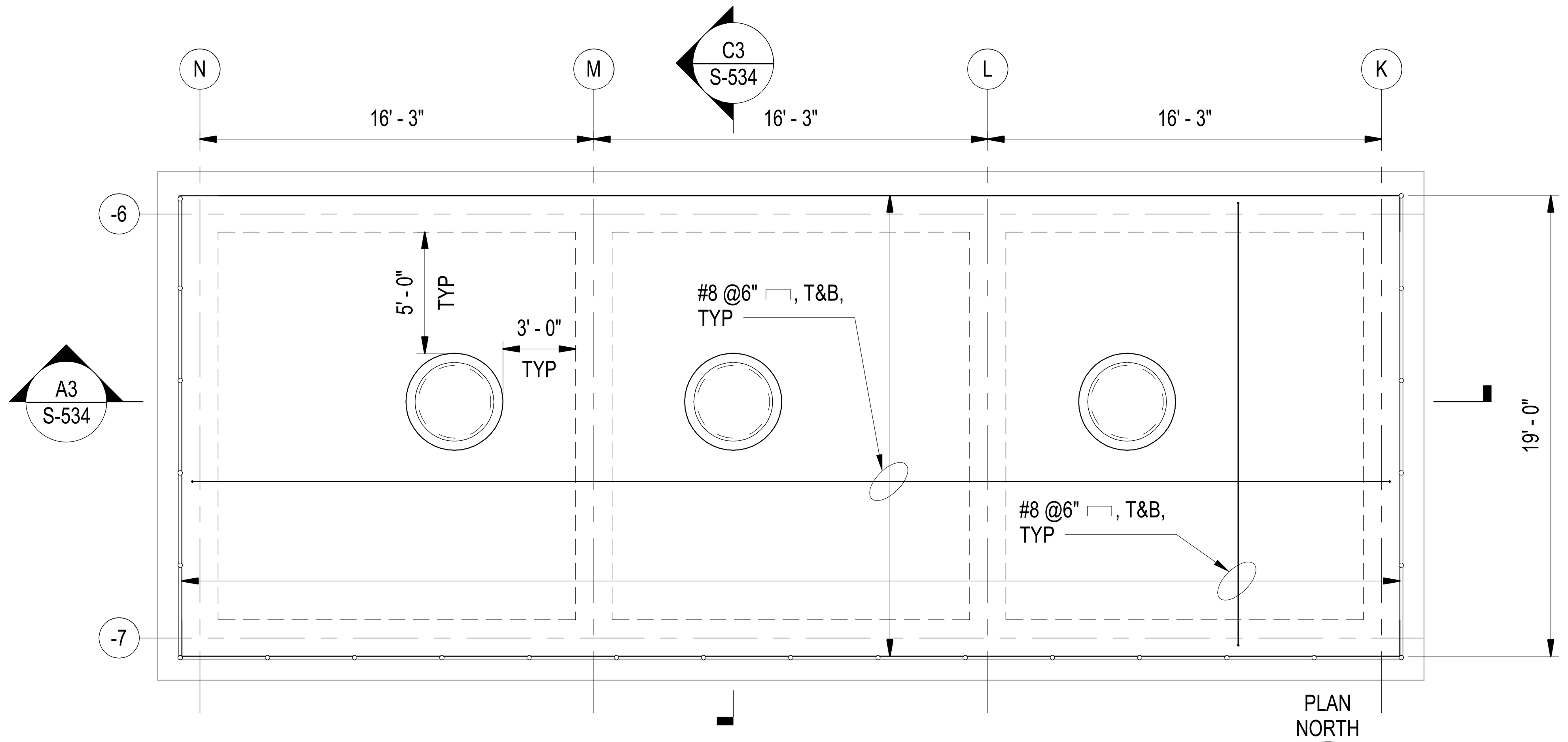
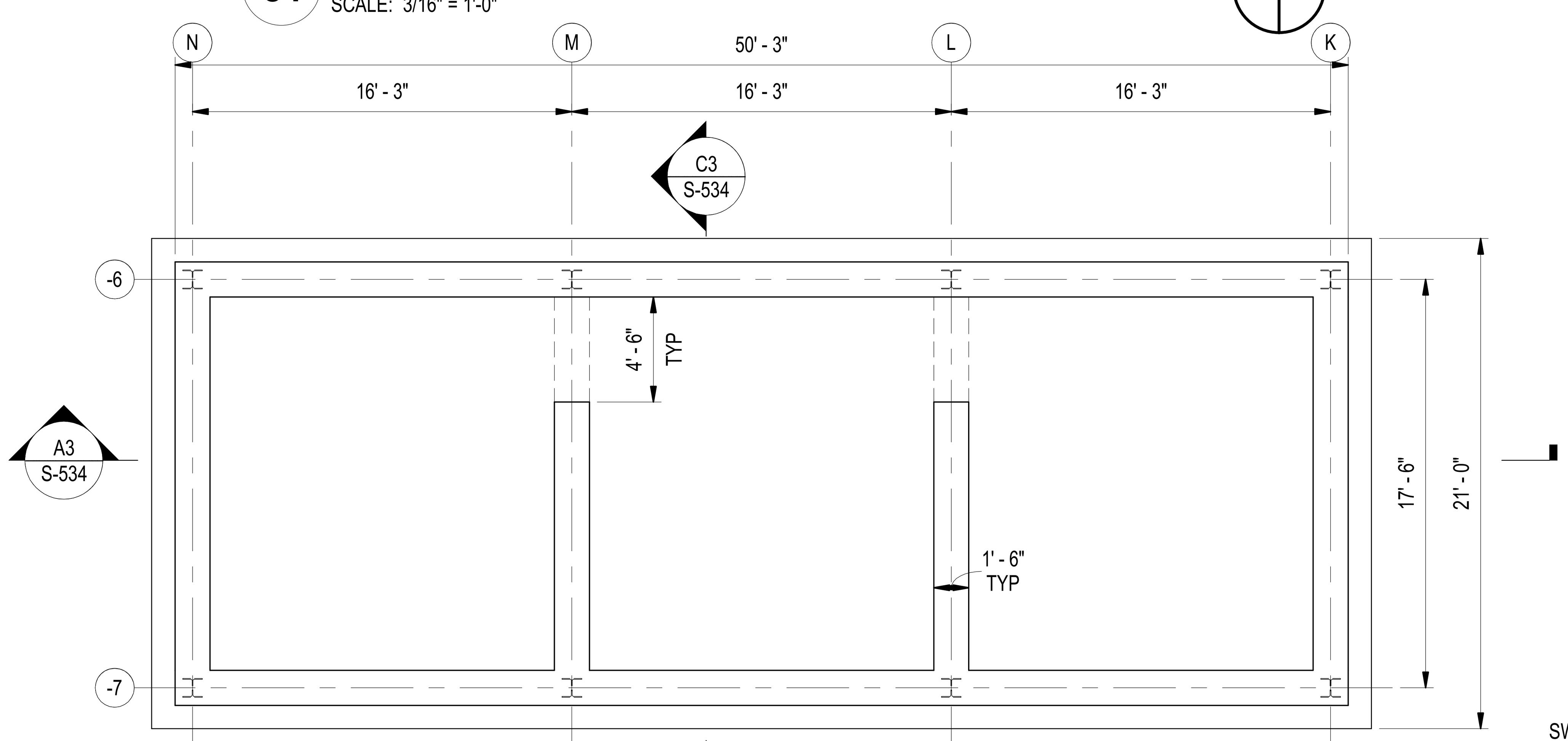
S-533
3/2/2026 7:37:24 PM
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt



C1 PART SWITCHGEAR DECK PLAN

SCALE: 3/16" = 1'-0"

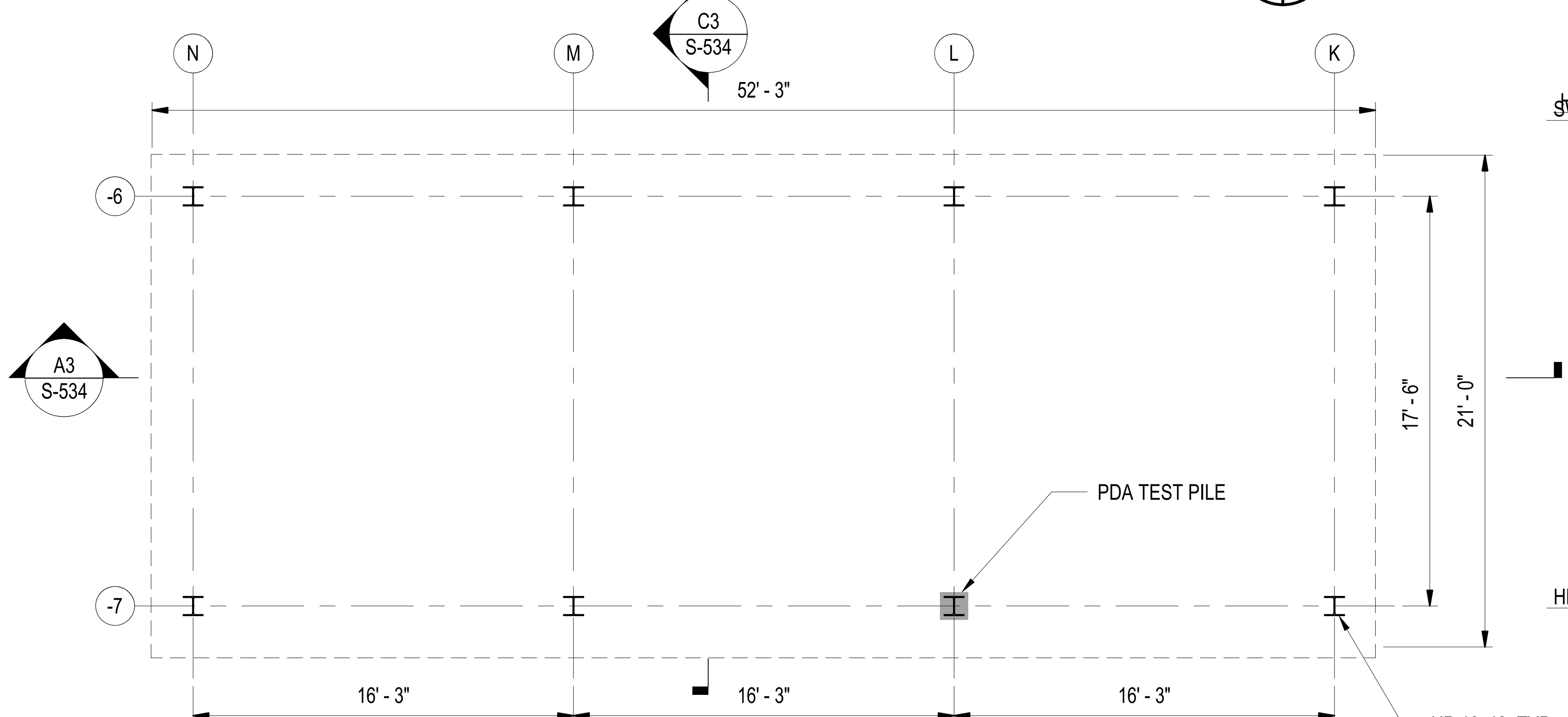
PLAN NORTH



B1 PART SWITCHGEAR FRAMING PLAN

SCALE: 3/16" = 1'-0"

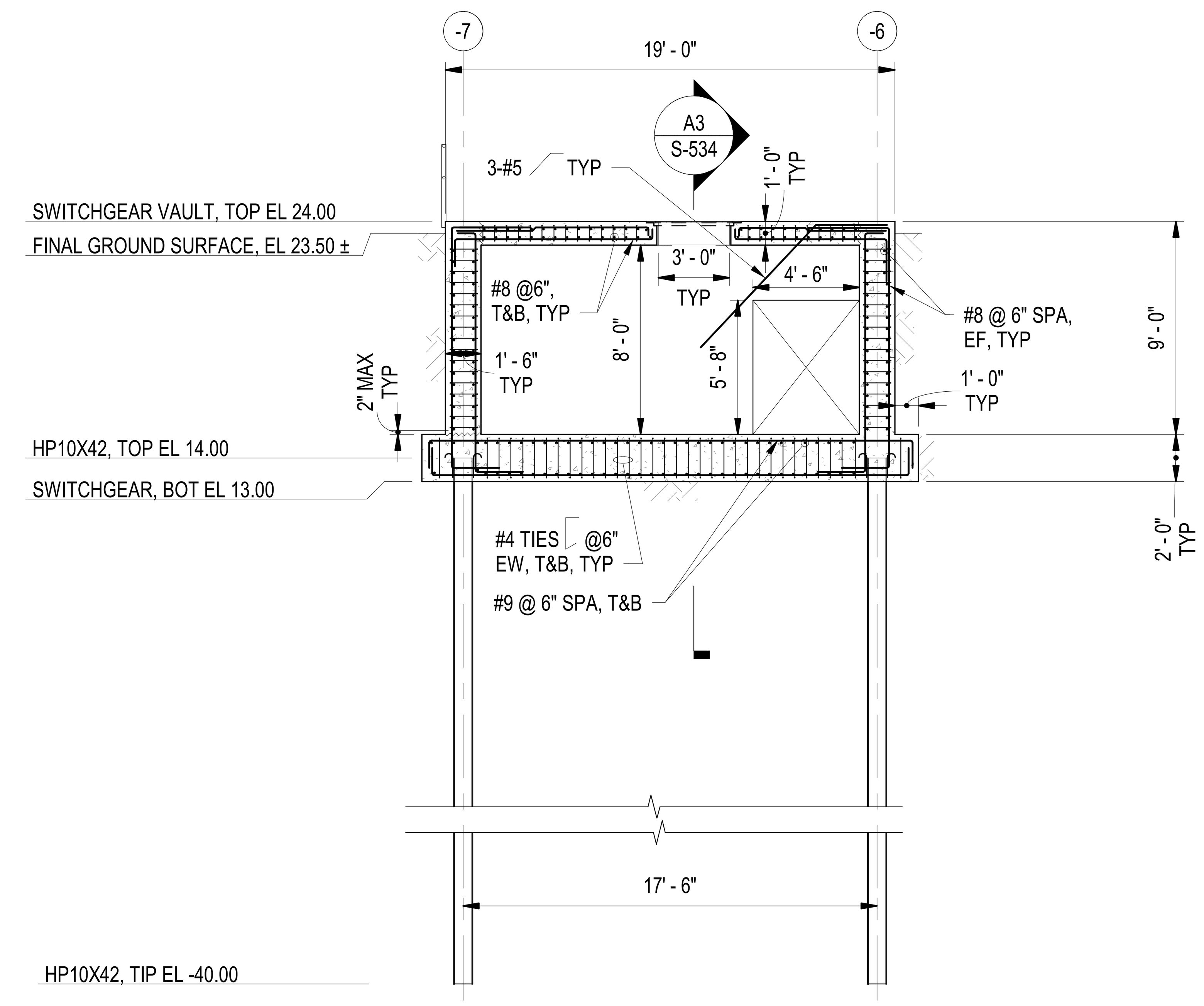
PLAN NORTH



A1 PART SWITCHGEAR FOUNDATION PLAN

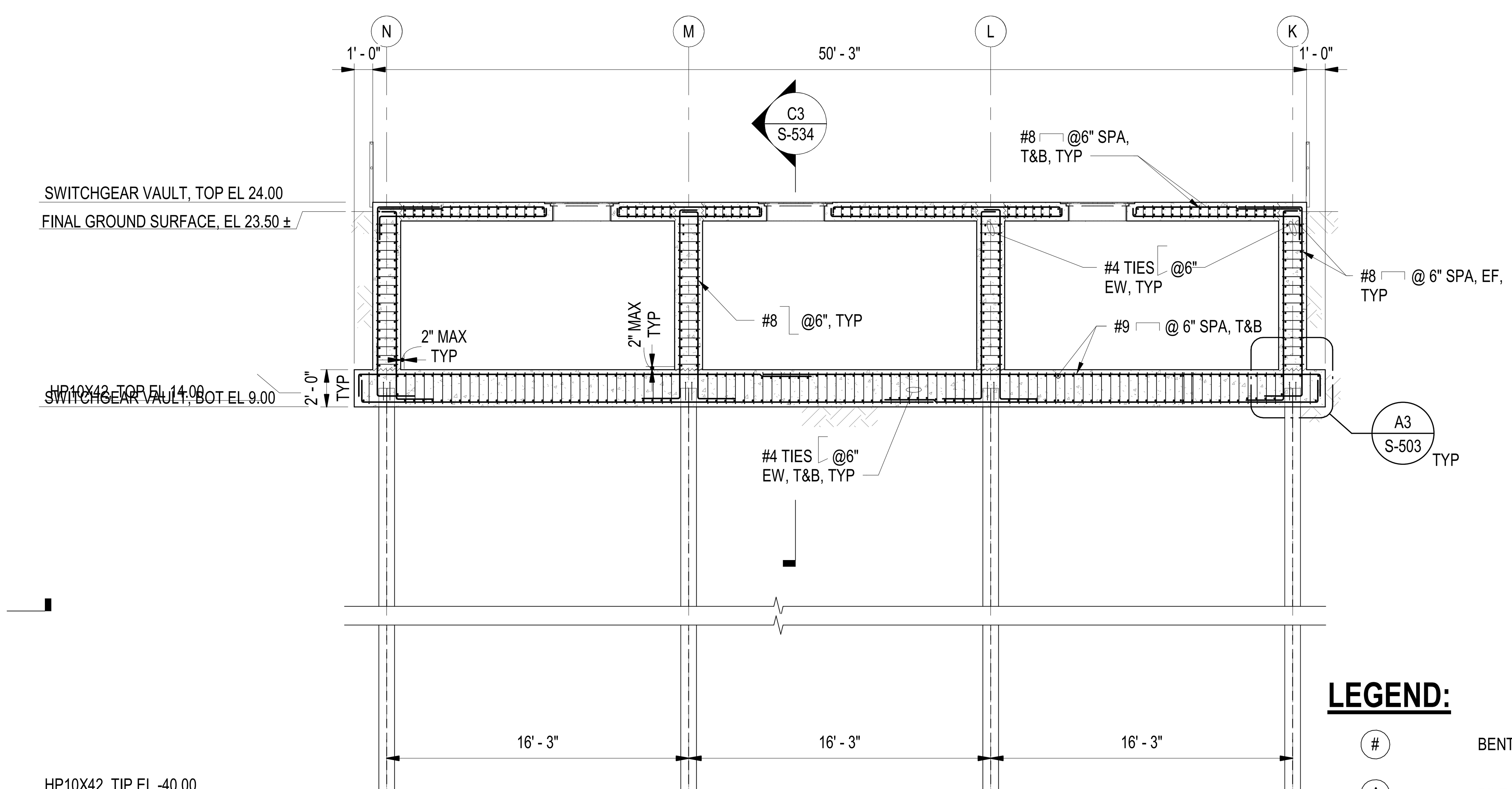
SCALE: 3/16" = 1'-0"

PLAN NORTH



C3 SWITCHGEAR SECTION

SCALE: 3/16" = 1'-0"



A3 SWITCHGEAR LONGITUDINAL SECTION

SCALE: 3/16" = 1'-0"

NOTES:

1. SEE DRAWING S-517 FOR REINFORCEMENT AROUND OPENING, TYPICAL.
2. SEE DRAWING S-522 FOR ADDITIONAL INFORMATION ON HANDRAIL.

LEGEND:

#	BENT DESIGNATION
A	PILE ROW DESIGNATION
I	HP 10x42
▥	HP 10x42 (PDA TEST PILE)

3/16" = 1' - 0" 0 5' 10' 20'

SYM DESCRIPTION DATE APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER JINQUAN ZHONG C 82242 CIVIL STATE OF CALIFORNIA

SEAL

Stantec

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 3/5/2026

ACTIVITY

Approved by Frank Cole USCG MASI

SATISFACTORY TO DATE 06 FEB 2026

DES JZ DRW AFC CHK MJF

PM/DM MRI/CWJ

BRANCH MANAGER NEK

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
15KV SWITCHGEAR STRUCTURE DETAILS

SCALE: AS NOTED

EPROJECT NO. 1826479

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12936731

SHEET 162 OF 247

S-534

DRAWFORM REVISION: 25 AUGUST 2020

3/2/2026 7:37:27 PM S-534
Autodesk Docs://224101525_USCG_OCR_Homeport_Initiative_OPC_Pier_1/PIER USCG-1737829-S.rvt

36" PILE SCHEDULE

PILE LOCATION	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL
A-1	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-34' - 5"
A-2	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-42' - 4"
A-3	36" Steel Pipe Pile	5' - 1"	-85' - 0"	90' - 1"	100'	-43' - 2"
A-4	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
A-5	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
A-6	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
A-7	36" Steel Pipe Pile	6' - 9"	-87' - 0"	93' - 9"	104'	-43' - 4"
A-8	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-45' - 9"
A-9	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-48' - 2"
A-10	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-50' - 7"
A-11	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-51' - 7"
A-12	36" Steel Pipe Pile	6' - 9"	-90' - 0"	96' - 9"	107'	-54' - 0"
A-13	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-56' - 5"
A-14	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-58' - 10"
A-15	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-61' - 3"
A-16	36" Steel Pipe Pile	6' - 9"	-91' - 0"	97' - 9"	108'	-63' - 8"
A-17	36" Steel Pipe Pile	6' - 9"	-96' - 0"	102' - 9"	113'	-66' - 1"
A-18	36" Steel Pipe Pile	6' - 9"	-96' - 0"	102' - 9"	113'	-66' - 6"
A-19	36" Steel Pipe Pile	6' - 9"	-98' - 0"	104' - 9"	115'	-68' - 4"
A-20	36" Steel Pipe Pile	6' - 9"	-100' - 0"	106' - 9"	117'	-70' - 1"
A-21	36" Steel Pipe Pile	6' - 9"	-100' - 0"	106' - 9"	117'	-70' - 11"
A-22	36" Steel Pipe Pile	6' - 9"	-102' - 0"	108' - 9"	119'	-72' - 9"
A-23	36" Steel Pipe Pile	6' - 9"	-102' - 0"	108' - 9"	119'	-74' - 7"
A-24	36" Steel Pipe Pile	6' - 9"	-104' - 0"	110' - 9"	121'	-76' - 5"
A-25	36" Steel Pipe Pile	6' - 9"	-106' - 0"	112' - 9"	123'	-78' - 2"
A-26	36" Steel Pipe Pile	6' - 9"	-107' - 0"	113' - 9"	124'	-80' - 0"
A-27	36" Steel Pipe Pile	6' - 9"	-109' - 0"	115' - 9"	126'	-81' - 10"
A-28	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-82' - 2"
A-29	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-82' - 7"
A-30	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-83' - 0"
A-31	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 2"
A-32	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 6"
A-33	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 10"
A-34	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 2"
A-35	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 7"
A-36	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 11"
A-37	36" Steel Pipe Pile	6' - 9"	-114' - 0"	120' - 9"	131'	-87' - 3"
A-38	36" Steel Pipe Pile	6' - 9"	-117' - 0"	123' - 9"	134'	-89' - 8"
A-39	36" Steel Pipe Pile	6' - 9"	-119' - 0"	125' - 9"	136'	-92' - 0"
A-40	36" Steel Pipe Pile	6' - 9"	-121' - 0"	127' - 9"	138'	-94' - 4"
A-41	36" Steel Pipe Pile	6' - 9"	-123' - 0"	129' - 9"	140'	-96' - 9"
A-42	36" Steel Pipe Pile	6' - 9"	-125' - 0"	131' - 9"	142'	-99' - 1"
A-43	36" Steel Pipe Pile	6' - 9"	-126' - 0"	132' - 9"	143'	-100' - 5"
B-1	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-34' - 5"
B-2	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-42' - 4"
B-3	36" Steel Pipe Pile	5' - 1"	-85' - 0"	90' - 1"	100'	-43' - 2"
B-4	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
B-5	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
B-6	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
B-7	36" Steel Pipe Pile	6' - 9"	-87' - 0"	93' - 9"	104'	-43' - 4"
B-8	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-45' - 9"
B-9	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-48' - 2"
B-10	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-50' - 7"
B-11	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-51' - 7"
B-12	36" Steel Pipe Pile	6' - 9"	-90' - 0"	96' - 9"	107'	-54' - 0"
B-13	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-56' - 5"
B-14	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-58' - 10"
B-15	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-61' - 3"
B-16	36" Steel Pipe Pile	6' - 9"	-91' - 0"	97' - 9"	108'	-63' - 8"
B-17*	36" Steel Pipe Pile	6' - 9"	-96' - 0"	102' - 9"	113'	-66' - 1"
B-18	36" Steel Pipe Pile	6' - 9"	-96' - 0"	102' - 9"	113'	-66' - 6"
B-19	36" Steel Pipe Pile	6' - 9"	-98' - 0"	104' - 9"	115'	-68' - 4"
B-20	36" Steel Pipe Pile	6' - 9"	-100' - 0"	106' - 9"	117'	-70' - 1"
B-21	36" Steel Pipe Pile	6' - 9"	-100' - 0"	106' - 9"	117'	-70' - 11"
B-22	36" Steel Pipe Pile	6' - 9"	-102' - 0"	108' - 9"	119'	-72' - 9"
B-23	36" Steel Pipe Pile	6' - 9"	-102' - 0"	108' - 9"	119'	-74' - 7"
B-24	36" Steel Pipe Pile	6' - 9"	-104' - 0"	110' - 9"	121'	-76' - 5"
B-25	36" Steel Pipe Pile	6' - 9"	-106' - 0"	112' - 9"	123'	-78' - 2"
B-26	36" Steel Pipe Pile	6' - 9"	-107' - 0"	113' - 9"	124'	-80' - 0"
B-27	36" Steel Pipe Pile	6' - 9"	-109' - 0"	115' - 9"	126'	-81' - 10"
B-28	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-82' - 2"
B-29	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-82' - 7"
B-30	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-83' - 0"
B-31	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 2"
B-32	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 6"
B-33	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 10"
B-34	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 2"
B-35	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 7"
B-36*	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 11"

36" PILE SCHEDULE

PILE LOCATION	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL
B-37	36" Steel Pipe Pile	6' - 9"	-114' - 0"	120' - 9"	131'	-87' - 3"
B-38	36" Steel Pipe Pile	6' - 9"	-117' - 0"	123' - 9"	134'	-89' - 8"
B-39	36" Steel Pipe Pile	6' - 9"	-119' - 0"	125' - 9"	136'	-92' - 0"
B-40	36" Steel Pipe Pile	6' - 9"	-121' - 0"	127' - 9"	138'	-94' - 4"
B-41	36" Steel Pipe Pile	6' - 9"	-123' - 0"	129' - 9"	140'	-96' - 9"
B-42	36" Steel Pipe Pile	6' - 9"	-125' - 0"	131' - 9"	142'	-99' - 1"
B-43	36" Steel Pipe Pile	6' - 9"	-126' - 0"	132' - 9"	143'	-100' - 5"
C-1	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-34' - 5"
C-2	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-42' - 4"
C-3	36" Steel Pipe Pile	5' - 1"	-85' - 0"	90' - 1"	100'	-43' - 2"
C-4	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
C-5	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
C-6	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
C-7	36" Steel Pipe Pile	6' - 9"	-87' - 0"	93' - 9"	104'	-43' - 4"
C-8	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-45' - 9"
C-9	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-48' - 2"
C-10	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-50' - 7"
C-11	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-51' - 7"
C-12	36" Steel Pipe Pile	6' - 9"	-90' - 0"	96' - 9"	107'	-54' - 0"
C-13	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-56' - 5"
C-14	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-58' - 10"
C-15	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-61' - 3"
C-16	36" Steel Pipe Pile	6' - 9"	-91' - 0"	97' - 9"	108'	-63' - 8"
C-17	36" Steel Pipe Pile	6' - 9"	-96' - 0"	102' - 9"	113'	-66' - 1"
C-18	36" Steel Pipe Pile	6' - 9"	-96' - 0"	102' - 9"	113'	-66' - 6"
C-19	36" Steel Pipe Pile	6' - 9"	-98' - 0"	104' - 9"	115'	-68' - 4"
C-20	36" Steel Pipe Pile	6' - 9"	-100' - 0"	106' - 9"	117'	-70' - 1"
C-21	36" Steel Pipe Pile	6' - 9"	-100' - 0"	106' - 9"	117'	-70' - 11"
C-22	36" Steel Pipe Pile	6' - 9"	-102' - 0"	108' - 9"	119'	-72' - 9"
C-23	36" Steel Pipe Pile	6' - 9"	-102' - 0"	108' - 9"	119'	-74' - 7"
C-24	36" Steel Pipe Pile	6' - 9"	-104' - 0"	110' - 9"	121'	-76' - 5"
C-25	36" Steel Pipe Pile	6' - 9"	-106' - 0"	112' - 9"	123'	-78' - 2"
C-26	36" Steel Pipe Pile	6' - 9"	-107' - 0"	113' - 9"	124'	-80' - 0"
C-27	36" Steel Pipe Pile	6' - 9"	-109' - 0"	115' - 9"	126'	-81' - 10"
C-28	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-82' - 2"
C-29	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-82' - 7"
C-30	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-83' - 0"
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C-35	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 7"
C-36	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 11"
C-37	36" Steel Pipe Pile	6' - 9"	-114' - 0"	120' - 9"	131'	-87' - 3"
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C-43	36" Steel Pipe Pile	6' - 9"	-126' - 0"	132' - 9"	143'	-100' - 5"
D-1	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-34' - 5"
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D-3	36" Steel Pipe Pile	5' - 1"	-85' - 0"	90' - 1"	100'	-43' - 2"
D-4	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
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D-9	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-48' - 2"
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D-15	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-61' - 3"
D-16	36" Steel Pipe Pile	6' - 9"	-91' - 0"	97' - 9"	108'	-63' - 8"
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D-27	36" Steel Pipe Pile	6' - 9"	-109' - 0"	115' - 9"	126'	-81' - 10"
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36" PILE SCHEDULE

PILE LOCATION	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL
D-30	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-83' - 0"
D-31	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 2"
D-32	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 6"
D-33	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 10"
D-34	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 2"
D-35	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 7"
D-36	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 11"
D-37	36" Steel Pipe Pile	6' - 9"	-114' - 0"	120' - 9"	131'	-87' - 3"
D-38	36" Steel Pipe Pile	6' - 9"	-117' - 0"	123' - 9"	134'	-89' - 8"
D-39	36" Steel Pipe Pile	6' - 9"	-119' - 0"	125' - 9"	136'	-92' - 0"
D-40	36" Steel Pipe Pile	6' - 9"	-121' - 0"	127' - 9"	138'	-94' - 4"
D-41	36" Steel Pipe Pile	6' - 9"	-123' - 0"	129' - 9"	140'	-96' - 9"
D-42	36" Steel Pipe Pile	6' - 9"	-125' - 0"	131' - 9"	142'	-99' - 1"
D-43	36" Steel Pipe Pile	6' - 9"	-126' - 0"	132' - 9"	143'	-100' - 5"
E-1	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-34' - 5"
E-2	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-42' - 4"
E-3	36" Steel Pipe Pile	5' - 1"	-85' - 0"	90' - 1"	100'	-43' - 2"
E-4	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
E-5	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
E-6	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
E-7*	36" Steel Pipe Pile	6' - 9"	-87' - 0"	93' - 9"	104'	-43' - 4"
E-8	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-45' - 9"
E-9	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-48' - 2"
E-10	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-50' - 7"
E-11	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-51' - 7"
E-12	36" Steel Pipe Pile	6' - 9"	-90' - 0"	96' - 9"	107'	-54' - 0"
E-13	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-56' - 5"
E-14	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-58' - 10"
E-15	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-61' - 3"
E-16	36" Steel Pipe Pile	6' - 9"	-91' - 0"	97' - 9"	108'	-63' - 8"
E-17	36" Steel Pipe Pile	6' - 9"	-96' - 0"	102' - 9"	113'	-66' - 1"
E-18	36" Steel Pipe Pile	6' - 9"	-96' - 0"	102' - 9"	113'	-66' - 6"
E-19	36" Steel Pipe Pile	6' - 9"	-98' - 0"	104' - 9"	115'	-68' - 4"
E-20	36" Steel Pipe Pile	6' - 9"	-100' - 0"	106' - 9"	117'	-70' - 1"
E-21	36" Steel Pipe Pile	6' - 9"	-100' - 0"	106' - 9"	117'	-70' - 11"
E-22	36" Steel Pipe Pile	6' - 9"	-102' - 0"	108' - 9"	119'	-72' - 9"
E-23	36" Steel Pipe Pile	6' - 9"	-102' - 0"	108' - 9"	119'	-74' - 7"
E-24	36" Steel Pipe Pile	6' - 9"	-104' - 0"	110' - 9"	121'	-76' - 5"
E-25	36" Steel Pipe Pile	6' - 9"	-106' - 0"	112' - 9"	123'	-78' - 2"
E-26	36" Steel Pipe Pile	6' - 9"	-107' - 0"	113' - 9"	124'	-80' - 0"
E-27	36" Steel Pipe Pile	6' - 9"	-109' - 0"	115' - 9"	126'	-81' - 10"
E-28*	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-82' - 2"
E-29	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-82' - 7"
E-30	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-83' - 0"
E-31	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 2"
E-32	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 6"
E-33	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 10"
E-34	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 2"
E-35	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 7"
E-36	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 11"
E-37	36" Steel Pipe Pile	6' - 9"	-114' - 0"	120' - 9"	131'	-87' - 3"
E-38	36" Steel Pipe Pile	6' - 9"	-117' - 0"	123' - 9"	134'	-89' - 8"
E-39	36" Steel Pipe Pile	6' - 9"	-119' - 0"	125' - 9"	136'	-92' - 0"
E-40	36" Steel Pipe Pile	6' - 9"	-121' - 0"	127' - 9"	138'	-94' - 4"
E-41	36" Steel Pipe Pile	6' - 9"	-123' - 0"	129' - 9"	140'	-96' - 9"
E-42*	36" Steel Pipe Pile	6' - 9"	-125' - 0"	131' - 9"	142'	-99' - 1"
E-43	36" Steel Pipe Pile	6' - 9"	-126' - 0"	132' - 9"	143'	-100' - 5"
F-1	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-34' - 5"
F-2	36" Steel Pipe Pile	3' - 9"	-85' - 0"	88' - 9"	99'	-42' - 4"
F-3	36" Steel Pipe Pile	5' - 1"	-85' - 0"	90' - 1"	100'	-43' - 2"
F-4	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
F-5	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
F-6	36" Steel Pipe Pile	6' - 9"	-85' - 0"	91' - 9"	102'	-43' - 2"
F-7	36" Steel Pipe Pile	6' - 9"	-87' - 0"	93' - 9"	104'	-43' - 4"
F-8	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-45' - 9"
F-9	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-48' - 2"
F-10	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-50' - 7"
F-11	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-51' - 7"
F-12	36" Steel Pipe Pile	6' - 9"	-90' - 0"	96' - 9"	107'	-54' - 0"
F-13	36" Steel Pipe Pile	6' - 9"	-88' - 0"	94' - 9"	105'	-56' - 5"
F-14	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-58' - 10"
F-15	36" Steel Pipe Pile	6' - 9"	-89' - 0"	95' - 9"	106'	-61' - 3"
F-16	36" Steel Pipe Pile	6' - 9"	-91' - 0"	97' - 9"	108'	-63' - 8"
F-17	36" Steel Pipe Pile	6' - 9"	-96' - 0"	102' - 9"	113'	-66' - 1"
F-18	36" Steel Pipe Pile	6' - 9"	-96' - 0"	102' - 9"	113'	-66' - 6"
F-19	36" Steel Pipe Pile	6' - 9"	-98' - 0"	104' - 9"	115'	-68' - 4"
F-20	36" Steel Pipe Pile	6' - 9"	-100' - 0"	106' - 9"	117'	-70' - 1"
F-21	36" Steel Pipe Pile	6' - 9"	-100' - 0"	106' - 9"	117'	-70' - 11"
F-22	36" Steel Pipe Pile	6' - 9"	-102' - 0"	108' - 9"	119'	-72' - 9"

36" PILE SCHEDULE

PILE LOCATION	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL
F-23	36" Steel Pipe Pile	6' - 9"	-102' - 0"	108' - 9"	119'	-74' - 7"
F-24	36" Steel Pipe Pile	6' - 9"	-104' - 0"	110' - 9"	121'	-76' - 5"
F-25	36" Steel Pipe Pile	6' - 9"	-106' - 0"	112' - 9"	123'	-78' - 2"
F-26	36" Steel Pipe Pile	6' - 9"	-107' - 0"	113' - 9"	124'	-80' - 0"
F-27	36" Steel Pipe Pile	6' - 9"	-109' - 0"	115' - 9"	126'	-81' - 10"
F-28	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-82' - 2"
F-29	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-82' - 7"
F-30	36" Steel Pipe Pile	6' - 9"	-110' - 0"	116' - 9"	127'	-83' - 0"
F-31	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 2"
F-32	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 6"
F-33	36" Steel Pipe Pile	6' - 9"	-111' - 0"	117' - 9"	128'	-83' - 10"
F-34	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 2"
F-35	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 7"
F-36	36" Steel Pipe Pile	6' - 9"	-112' - 0"	118' - 9"	129'	-84' - 11"
F-37	36" Steel Pipe Pile	6' - 9"	-114' - 0"	120' - 9"	131'	-87' - 3"
F-38	36" Steel Pipe Pile	6' - 9"	-117' - 0"	123' - 9"	134'	-89' - 8"
F-39	36" Steel Pipe Pile	6' - 9"	-119' - 0"	125' - 9"	136'	-92' - 0"
F-40	36" Steel Pipe Pile	6' - 9"	-121' - 0"	127' - 9"	138'	-94' - 4"
F-41	36" Steel Pipe Pile	6' - 9"	-123' - 0"	129' - 9"	140'	-96' - 9"
F-42	36" Steel Pipe Pile	6' - 9"	-125' - 0"	131' - 9"	142'	-99' - 1"
F-43	36" Steel Pipe Pile	6' - 9"	-126' - 0"	132' - 9"	143'	-100' - 5"

30" PILE SCHEDULE

PILE LOCATION	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL
A'-0	30" Steel Pipe Pile	4' - 9"	-85' - 0"	89' - 9"	100'	-24' - 0"
B.1-0	30" Steel Pipe Pile	4' - 9"	-85' - 0"	89' - 9"	100'	-24' - 0"
B.8-0	30" Steel Pipe Pile	4' - 9"	-85' - 0"	89' - 9"	100'	-24' - 0"
C.3-0	30" Steel Pipe Pile	4' - 9"	-85' - 0"	89' - 9"	100'	-24' - 0"
D.9-0	30" Steel Pipe Pile	4' - 9"	-85' - 0"	89' - 9"	100'	-24' - 0"
F-0	30" Steel Pipe Pile	2' - 9"	-85' - 0"	87' - 9"	98'	-24' - 0"

HP14X102 PILE SCHEDULE

PILE LOCATION	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL
-1-G	HP14X102	7' - 4"	-35' - 0"	42' - 4"	52'	-24' - 0"
-1-H	HP14X102	7' - 4"	-35' - 0"	42' - 4"	52'	-24' - 0"
-1-I	HP14X102	7' - 4"	-35' - 0"	42' - 4"	52'	-24' - 0"
-1-J	HP14X102	7' - 4"	-35' - 0"	42' - 4"	52'	-24' - 0"
-2-G	HP14X102	7' - 4"	-35' - 0"	42' - 4"	52'	-24' - 0"
-2-H	HP14X102	7' - 4"	-35' - 0"	42' - 4"	52'	-24' - 0"
-2-I*	HP14X102	7' - 4"	-35' - 0"	42' - 4"	52'	-24' - 0"
-2-J	HP14X102	7' - 4"	-35' - 0"	42' - 4"	52'	-24' - 0"

HP10X42 PILE SCHEDULE

PILE LOCATION	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL
-6-K	HP10X42	14' - 0"	-40' - 0"	54' - 0"	64'	-24' - 0"
-6-L	HP10X42	14' - 0"	-40' - 0"	54' - 0"	64'	-24' - 0"
-6-M	HP10X42	14' - 0"	-40' - 0"	54' - 0"	64'	-24' - 0"
-6-N	HP10X42	14' - 0"	-40' - 0"	54' - 0"	64'	-24' - 0"
-7-K	HP10X42	14' - 0"	-40' - 0"	54' - 0"	64'	-24' - 0"
-7-L*	HP10X42	14' - 0"	-40' - 0"	54' - 0"	64'	-24' - 0"
-7-M	HP10X42	14' - 0"	-40' - 0"	54' - 0"	64'	-24' - 0"
-7-N	HP10X42	14' - 0"	-40' - 0"	54' - 0"	64'	-24' - 0"

NOTE:

* INDICATES PILES TO BE PDA TESTED PER THE PROJECT SPECIFICATIONS.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA

Stantec

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES JZ DRW SA CHK MJF
PM/DM MRI/CWJ
BRANCH MANAGER NEK
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS
NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT
NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
STEEL PILE SCHEDULE

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936732
SHEET 163 OF 247

S-601

FENDER PILE SCHEDULE

PILE NUMBER	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL (NAVD88)	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL	ESTIMATED TOP OF GLACIAL TILL EL
1	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
2	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
3	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
4	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
5	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
6	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
7	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 2"	-43' - 1"
8	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 2"	-43' - 1"
9	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 2"	-43' - 1"
10	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 3"	-43' - 1"
11	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 5"	-43' - 5"
12	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 8"	-43' - 5"
13	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 10"	-43' - 8"
14	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-44' - 1"	-43' - 11"
15	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-44' - 8"	-44' - 5"
16	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-45' - 3"	-44' - 11"
17	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-45' - 9"	-45' - 5"
18	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-46' - 7"	-46' - 1"
19	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-47' - 4"	-46' - 10"
20	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-48' - 2"	-47' - 7"
21	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-48' - 9"	-48' - 0"
22	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-49' - 3"	-48' - 6"
23	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-49' - 9"	-49' - 0"
24	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-49' - 11"	-49' - 3"
25	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-50' - 1"	-49' - 7"
26	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-50' - 3"	-49' - 7"
27	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-50' - 5"	-49' - 9"
28	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-50' - 7"	-49' - 9"
29	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-50' - 9"	-49' - 9"
30	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-51' - 0"	-50' - 1"
31	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-51' - 3"	-50' - 5"
32	16" Fender Piles	11' - 10"	-61' - 0"	72' - 10"	83'	-51' - 8"	-50' - 8"
33	16" Fender Piles	11' - 10"	-62' - 0"	73' - 10"	84'	-52' - 6"	-51' - 4"
34	16" Fender Piles	11' - 10"	-63' - 0"	74' - 10"	85'	-53' - 3"	-52' - 1"
35	16" Fender Piles	11' - 10"	-63' - 0"	74' - 10"	85'	-54' - 1"	-52' - 10"
36	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-54' - 11"	-53' - 6"
37	16" Fender Piles	11' - 10"	-65' - 0"	76' - 10"	87'	-55' - 8"	-54' - 3"
38	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-56' - 6"	-55' - 0"
39	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-57' - 0"	-55' - 5"
40	16" Fender Piles	11' - 10"	-67' - 0"	78' - 10"	89'	-57' - 7"	-55' - 11"
41	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-58' - 2"	-56' - 5"
42	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-58' - 5"	-56' - 8"
43	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-58' - 7"	-57' - 0"
44	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-58' - 10"	-57' - 0"
45	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-59' - 0"	-57' - 0"
46	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-59' - 3"	-57' - 3"
47	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-59' - 5"	-57' - 6"
48	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-59' - 8"	-57' - 10"
49	16" Fender Piles	11' - 10"	-69' - 0"	80' - 10"	91'	-60' - 3"	-58' - 3"
50	16" Fender Piles	11' - 10"	-69' - 0"	80' - 10"	91'	-60' - 9"	-58' - 9"
51	16" Fender Piles	11' - 10"	-70' - 0"	81' - 10"	92'	-61' - 4"	-59' - 3"
52	16" Fender Piles	11' - 10"	-72' - 0"	83' - 10"	94'	-63' - 9"	-61' - 5"
53	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-64' - 3"	-61' - 11"
54	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-64' - 10"	-62' - 5"
55	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-65' - 5"	-62' - 11"
56	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-65' - 8"	-63' - 2"
57	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-65' - 10"	-63' - 5"
58	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-66' - 1"	-63' - 5"
59	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-66' - 1"	-63' - 6"
60	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-66' - 1"	-63' - 6"
61	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-66' - 1"	-63' - 6"
62	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-66' - 1"	-63' - 6"
63	16" Fender Piles	11' - 10"	-74' - 0"	85' - 10"	96'	-66' - 1"	-63' - 6"
64	16" Fender Piles	11' - 10"	-74' - 0"	85' - 10"	96'	-66' - 1"	-63' - 6"
65	16" Fender Piles	11' - 10"	-74' - 0"	85' - 10"	96'	-66' - 6"	-63' - 8"
66	16" Fender Piles	11' - 10"	-75' - 0"	86' - 10"	97'	-66' - 11"	-63' - 11"
67	16" Fender Piles	11' - 10"	-75' - 0"	86' - 10"	97'	-67' - 4"	-64' - 1"
68	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-67' - 9"	-64' - 3"
69	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-67' - 11"	-64' - 4"
70	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 1"	-64' - 6"
71	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 3"	-64' - 6"
72	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 5"	-64' - 7"
73	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 7"	-64' - 7"
74	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 9"	-64' - 8"
75	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 11"	-64' - 9"
76	16" Fender Piles	11' - 10"	-76' - 0"	87' - 10"	98'	-69' - 4"	-65' - 0"
77	16" Fender Piles	11' - 10"	-76' - 0"	87' - 10"	98'	-69' - 9"	-65' - 2"
78	16" Fender Piles	11' - 10"	-76' - 0"	87' - 10"	98'	-70' - 2"	-65' - 4"
79	16" Fender Piles	11' - 10"	-76' - 0"	87' - 10"	98'	-70' - 6"	-65' - 6"

FENDER PILE SCHEDULE

PILE NUMBER	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL (NAVD88)	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL	ESTIMATED TOP OF GLACIAL TILL EL
80	16" Fender Piles	11' - 10"	-76' - 0"	87' - 10"	98'	-70' - 11"	-65' - 8"
81	16" Fender Piles	11' - 10"	-77' - 0"	88' - 10"	99'	-71' - 4"	-65' - 11"
82	16" Fender Piles	11' - 10"	-77' - 0"	88' - 10"	99'	-71' - 9"	-66' - 1"
83	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-72' - 2"	-66' - 3"
84	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-72' - 5"	-66' - 4"
85	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-72' - 6"	-66' - 6"
86	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-72' - 8"	-66' - 6"
87	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-72' - 10"	-66' - 7"
88	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-73' - 0"	-66' - 7"
89	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-73' - 2"	-66' - 8"
90	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-73' - 4"	-66' - 10"
91	16" Fender Piles	11' - 10"	-78' - 0"	89' - 10"	100'	-73' - 9"	-67' - 0"
92	16" Fender Piles	11' - 10"	-78' - 0"	89' - 10"	100'	-74' - 2"	-67' - 2"
93	16" Fender Piles	11' - 10"	-78' - 0"	89' - 10"	100'	-74' - 7"	-67' - 4"
94	16" Fender Piles	11' - 10"	-78' - 0"	89' - 10"	100'	-74' - 10"	-67' - 6"
95	16" Fender Piles	11' - 10"	-80' - 0"	91' - 10"	102'	-78' - 3"	-69' - 0"
96	16" Fender Piles	11' - 10"	-80' - 0"	91' - 10"	102'	-78' - 8"	-69' - 2"
97	16" Fender Piles	11' - 10"	-80' - 0"	91' - 10"	102'	-79' - 1"	-69' - 5"
98	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-79' - 6"	-69' - 7"
99	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-79' - 8"	-69' - 8"
100	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-79' - 10"	-69' - 9"
101	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-80' - 0"	-69' - 9"
102	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-80' - 1"	-69' - 11"
103	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-80' - 3"	-69' - 11"
104	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-80' - 5"	-70' - 0"
105	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-80' - 8"	-70' - 1"
106	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-81' - 0"	-70' - 3"
107	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-81' - 5"	-70' - 6"
108	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-81' - 10"	-70' - 8"
109	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-82' - 2"	-70' - 10"
110	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-82' - 2"	-70' - 10"
111	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-82' - 2"	-70' - 10"
112	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 5"	-71' - 0"
113	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 6"	-71' - 0"
114	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 6"	-71' - 0"
115	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 7"	-71' - 0"
116	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 7"	-71' - 1"
117	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 8"	-71' - 1"
118	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 8"	-71' - 1"
119	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 8"	-71' - 1"
120	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-82' - 9"	-71' - 2"
121	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-82' - 10"	-71' - 3"
122	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 0"	-71' - 3"
123	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 0"	-71' - 4"
124	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 2"	-71' - 5"
125	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 3"	-71' - 6"
126	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 4"	-71' - 6"
127	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 6"	-71' - 7"
128	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 7"	-71' - 8"
129	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 8"	-71' - 9"
130	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 9"	-71' - 9"
131	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 9"	-71' - 10"
132	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 9"	-71' - 10"
133	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 10"	-71' - 10"
134	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 10"	-71' - 10"
135	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 11"	-71' - 10"
136	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 11"	-71' - 11"
137	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 0"	-71' - 11"
138	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 1"	-72' - 0"
139	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 2"	-72' - 1"
140	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 4"	-72' - 2"
141	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 5"	-72' - 3"
142	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 7"	-72' - 3"
143	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 7"	-72' - 4"
145	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-85' - 0"	-72' - 5"
145	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 7"	-72' - 4"
146	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-85' - 6"	-72' - 7"
147	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-86' - 0"	-72' - 9"
148	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-86' - 7"	-72' - 11"
149	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-86' - 10"	-73' - 0"
150	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-87' - 0"	-73' - 1"
151	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-87' - 3"	-73' - 3"
152	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-87' - 5"	-73' - 2"
153	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-87' - 8"	-73' - 2"
154	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-87' - 10"	-73' - 4"
155	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-88' - 1"	-73' - 5"
156	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-88' - 7"	-73' - 7"
157	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-89' - 2"	-73' - 9"
158	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-89' - 8"	-73' - 11"

FENDER PILE SCHEDULE

PILE NUMBER	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL (NAVD88)	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL	ESTIMATED TOP OF GLACIAL TILL EL
159	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-90' - 3"	-74' - 1"
160	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-90' - 9"	-74' - 3"
161	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-91' - 4"	-74' - 5"
162	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-91' - 7"	-74' - 6"
163	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-91' - 9"	-74' - 7"
164	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-91' - 11"	-74' - 7"
165	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-92' - 2"	-74' - 7"
166	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-92' - 4"	-74' - 8"
167	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-92' - 7"	-74' - 9"
168	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-92' - 10"	-74' - 10"
169	16" Fender Piles	11' - 10"	-86' - 0"	97' - 10"	108'	-93' - 4"	-75' - 0"
170	16" Fender Piles	11' - 10"	-86' - 0"	97' - 10"	108'	-93' - 11"	-75' - 2"
171	16" Fender Piles	11' - 10"	-86' - 0"	97' - 10"	108'	-94' - 5"	-75' - 4"
172	16" Fender Piles	11' - 10"	-86' - 0"	97' - 10"	108'	-94' - 11"	-75' - 6"
173	16" Fender Piles	11' - 10"	-86' - 0"	97' - 10"	108'	-95' - 6"	-75' - 8"
174	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-96' - 0"	-75' - 10"
175	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-96' - 3"	-75' - 11"
176	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-96' - 6"	-76' - 0"
177	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-96' - 8"	-76' - 0"
178	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-96' - 10"	-76' - 2"
179	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-97' - 1"	-76' - 2"
180	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-97' - 3"	-76' - 3"
181	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-97' - 6"	-76' - 4"
182	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-98' - 1"	-76' - 6"
183	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-98' - 7"	-76' - 8"
184	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-99' - 2"	-76' - 10"
185	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
186	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
187	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
188	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
189	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
190	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
191	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
192	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
193	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
194	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
195	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
196	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
197	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
198	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
199	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
200	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
201	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
202	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
203	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
204	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
205	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
206	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
207	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
208	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
209	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
210	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
211	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
212	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
213	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
214	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
215	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
216	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
217	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
218	16" Fender Piles	11' - 10"	-88' - 0"	99' - 10"	110'	-100' - 5"	-77' - 2"
219	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-99' - 2"	-76' - 10"
220	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-98' - 7"	-76' - 8"
221	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-98' - 1"	-76' - 6"
222	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-97' - 6"	-76' - 4"
223	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-97' - 3"	-76' - 3"
224	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-97' - 1"	-76' - 2"
225	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-96' - 10"	-76' - 2"
226	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-96' - 8"	-76' - 0"
227	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-96' - 6"	-76' - 0"
228	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-96' - 3"	-75' - 11"
229	16" Fender Piles	11' - 10"	-98' - 0"	109' - 10"	120'	-96' - 0"	-75' - 10"
230	16" Fender Piles	11' - 10"	-86' - 0"	97' - 10"	108'	-95' - 6"	-75' - 8"

NOTE:

* INDICATES PILES TO BE PDA TESTED PER THE PROJECT SPECIFICATIONS.

SYM | DESCRIPTION | DATE | APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA
SEAL

Stantec

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES JZ DRW SA CHK MJF
PM/DM MRI/CWJ
BRANCH MANAGER NEK
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
FENDER PILE SCHEDULE - SHEET 1 OF 2

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936733
SHEET 164 OF 247
S-602

FENDER PILE SCHEDULE

PILE NUMBER	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL (NAVD88)	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL	ESTIMATED TOP OF GLACIAL TILL EL
231	16" Fender Piles	11' - 10"	-86' - 0"	97' - 10"	108'	-94' - 11"	-75' - 6"
232	16" Fender Piles	11' - 10"	-86' - 0"	97' - 10"	108'	-94' - 5"	-75' - 4"
233	16" Fender Piles	11' - 10"	-86' - 0"	97' - 10"	108'	-93' - 10"	-75' - 2"
234	16" Fender Piles	11' - 10"	-86' - 0"	97' - 10"	108'	-93' - 4"	-75' - 0"
235	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-92' - 9"	-74' - 10"
236	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-92' - 6"	-74' - 9"
237	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-92' - 4"	-74' - 8"
238*	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-92' - 2"	-74' - 7"
239	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-91' - 11"	-74' - 7"
240	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-91' - 9"	-74' - 7"
241	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-91' - 7"	-74' - 6"
242	16" Fender Piles	11' - 10"	-93' - 0"	104' - 10"	115'	-91' - 4"	-74' - 5"
243	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-90' - 10"	-74' - 3"
244	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-90' - 3"	-74' - 1"
245	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-89' - 8"	-73' - 11"
246	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-89' - 2"	-73' - 9"
247	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-88' - 7"	-73' - 7"
248	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-88' - 1"	-73' - 5"
249	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-87' - 10"	-73' - 4"
250	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-87' - 8"	-73' - 2"
251	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-87' - 5"	-73' - 2"
252	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-87' - 3"	-73' - 1"
253	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-87' - 0"	-73' - 1"
254	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-86' - 10"	-73' - 0"
255	16" Fender Piles	11' - 10"	-89' - 0"	100' - 10"	111'	-86' - 7"	-72' - 11"
256	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-86' - 0"	-72' - 9"
257	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-85' - 6"	-72' - 7"
258	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-85' - 0"	-72' - 5"
259	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 7"	-72' - 4"
260	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 7"	-72' - 4"
261	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 7"	-72' - 3"
262	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 5"	-72' - 3"
263	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 4"	-72' - 2"
264	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 2"	-72' - 1"
265	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 1"	-72' - 0"
266	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-84' - 0"	-71' - 11"
267	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 11"	-71' - 11"
268	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 11"	-71' - 10"
269	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 10"	-71' - 10"
270	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 10"	-71' - 10"
271	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 9"	-71' - 10"
272	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 9"	-71' - 9"
273	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 9"	-71' - 9"
274	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-83' - 8"	-71' - 9"
275	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 7"	-71' - 8"
276*	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 6"	-71' - 7"
277	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 4"	-71' - 6"
278	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 3"	-71' - 6"
279	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 2"	-71' - 5"
280	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 0"	-71' - 4"
281	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-83' - 0"	-71' - 3"
282	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-82' - 10"	-71' - 3"
283	16" Fender Piles	11' - 10"	-82' - 0"	93' - 10"	104'	-82' - 9"	-71' - 2"
284	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 8"	-71' - 1"
285	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 8"	-71' - 1"
286	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 8"	-71' - 1"
287	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 7"	-71' - 0"
288	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 7"	-71' - 0"
289	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 6"	-71' - 0"
290	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 6"	-71' - 0"
291	16" Fender Piles	11' - 10"	-83' - 0"	94' - 10"	105'	-82' - 5"	-71' - 0"
292	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-82' - 2"	-70' - 10"
293	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-82' - 2"	-70' - 10"
294	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-82' - 2"	-70' - 10"
295	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-81' - 10"	-70' - 8"
296	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-81' - 5"	-70' - 6"
297	16" Fender Piles	11' - 10"	-81' - 0"	92' - 10"	103'	-81' - 0"	-70' - 3"
298	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-80' - 8"	-70' - 1"
299	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-80' - 5"	-70' - 0"
300	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-80' - 3"	-69' - 11"
301	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-80' - 1"	-69' - 11"
302	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-80' - 0"	-69' - 9"
303	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-79' - 10"	-69' - 9"
304	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-79' - 8"	-69' - 8"
305	16" Fender Piles	11' - 10"	-91' - 0"	102' - 10"	113'	-79' - 6"	-69' - 7"
306	16" Fender Piles	11' - 10"	-80' - 0"	91' - 10"	102'	-79' - 1"	-69' - 5"
307	16" Fender Piles	11' - 10"	-80' - 0"	91' - 10"	102'	-78' - 8"	-69' - 2"
308	16" Fender Piles	11' - 10"	-80' - 0"	91' - 10"	102'	-78' - 3"	-69' - 0"

FENDER PILE SCHEDULE

PILE NUMBER	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL (NAVD88)	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL	ESTIMATED TOP OF GLACIAL TILL EL
309	16" Fender Piles	11' - 10"	-78' - 0"	89' - 10"	100'	-74' - 10"	-67' - 6"
310	16" Fender Piles	11' - 10"	-78' - 0"	89' - 10"	100'	-74' - 7"	-67' - 4"
311	16" Fender Piles	11' - 10"	-78' - 0"	89' - 10"	100'	-74' - 2"	-67' - 2"
312	16" Fender Piles	11' - 10"	-78' - 0"	89' - 10"	100'	-73' - 9"	-67' - 0"
313	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-73' - 4"	-66' - 10"
314	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-73' - 2"	-66' - 8"
315	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-73' - 0"	-66' - 7"
316*	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-72' - 10"	-66' - 6"
317	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-72' - 8"	-66' - 6"
318	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-72' - 6"	-66' - 6"
319	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-72' - 5"	-66' - 4"
320	16" Fender Piles	11' - 10"	-87' - 0"	98' - 10"	109'	-72' - 2"	-66' - 3"
321	16" Fender Piles	11' - 10"	-77' - 0"	88' - 10"	99'	-71' - 9"	-66' - 1"
322	16" Fender Piles	11' - 10"	-77' - 0"	88' - 10"	99'	-71' - 4"	-65' - 11"
323	16" Fender Piles	11' - 10"	-76' - 0"	87' - 10"	98'	-70' - 11"	-65' - 8"
324	16" Fender Piles	11' - 10"	-76' - 0"	87' - 10"	98'	-70' - 6"	-65' - 6"
325	16" Fender Piles	11' - 10"	-76' - 0"	87' - 10"	98'	-70' - 2"	-65' - 4"
326	16" Fender Piles	11' - 10"	-76' - 0"	87' - 10"	98'	-69' - 9"	-65' - 2"
327	16" Fender Piles	11' - 10"	-76' - 0"	87' - 10"	98'	-69' - 4"	-65' - 0"
328	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 11"	-64' - 9"
329	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 9"	-64' - 8"
330	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 7"	-64' - 7"
331	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 5"	-64' - 7"
332	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 3"	-64' - 6"
333	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-68' - 1"	-64' - 6"
334	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-67' - 11"	-64' - 4"
335	16" Fender Piles	11' - 10"	-85' - 0"	96' - 10"	107'	-67' - 9"	-64' - 3"
336	16" Fender Piles	11' - 10"	-75' - 0"	86' - 10"	97'	-67' - 4"	-64' - 1"
337	16" Fender Piles	11' - 10"	-75' - 0"	86' - 10"	97'	-66' - 11"	-63' - 11"
338	16" Fender Piles	11' - 10"	-74' - 0"	85' - 10"	96'	-66' - 6"	-63' - 8"
339	16" Fender Piles	11' - 10"	-74' - 0"	85' - 10"	96'	-66' - 1"	-63' - 6"
340	16" Fender Piles	11' - 10"	-74' - 0"	85' - 10"	96'	-66' - 1"	-63' - 6"
341	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-66' - 1"	-63' - 6"
342	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-66' - 1"	-63' - 6"
343	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-66' - 1"	-63' - 6"
344	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-66' - 1"	-63' - 5"
345	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-66' - 1"	-63' - 5"
346	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-65' - 10"	-63' - 5"
347	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-65' - 8"	-63' - 2"
348	16" Fender Piles	11' - 10"	-84' - 0"	95' - 10"	106'	-65' - 5"	-62' - 11"
349	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-64' - 10"	-62' - 5"
350	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-64' - 3"	-61' - 11"
351	16" Fender Piles	11' - 10"	-72' - 0"	83' - 10"	94'	-63' - 9"	-61' - 5"
352	16" Fender Piles	11' - 10"	-70' - 0"	81' - 10"	92'	-61' - 4"	-59' - 3"
353	16" Fender Piles	11' - 10"	-69' - 0"	80' - 10"	91'	-60' - 9"	-58' - 9"
354	16" Fender Piles	11' - 10"	-69' - 0"	80' - 10"	91'	-60' - 3"	-58' - 3"
355	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-59' - 8"	-57' - 10"
356	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-59' - 5"	-57' - 6"
357	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-59' - 3"	-57' - 3"
358	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-59' - 0"	-57' - 0"
359	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-58' - 10"	-57' - 0"
360	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-58' - 7"	-57' - 0"
361	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-58' - 5"	-56' - 8"
362	16" Fender Piles	11' - 10"	-73' - 0"	84' - 10"	95'	-58' - 2"	-56' - 5"
363	16" Fender Piles	11' - 10"	-67' - 0"	78' - 10"	89'	-57' - 7"	-55' - 11"
364	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-57' - 0"	-55' - 5"
365	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-56' - 6"	-55' - 0"
366	16" Fender Piles	11' - 10"	-65' - 0"	76' - 10"	87'	-55' - 8"	-54' - 3"
367	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-54' - 11"	-53' - 6"
368	16" Fender Piles	11' - 10"	-63' - 0"	74' - 10"	85'	-54' - 1"	-52' - 10"
369	16" Fender Piles	11' - 10"	-63' - 0"	74' - 10"	85'	-53' - 3"	-52' - 1"
370	16" Fender Piles	11' - 10"	-62' - 0"	73' - 10"	84'	-52' - 6"	-51' - 4"
371	16" Fender Piles	11' - 10"	-61' - 0"	72' - 10"	83'	-51' - 8"	-50' - 8"
372	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-51' - 3"	-50' - 5"
373	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-51' - 0"	-50' - 1"
374	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-50' - 9"	-50' - 1"
375	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-50' - 7"	-49' - 9"
376	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-50' - 5"	-49' - 9"
377	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-50' - 3"	-49' - 7"
378	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-50' - 1"	-49' - 3"
379	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-49' - 11"	-49' - 3"
380	16" Fender Piles	11' - 10"	-66' - 0"	77' - 10"	88'	-49' - 9"	-49' - 0"
381	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-49' - 3"	-48' - 6"
382	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-48' - 9"	-48' - 0"
383	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-48' - 2"	-47' - 7"
384	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-47' - 4"	-46' - 10"
385	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-46' - 7"	-46' - 1"
386	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-45' - 9"	-45' - 5"

FENDER PILE SCHEDULE

PILE NUMBER	PILE TYPE	PILE CUTOFF EL (NAVD88)	ESTIMATED PILE TIP EL (NAVD88)	MINIMUM LENGTH	BID LENGTH (VLF)	ESTIMATED TOP OF HWR EL	ESTIMATED TOP OF GLACIAL TILL EL
387	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-45' - 3"	-44' - 11"
388	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-44' - 8"	-44' - 5"
389	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-44' - 1"	-43' - 11"
390	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 10"	-43' - 8"
391	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 8"	-43' - 5"
392	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 5"	-43' - 5"
393	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 3"	-43' - 1"
394	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 2"	-43' - 1"
395	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 2"	-43' - 1"
396	16" Fender Piles	11' - 10"	-64' - 0"	75' - 10"	86'	-43' - 2"	-43' - 1"
397	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
398	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
399	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
400	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
401	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"
402	16" Fender Piles	11' - 10"	-59' - 0"	70' - 10"	81'	-43' - 2"	-43' - 1"

NOTE:

* INDICATES PILES TO BE PDA TESTED PER THE PROJECT SPECIFICATIONS.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

REGISTERED PROFESSIONAL ENGINEER
JINQUAN ZHONG
C 82242
CIVIL
STATE OF CALIFORNIA
SEAL

Stantec

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 3/5/2026
ACTIVITY
Approved by Frank Cole USCG MASI
SATISFACTORY TO DATE 06 FEB 2026
DES JZ DRW SA CHK MJF
PM/DM MRI/CWJ
BRANCH MANAGER NEK
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC
CI CORE - HAMPTON ROADS NAVAL STATION - NORFOLK, VA
NAVAL STATION NEWPORT NEWPORT, RI
USCG OCR HOMEPORT INITIATIVE OPC PIER
FENDER PILE SCHEDULE - SHEET 2 OF 2

SCALE: AS NOTED
EPROJECT NO. 1826479
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12936734
SHEET 165 OF 247
S-603

DRAWFORM REVISION: 25 AUGUST 2020

S-603
3/2/2026 7:37:33 PM
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